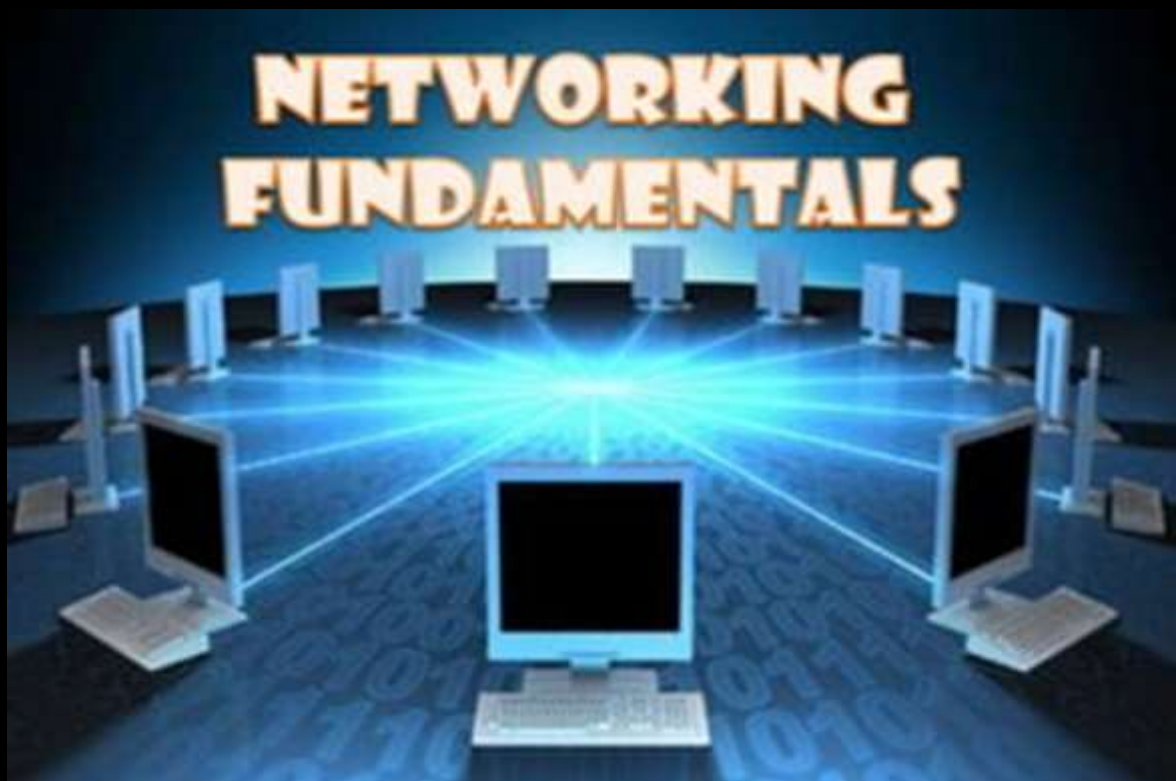




DEBORAH BINYANYA

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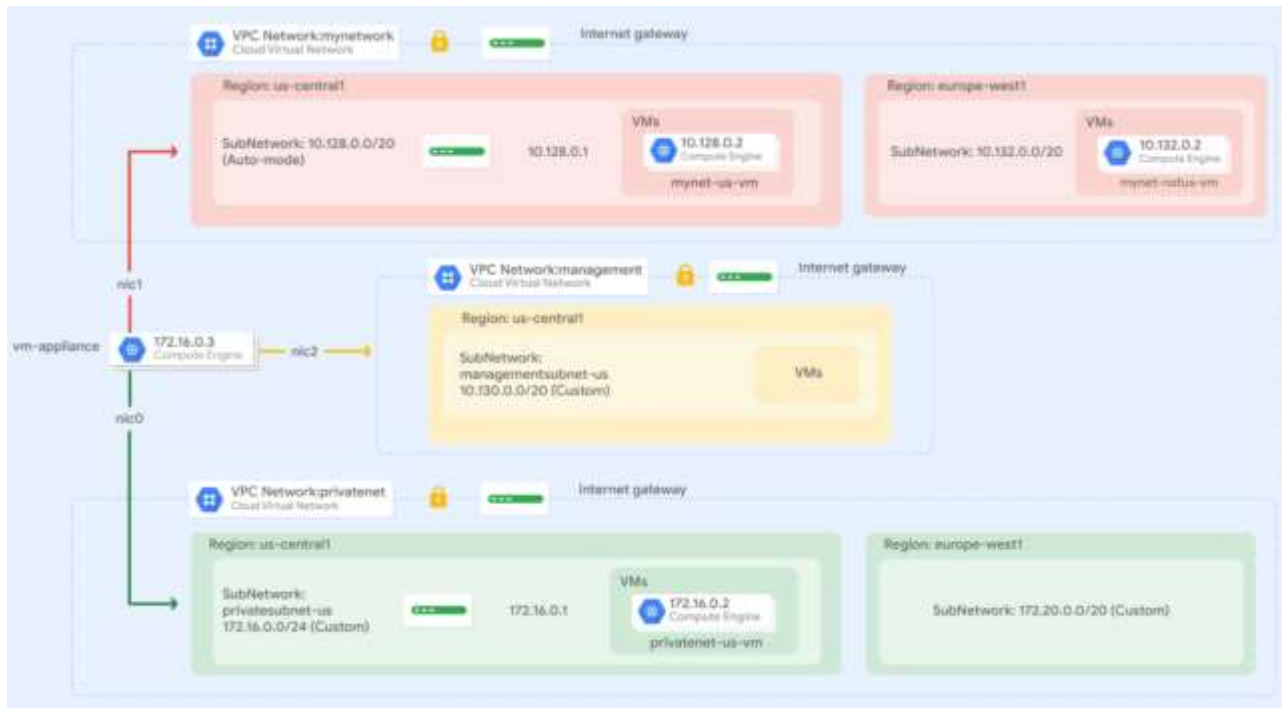
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LAB 1: WORKING WITH MULTIPLE VPC NETWORKS

Overview

In this lab, you create several VPC networks and VM instances and test connectivity across networks. Specifically, you create two custom mode networks (**managementnet** and **privatenet**) with firewall rules and VM instances, as shown in this network diagram:



The **mynetwork** network, its firewall rules, and two VM instances (**mynet-notus-vm** and **mynet-us-vm**) have already been created for you in this Qwiklabs project.

Objectives

In this lab, you learn how to perform the following tasks:

- Create custom mode VPC networks with firewall rules
- Create VM instances using Compute Engine
- Explore the connectivity for VM instances across VPC networks
- Create a VM instance with multiple network interfaces

Setup and requirements

For each lab, you get a new Google Cloud project and set of resources for a fixed time at no cost.

1. Click the **Start Lab** button. If you need to pay for the lab, a pop-up opens for you to select your payment method. On the left is the **Lab Details** panel with the following:
 - The **Open Google Cloud console** button
 - Time remaining

- The temporary credentials that you must use for this lab
 - Other information, if needed, to step through this lab
2. Click **Open Google Cloud console** (or right-click and select **Open Link in Incognito Window** if you are running the Chrome browser).

The lab spins up resources, and then opens another tab that shows the **Sign in** page.

Tip: Arrange the tabs in separate windows, side-by-side.

Note: If you see the **Choose an account** dialog, click **Use Another Account**.

3. If necessary, copy the **Username** below and paste it into the **Sign in** dialog.

"Username"

You can also find the **Username** in the **Lab Details** panel.

4. Click **Next**.
5. Copy the **Password** below and paste it into the **Welcome** dialog.

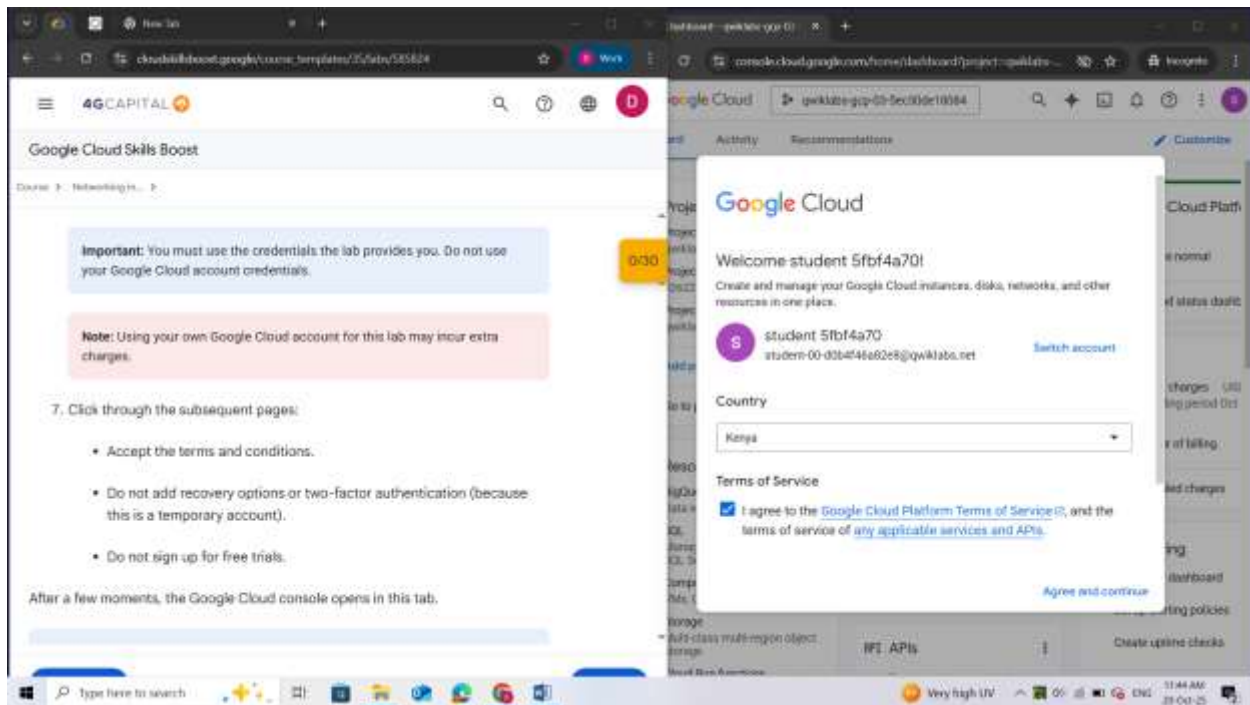
"Password"

You can also find the **Password** in the **Lab Details** panel.

6. Click **Next**.

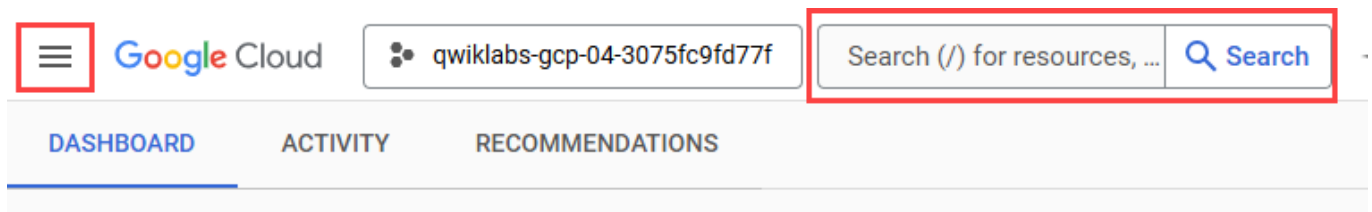
Important: You must use the credentials the lab provides you. Do not use your Google Cloud account credentials. **Note:** Using your own Google Cloud account for this lab may incur extra charges.

7. Click through the subsequent pages:
 - Accept the terms and conditions.
 - Do not add recovery options or two-factor authentication (because this is a temporary account).
 - Do not sign up for free trials.



After a few moments, the Google Cloud console opens in this tab.

Note: To view a menu with a list of Google Cloud products and services, click the **Navigation menu** at the top-left, or type the service or product name in the **Search** field.



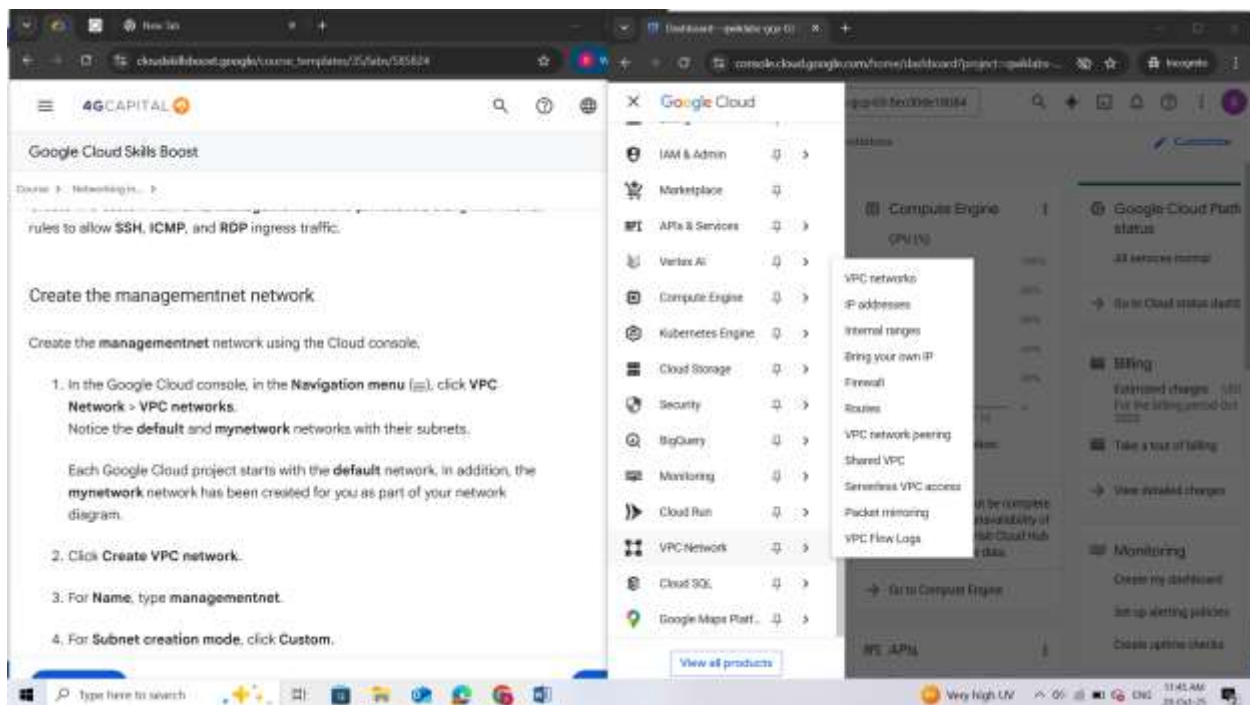
Task 1. Create custom mode VPC networks with firewall rules

Create two custom networks, **managementnet** and **privatenet**, along with firewall rules to allow **SSH**, **ICMP**, and **RDP** ingress traffic.

Create the managementnet network

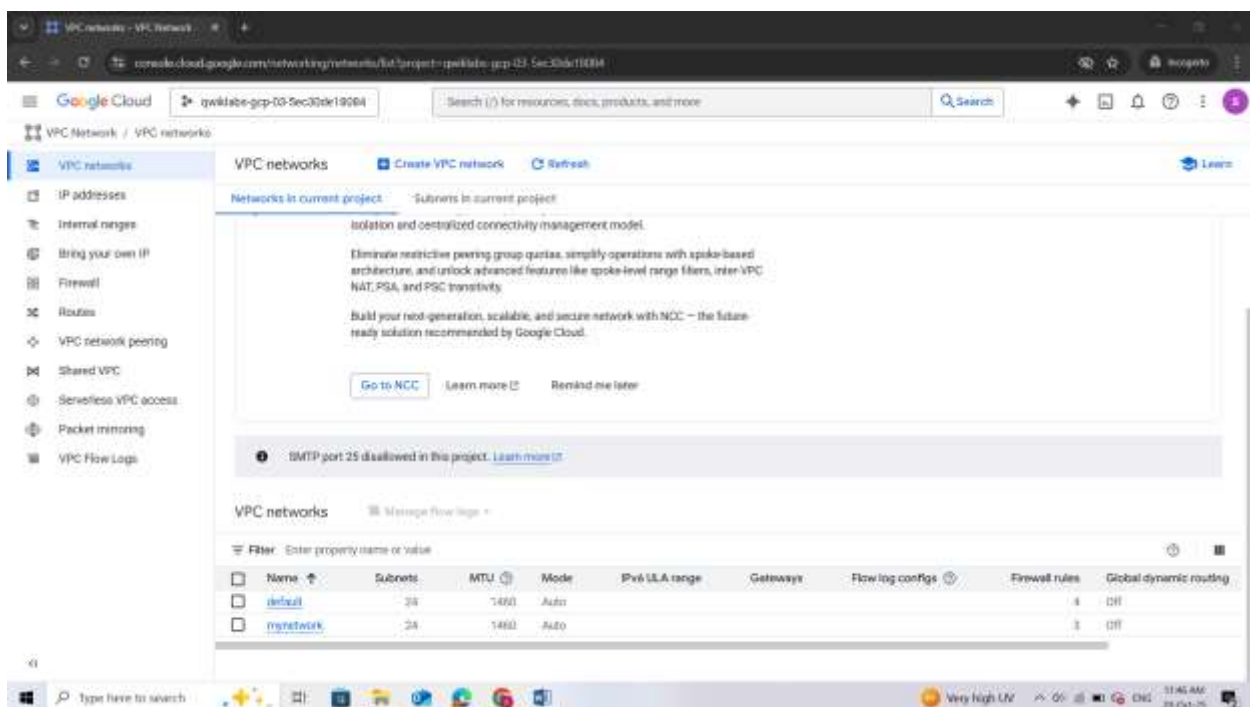
Create the **managementnet** network using the Cloud console.

1. In the Google Cloud console, in the **Navigation menu** (≡), click **VPC Network** > **VPC networks**.



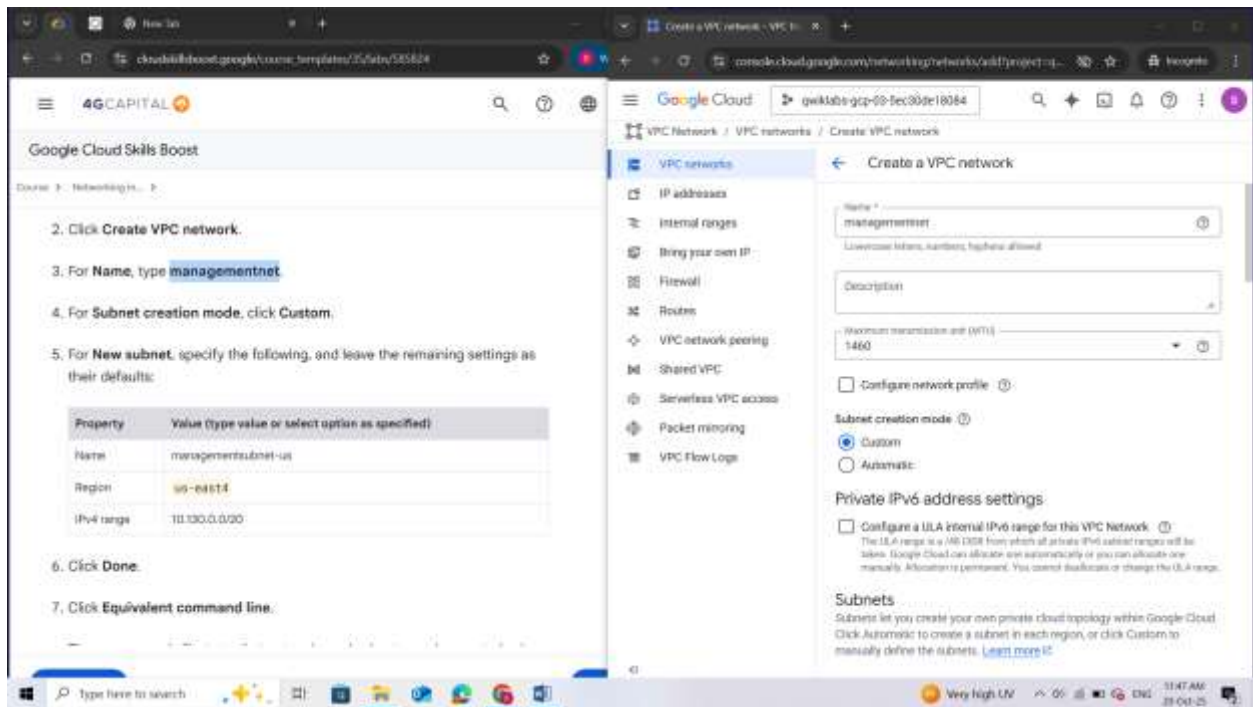
Notice the **default** and **mynetwork** networks with their subnets.

Each Google Cloud project starts with the **default** network. In addition, the **mynetwork** network has been created for you as part of your network diagram.



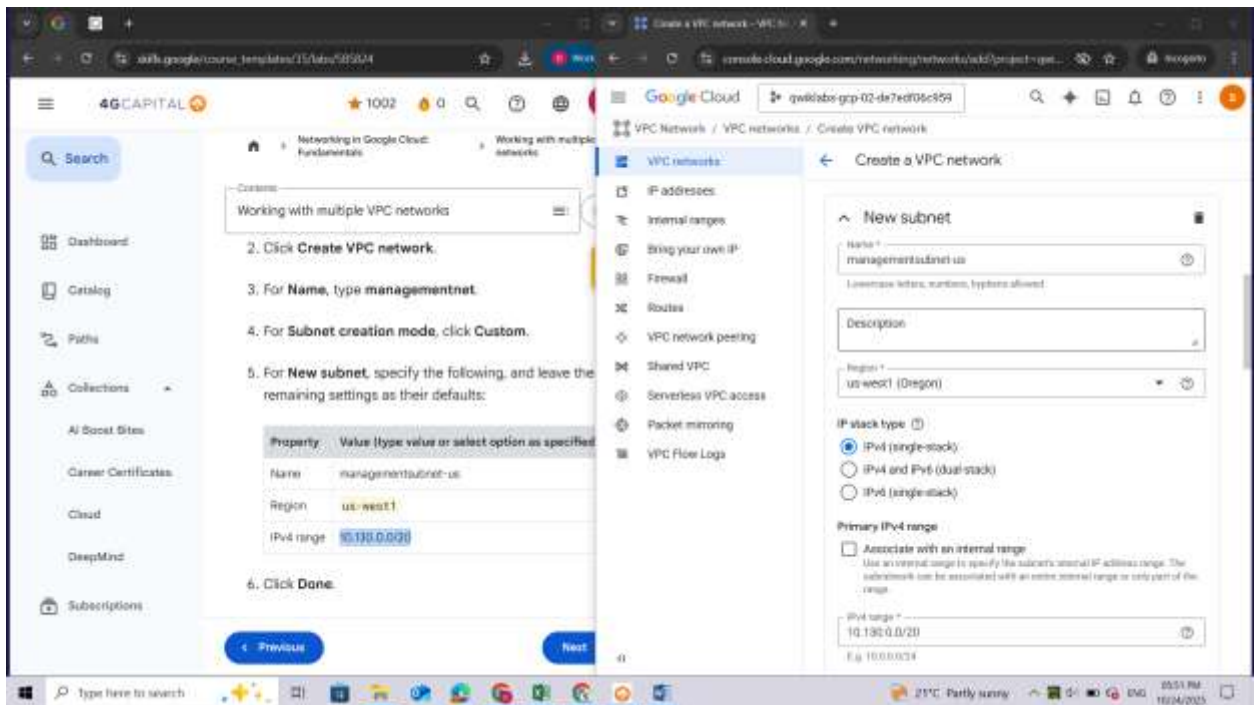
2. Click **Create VPC network**.
3. For **Name**, type **managementnet**.

4. For **Subnet creation mode**, click **Custom**.

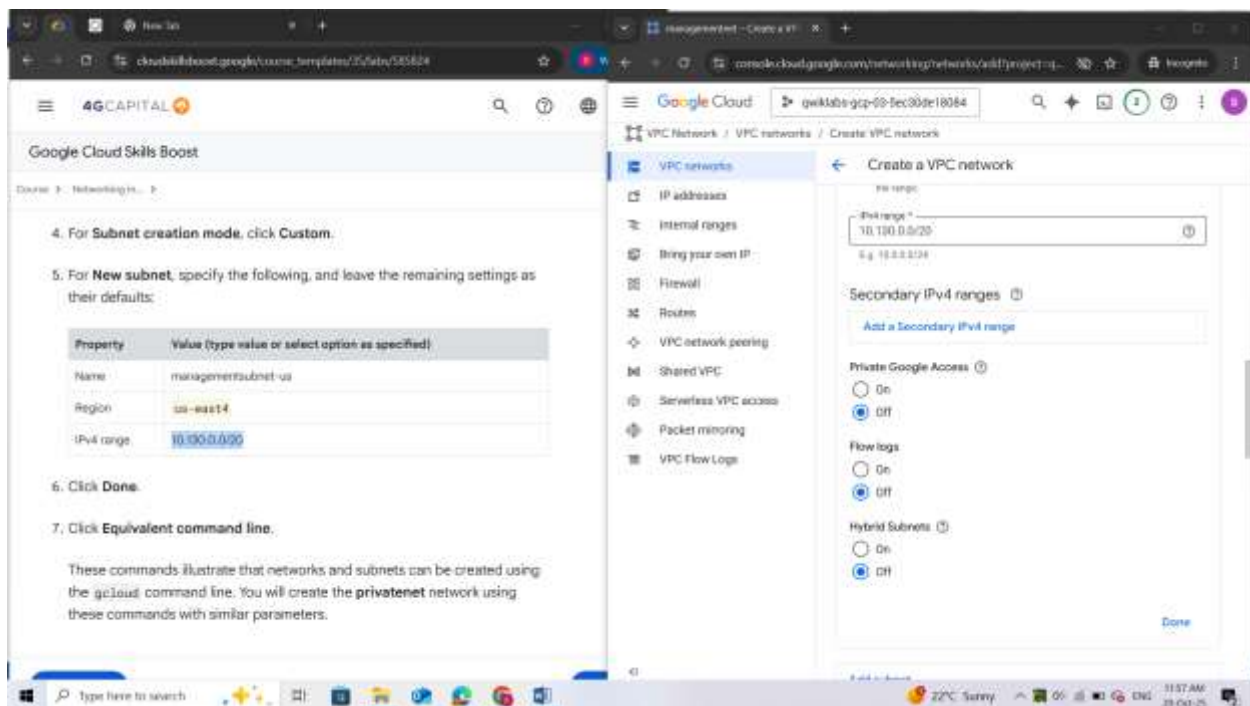


5. For **New subnet**, specify the following, and leave the remaining settings as their defaults:

Property	Value (type value or select option as specified)
Name	managementsubnet-us
Region	<filled at lab start>
IPv4 range	10.130.0.0/20



6. Click **Done**.



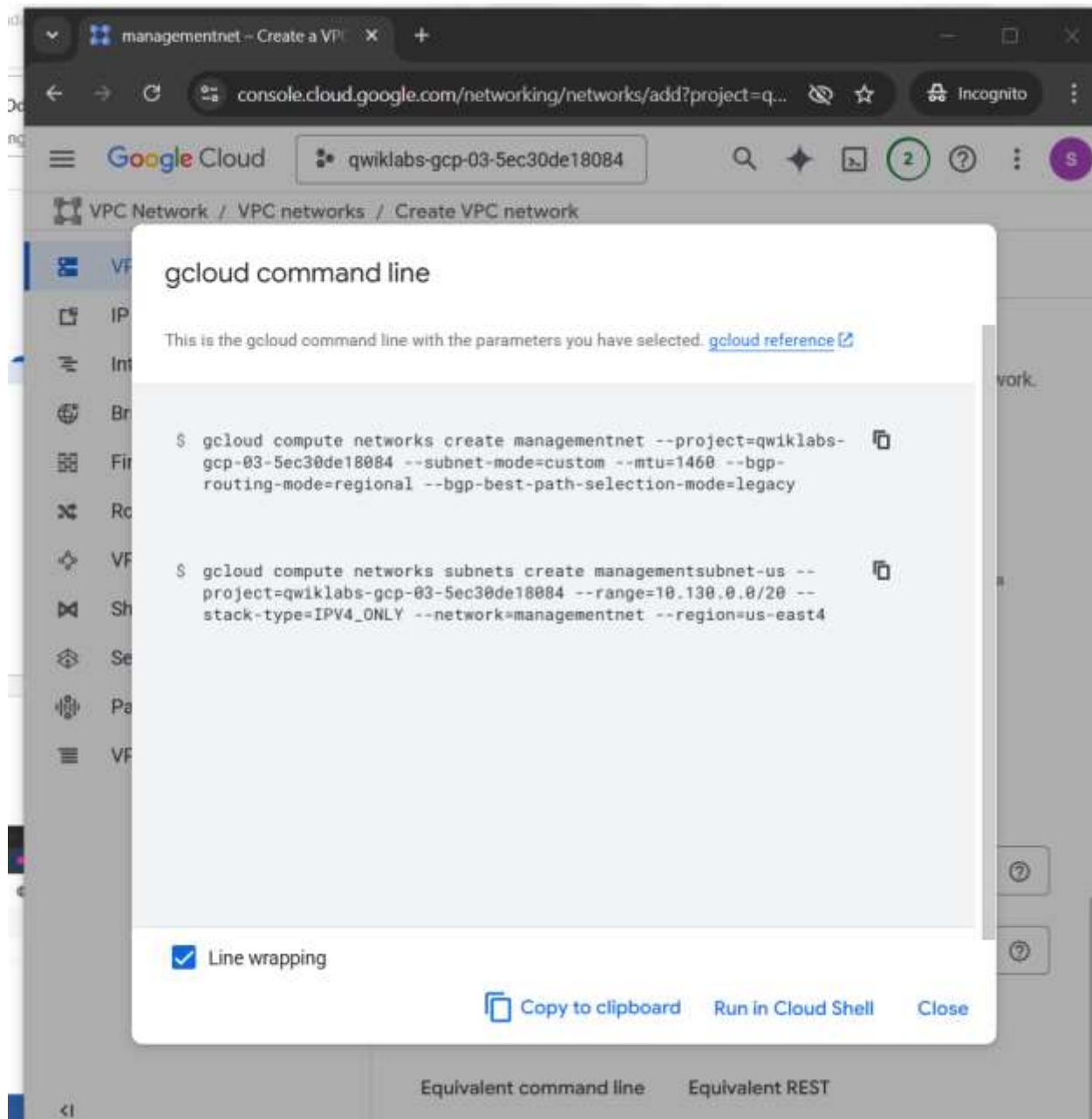
7. Click **Equivalent command line**.

The screenshot displays two browser windows. The left window, titled 'Google Cloud Skills Boost', shows a course page for 'Networking' with steps 4, 5, 6, and 7. Step 5 includes a table for subnet configuration:

Property	Value (type value or select option as specified)
Name	management-subnet-us
Region	us-east14
IPv4 range	10.100.0.0/20

The right window, titled 'Google Cloud', shows the 'Create a VPC network' page. The 'VPC networks' sidebar is active. The main configuration area shows 'Advanced dynamic routing configuration' with 'Dynamic routing mode' set to 'Regional' and 'Best path selection mode' set to 'Legacy (default)'. The 'DNS configuration (optional)' section is visible at the bottom.

These commands illustrate that networks and subnets can be created using the `gcloud` command line. You will create the **privatenet** network using these commands with similar parameters.



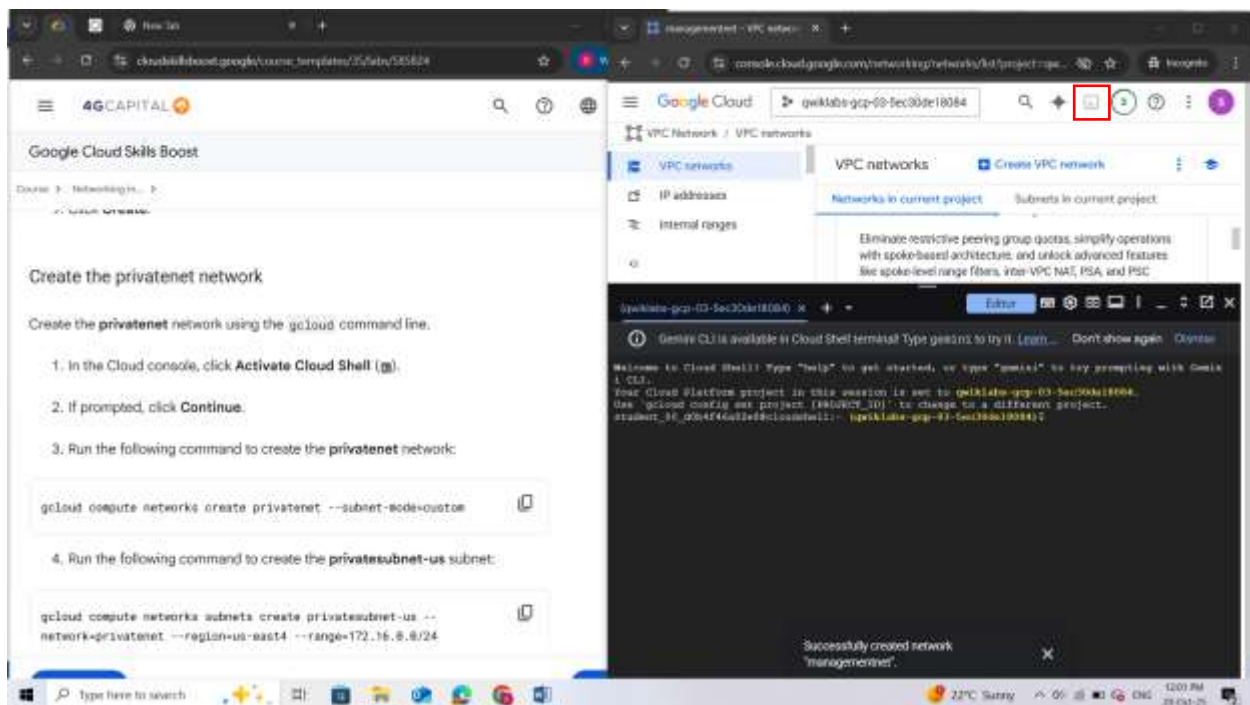
8. Click **Close**.

9. Click **Create**.

Create the privatenet network

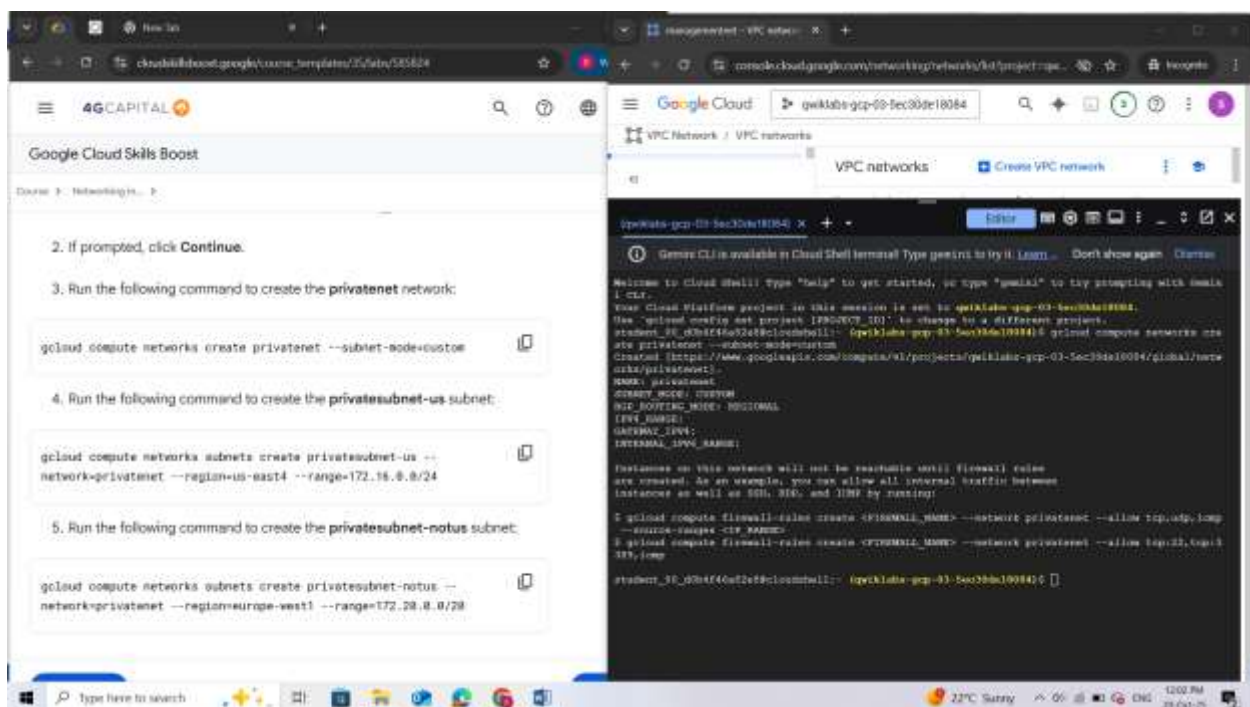
Create the **privatenet** network using the gcloud command line.

1. In the Cloud console, click **Activate Cloud Shell** ().
2. If prompted, click **Continue**.



3. Run the following command to create the **privatenet** network:

`gcloud compute networks create privatenet --subnet-mode=custom`

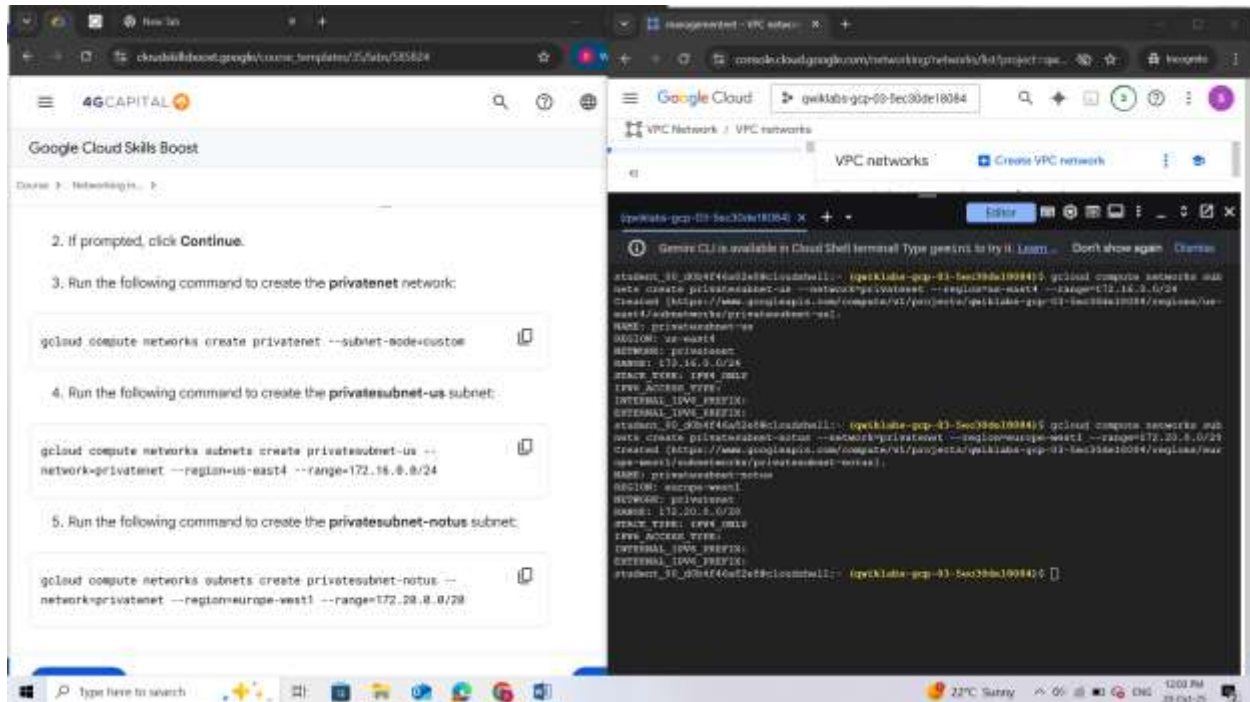


4. Run the following command to create the **privatesubnet-us** subnet:

`gcloud compute networks subnets create privatesubnet-us --network=privatenet --region="filled at lab start" --range=172.16.0.0/24`

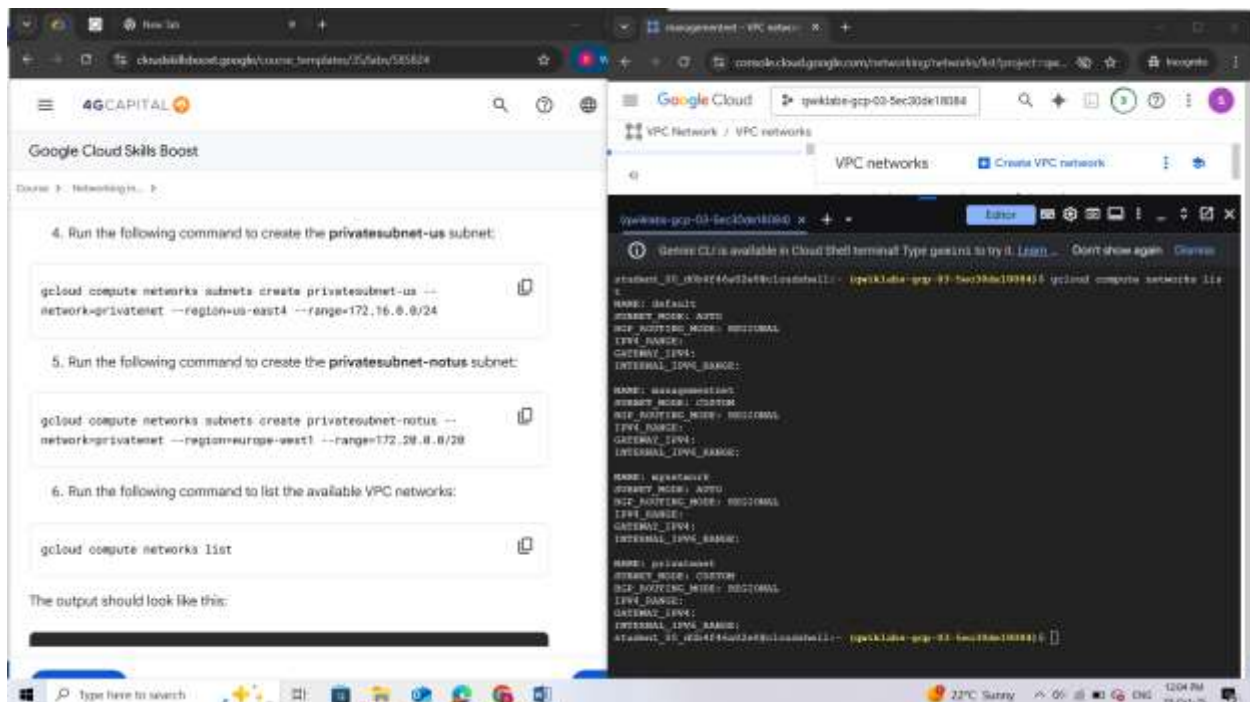
5. Run the following command to create the **privatesubnet-notus** subnet:

gcloud compute networks subnets create privatesubnet-notus --network=privatenet --region="filled at lab start" --range=172.20.0.0/20



6. Run the following command to list the available VPC networks:

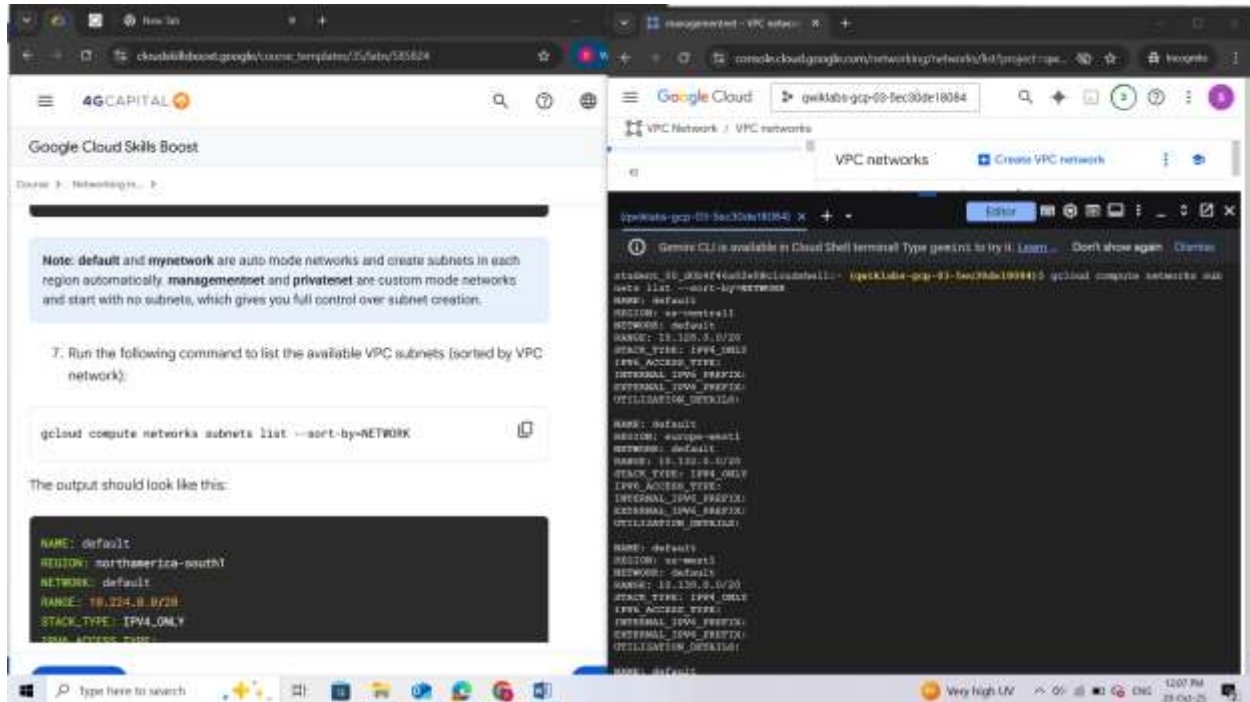
gcloud compute networks list



Note: *default* and *mynetwork* are auto mode networks and create subnets in each region automatically. *managementnet* and *privatenet* are custom mode networks and start with no subnets, which gives you full control over subnet creation.

7. Run the following command to list the available VPC subnets (sorted by VPC network):

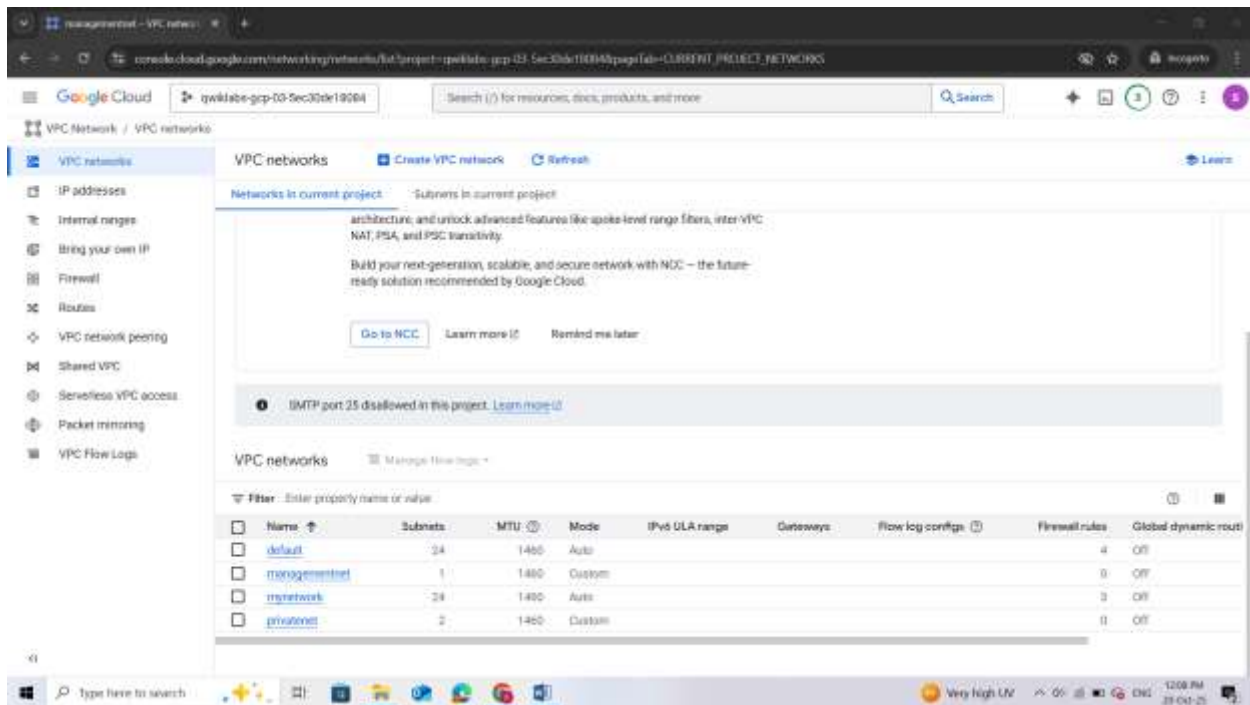
`gcloud compute networks subnets list --sort-by=NETWORK`



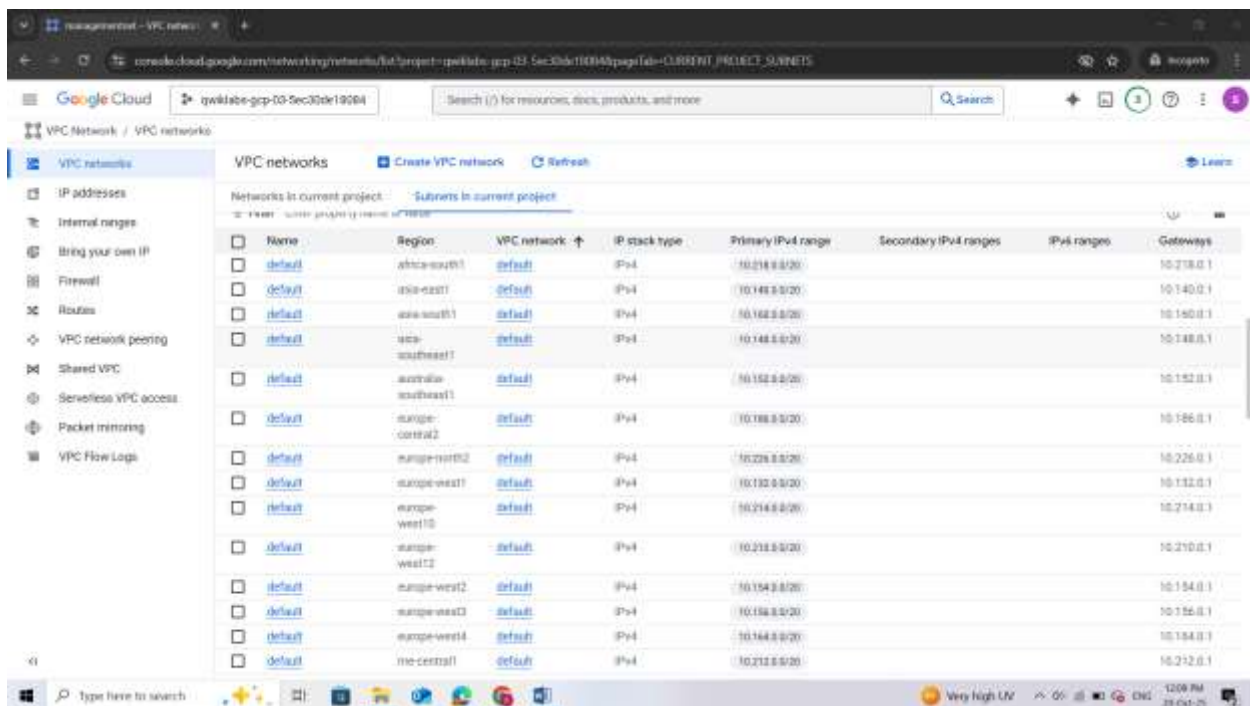
Note: As expected, the **default** and **mynetwork** networks have subnets in each region, because they are auto mode networks. The **managementnet** and **privatenet** networks only have the subnets that you created, because they are custom mode networks.

8. In the Cloud console, in the **Navigation menu** (≡), click **VPC Network > VPC networks**.
Verify that the same **networks** and **subnets** are listed in the Cloud console.

Networks:



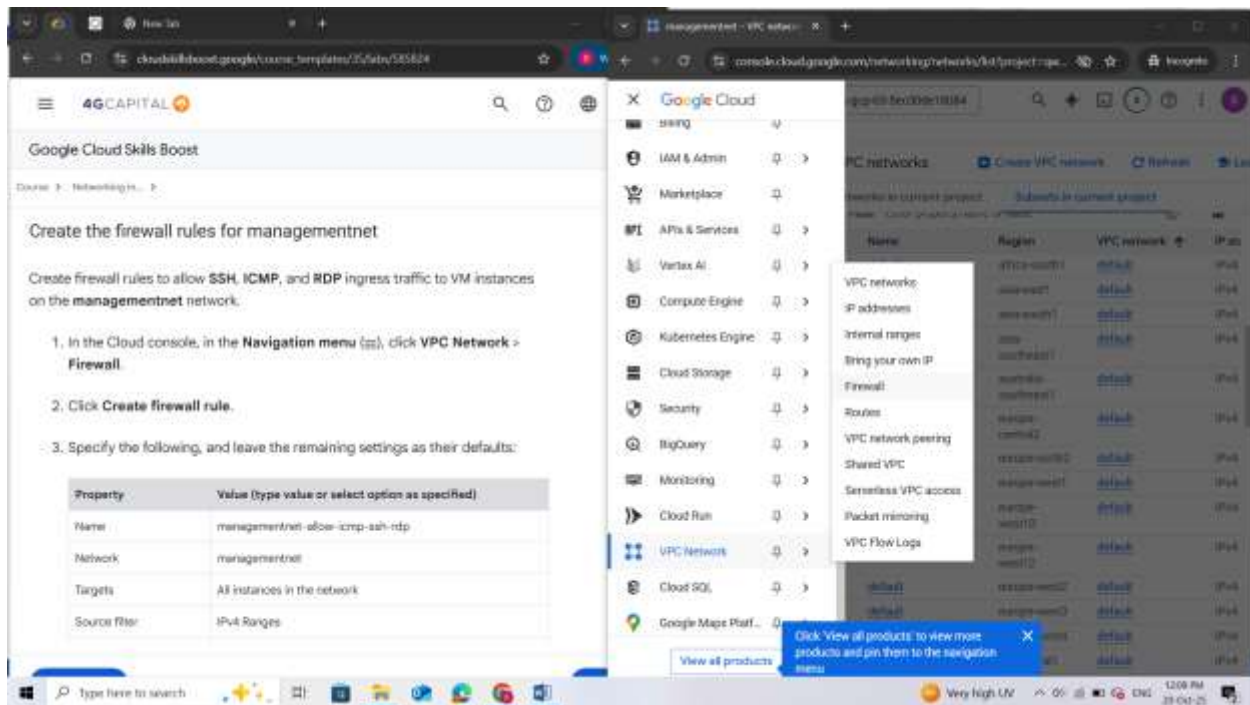
Subnets:



Create the firewall rules for managementnet

Create firewall rules to allow **SSH**, **ICMP**, and **RDP** ingress traffic to VM instances on the **managementnet** network.

1. In the Cloud console, in the **Navigation menu** () , click **VPC Network > Firewall**.



2. Click **Create firewall rule**.
3. Specify the following, and leave the remaining settings as their defaults:

Property	Value (type value or select option as specified)
Name	managementnet-allow-icmp-ssh-rdp
Network	managementnet
Targets	All instances in the network
Source filter	IPv4 Ranges
Source IPv4 ranges	0.0.0.0/0
Protocols and ports	Specified protocols and ports

Google Cloud

qwiklabs-gcp-03-5ec30de18084

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3

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S

Network Security

Create a firewall rule

Cloud Armor

DDoS Dashboard

Cloud Armor policies

Adaptive Protection

Cloud Armor Service Tier

Cloud IDS

IDS Dashboard

IDS Endpoints

IDS Threats

Cloud NGFW

Dashboard

Firewall policies

Threats

Firewall endpoints

Secure Access Connect

Realms Preview

Attachments

Firewall rules control incoming and outgoing traffic to an instance. By default, all incoming traffic to your network is blocked. [Learn more](#)

Name *
managementnet-allow-icmp-ssh-rdp
Lowercase letters, numbers, hyphens allowed

Description

Logs
Turning on firewall logs can generate a large number of logs which can increase costs in Logging. [Learn more](#)

☐ On
☒ Off

Network *
managementnet

Priority *
1000
[Compare](#)
Priority can be 0 - 65535

Direction of traffic
☒ Ingress
☐ Egress

Action on match

Google Cloud qwiklabs-gcp-03-5ec30de18084

Network Security

Create a firewall rule

Cloud Armor

- DDoS Dashboard
- Cloud Armor policies
- Adaptive Protection
- Cloud Armor Service Tier

Cloud IDS

- IDS Dashboard
- IDS Endpoints
- IDS Threats

Cloud NGFW

- Dashboard
- Firewall policies**
- Threats
- Firewall endpoints

Secure Access Connect

- Realms **Preview**
- Attachments

Direction of traffic ?

☒ Ingress

☐ Egress

Action on match ?

☒ Allow

☐ Deny

Targets

All instances in the network

Source filter

IPv4 ranges

Source IPv4 ranges *

0.0.0.0/0 X for example, 0.0.0.0/0, 192.168.2.0/24

Second source filter

None

Destination filter

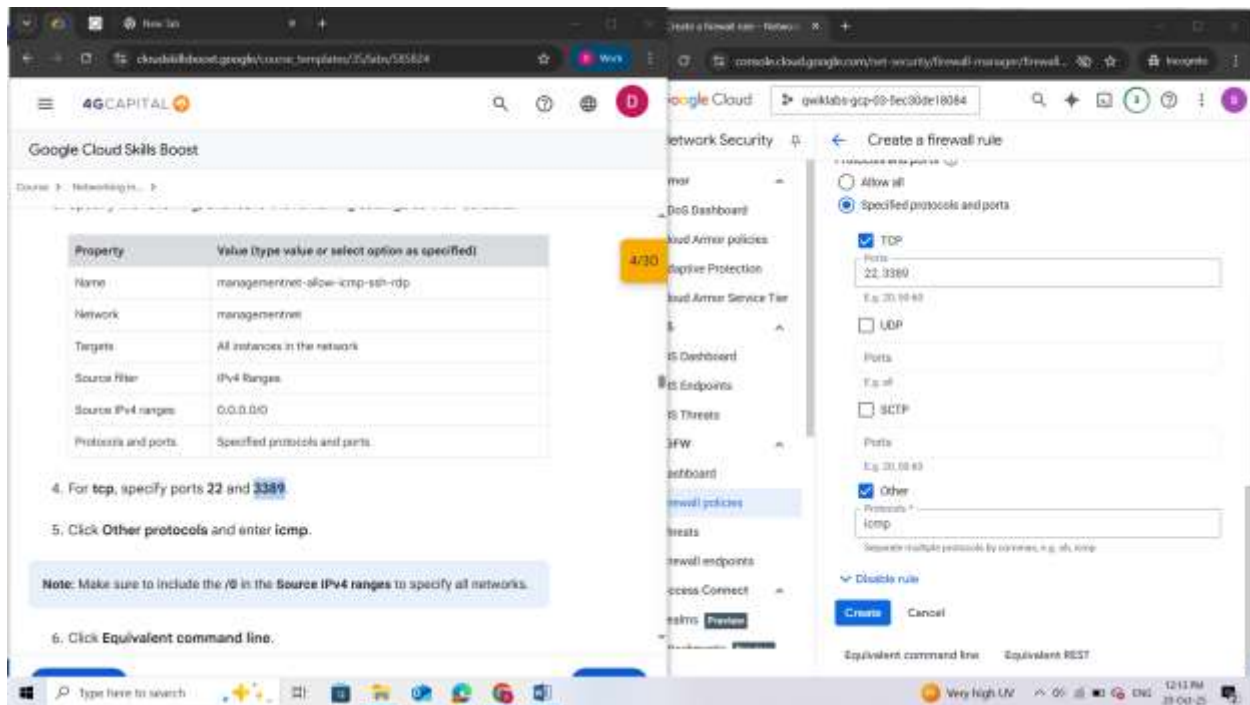
None

Protocols and ports ?

☐ Allow all

☒ Specified protocols and ports

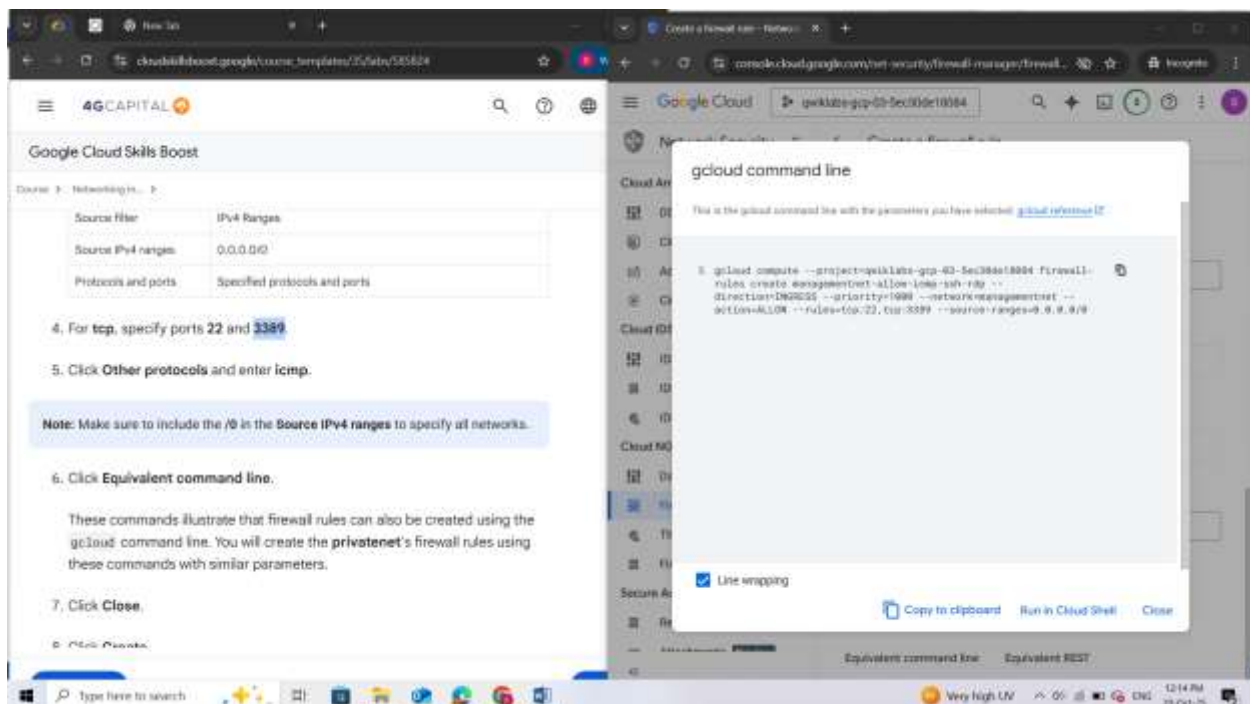
- For **tcp**, specify ports **22** and **3389**.
- Click **Other protocols** and enter **icmp**.



Note: Make sure to include the /0 in the **Source IPv4 ranges** to specify all networks.

6. Click **Equivalent command line**.

These commands illustrate that firewall rules can also be created using the gcloud command line. You will create the **privatenet**'s firewall rules using these commands with similar parameters.



7. Click **Close**.

8. Click **Create**.

Create the firewall rules for privatenet

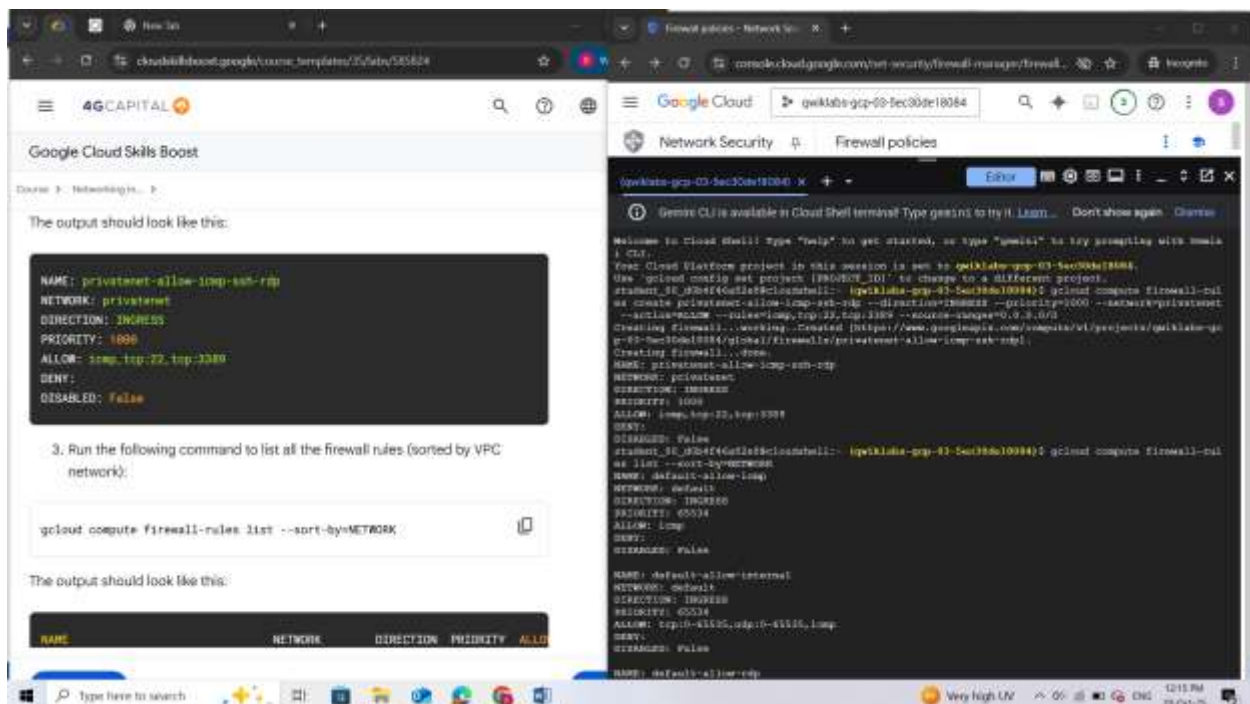
Create the firewall rules for **privatenet** network using the gcloud command line.

1. Return to **Cloud Shell**. If necessary, click **Activate Cloud Shell** ().
2. Run the following command to create the **privatenet-allow-icmp-ssh-rdp** firewall rule:

```
gcloud compute firewall-rules create privatenet-allow-icmp-ssh-rdp --direction=INGRESS --priority=1000 --network=privatenet --action=ALLOW --rules=icmp,tcp:22,tcp:3389 --source-ranges=0.0.0.0/0
```

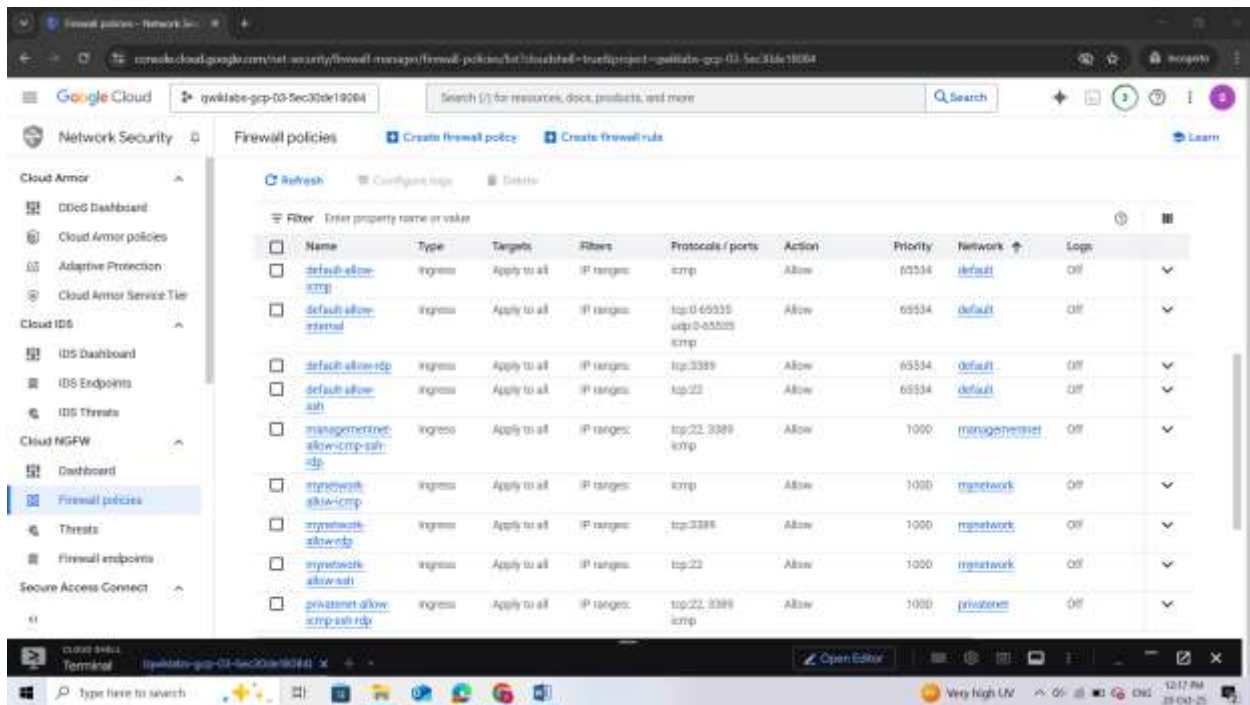
3. Run the following command to list all the firewall rules (sorted by VPC network):

```
gcloud compute firewall-rules list --sort-by=NETWORK
```



The firewall rules for **mynetwork** network have been created for you. You can define multiple protocols and ports in one firewall rule (**privatenet** and **managementnet**) or spread them across multiple rules (**default** and **mynetwork**).

4. In the Cloud console, on the **Navigation menu** (), click **VPC Network > Firewall**. Verify that the same firewall rules are listed in the Cloud console.



Task 2. Create VM instances

Create two VM instances:

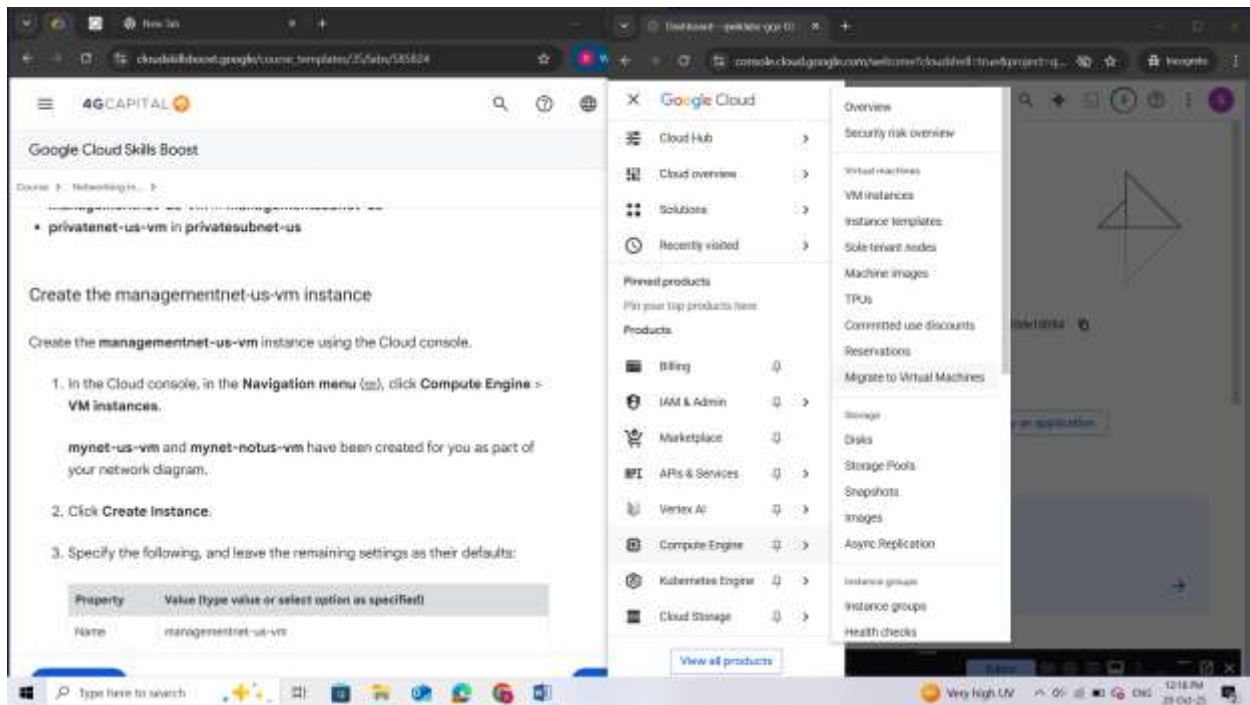
- **managementnet-us-vm** in **managementsubnet-us**
- **privatenet-us-vm** in **privatesubnet-us**

Create the **managementnet-us-vm** instance

Create the **managementnet-us-vm** instance using the Cloud console.

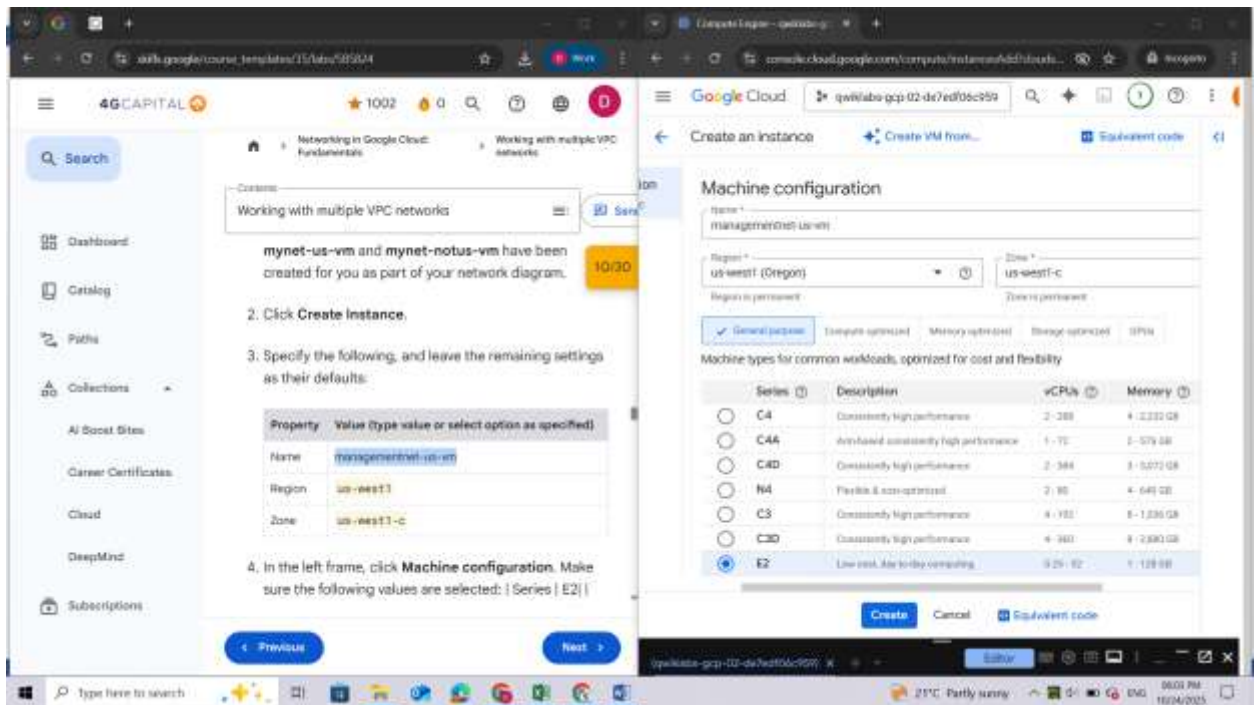
1. In the Cloud console, in the **Navigation menu** () , click **Compute Engine > VM instances**.

mynet-us-vm and **mynet-notus-vm** have been created for you as part of your network diagram.

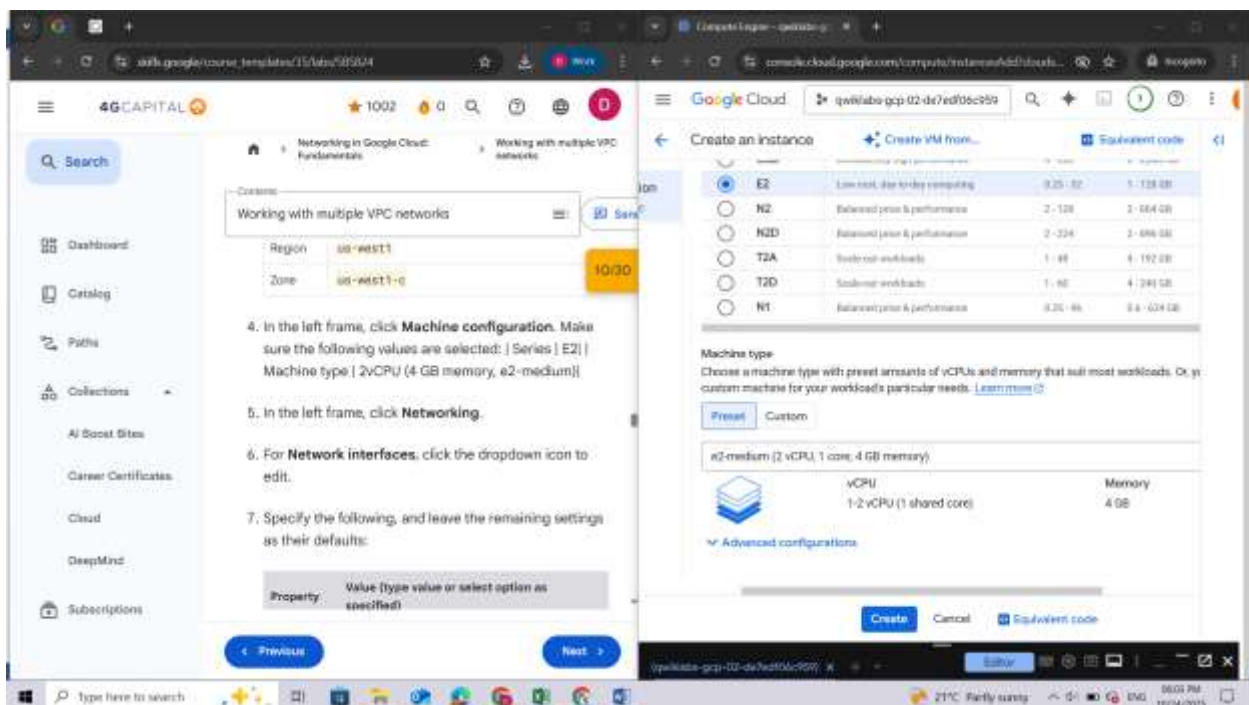


2. Click **Create Instance**.
3. Specify the following, and leave the remaining settings as their defaults:

Property	Value (type value or select option as specified)
Name	managementnet-us-vm
Region	<filled at lab start>
Zone	<filled at lab start>



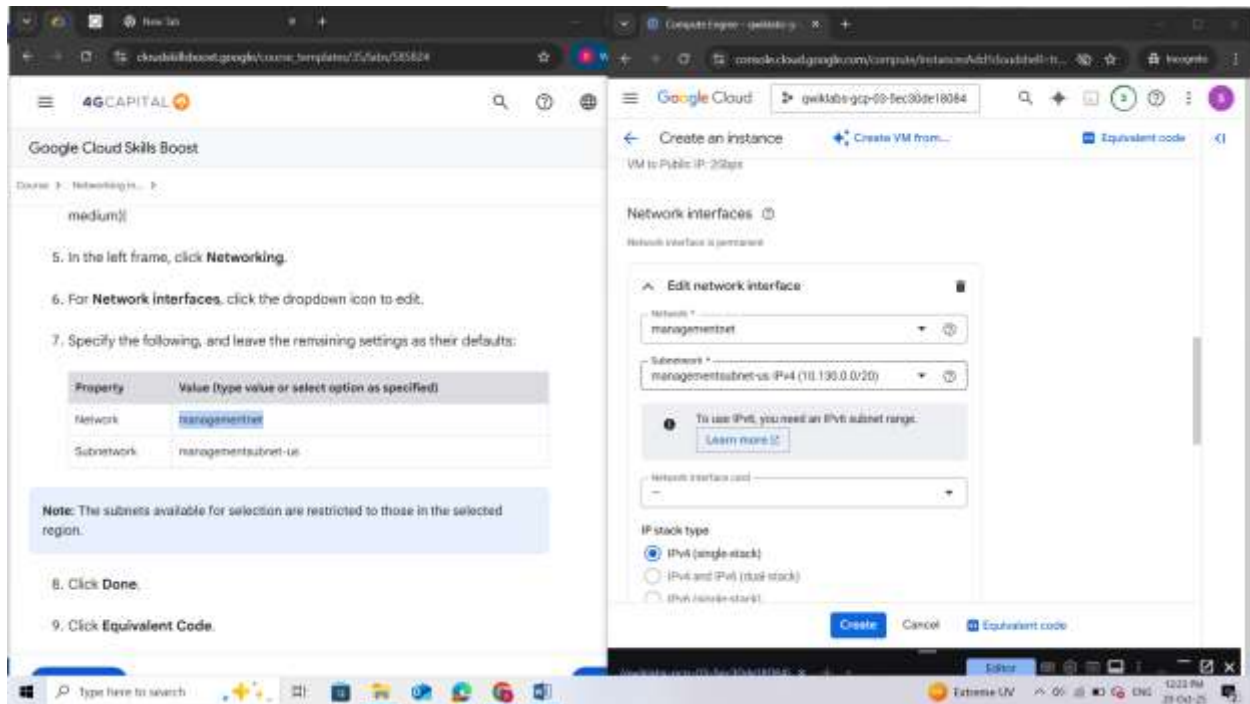
4. In the left frame, click **Machine configuration**. Make sure the following values are selected: | Series | E2 | | Machine type | 2vCPU (4 GB memory, e2-medium) |



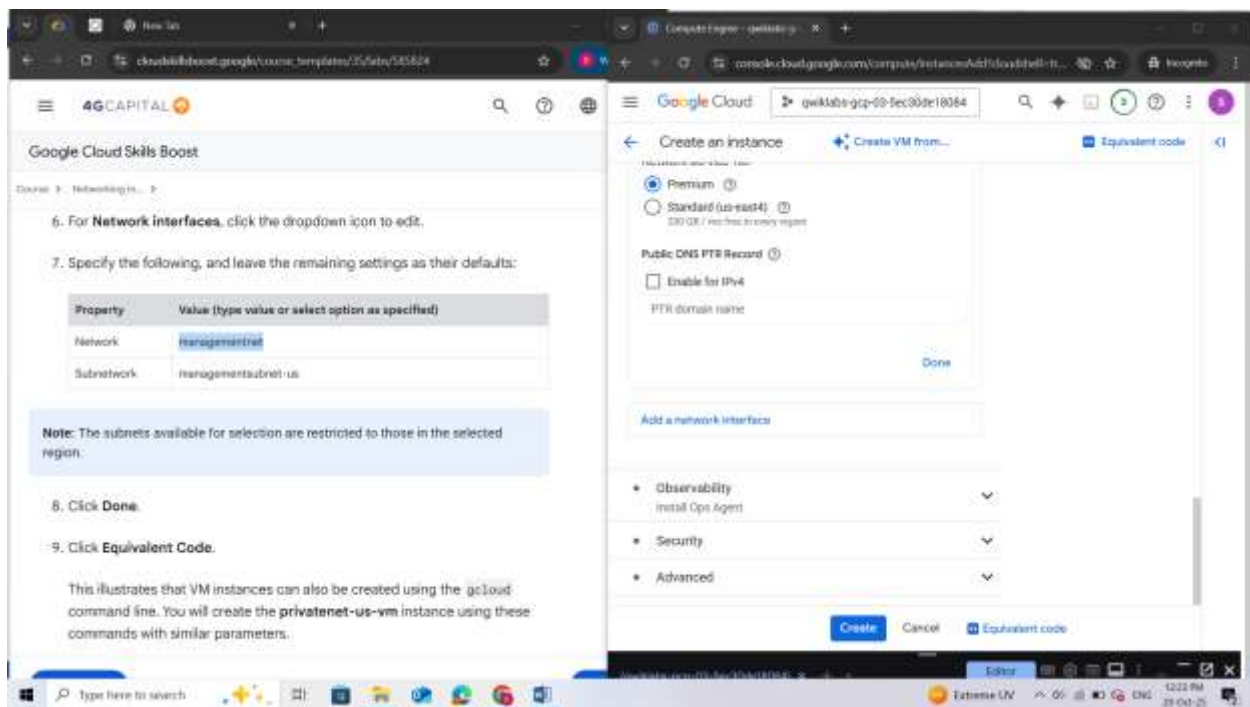
5. In the left frame, click **Networking**.
6. For **Network interfaces**, click the dropdown icon to edit.
7. Specify the following, and leave the remaining settings as their defaults:

Property	Value (type value or select option as specified)
Network	managementnet
Subnetwork	managementsubnet-us

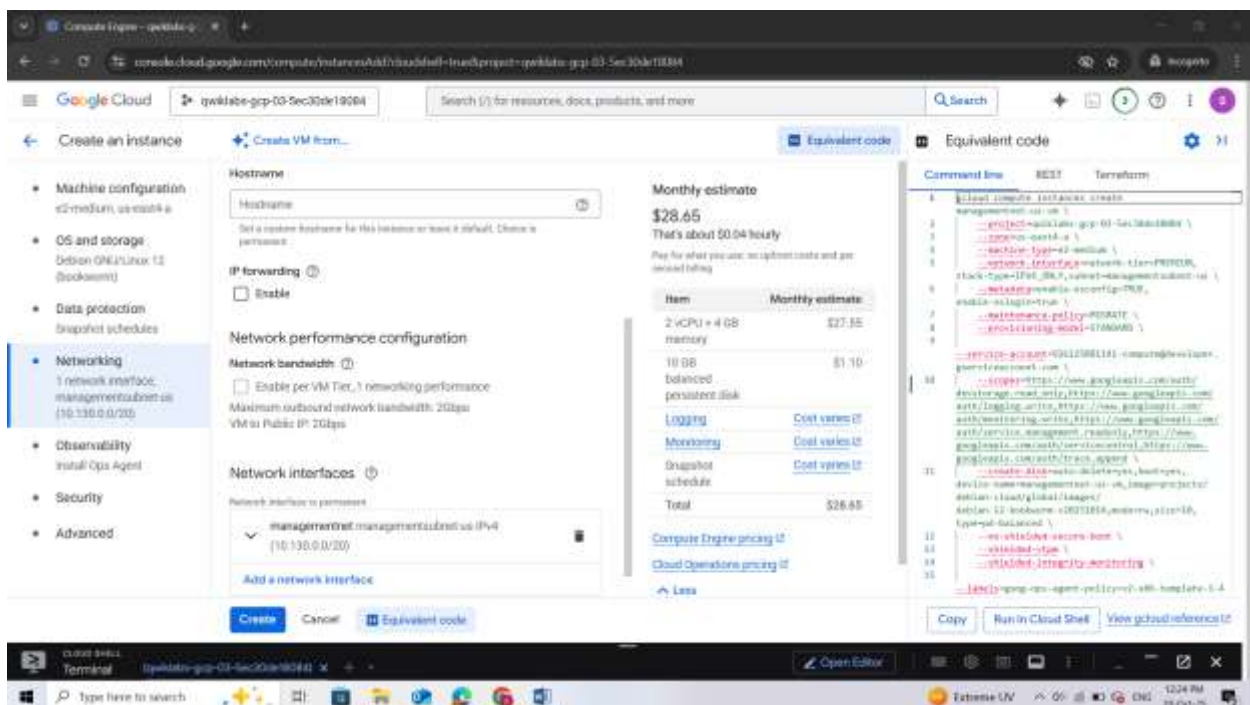
Note: The subnets available for selection are restricted to those in the selected region.



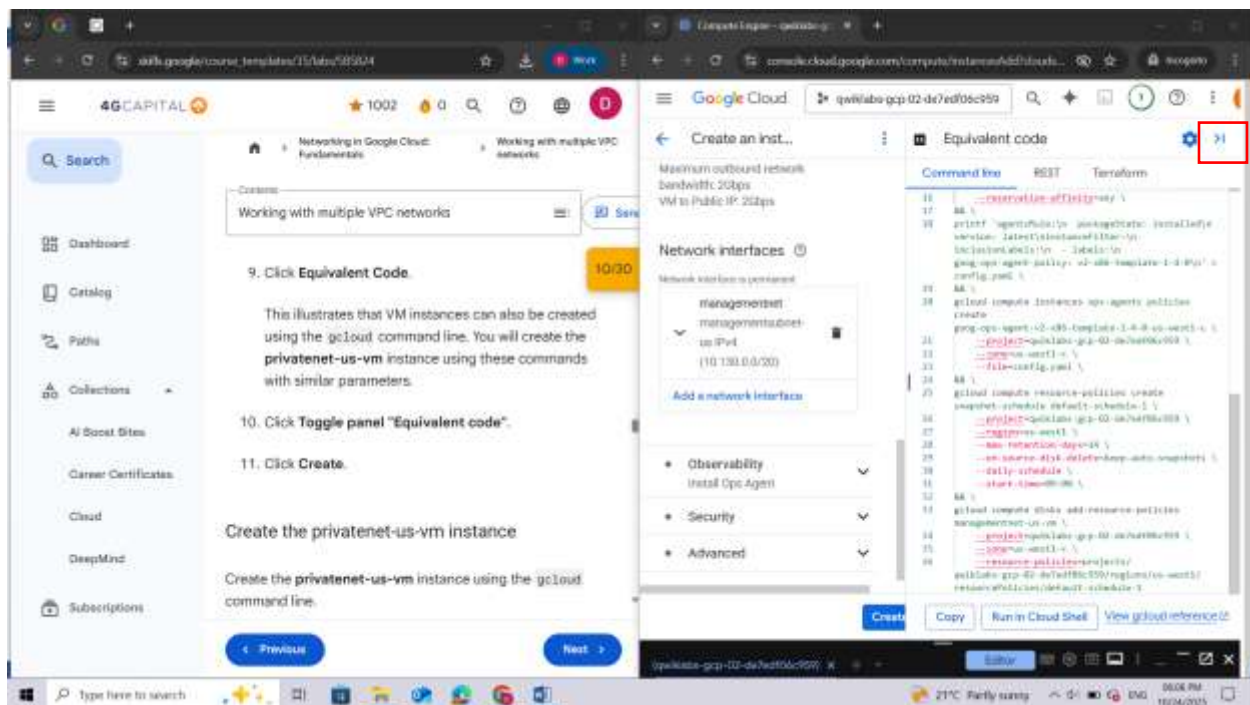
8. Click **Done**.
9. Click **Equivalent Code**.



This illustrates that VM instances can also be created using the gcloud command line. You will create the **privatenet-us-vm** instance using these commands with similar parameters.



10. Click Toggle panel "Equivalent code".



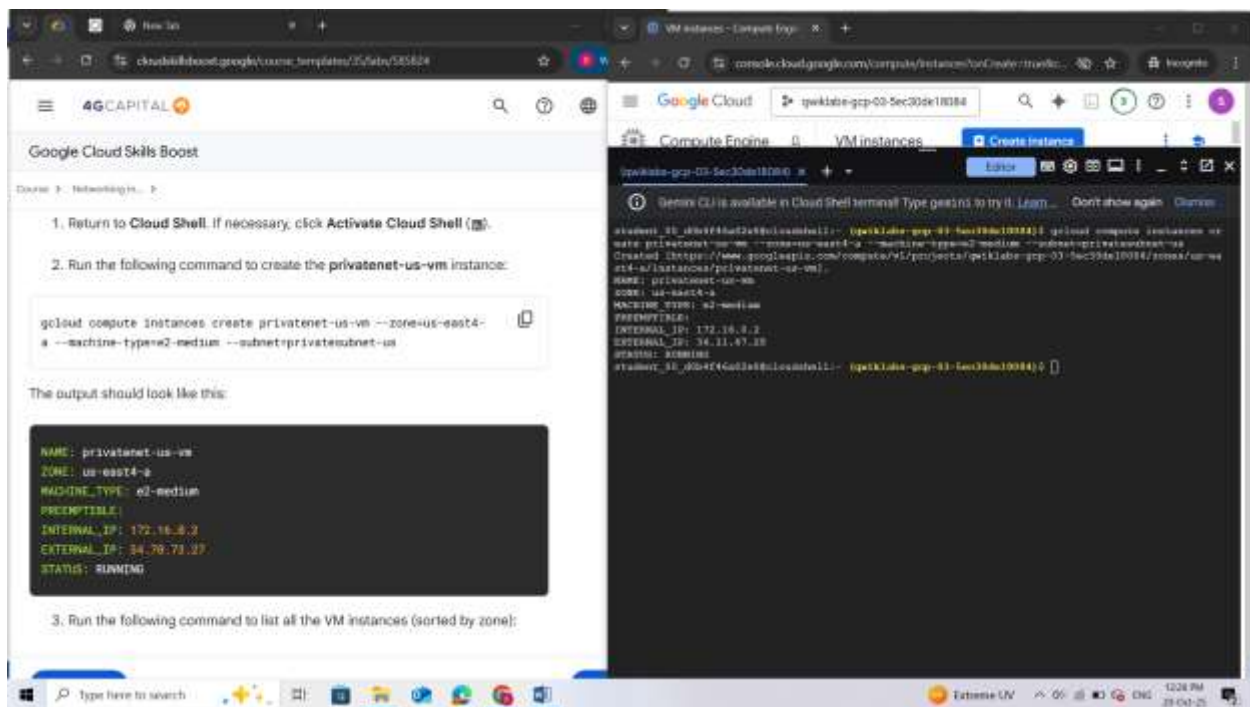
11. Click **Create**.

Create the privatenet-us-vm instance

Create the **privatenet-us-vm** instance using the gcloud command line.

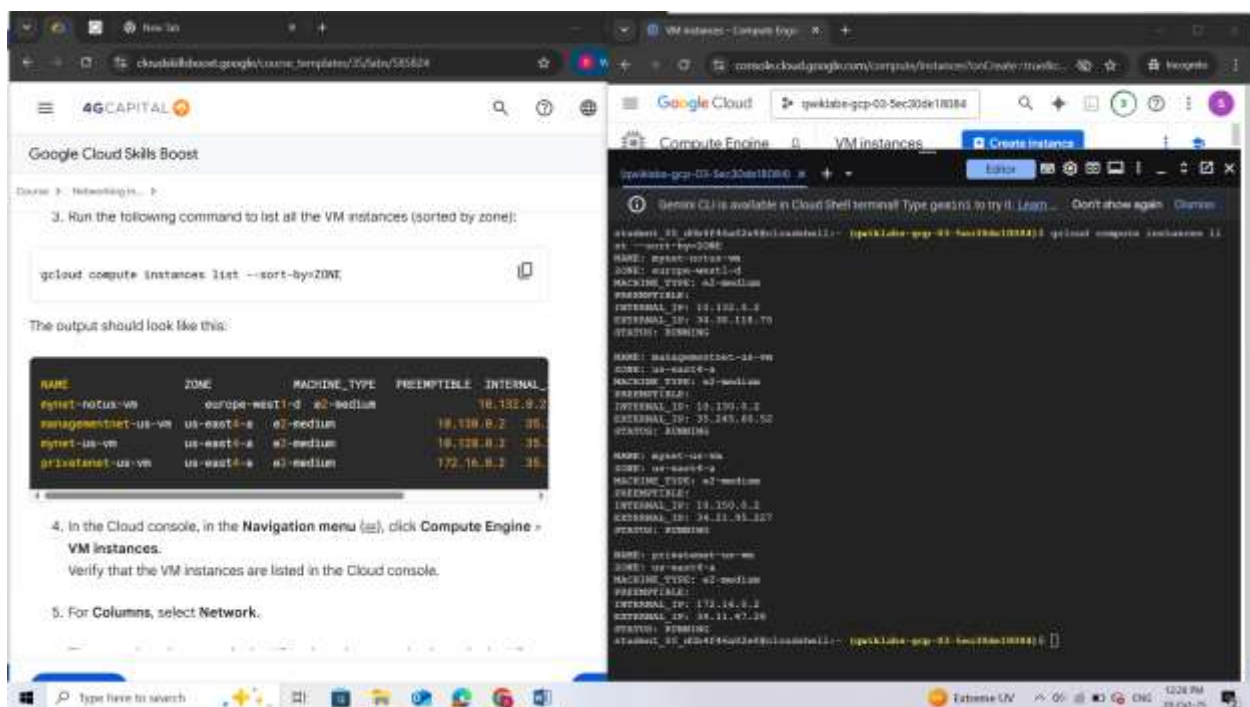
1. Return to **Cloud Shell**. If necessary, click **Activate Cloud Shell** (▶).
2. Run the following command to create the **privatenet-us-vm** instance:

gcloud compute instances create privatenet-us-vm --zone="filled at lab start" --machine-type=e2-medium --subnet=privatesubnet-us

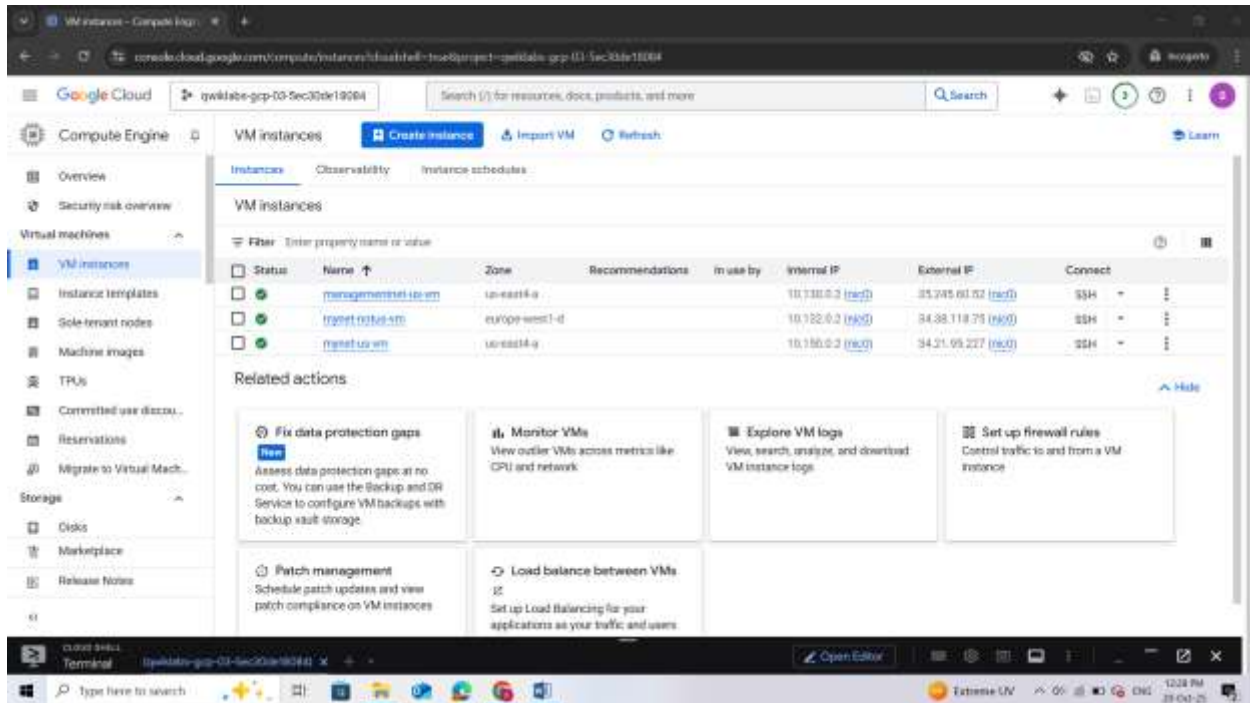


3. Run the following command to list all the VM instances (sorted by zone):

`gcloud compute instances list --sort-by=ZONE`



4. In the Cloud console, in the **Navigation menu** () , click **Compute Engine > VM instances**.
Verify that the VM instances are listed in the Cloud console.



5. For **Columns**, select **Network**.

There are three instances in the US and one instance that is not in the US. These instances are spread across three VPC networks (**managementnet**, **mynet**, and **privatenet**), with no instance in the same zone and network as another. In the next task, you explore the effect this has on internal connectivity.

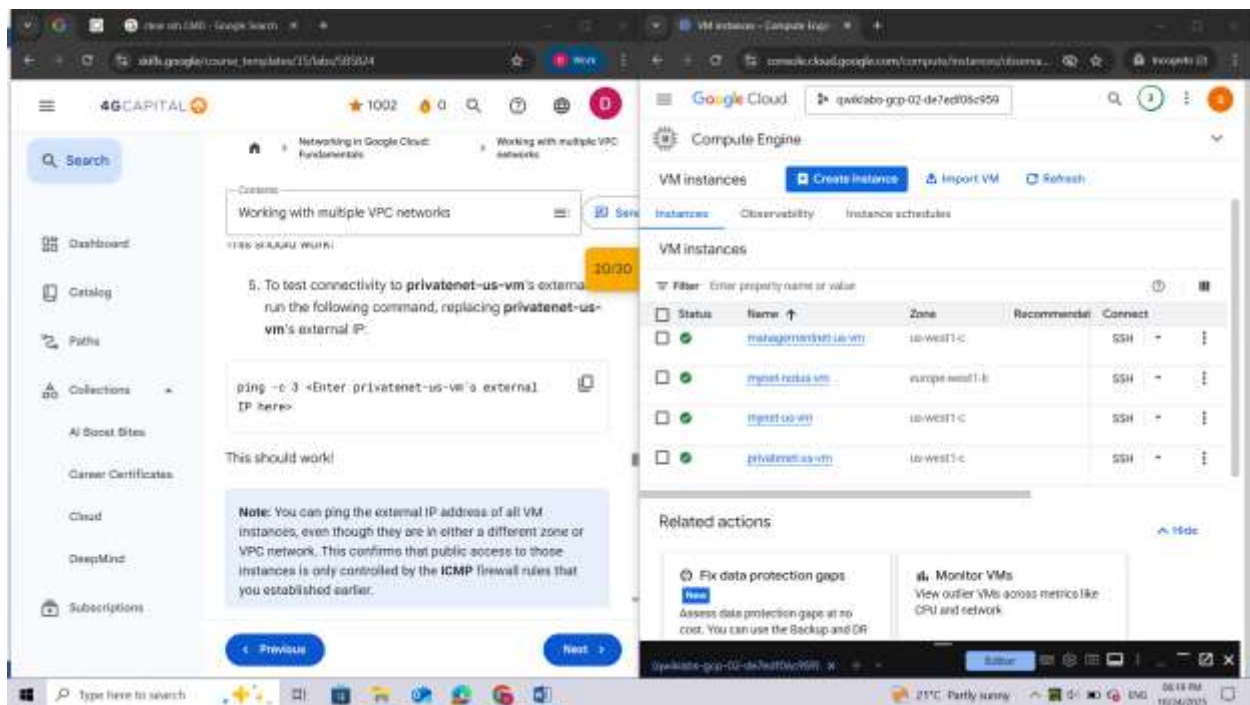
Task 3. Explore the connectivity between VM instances

Explore the connectivity between the VM instances. Specifically, determine the effect of having VM instances in the same zone versus having instances in the same VPC network.

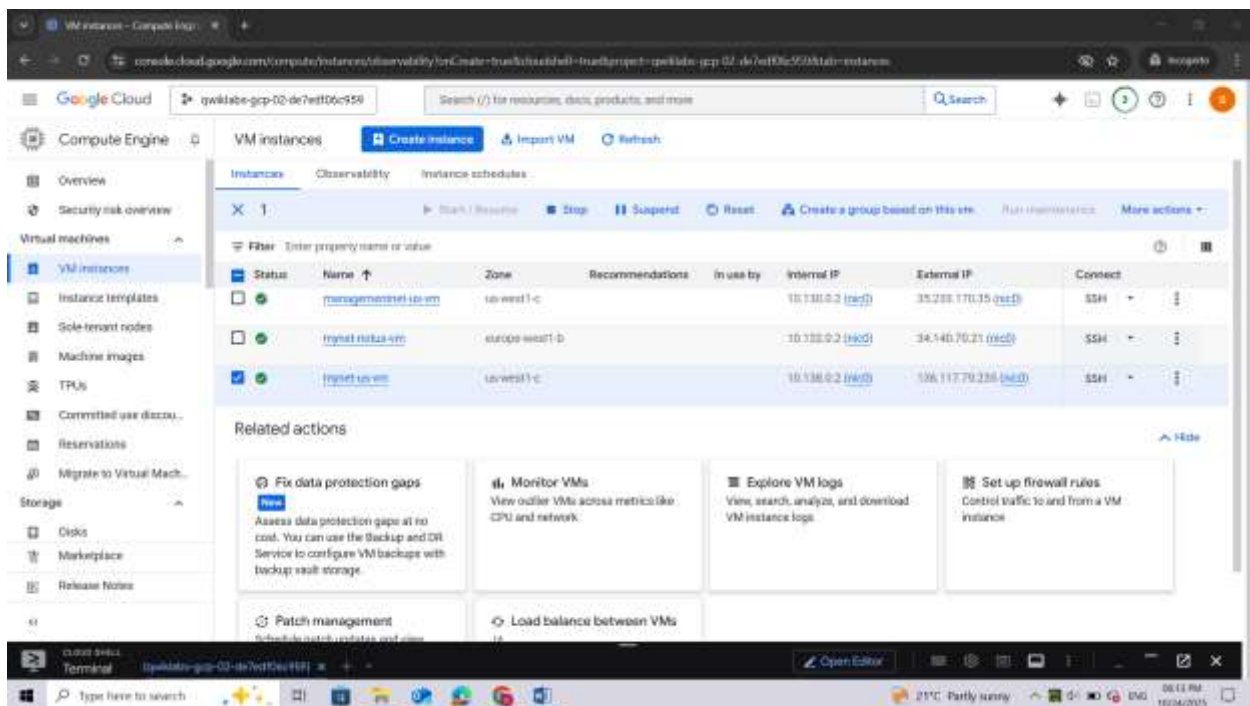
Ping the external IP addresses

Ping the external IP addresses of the VM instances to determine whether you can reach the instances from the public internet.

1. In the Cloud console, in the **Navigation menu**, click **Compute Engine > VM instances**. Note the external IP addresses for **mynet-notus-vm**, **managementnet-us-vm**, and **privatenet-us-vm**.



2. For **mynet-us-vm**, click **SSH** to launch a terminal and connect.



3. To test connectivity to **mynet-notus-vm**'s external IP, run the following command, replacing **mynet-notus-vm**'s external IP:

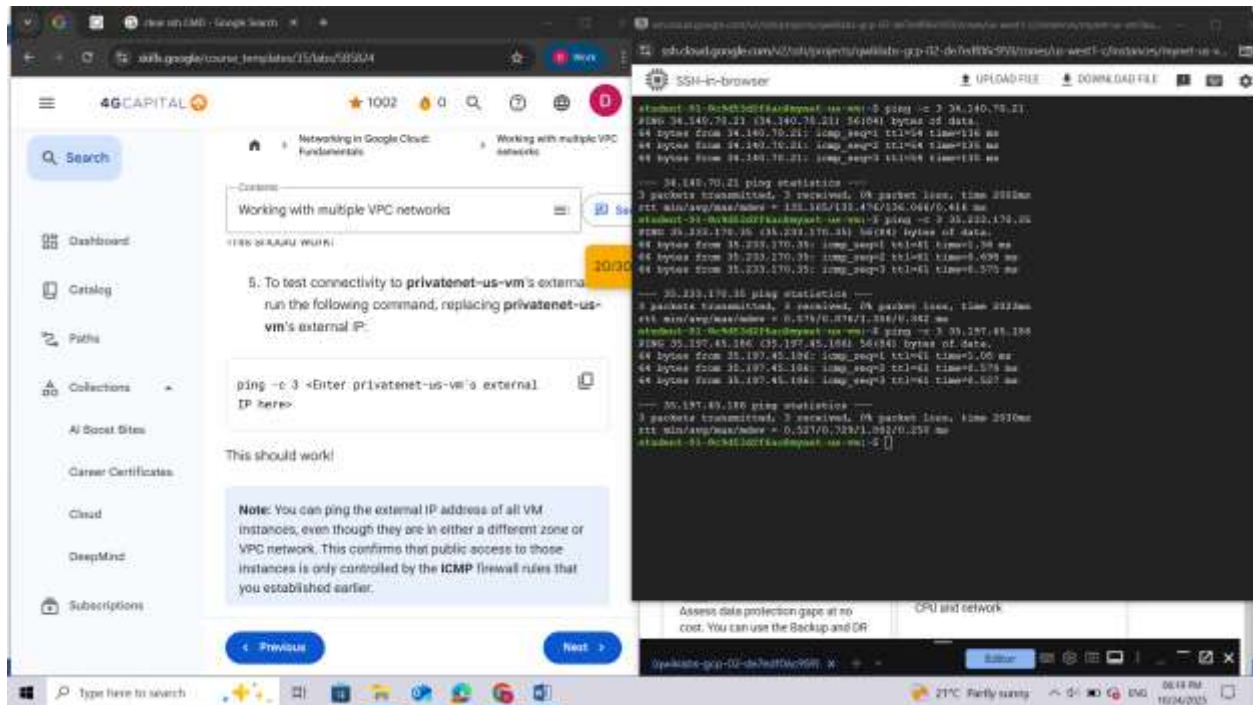
`ping -c 3 <Enter mynet-notus-vm's external IP here>`

- To test connectivity to **managementnet-us-vm**'s external IP, run the following command, replacing **managementnet-us-vm**'s external IP:

ping -c 3 <Enter managementnet-us-vm's external IP here>

- To test connectivity to **privatenet-us-vm**'s external IP, run the following command, replacing **privatenet-us-vm**'s external IP:

ping -c 3 <Enter privatenet-us-vm's external IP here>



Note: You can ping the external IP address of all VM instances, even though they are in either a different zone or VPC network. This confirms that public access to those instances is only controlled by the **ICMP** firewall rules that you established earlier.

Ping the internal IP addresses

Ping the internal IP addresses of the VM instances to determine whether you can reach the instances from within a VPC network.

Which instance(s) should you be able to ping from mynet-us-vm using internal IP addresses?

☐

managementnet-us-vm

☒

mynet-notus-vm

☐

privatenet-us-vm

1. In the Cloud console, in the **Navigation menu**, click **Compute Engine > VM instances**.
Note the internal IP addresses for **mynet-notus-vm**, **managementnet-us-vm**, and **privatenet-us-vm**.
2. Return to the **SSH** terminal for **mynet-us-vm**.
3. To test connectivity to **mynet-notus-vm**'s internal IP, run the following command, replacing **mynet-notus-vm**'s internal IP:

ping -c 3 <Enter mynet-notus-vm's internal IP here>

Note: You can ping the internal IP address of **mynet-notus-vm** because it is on the same VPC network as the source of the ping (**mynet-us-vm**), even though both VM instances are in separate zones, regions, and continents!

4. To test connectivity to **managementnet-us-vm**'s internal IP, run the following command, replacing **managementnet-us-vm**'s internal IP:

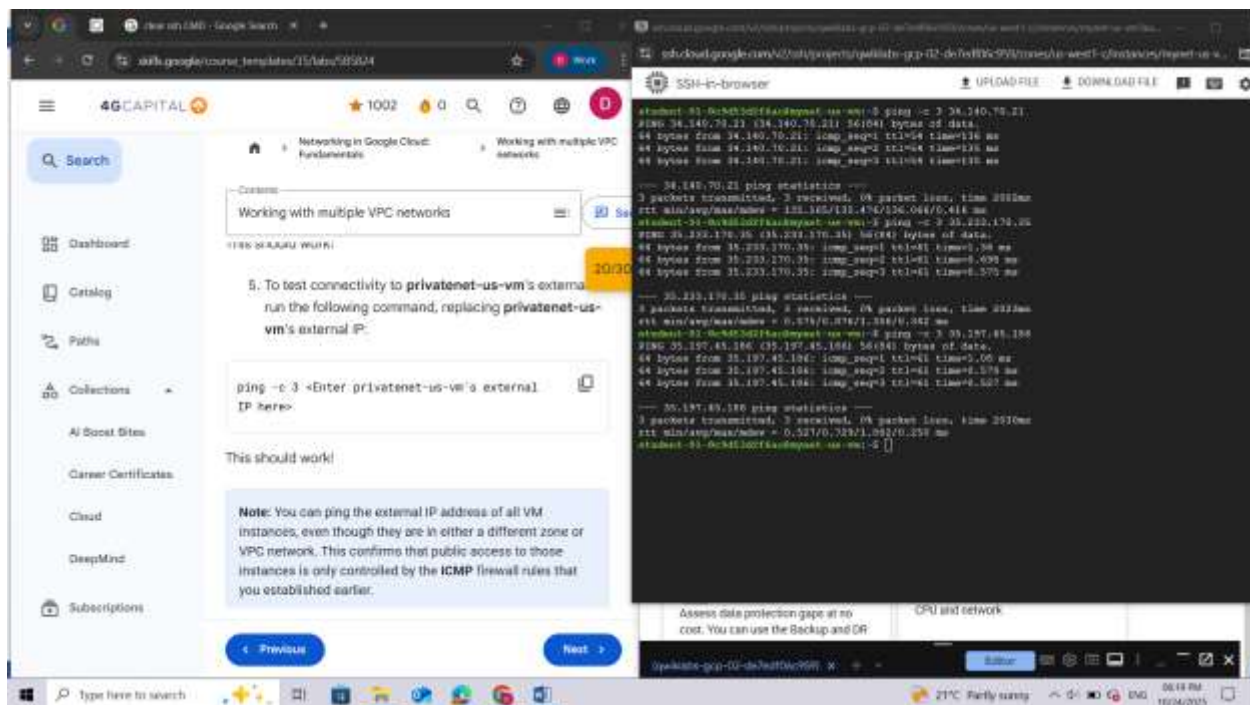
ping -c 3 <Enter managementnet-us-vm's internal IP here>

Note: This should not work, as indicated by a 100% packet loss!

5. To test connectivity to **privatenet-us-vm**'s internal IP, run the following command, replacing **privatenet-us-vm**'s internal IP:

ping -c 3 <Enter privatenet-us-vm's internal IP here>

Note: This should not work either, as indicated by a 100% packet loss! You cannot ping the internal IP address of **managementnet-us-vm** and **privatenet-us-vm** because they are in separate VPC networks from the source of the ping (**mynet-us-vm**), even though they are all in the same zone.



VPC networks are by default isolated private networking domains. However, no internal IP address communication is allowed between networks, unless you set up mechanisms such as VPC peering or VPN.

Task 4. Create a VM instance with multiple network interfaces

Every instance in a VPC network has a default network interface. You can create additional network interfaces attached to your VMs. Multiple network interfaces enable you to create configurations in which an instance connects directly to several VPC networks (up to 8 interfaces, depending on the instance's type).

Create the VM instance with multiple network interfaces

Create the **vm-appliance** instance with network interfaces in **privatesubnet-us**, **managementsubnet-us**, and **mynetwork**. The CIDR ranges of these subnets do not overlap, which is a requirement for creating a VM with multiple network interface controllers (NICs).

1. In the Cloud console, in the **Navigation menu**, click **Compute Engine > VM instances**.
2. Click **Create instance**.
3. Specify the following, and leave the remaining settings as their defaults:

Property	Value (type value or select option as specified)
Name	vm-appliance
Region	<filled at lab start>

Zone	<filled at lab start>
------	-----------------------

qwiklabs-gcp-02-de7edf06c959
7

[← Create an instance](#)
[Create VM from...](#)
[↔ Equivalent code](#)

- Machine configuration**
e2-standard-4, us-west1-c
- OS and storage
Debian GNU/Linux 12 (bookworm)
- Data protection
Snapshot schedules
- Networking
1 network interface
- Observability
Install Ops Agent

Machine configuration

Name *
vm-appliance

Region *
us-west1 (Oregon) ?

Zone *
us-west1-c ?

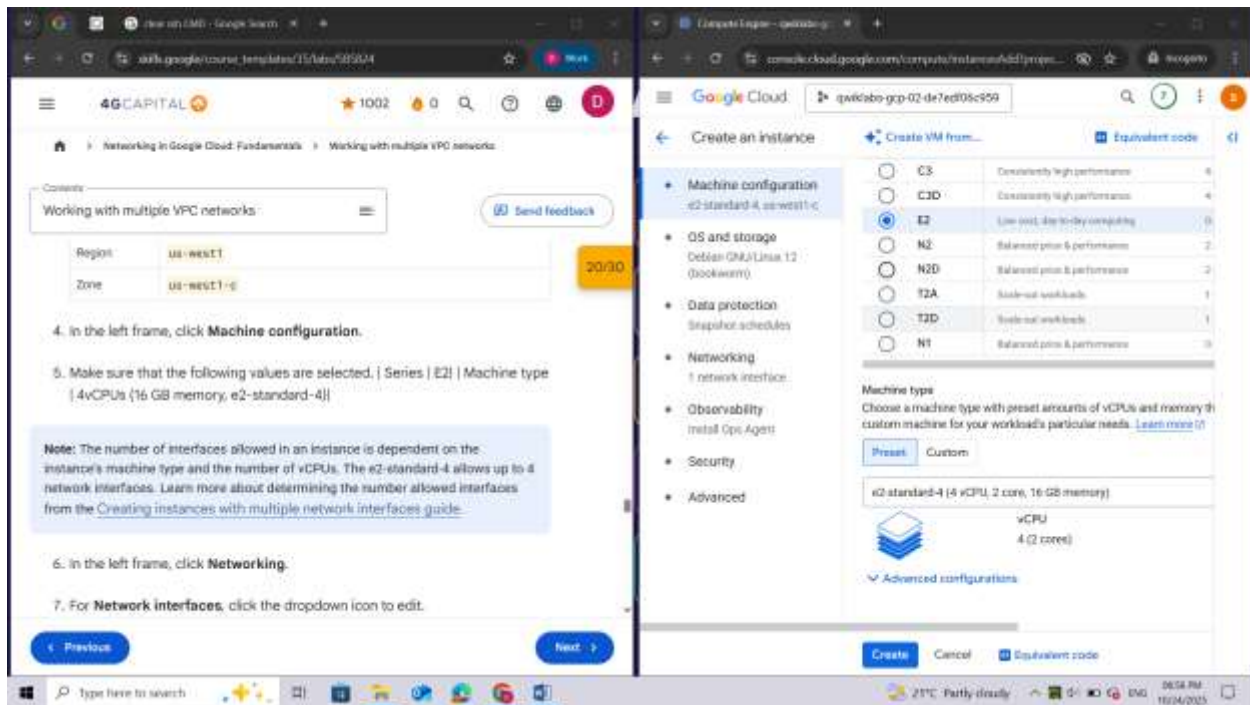
Region is permanent Zone is permanent

☒ General purpose
 ☐ Compute optimized
 ☐ Memory optimized
 ☐ Storage optimized

Machine types for common workloads, optimized for cost and flexibility

Series ?	Description	VM size
☐ C4	Consistently high performance	2 vCPUs (8 GB memory)
☐ E2	Cost-optimized	2 vCPUs (16 GB memory)

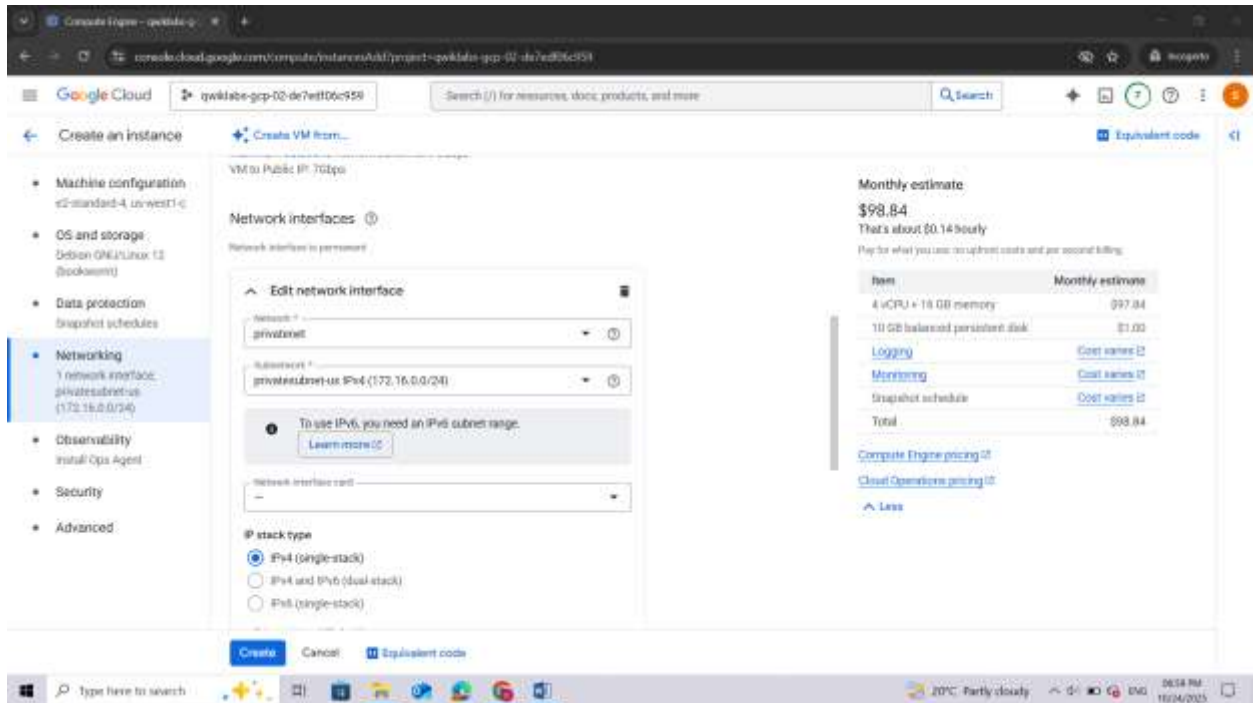
- In the left frame, click **Machine configuration**.
- Make sure that the following values are selected. | Series | E2 | | Machine type | 4vCPUs (16 GB memory, e2-standard-4) |



Note: The number of interfaces allowed in an instance is dependent on the instance's machine type and the number of vCPUs. The e2-standard-4 allows up to 4 network interfaces. Learn more about determining the number allowed interfaces from the [Creating instances with multiple network interfaces guide](#).

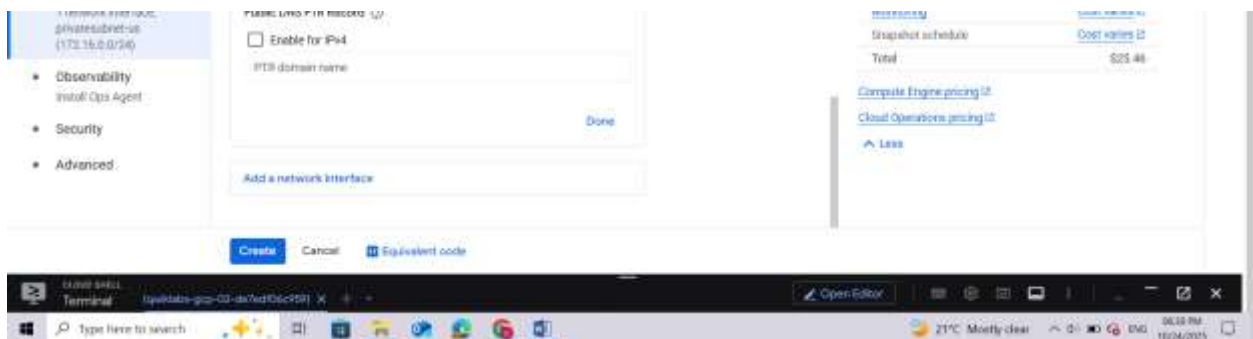
6. In the left frame, click **Networking**.
7. For **Network interfaces**, click the dropdown icon to edit.
8. Specify the following, and leave the remaining settings as their defaults:

Property	Value (type value or select option as specified)
Network	privatenet
Subnetwork	privatesubnet-us



9. Click **Done**.

10. Click **Add a network interface**.



11. Specify the following, and leave the remaining settings as their defaults:

Property	Value (type value or select option as specified)
Network	managementnet
Subnetwork	managementsubnet-us

12. Click **Done**.

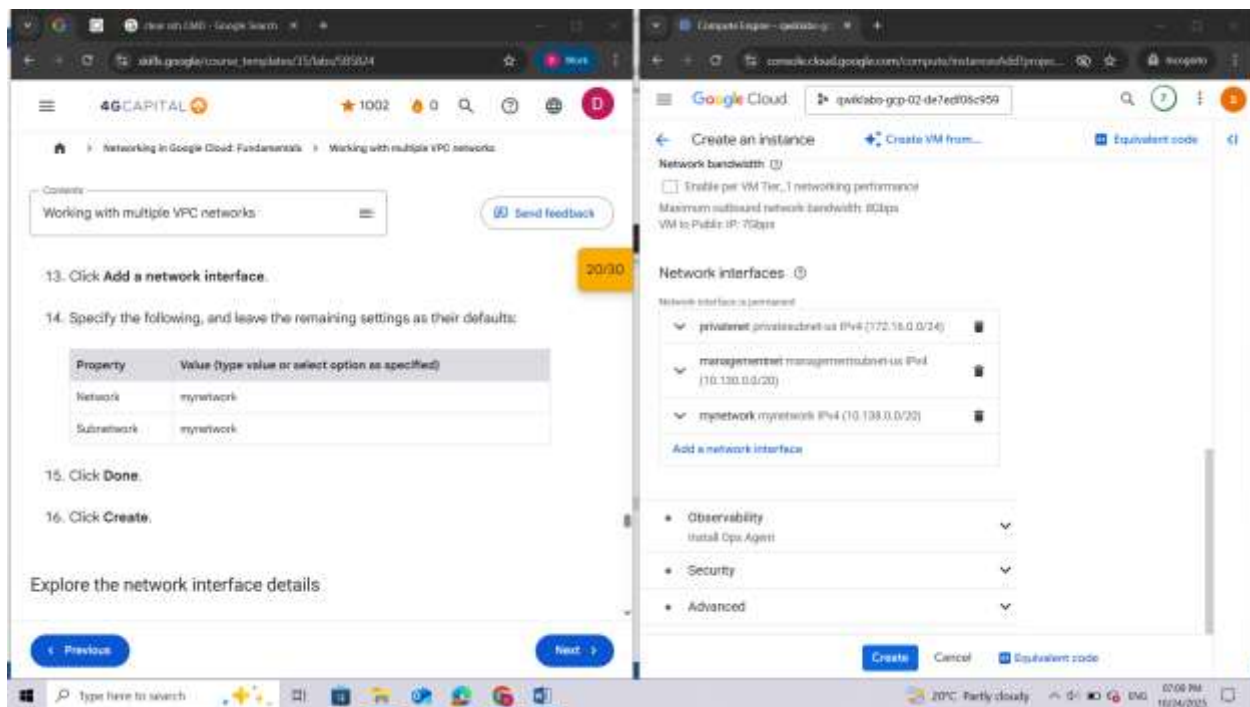
13. Click **Add a network interface**.

14. Specify the following, and leave the remaining settings as their defaults:

Property	Value (type value or select option as specified)
Network	mynetwork
Subnetwork	mynetwork

15. Click **Done**.

16. Click **Create**.



Explore the network interface details

Explore the network interface details of **vm-appliance** within the Cloud console and within the VM's terminal.

1. In the Cloud console, in the **Navigation menu**, click **Compute Engine > VM instances**.
2. To open the **Network interface details** page, in the **Internal IP** address of **vm-appliance**, click **nic0**.
3. Verify that **nic0** is attached to **privatesubnet-us**, is assigned an internal IP address within that subnet (172.16.0.0/24), and has applicable firewall rules.

The screenshot shows the Google Cloud Platform console for a Network Interface (NIC) named **nic0**. The left sidebar lists various VPC Network components. The main content area displays details for the selected network interface.

Selected network interface: **nic0**

Network interface details

Name	Network	Subnetwork	Primary internal IP address	Alias IP ranges	IP stack type	External IP address	Network Service Tier
nic0	privatenet	privatesubnet-us	172.16.0.4	—	IPv4	34.137.19.82	Premium

VM instance details

Name	Zone	Network tags	Service account	IP forwarding
vm-appliance	us-west1-c	None	275665775667-compute@developer.gserviceaccount.com	Off

Firewall and routes details

Firewalls

Name	Enforcement order	Type	Deployment scope	Rule priority	Source	Destination	Protocols and ports	Action
firewall	1	Hierarchical firewall policy	Global					
vpc-firewall-rules	2	VPC firewall rules	Global					

4. Click **nic0** and select **nic1**.

5. Verify that **nic1** is attached to **managementsubnet-us**, is assigned an internal IP address within that subnet (10.130.0.0/20), and has applicable firewall rules.

The screenshot shows the Google Cloud Platform console for a Network Interface (NIC) named **nic1**. The left sidebar lists various VPC Network components. The main content area displays details for the selected network interface.

Selected network interface: **nic1**

Network interface details

Name	Network	Subnetwork	Primary internal IP address	Alias IP ranges	IP stack type	External IP address	Network Service Tier
nic1	managenetnet	managementsubnet-us	10.130.0.3	—	IPv4	35.185.187.156	Premium

VM instance details

Name	Zone	Network tags	Service account	IP forwarding
vm-appliance	us-west1-c	None	275665775667-compute@developer.gserviceaccount.com	Off

Firewall and routes details

Firewalls

Name	Enforcement order	Type	Deployment scope	Rule priority	Source	Destination	Protocols and ports	Action
firewall	1	Hierarchical firewall policy	Global					
vpc-firewall-rules	2	VPC firewall rules	Global					

6. Click **nic1** and select **nic2**.

7. Verify that **nic2** is attached to **mynetwork**, is assigned an internal IP address within that subnet (10.128.0.0/20), and has applicable firewall rules. The subnet shown may differ depending on the region selected during VM creation.

The screenshot shows the Google Cloud console interface for the 'Network interface details' of a VM instance named 'vm-appliance'. The selected network interface is 'nic2'. The page displays the following details:

Network interface details

Name	Network	Subnetwork	Primary internal IP address	Alias IP ranges	IP stack type	External IP address	Network Service Tier
nic2	mynetwork	mynetwork	10.128.0.4	—	IPv4	34.160.82.52	Premium

VM instance details

Name	Zone	Network tag	Service account	IP forwarding
vm-appliance	us-west1-c	None	275865775867-compute@developer.gserviceaccount.com	Off

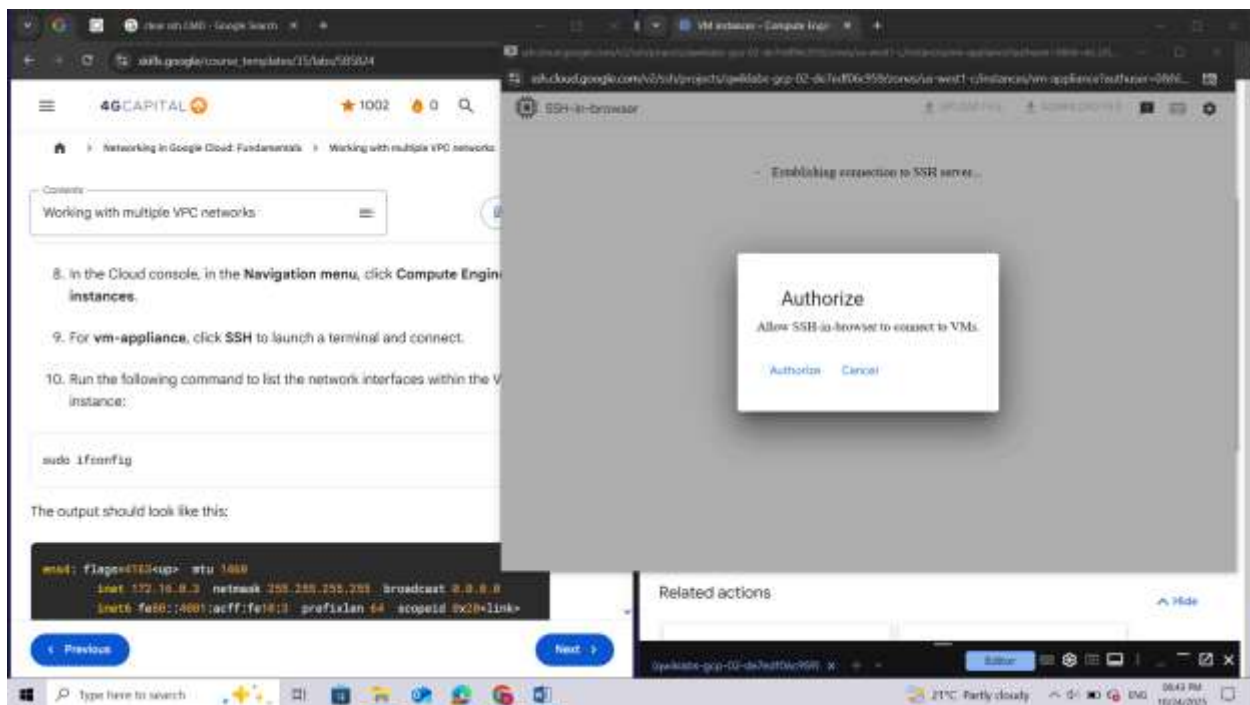
Firewall and routes details

Filter: Enter property name or value

Name	Enforcement order	Type	Deployment scope	Rule priority	Source	Destination	Protocols and ports	Action
firewall	1	Hierarchical firewall policy	Global					
vpc-firewall-rules	2	VPC firewall rules	Global					

Note: Each network interface has its own internal IP address so that the VM instance can communicate with those networks.

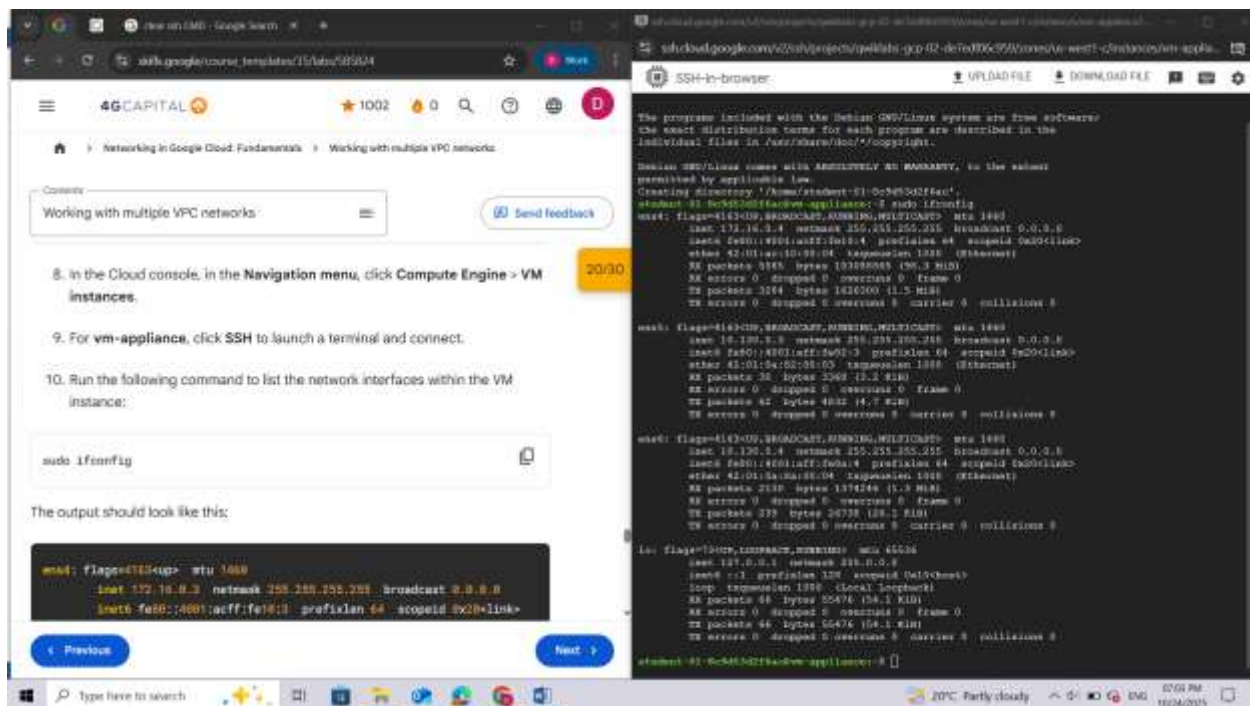
8. In the Cloud console, in the **Navigation menu**, click **Compute Engine > VM instances**.
9. For **vm-appliance**, click **SSH** to launch a terminal and connect.



10. Run the following command to list the network interfaces within the VM instance:

`sudo ifconfig`

Note: The `sudo ifconfig` command lists a Linux VM's network interfaces with the internal IP addresses for each interface.



Explore the network interface connectivity

Demonstrate that the **vm-appliance** instance is connected to **privatesubnet-us**, **managementsubnet-us**, and **mynetwork** by pinging VM instances on those subnets.

Which instance(s) should you be able to ping from vm-appliance using internal IP addresses?

☐

mynet-notus-vm

☒

mynet-us-vm

☒

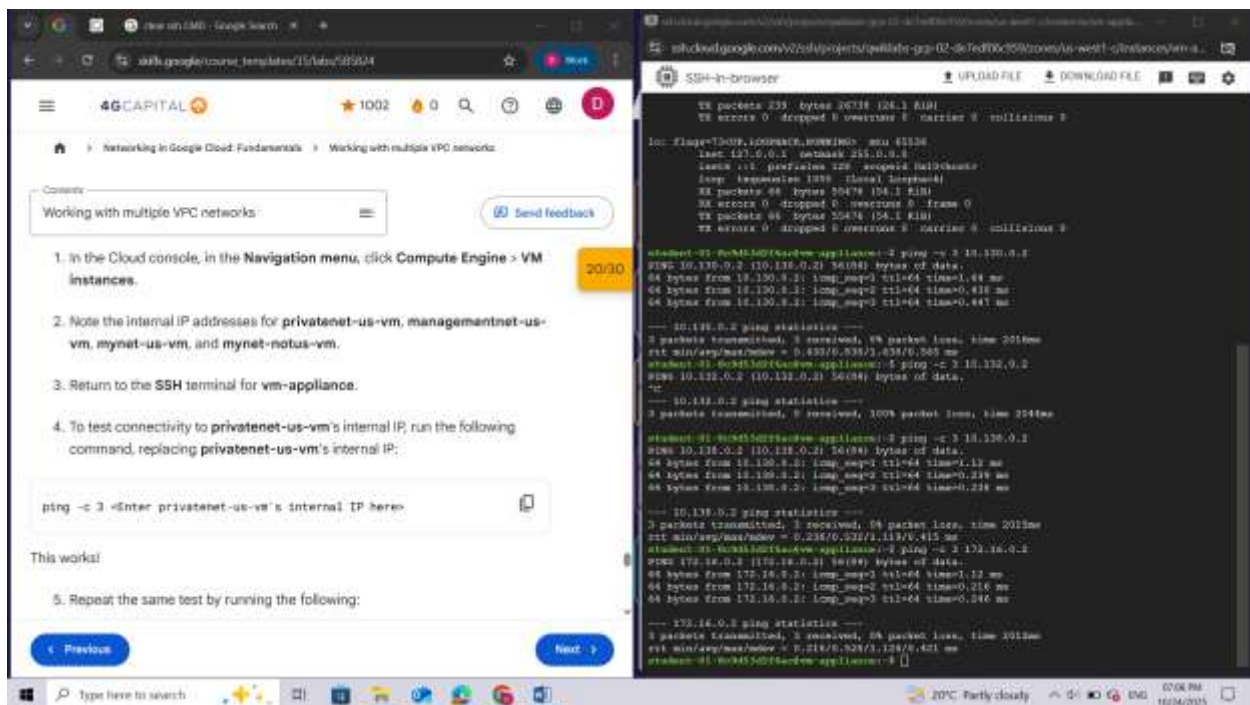
privatenet-us-vm

☒

managementnet-us-vm

1. In the Cloud console, in the **Navigation menu**, click **Compute Engine > VM instances**.
2. Note the internal IP addresses for **privatenet-us-vm**, **managementnet-us-vm**, **mynet-us-vm**, and **mynet-notus-vm**.
3. Return to the **SSH** terminal for **vm-appliance**.
4. To test connectivity to **privatenet-us-vm**'s internal IP, run the following command, replacing **privatenet-us-vm**'s internal IP:

ping -c 3 <Enter privatenet-us-vm's internal IP here>



5. Repeat the same test by running the following:

`ping -c 3 privatenet-us-vm`

Note: You can ping **privatenet-us-vm** by its name because VPC networks have an internal DNS service that allows you to address instances by their DNS names instead of their internal IP addresses. When an internal DNS query is made with the instance hostname, it resolves to the primary interface (nic0) of the instance. Therefore, this only works for **privatenet-us-vm** in this case.

```
student-01-8c9d53d2f6ac@vm-appliance:~$ ping -c 3 privatenet-us-vm
PING privatenet-us-vm.us-west1-c.c.qwiklabs-gcp-02-de7edf06c959.internal (172.16.0.2) 56(84)
bytes of data.
64 bytes from privatenet-us-vm.us-west1-c.c.qwiklabs-gcp-02-de7edf06c959.internal (172.16.0.
2): icmp_seq=1 ttl=64 time=0.924 ms
64 bytes from privatenet-us-vm.us-west1-c.c.qwiklabs-gcp-02-de7edf06c959.internal (172.16.0.
2): icmp_seq=2 ttl=64 time=0.237 ms
64 bytes from privatenet-us-vm.us-west1-c.c.qwiklabs-gcp-02-de7edf06c959.internal (172.16.0.
2): icmp_seq=3 ttl=64 time=0.304 ms
--- privatenet-us-vm.us-west1-c.c.qwiklabs-gcp-02-de7edf06c959.internal ping statistics ---
3 packets transmitted, 3 received, 0% packet loss, time 2004ms
rtt min/avg/max/mdev = 0.237/0.528/1.139/0.412 ms
student-01-8c9d53d2f6ac@vm-appliance:~$
```

6. To test connectivity to **managementnet-us-vm**'s internal IP, run the following command, replacing **managementnet-us-vm**'s internal IP:

`ping -c 3 <Enter managementnet-us-vm's internal IP here>`

7. To test connectivity to **myynet-us-vm**'s internal IP, run the following command, replacing **myynet-us-vm**'s internal IP:

`ping -c 3 <Enter myynet-us-vm's internal IP here>`

- To test connectivity to **mynet-notus-vm**'s internal IP, run the following command, replacing **mynet-notus-vm**'s internal IP:

ping -c 3 <Enter mynet-notus-vm's internal IP here>

Note: This does not work! In a multiple interface instance, every interface gets a route for the subnet that it is in. In addition, the instance gets a single default route that is associated with the primary interface `ens4`. Unless manually configured otherwise, any traffic leaving an instance for any destination other than a directly connected subnet will leave the instance via the default route on `ens4`.

- To list the routes for **vm-appliance** instance, run the following command:

ip route

```
student-01-8c9d53d2f6ac@vm-appliance:~$ ip route
default via 172.16.0.1 dev ens4 proto dhcp src 172.16.0.4 metric 100
10.130.0.0/20 via 10.130.0.1 dev ens5 proto dhcp src 10.130.0.3 metric 100
10.130.0.1 dev ens5 proto dhcp scope link src 10.130.0.3 metric 100
10.138.0.0/20 via 10.138.0.1 dev ens6 proto dhcp src 10.138.0.4 metric 100
10.138.0.1 dev ens6 proto dhcp scope link src 10.138.0.4 metric 100
169.254.169.254 via 172.16.0.1 dev ens4 proto dhcp src 172.16.0.4 metric 100
172.16.0.0/24 via 172.16.0.1 dev ens4 proto dhcp src 172.16.0.4 metric 100
172.16.0.1 dev ens4 proto dhcp scope link src 172.16.0.4 metric 100
student-01-8c9d53d2f6ac@vm-appliance:~$
```

Note: The primary interface `ens4` gets the default route (default via 172.16.0.1 dev `ens4`), and all three interfaces, `ens4`, `ens5`, and `ens6`, get routes for their respective subnets. Because the subnet of **mynet-notus-vm** (10.132.0.0/20) is not included in this routing table, the ping to that instance leaves **vm-appliance** on `ens4` (which is on a different VPC network).

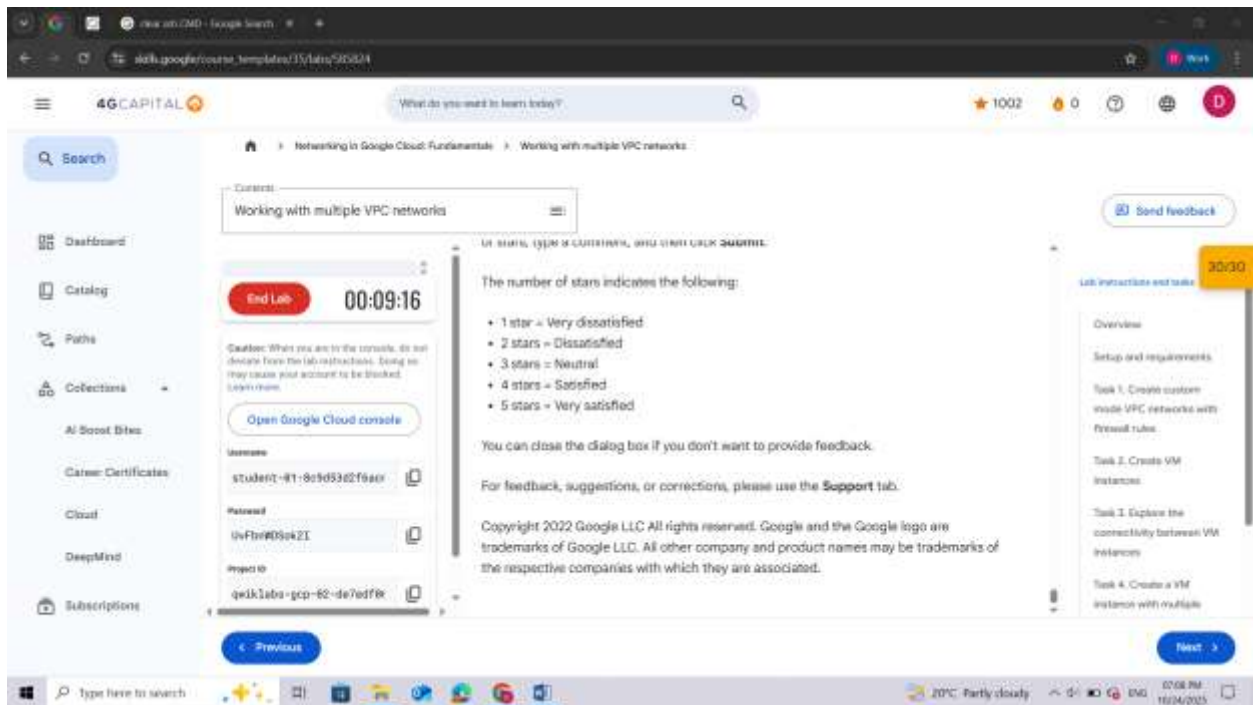
Learn more about how you can change this behavior by configuring policy routing from the [Creating instances with multiple network interfaces guide](#).

Review

In this lab, you created several custom mode VPC networks, firewall rules, and VM instances using the Cloud console and the `gcloud` command line. Then you tested the connectivity across VPC networks, which worked when pinging external IP addresses but not when pinging internal IP addresses. Thus you created a VM instance with three network interfaces and verified internal connectivity for VM instances that are on the subnets that are attached to the multiple interface VM.

End your lab

When you have completed your lab, click **End Lab**. Google Cloud Skills Boost removes the resources you've used and cleans the account for you.



QUIZ

- ✓ 1. You are designing a virtual machine in the cloud to act as a network gateway between an external public network and a private internal network. To ensure strong security and traffic separation, what technology can you implement?
- ☐ Cloud VPN
 - ✓ ☒ Multiple Network Interface Cards (NICs)
 - ☐ VLAN tagging within a single NIC
 - ☐ Premium Tier IPs

Correct! Multiple NICs attached to separate VPC networks achieve the strongest traffic isolation and control for the gateway scenario.

- ✓ 2. You want to lower cloud networking cost and have no problem leveraging the public internet for cross-region traffic. Which network service tier is best for you?

☐ Premium tier

☐ Pro version

✓ ☒ Standard tier

☐ Prime tier

Correct! Most cost-effective, suitable for cost-conscious users willing to use public internet routing.

- ✓ 3. You want to improve network performance. You are not comfortable using the public internet to route traffic. Which service tier is the best fit?

☐ Standard tier

☐ Pro version

☐ Prime tier

✓ ☒ Premium tier

Correct! Premium Tier improves performance and leverages the Google network

LAB 2: CONFIGURING VPC NETWORK PEERING

Overview

In this lab, you configure VPC network peering between two networks. Then you verify private communication between two VMs in those networks, as illustrated in this diagram.



VPC network peering allows you to build SaaS (Software as a service) ecosystems in Google Cloud, which makes services available privately across different VPC networks within and across organizations. This allows workloads to communicate in private RFC 1918 space.

VPC network peering gives you several advantages over using external IP addresses or VPNs to connect networks, including:

- **Network latency:** Public IP networking results in higher latency than private networking.
- **Network security:** Service owners do not need to have their services exposed to the public internet and deal with its associated risks.
- **Network cost:** Google Cloud charges [egress bandwidth pricing](#) for networks using external IPs to communicate, even if the traffic is within the same zone. If, however, the networks are peered, they can use internal IPs to communicate and save on those egress costs. [Regular network pricing](#) still applies to all traffic.

Objectives

In this lab, you will learn how to perform the following tasks:

- Explore connectivity between non-peered VPC networks.
- Configure VPC network peering.
- Verify private communication between peered VPC networks.
- Delete VPC network peering.

Setup and requirements

For each lab, you get a new Google Cloud project and set of resources for a fixed time at no cost.

1. Click the **Start Lab** button. If you need to pay for the lab, a pop-up opens for you to select your payment method. On the left is the **Lab Details** panel with the following:

- The **Open Google Cloud console** button
 - Time remaining
 - The temporary credentials that you must use for this lab
 - Other information, if needed, to step through this lab
2. Click **Open Google Cloud console** (or right-click and select **Open Link in Incognito Window** if you are running the Chrome browser).

The lab spins up resources, and then opens another tab that shows the **Sign in** page.

Tip: Arrange the tabs in separate windows, side-by-side.

Note: If you see the **Choose an account** dialog, click **Use Another Account**.

3. If necessary, copy the **Username** below and paste it into the **Sign in** dialog.

You can also find the **Username** in the **Lab Details** panel.

4. Click **Next**.

5. Copy the **Password** below and paste it into the **Welcome** dialog.

You can also find the **Password** in the **Lab Details** panel.

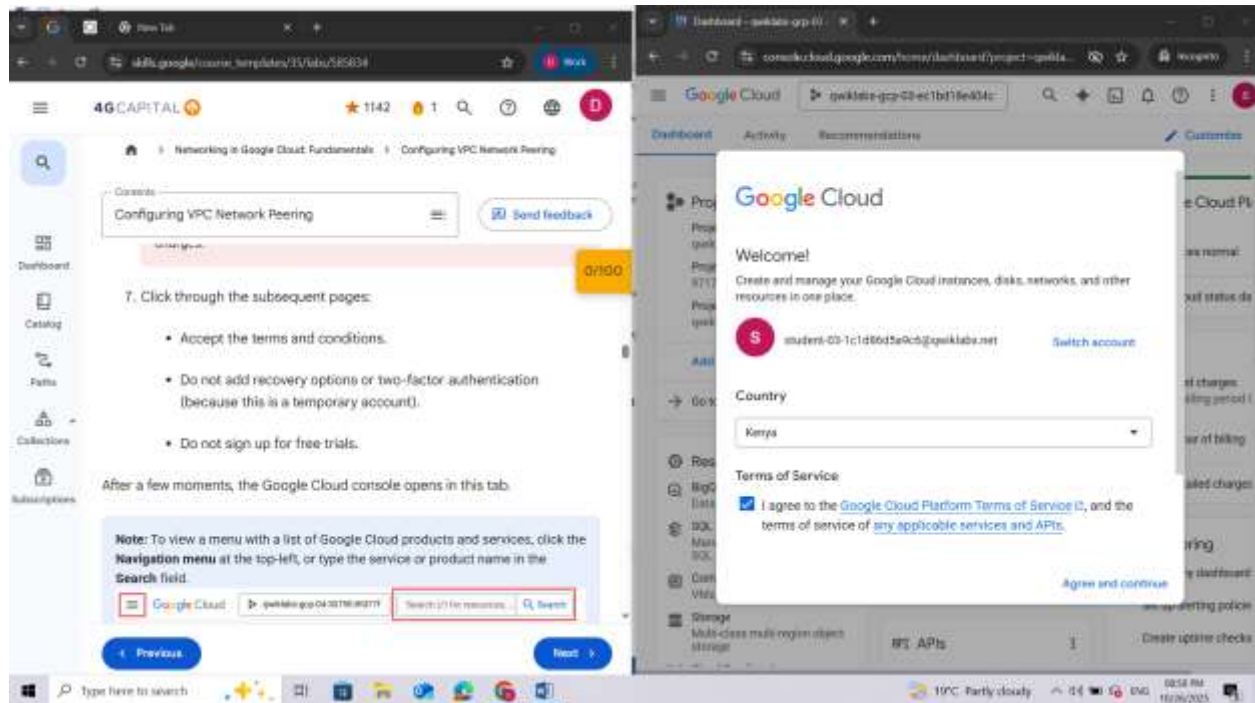
6. Click **Next**.

Important: You must use the credentials the lab provides you. Do not use your Google Cloud account credentials.**Note:** Using your own Google Cloud account for this lab may incur extra charges.

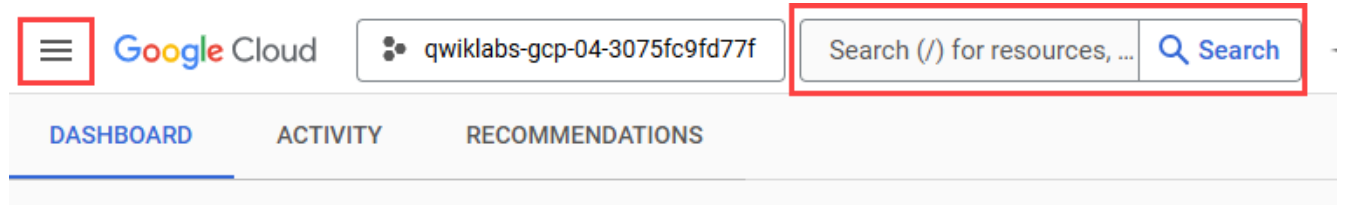
7. Click through the subsequent pages:

- Accept the terms and conditions.
- Do not add recovery options or two-factor authentication (because this is a temporary account).
- Do not sign up for free trials.

After a few moments, the Google Cloud console opens in this tab.



Note: To view a menu with a list of Google Cloud products and services, click the **Navigation menu** at the top-left, or type the service or product name in the **Search** field.



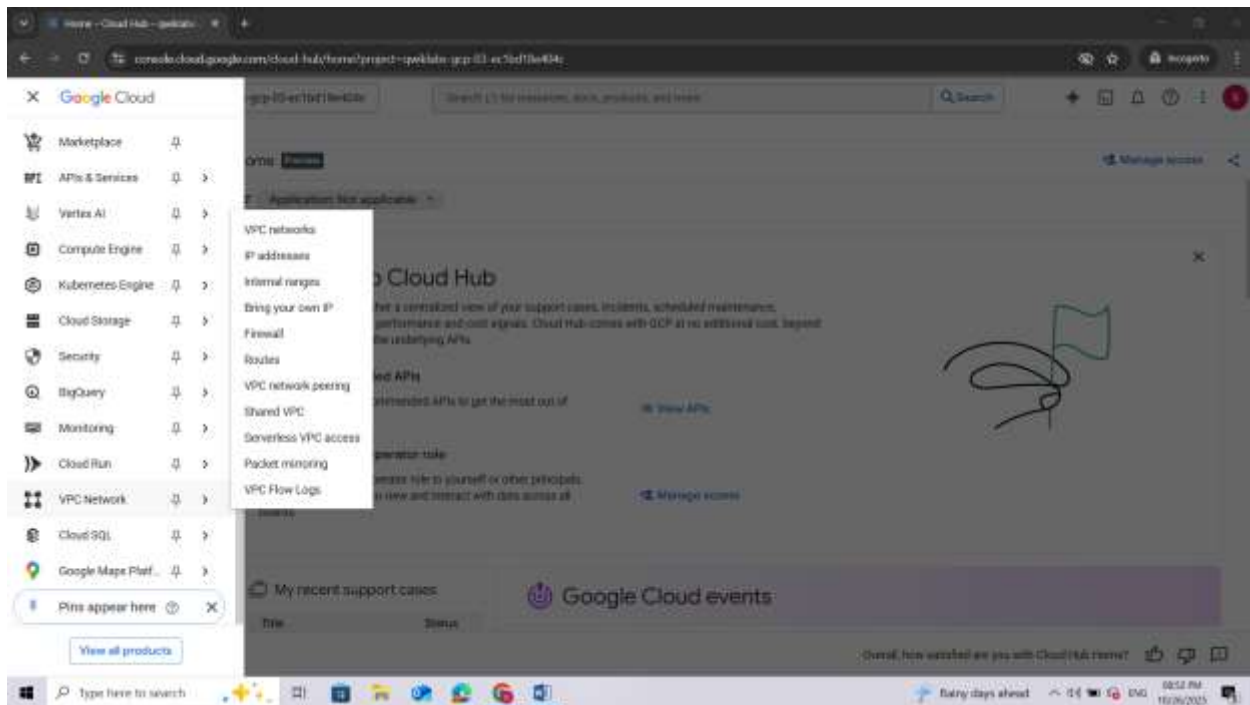
Task 1. Explore connectivity between non-peered VPC networks

Each Google Cloud project starts with the **default** network. In addition, **mynetwork**, **privatenet**, and **managementnet** have been created for you along with firewall rules to allow ICMP-SSH-RDP traffic and four VM instances.

Verify VPC network peering requirements

In a peered VPC network, no subnet IP range can overlap with another subnet IP range. Therefore, verify that the CIDR blocks of the subnets of **mynetwork** and **privatenet** are non-overlapping.

1. In the Google Cloud console, in the **Navigation menu** (≡), click **VPC network** > **VPC networks**.



2. Examine the **IP addresses ranges** for the subnets of **mynetwork**.

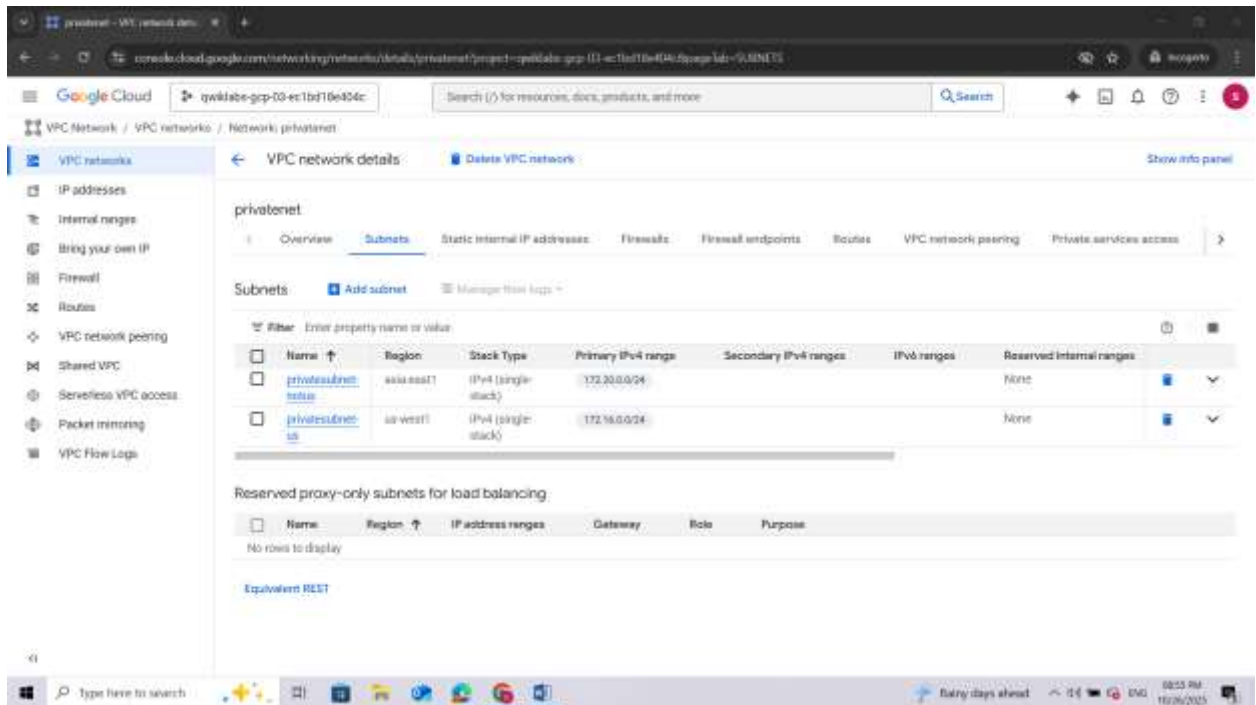
The subnets of **mynetwork** fit within the $10.128.0.0/9$ CIDR block. As new Google Cloud regions become available, new subnets in those regions are automatically added to this auto mode network using an IP range from that block.

The screenshot shows the 'Subnets' tab for the 'mynetwork' VPC network. The table lists subnets across various regions, each with a specific IP range. The subnets are all of type 'IPv4 (single-stack)' and are part of the 'mynetwork' project.

Name	Region	Stack Type	Primary IPv4 range	Secondary IPv4 ranges	IPv6 ranges	Reserved internal ranges
mynetwork	afro-south1	IPv4 (single-stack)	10.218.0.0/20		None	
mynetwork	asia-east1	IPv4 (single-stack)	10.140.0.0/20		None	
mynetwork	asia-south1	IPv4 (single-stack)	10.160.0.0/20		None	
mynetwork	asia-southeast1	IPv4 (single-stack)	10.140.0.0/20		None	
mynetwork	australia-southeast1	IPv4 (single-stack)	10.152.0.0/20		None	
mynetwork	centralus-east2	IPv4 (single-stack)	10.160.0.0/20		None	
mynetwork	eu-central-2	IPv4 (single-stack)	10.220.0.0/20		None	
mynetwork	eu-west-1	IPv4 (single-stack)	10.192.0.0/20		None	

3. Examine the **IP addresses ranges** for the subnets of **privatenet**.

The subnets of **privatenet** fit within the 172.16.0.0/24 and 172.20.0.0/24 CIDR blocks. They do not overlap with the 10.128.0.0/9 CIDR block of **mynetwork**.



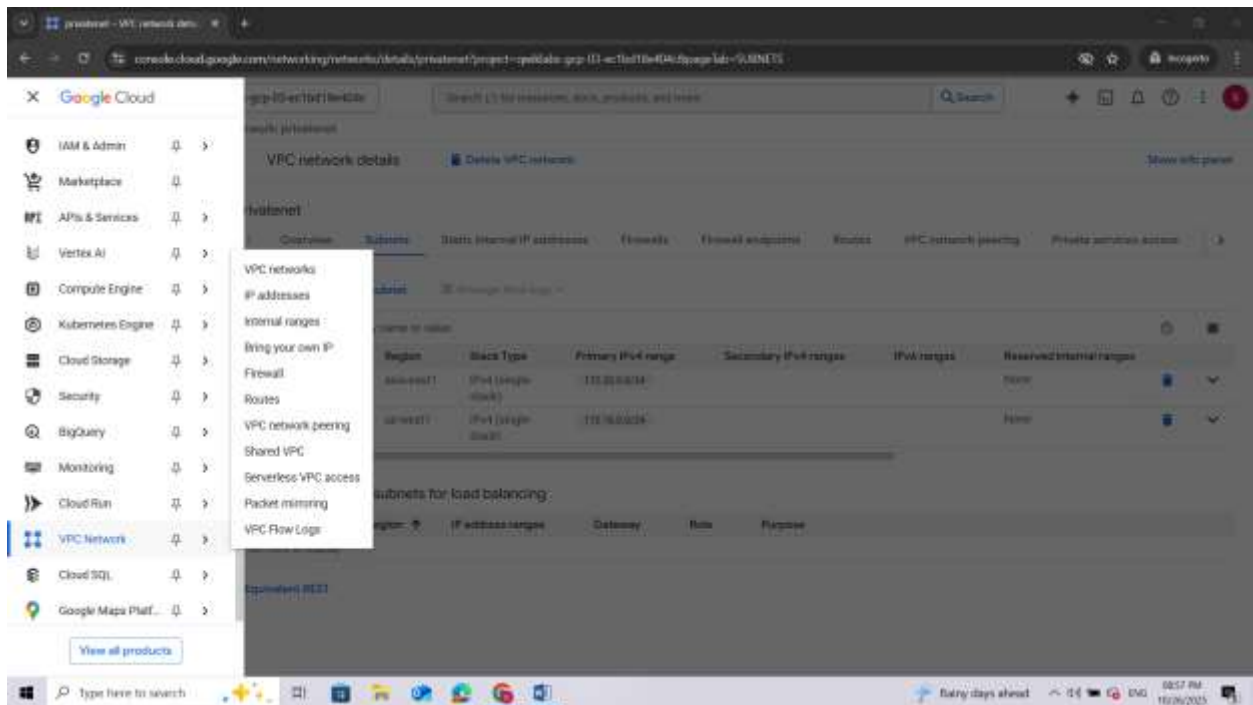
Note: You can configure VPC network peering between **mynetwork** and **privatenet** because their subnets' CIDR blocks are non-overlapping.

Explore the connectivity between **mynetwork** and **privatenet**

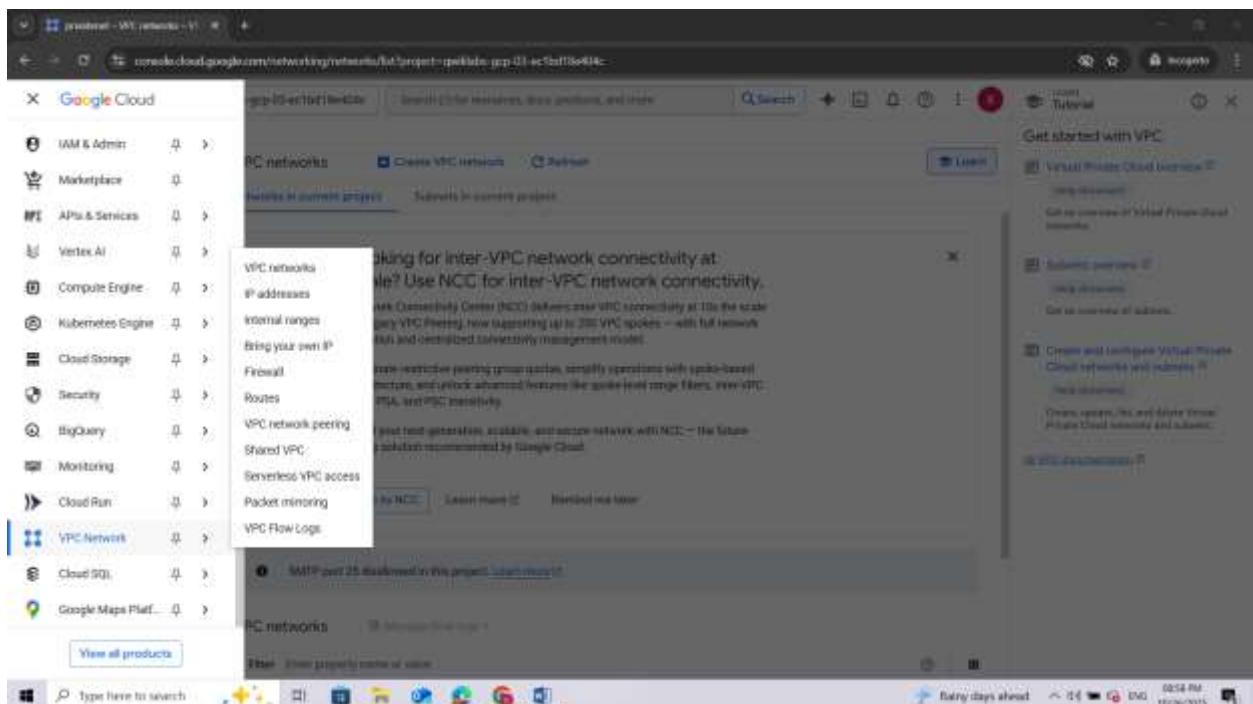
Before configuring VPC network peering, explore the current connectivity between **mynetwork** and **privatenet**.

1. On the **Navigation menu** () , click **VPC network > VPC network peering**. Notice that there is no peering connection.

You will return to this page to configure the VPC network peering connections.



2. On the Navigation menu, click **VPC network > Routes**.



3. Specify the following:

Property	Value (type value or select option as specified)
Network	mynetwork
Region	<filled at lab start>

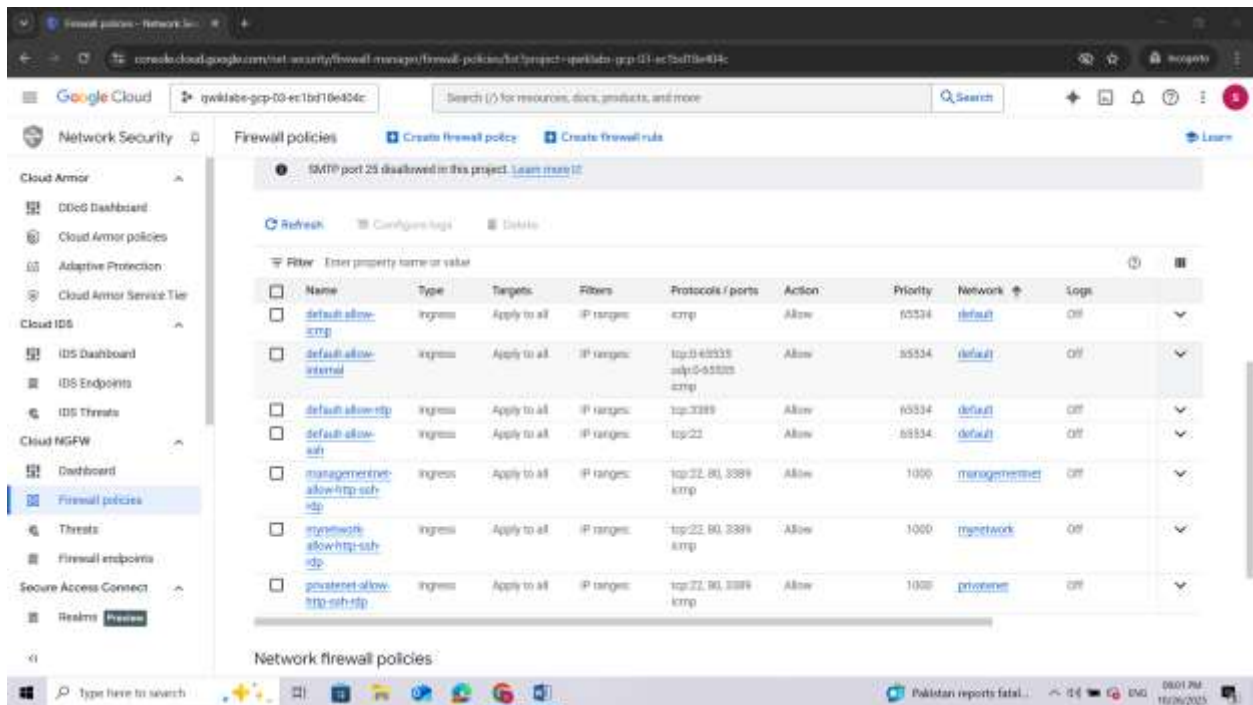
- Click **View**. Notice that none of the routes have a peering connection as the **Next hop**.

You will return to this page after configuring the VPC network peering connection.

The screenshot shows the Google Cloud Platform console interface. On the left is a navigation menu with options like 'VPC networks', 'IP addresses', 'Internal ranges', 'Bring your own IP', 'Firewall', 'Routes', 'VPC network peering', 'Shared VPC', 'Serverless VPC access', 'Packet mirroring', and 'VPC Flow Logs'. The 'Routes' option is selected. The main panel is titled 'Routes' and shows a table of routes for the network 'mynetwork' in the 'us-west1' region. The table has columns: Status, Name, Type, IP version, Destination IP range, Priority, Scope limits, and Next hop. The routes are managed by 'default route' instances. The 'Next hop' column shows various network destinations, including 'Default internet gateway' and 'Network mynetwork'.

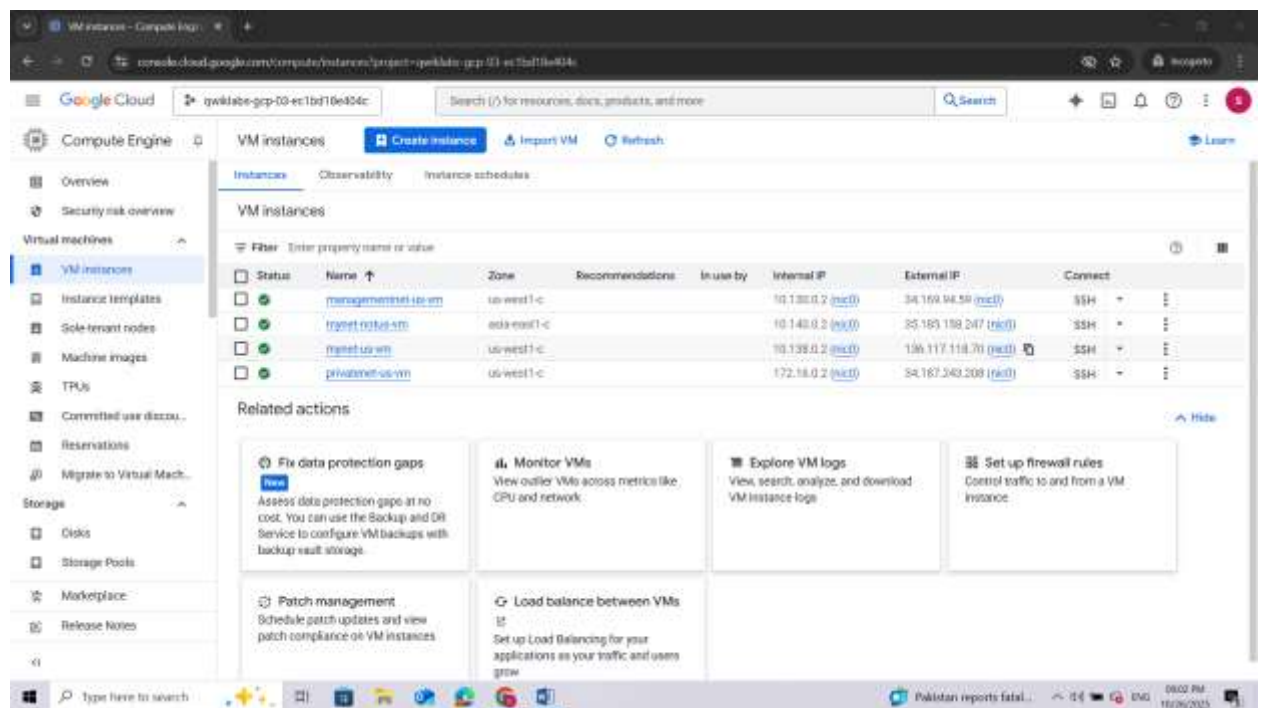
Status	Name	Type	IP version	Destination IP range	Priority	Scope limits	Next hop
✓	default route-e50256c194701e1b1	Static	IPv4	0.0.0.0/0	1000	—	Default internet gateway
✓	default route+03aea28189M28c3	Subnet	IPv4	10.132.0.0/20	0	—	Network mynetwork
✓	default route+0483460c4722c88	Subnet	IPv4	10.219.0.0/20	0	—	Network mynetwork
✓	default route+06c7e3488667e92	Subnet	IPv4	10.214.0.0/20	0	—	Network mynetwork
✓	default route+1933254505e4251	Subnet	IPv4	10.140.0.0/20	0	—	Network mynetwork
✓	default route+146145441979d288	Subnet	IPv4	10.189.0.0/20	0	—	Network mynetwork
✓	default route+1e740e0a0504d37	Subnet	IPv4	10.138.0.0/20	0	—	Network mynetwork
✓	default route+279cbe13500f32b	Subnet	IPv4	10.149.0.0/20	0	—	Network mynetwork
✓	default route+43ae42932f6c17bd	Subnet	IPv4	10.182.0.0/20	0	—	Network mynetwork
✓	default route+5c2678103197144	Subnet	IPv4	10.182.0.0/20	0	—	Network mynetwork
✓	default route+788dfe72a1371ds	Subnet	IPv4	10.190.0.0/20	0	—	Network mynetwork
✓	default route+7a8e9c1d4880118d	Subnet	IPv4	10.164.0.0/20	0	—	Network mynetwork
✓	default route+a08a2692aacb1a22	Subnet	IPv4	10.210.0.0/20	0	—	Network mynetwork
✓	default route+g1f1caedac0326e	Subnet	IPv4	10.224.0.0/20	0	—	Network mynetwork
✓	default route+ac1e1142011812d4d4	Subnet	IPv4	10.180.0.0/20	0	—	Network mynetwork

- On the **Navigation menu**, click **VPC network > Firewall**. Notice the allow **SSH** and **ICMP** firewall rules under Protocol / ports for **mynetwork** and **privatenet**. These firewall rules have been created for you.



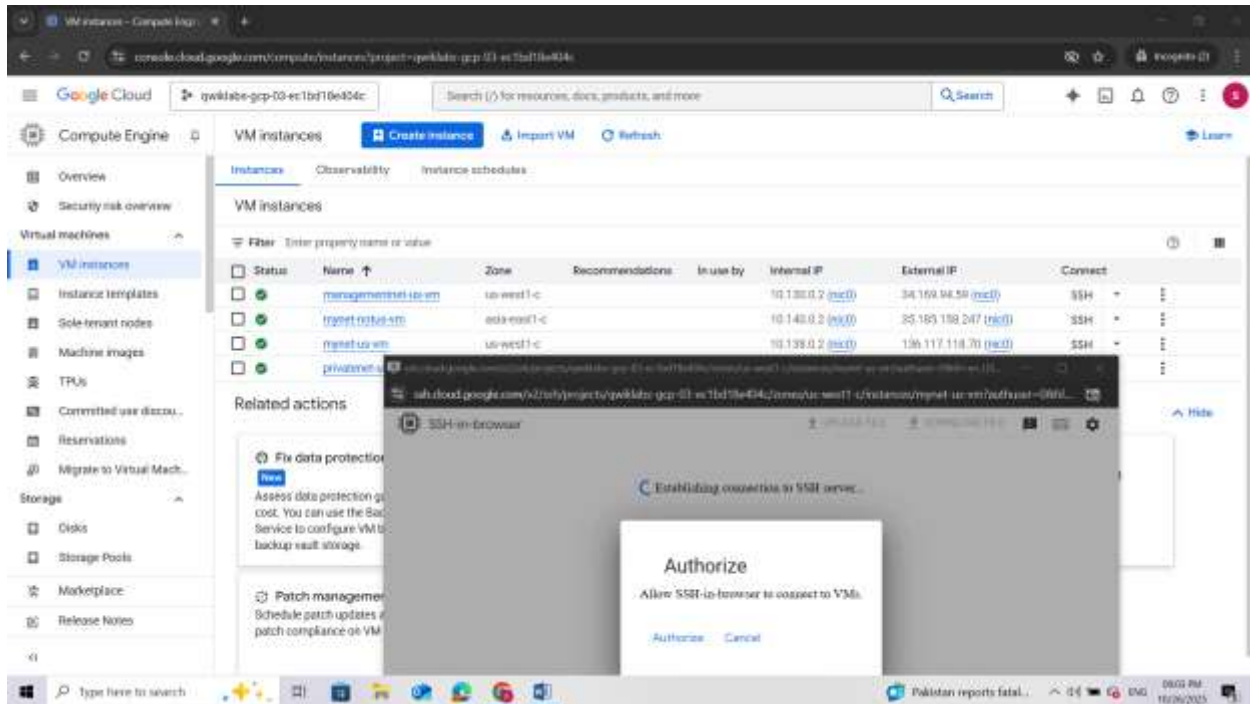
- On the **Navigation menu**, click **Compute Engine > VM instances**. Notice the **mytenetnet-us-vm**, **mytenet-us-vm**, **privatenet-us-vm**, and **managementnet-us-vm** instances.

These VM instances have been created for you.



7. Note the internal and external IP addresses for **privatenet-us-vm**.
172.16.0.2
34.187.243.208

8. For **myinet-us-vm**, click **SSH** to launch a terminal and connect.



9. To test connectivity to **privatenet-us-vm**'s external IP, run the following command, replacing **privatenet-us-vm**'s external IP:

```
ping -c 3 <Enter privatenet-us-vm's external IP here>
```

This should work!

10. To test connectivity to **privatenet-us-vm**'s internal IP, run the following command, replacing **privatenet-us-vm**'s internal IP:

```
ping -c 3 <Enter privatenet-us-vm's internal IP here>
```

Note: This should not work, as indicated by a 100% packet loss! But why?

exit

Task 2. Configure VPC network peering

VPC network peering can be configured for different VPC networks within and across organizations. Configure the following peering connections in this project:

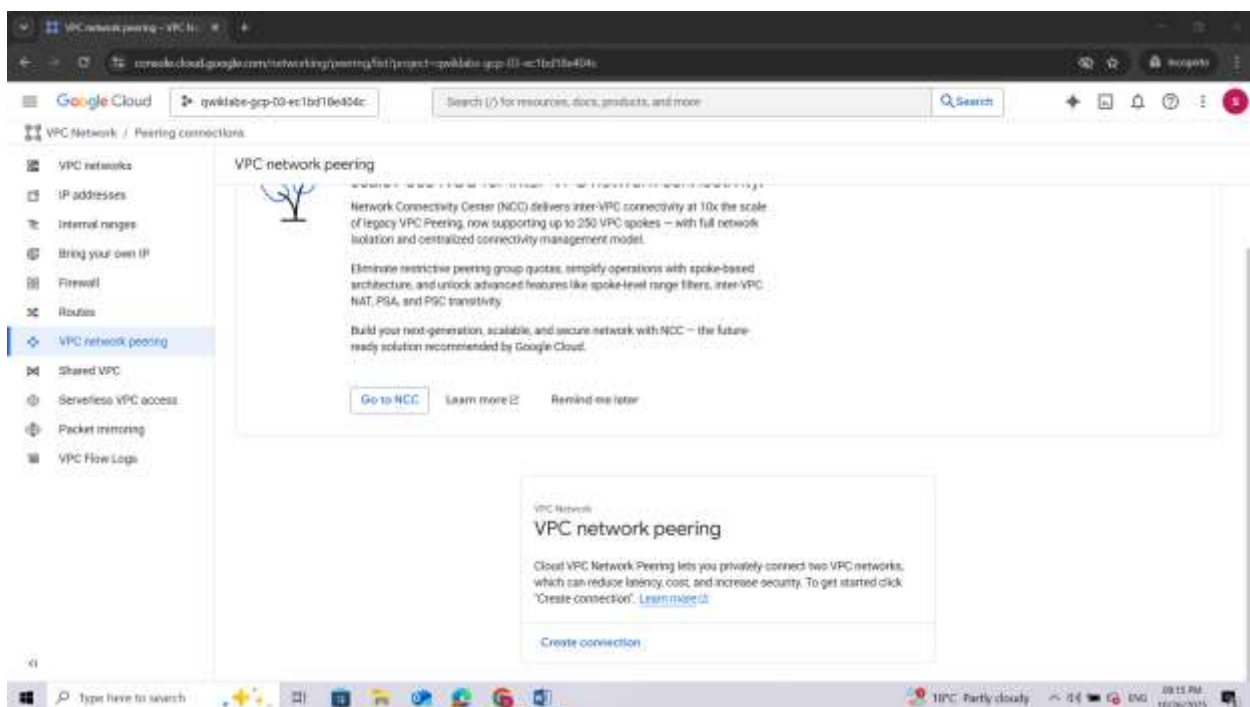
- **peering-1-2:** Peer **mynetwork** with **privatenet**
- **peering-2-1:** Peer **privatenet** with **mynetwork**

Each side of a peering association is set up independently. Peering is active only when the configuration from both sides matches.

Create peering 1-2

Peer **mynetwork** with **privatenet**.

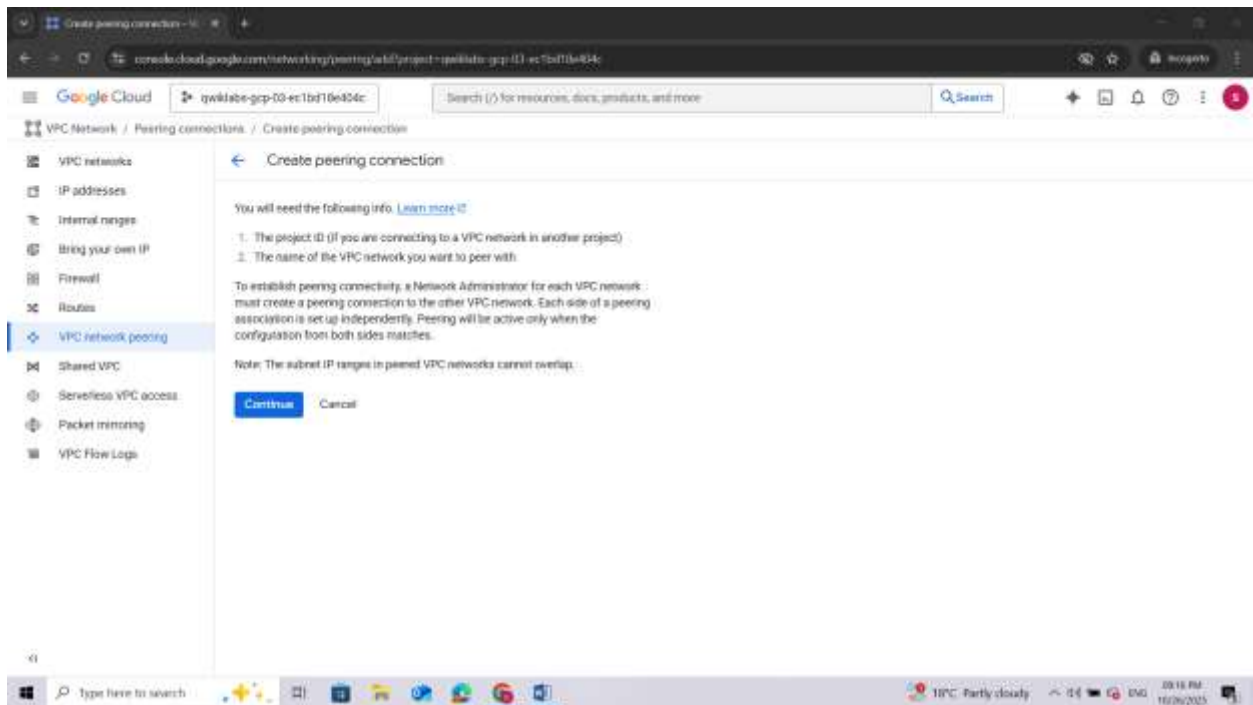
1. In the Cloud console, in the **Navigation menu** () , click **VPC network** > **VPC network peering**.
2. Click **Create connection**.



3. Read through the requirements.

Note: You won't need the **project ID** because you are connecting to a VPC network within the same project.

4. Click **Continue**.

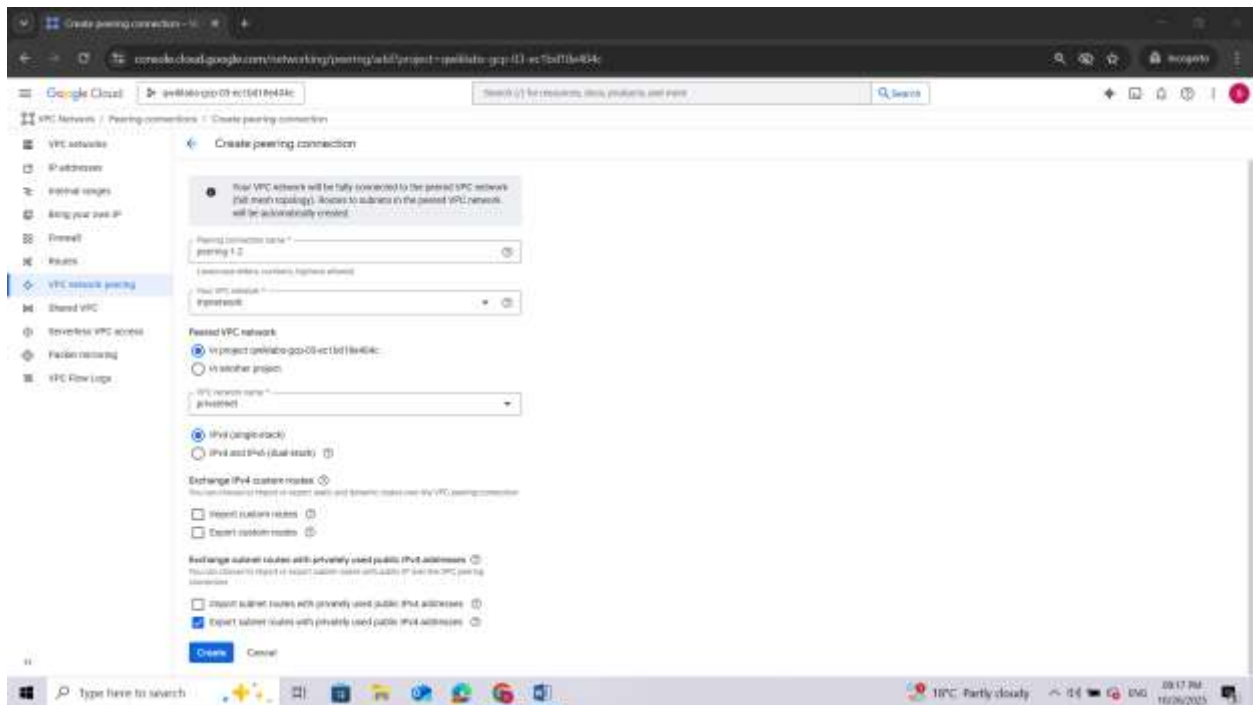


5. Specify the following, and leave the remaining settings as their defaults:

Property	Value (type value or select option as specified)
Name	peering-1-2
Your VPC Network	mynetwork
VPC network name	privatenet

Note: If you peered with a VPC network in another project, you would need the **Network admin** or **Project owner/editor** IAM role for that project to create a peering connection.

6. Click **Create**.



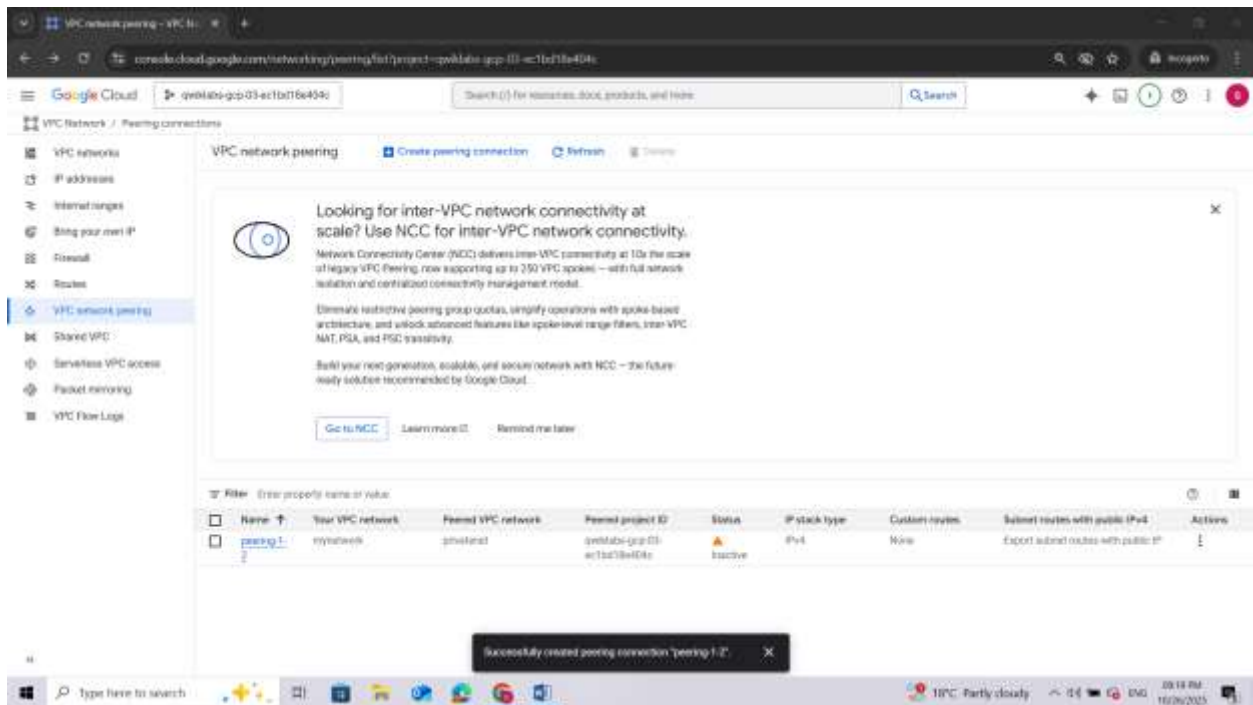
Note: At this point, the peering status remains **INACTIVE** because the other side has not been configured, the networks are not yet peered.

Filter: Enter property name or value								
☐	Name ↑	Your VPC network	Peered VPC network	Peered project ID	Status	IP stack type	Custom routes	Subnet routes with public IPv4
☐	peering-1-2	mynetwork	private1net	qwkljzbs-gcp-03-ec1bd1be454c	Inactive	IPv4	None	Export subnet routes with public IP

Create peering 2-1

Peer **privatenet** with **mynetwork**.

1. In the Cloud console, return to the **VPC network peering** page.
2. Click **Create peering connection**.

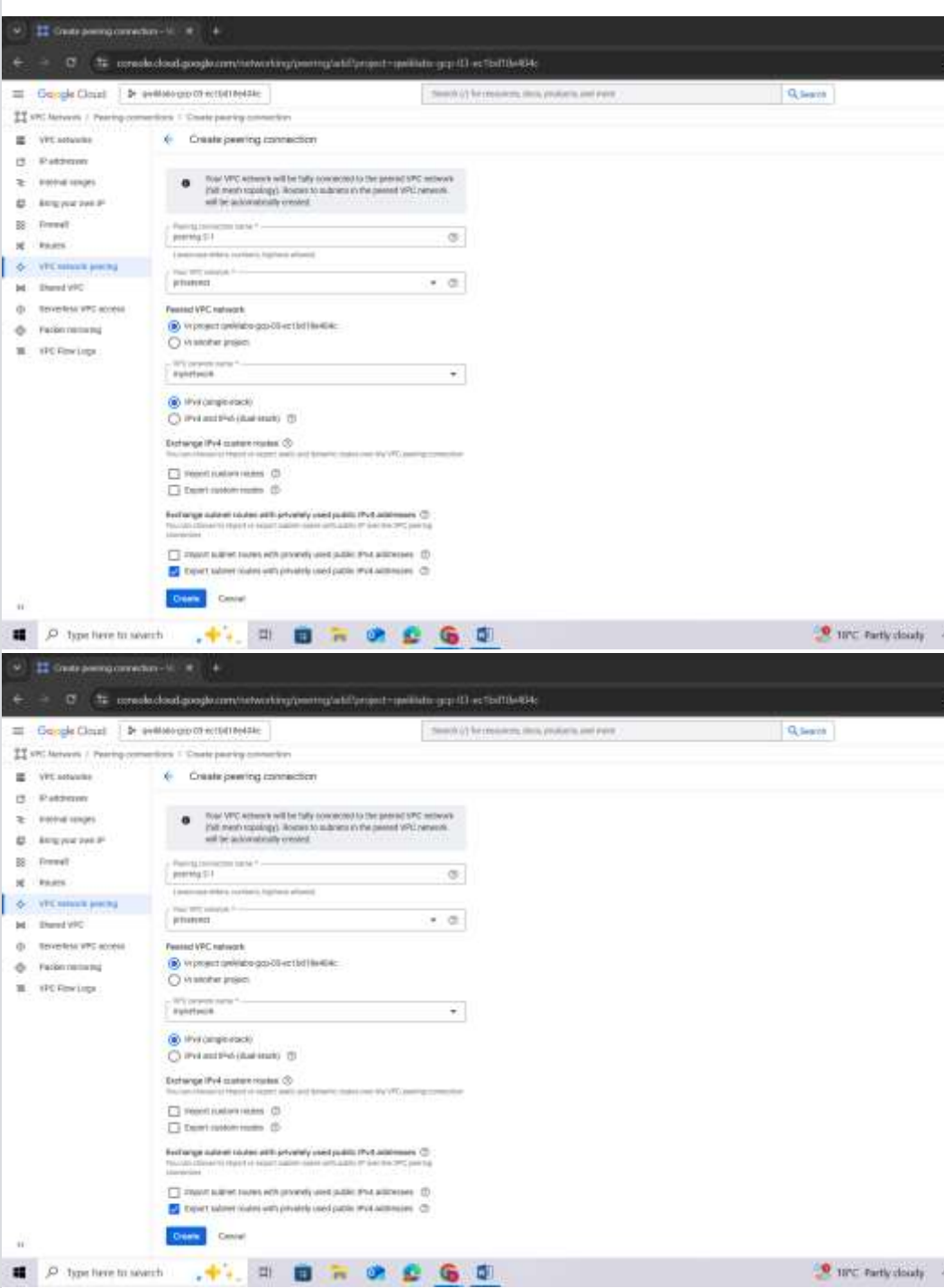


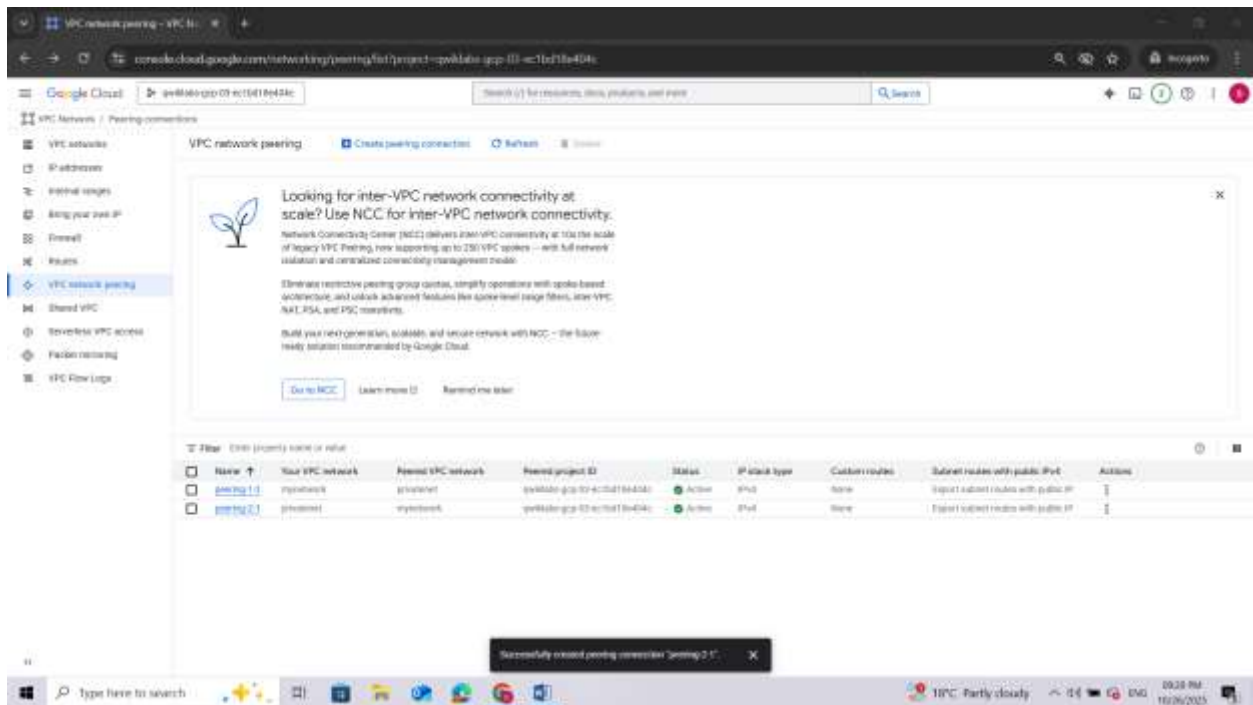
3. Click **Continue**.

4. Specify the following, and leave the remaining settings as their defaults:

Property	Value (type value or select option as specified)
----------	--

Name	peering-2-1
Your VPC Network	privatenet





Task 3. Verify private communication between peered VPC networks

Verify private RFC 1918 connectivity across **mynetwork** and **privatenet**.

Verify routes between networks

Verify that routes have been established between **mynetwork** and **privatenet**.

1. In the Cloud console, in the **Navigation menu** () , click **VPC network > Routes**.
2. For Network, select **mynetwork**.
3. For Region, select **<filled at lab start>**.



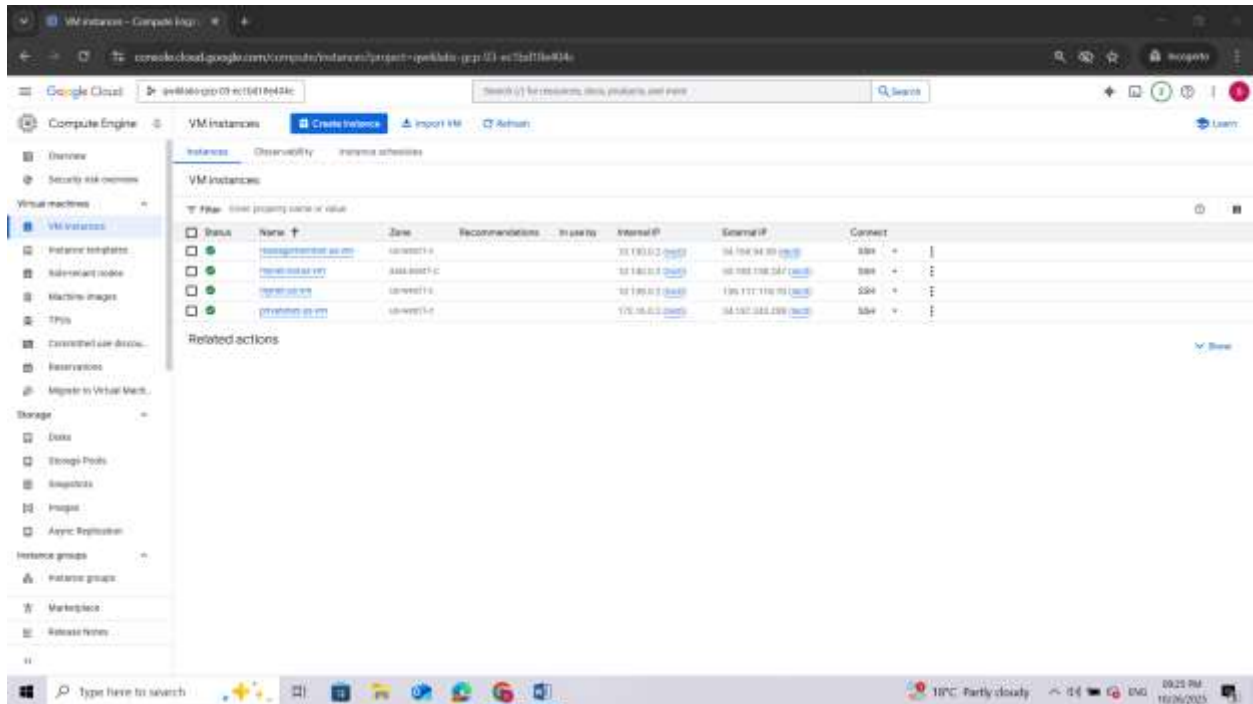
Notice that there is a route for each subnet in **mynetwork** with **peering-1-2** as the **Next hop**.

Note: User-configured routes are not propagated across peered networks. If you configure a route in a network to a destination in a VPC network, that destination will not be reachable from a peered network.

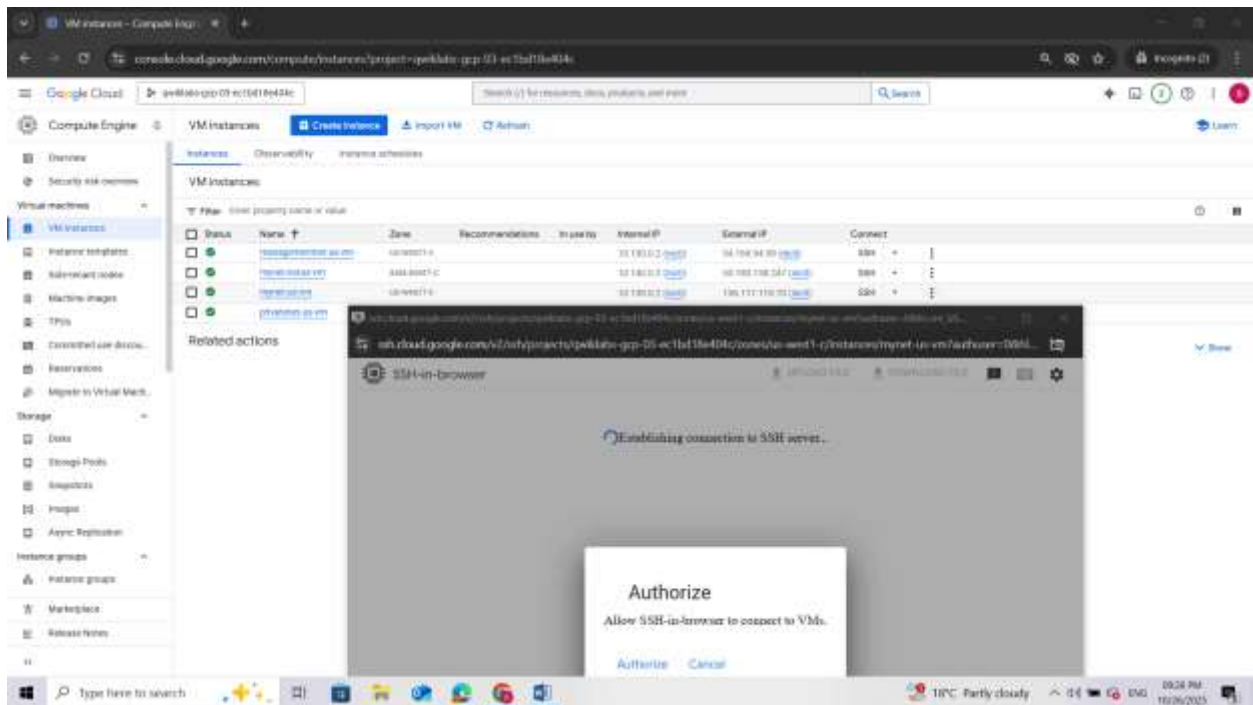
Ping from mynetwork to privatenet

Try to ping the internal IP of **privatenet-us-vm** from **myinet-us-vm**.

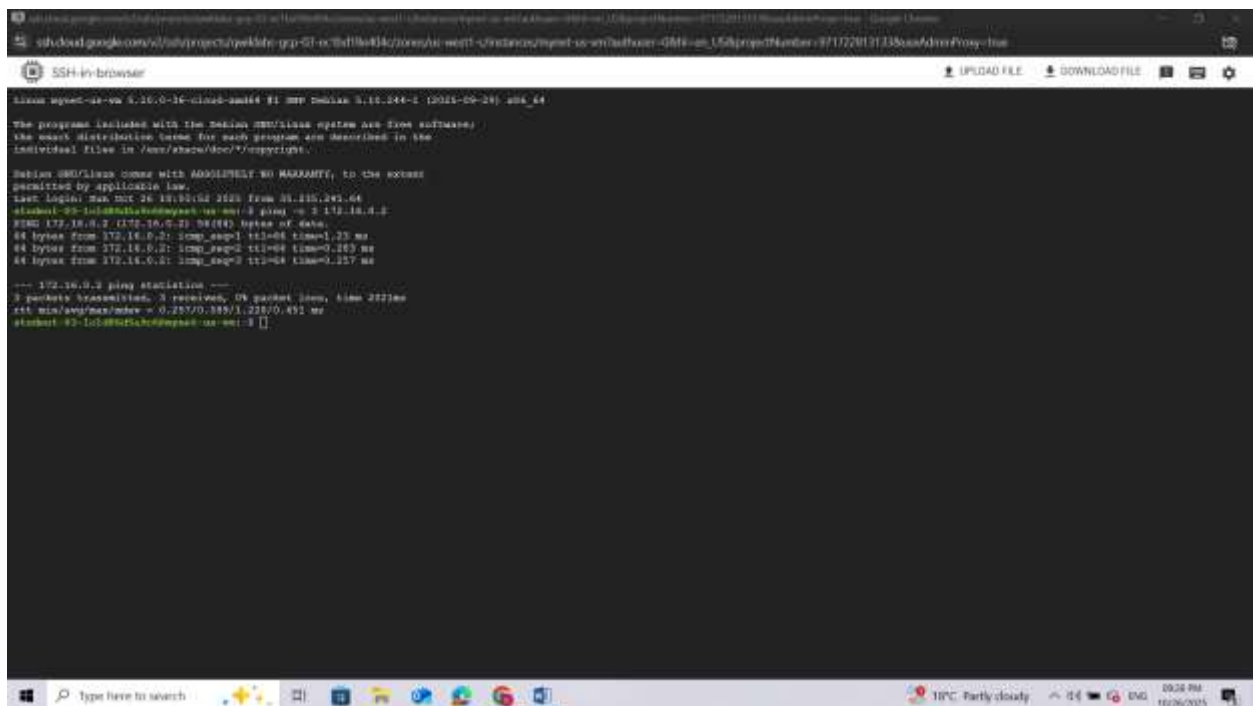
1. On the **Navigation menu**, click **Compute Engine > VM instances**.
2. Note the internal IP address for **privatenet-us-vm**.
172.16.0.2



3. For **myinet-us-vm**, click **SSH** to launch a terminal and connect.



- To test connectivity to **privatenet-us-vm's** internal IP, run the following command, replacing **privatenet-us-vm's** internal IP:
`ping -c 3 <Enter privatenet-us-vm's internal IP here>`
 This should work because of the route that was established by the peering connection.



- Please close the SSH terminal to **mytenant-us-vm** before verifying objective:

```
exit
```

Ping from privatenet to mynetwork

Similarly, try to ping the internal IP of **mynet-us-vm** from **privatenet-us-vm**.

1. Note the internal IP address for **mynet-us-vm**.
10.138.0.2
2. For **privatenet-us-vm**, click **SSH** to launch a terminal and connect.
3. To test connectivity to **mynet-us-vm**'s internal IP, run the following command, replacing **mynet-us-vm**'s internal IP:

```
ping -c 3 <Enter mynet-us-vm's internal IP here>
```

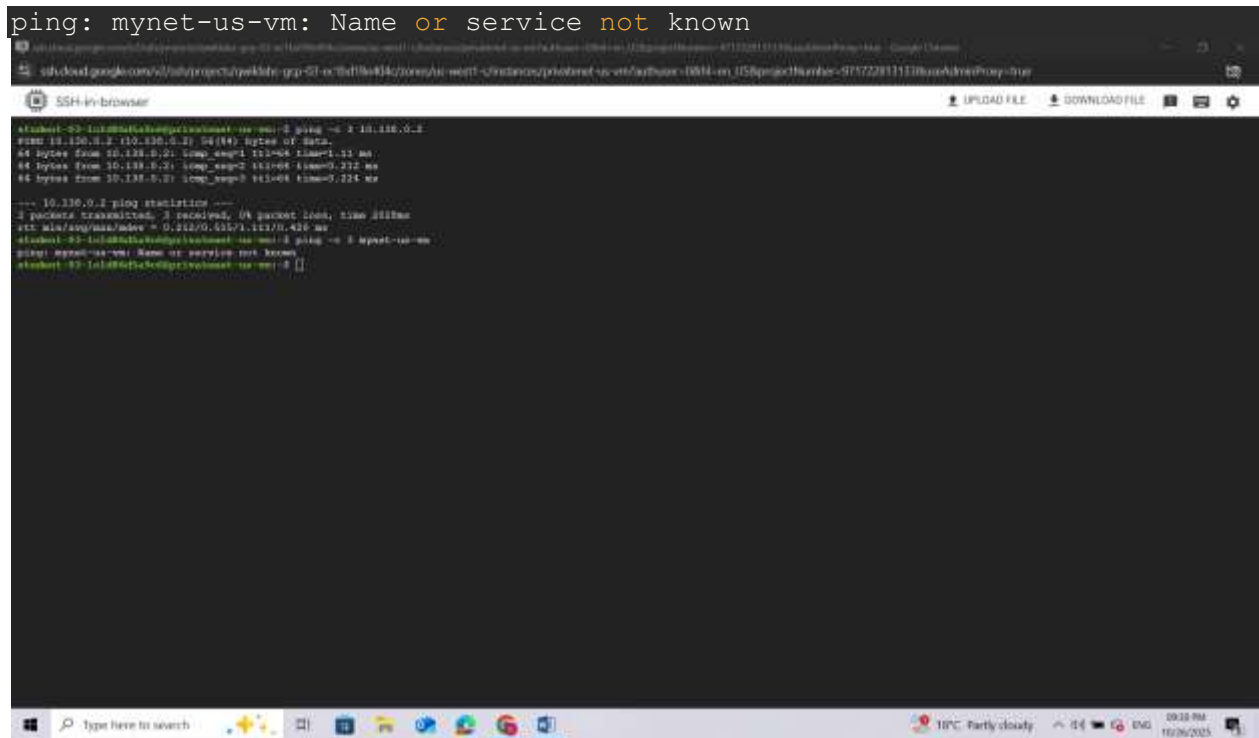
This should also work because of the route that was established by the peering connection.

4. To test Compute Engine DNS across peered networks, run the following command:

```
ping -c 3 mynet-us-vm
```

Output:

```
ping: mynet-us-vm: Name or service not known
```



Note: Compute Engine internal DNS names created in a network are not accessible to peered networks. The IP address of the VM should be used to reach the VM instances in peered network.

5. Please close the SSH terminal to **privatenet-us-vm** before verifying objective:

```
exit
```

Note: Now that you've verified the VPC Peering connection, you could stop both instances to remove their external IP addresses. This helps secure your instances and reduces your networking costs. You can still SSH to an instance without a public IP address using [Cloud IAP tunnels](#).

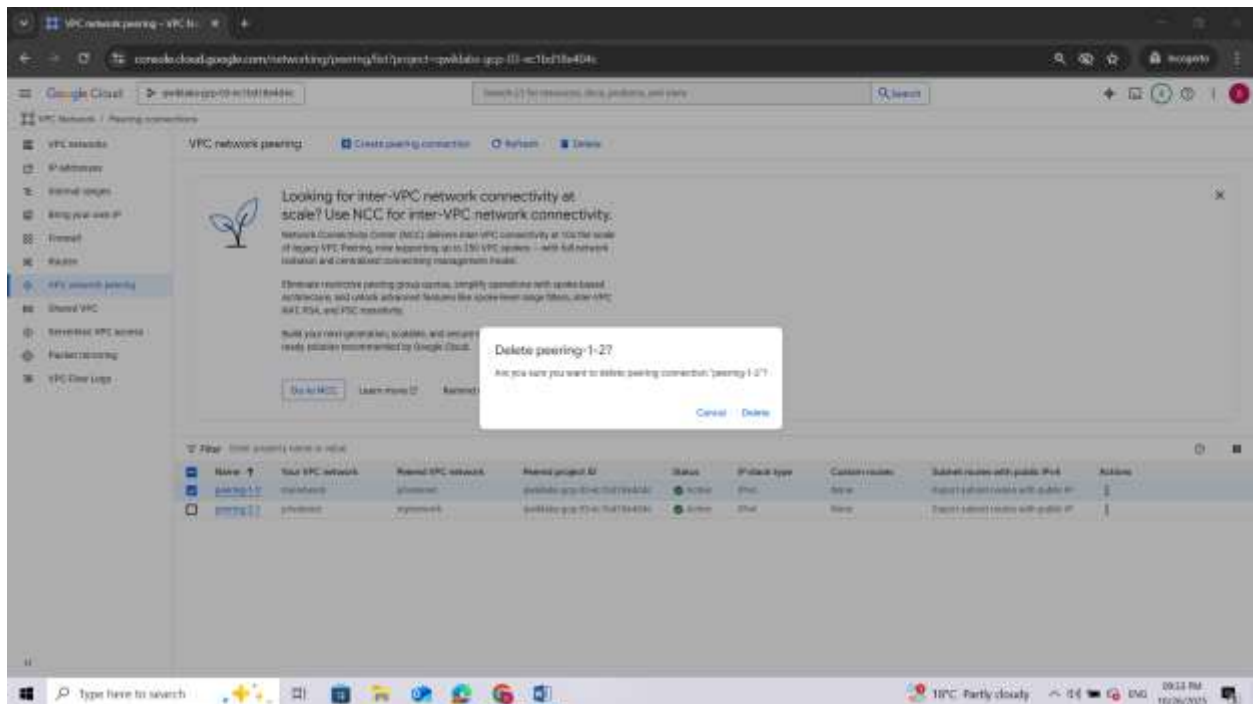
Task 4. Delete the VPC peering connection

Delete the VPC Peering connection and verify the deletion.

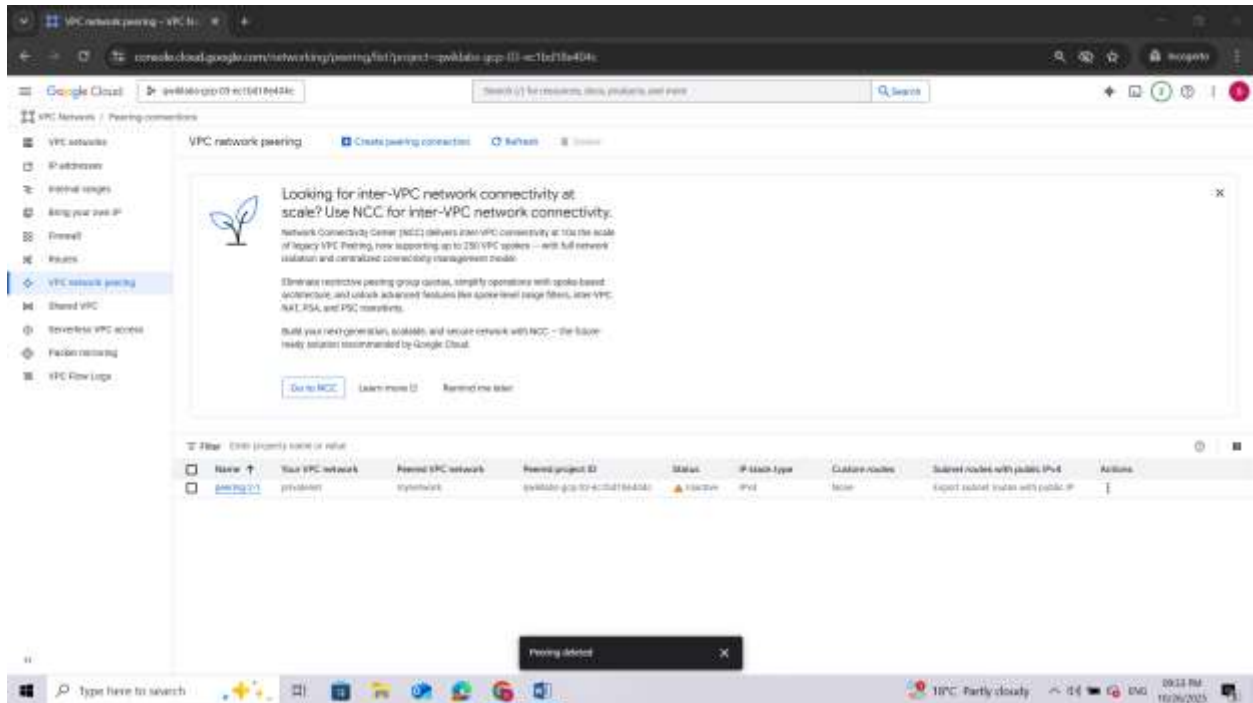
Delete the peering connection

Delete the **peering-1-2** connection.

1. On the **Navigation menu**, click **VPC network > VPC network peering**.
2. Select the **peering-1-2** connection.
3. Click **Delete**.



4. Click **Delete** to confirm the deletion. When the connection is deleted, notice the **INACTIVE** status for **peering-2-1**.



Note: Deleting one side of the peering connection terminates the overall peering connection.

Verify the peering deletion

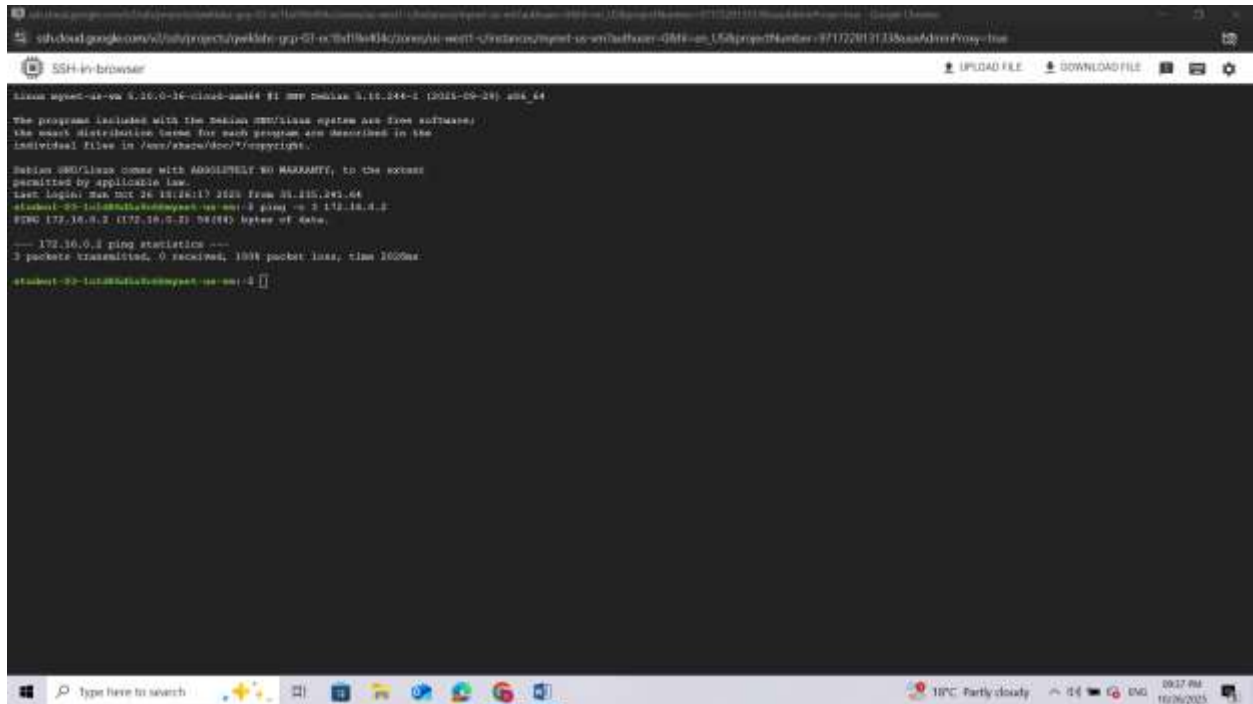
Verify that **routes** no longer exist for the peering connection and that there is no private RFC 1918 connectivity across **mynetwork** and **privatenet**.

1. On the **Navigation menu**, click **VPC network > Routes**. Notice that the VPC Peering routes have disappeared.

Note: Deleting one side of the VPC Peering connection removes all peering routes.

ping -c 3 <Enter privatenet-us-vm's internal IP here>

Note: This should not work as indicated by a 100% packet loss!

A screenshot of an SSH terminal window titled "SSH in browser". The terminal shows a user at a Debian 9.10.0-16-cloud-amd64 prompt. They run the command "ping -c 3 172.16.0.2". The output shows three ping attempts, all failing with "100% packet loss, time 1600ms". The terminal window has a standard Linux desktop environment at the bottom with a taskbar and system tray.

```
sshcloud.google.com/v2/auth/project/pktoken: gcp-SI-0t8tHk4k6/zvzw/ku-west-1/instances/mynet-us-vm/ssh/ssh-0t8tHk4k6-us-150/projectNumber=971722819133/ssh/sshKey=ssh-0t8tHk4k6-us-150
SSH in browser
UPLOADED FILE
DOWNLOAD FILE

klaus@debian-9-vm:~$ ping -c 3 172.16.0.2
The programs included with the Debian GNU/Linux system are free software;
the exact distribution terms for each program are described in the
individual files in /usr/share/doc/*/*copyright.

Debian GNU/Linux comes with ABSOLUTELY NO WARRANTY, to the extent
permitted by applicable law.
last login: Sat Mar 26 10:28:13 2022 from 31.225.241.64
klaus@172-16-0-2:~$ ping -c 3 172.16.0.2
PING 172.16.0.2 (172.16.0.2) 56(84) bytes of data:
--- 172.16.0.2 ping statistics ---
3 packets transmitted, 0 received, 100% packet loss, time 1600ms
klaus@172-16-0-2:~$
```

7. Please close the SSH terminal to **mynet-us-vm** before verifying objective:

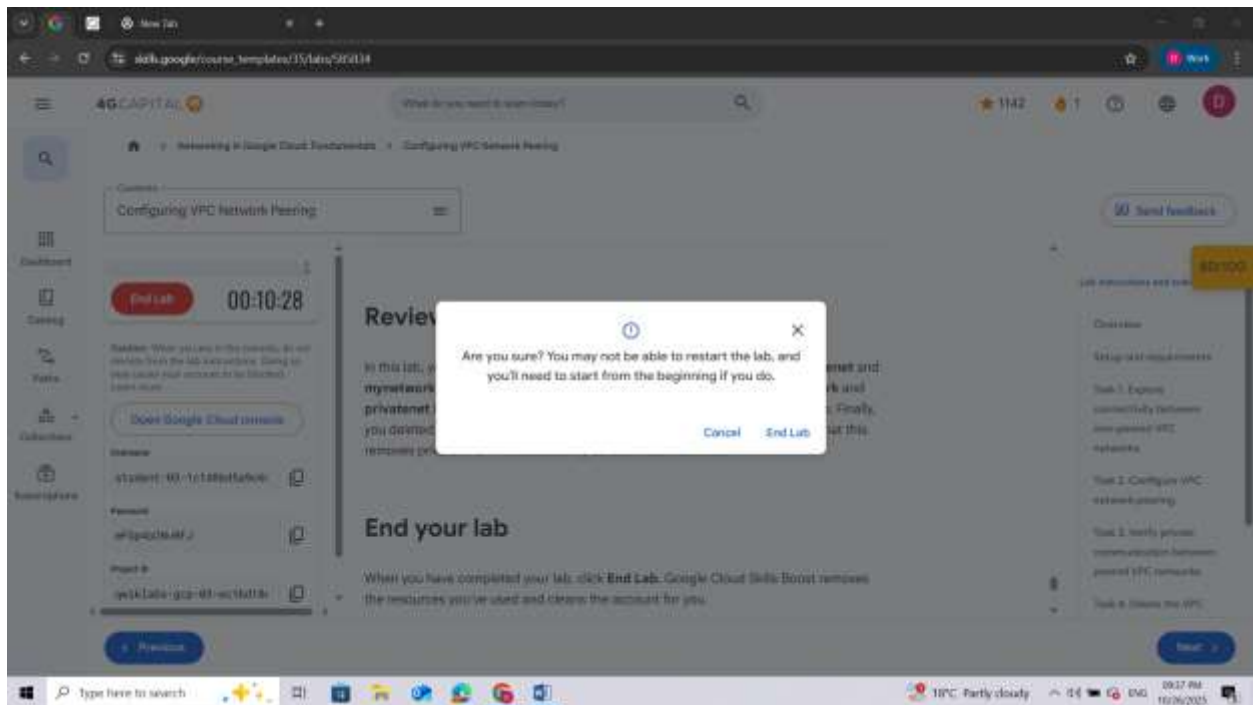
exit

Review

In this lab, you configured VPC network peering between two networks (**privatenet** and **mynetwork**). Then you verified private RFC 1918 connectivity across **mynetwork** and **privatenet** by pinging VMs on their internal IP addresses within those networks. Finally, you deleted one side of the VPC network peering connection to demonstrate that this removes private RFC 1918 connectivity across those networks.

End your lab

When you have completed your lab, click **End Lab**. Google Cloud Skills Boost removes the resources you've used and cleans the account for you.



QUIZ

- ✓ 1. Which of the following statements about VPC Network Peering is correct?
- ☐ Subnet IP ranges can overlap across peered VPC networks.
 - ✓ ☒ Transitive peering is not supported.
 - ☐ Both sides of a peering association are set up in one single step.
 - ☐ Peered VPC networks do not remain administratively separate.

This statement is correct. VPC Network Peering establishes a direct connection between two VPC networks, but it does not extend to other VPC networks they might be peered with.

- ✓ 3. How does VPC Peering exchange routing information between two peered VPCs?
- ☐ It uses Border Gateway Protocol (BGP) to dynamically exchange routes.
 - ✓ ☒ It automatically discovers and propagates routes through the Google Cloud Router.
 - ☐ It requires manual configuration of static routes in each VPC's route table.
 - ☐ It relies on the default route table to forward traffic between the VPCs.

Correct. VPC Peering automatically propagates custom dynamic routes through the Google Cloud Router, allowing each VPC to learn the routes of the other peered VPC.

- ✓ 2. Which of the following approaches to multi-project networking uses a centralized network administration model?
- ✓ ☒ Shared VPC
 - ☐ Cloud VPN and Shared VPC
 - ☐ Cloud VPN
 - ☐ VPC Network Peering

This is correct. Shared VPC allows multiple projects to share a common VPC network, with a centralized host project managing the network resources.

LAB 3: RESOURCE MONITORING

Overview

In this lab, you learn how to use Cloud Monitoring to gain insight into applications that run on Google Cloud.

Objectives

In this lab, you learn how to perform the following tasks:

- Explore Cloud Monitoring by creating a custom dashboard with charts
- Create alerts with multiple conditions
- Create resource groups
- Create uptime checks

Setup and Requirements

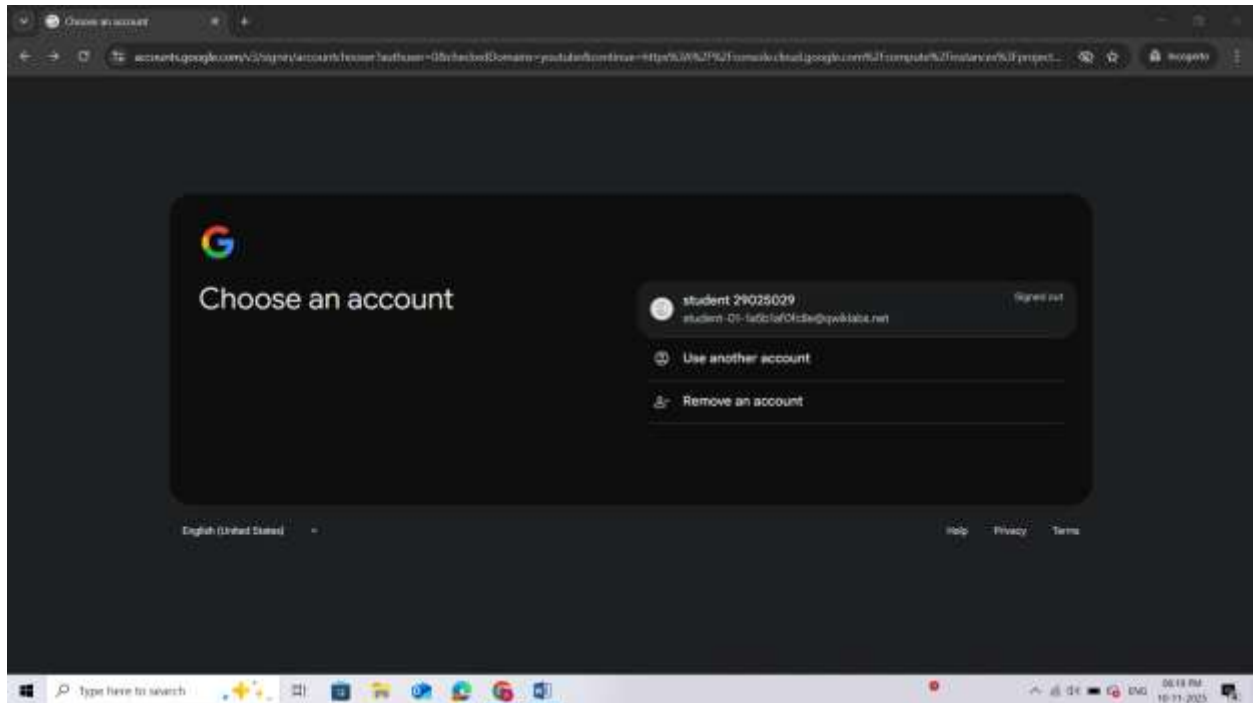
For each lab, you get a new Google Cloud project and set of resources for a fixed time at no cost.

1. Click the **Start Lab** button. If you need to pay for the lab, a pop-up opens for you to select your payment method. On the left is the **Lab Details** panel with the following:
 - The **Open Google Cloud console** button
 - Time remaining
 - The temporary credentials that you must use for this lab
 - Other information, if needed, to step through this lab
2. Click **Open Google Cloud console** (or right-click and select **Open Link in Incognito Window** if you are running the Chrome browser).

The lab spins up resources, and then opens another tab that shows the **Sign in** page.

***Tip:** Arrange the tabs in separate windows, side-by-side.*

***Note:** If you see the **Choose an account** dialog, click **Use Another Account**.*



3. If necessary, copy the **Username** below and paste it into the **Sign in** dialog.

"Username"

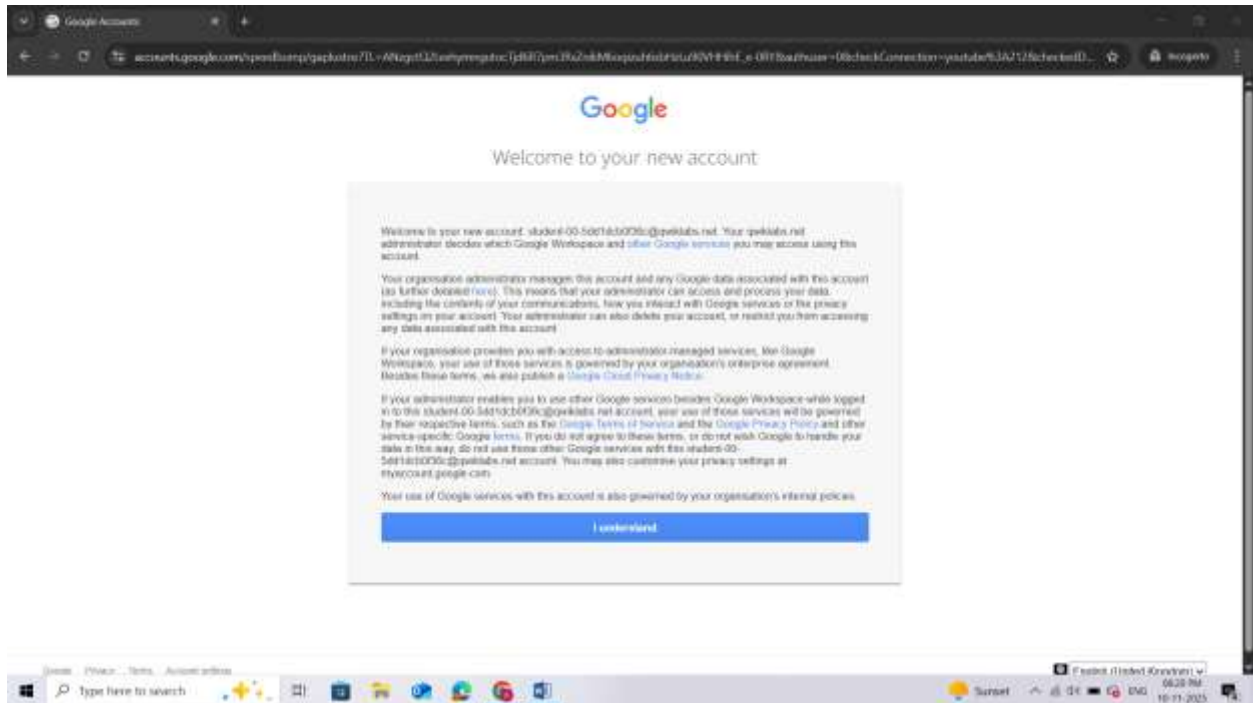
You can also find the **Username** in the **Lab Details** panel.

4. Click **Next**.
5. Copy the **Password** below and paste it into the **Welcome** dialog.

"Password"

You can also find the **Password** in the **Lab Details** panel.

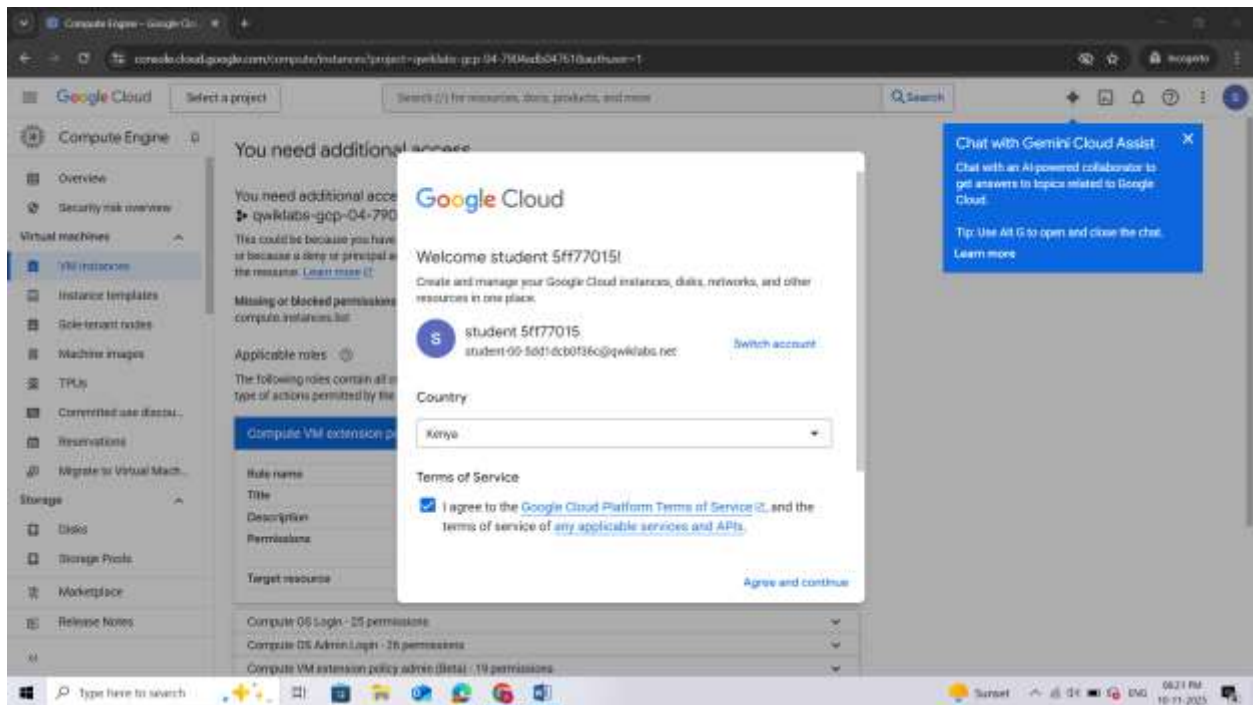
6. Click **Next**.



Important: You must use the credentials the lab provides you. Do not use your Google Cloud account credentials. **Note:** Using your own Google Cloud account for this lab may incur extra charges.

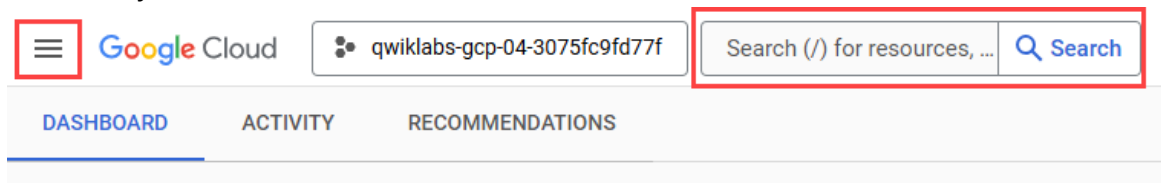
7. Click through the subsequent pages:

- Accept the terms and conditions.
- Do not add recovery options or two-factor authentication (because this is a temporary account).
- Do not sign up for free trials.



After a few moments, the Google Cloud console opens in this tab.

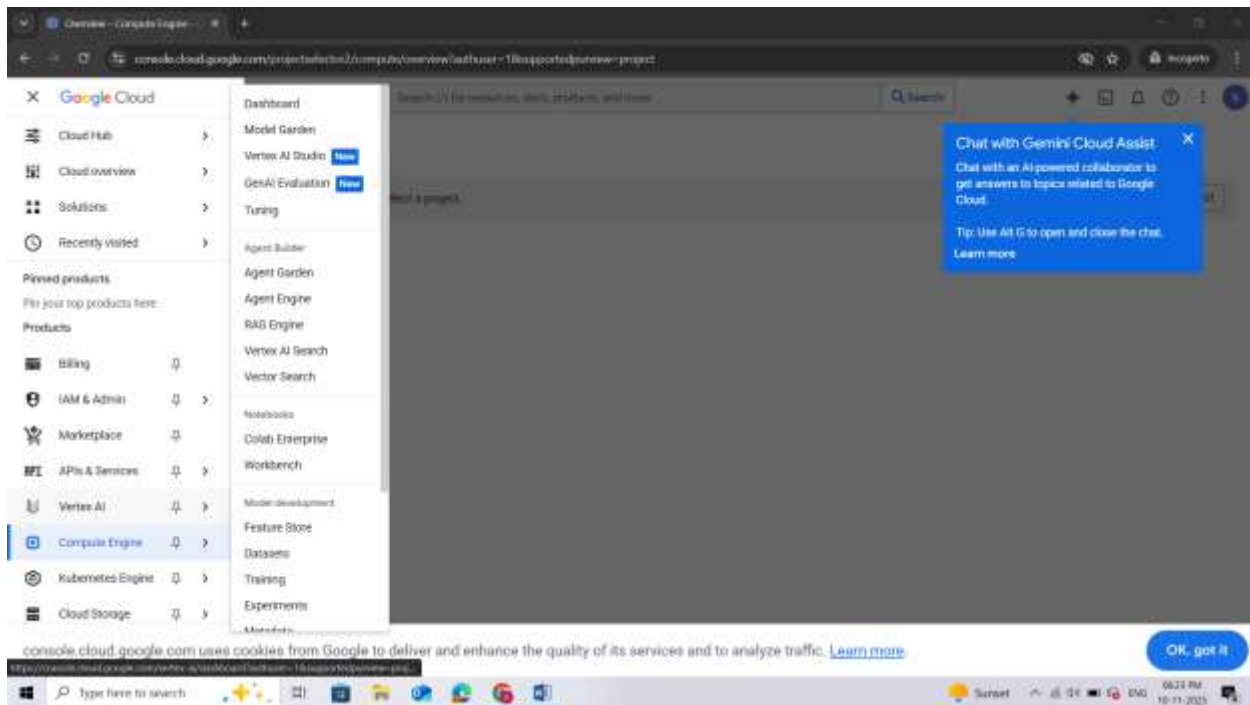
Note: To view a menu with a list of Google Cloud products and services, click the **Navigation menu** at the top-left, or type the service or product name in the **Search** field.



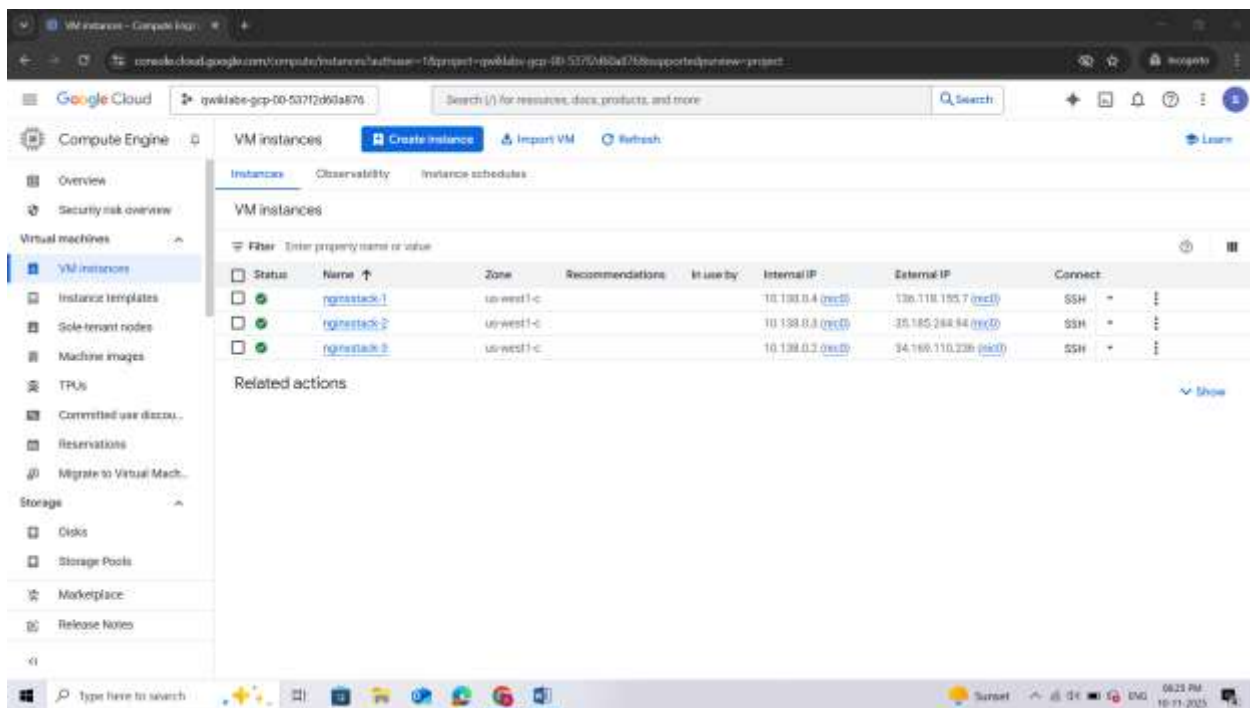
Task 1. Verify resources

In this task, you verify that three VM instances have been created. You monitor these in the tasks that follow.

1. In the Google Cloud console, in the **Navigation menu** (≡), click **Compute Engine > VM instances**.



2. Verify that the **nginxstack-1**, **nginxstack-2** and **nginxstack-3** instances are visible.

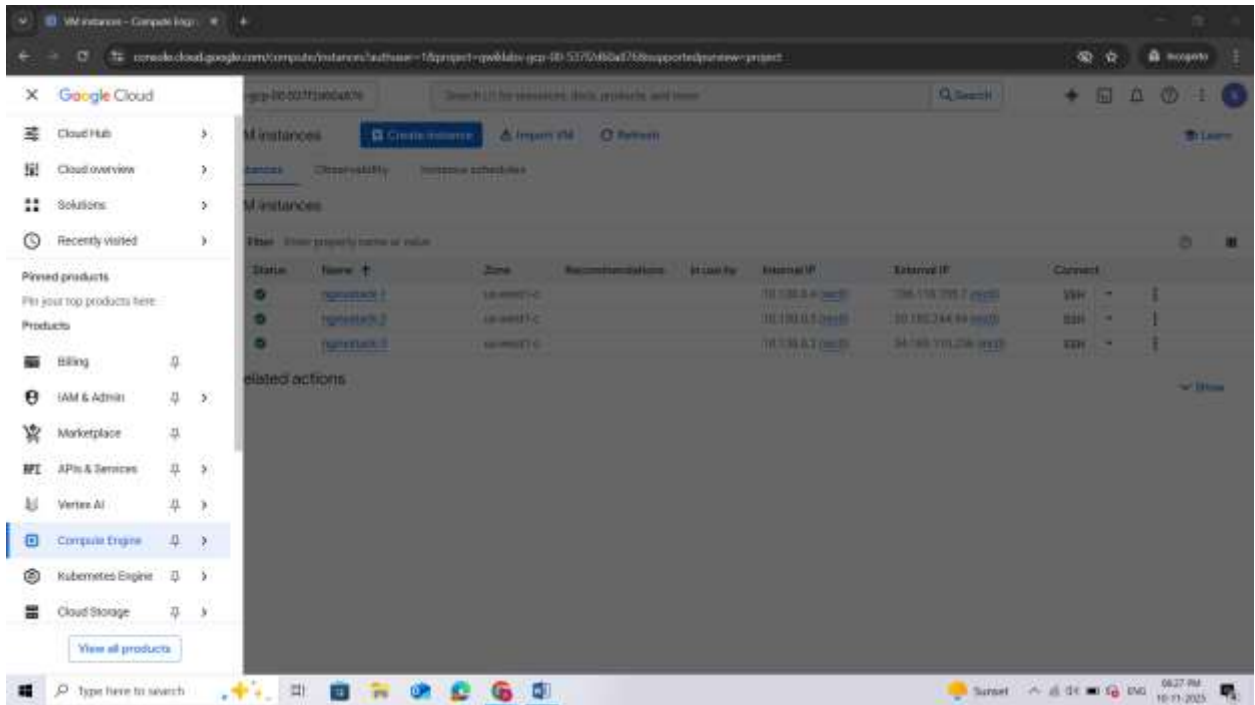


Task 2. Custom dashboards

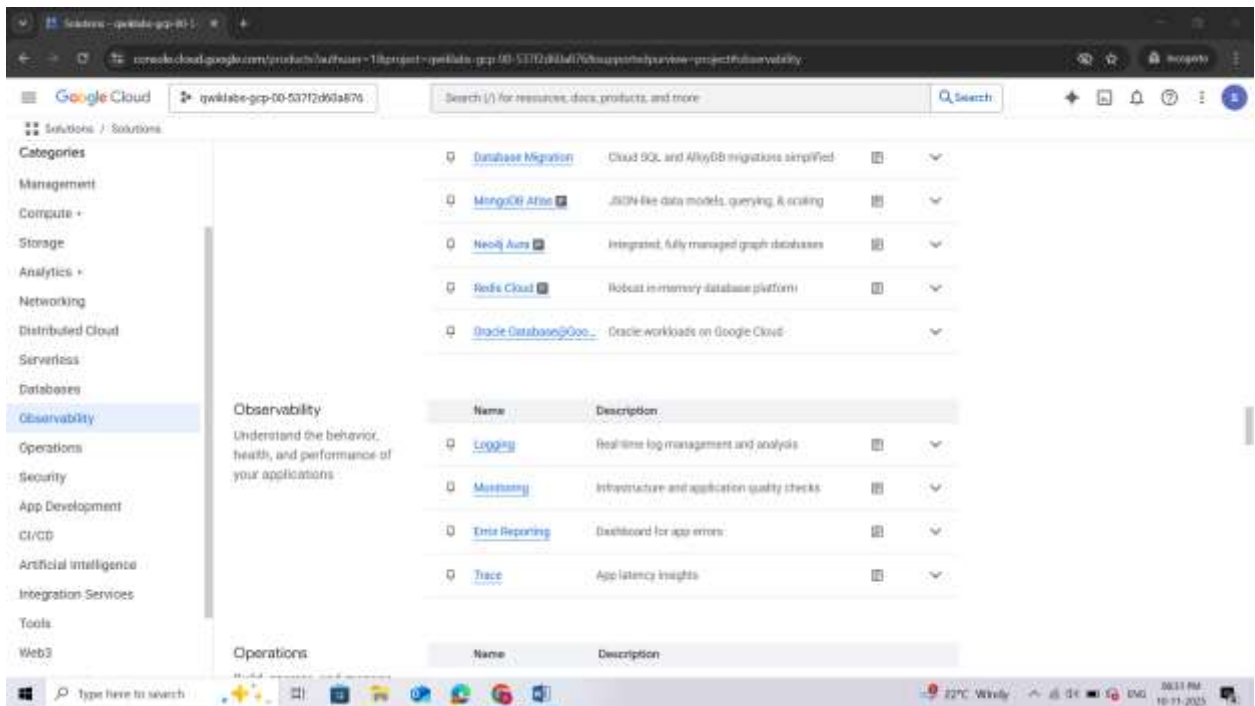
In this task, you create a custom dashboard in Cloud Monitoring and examine resource usage data.

Create a dashboard

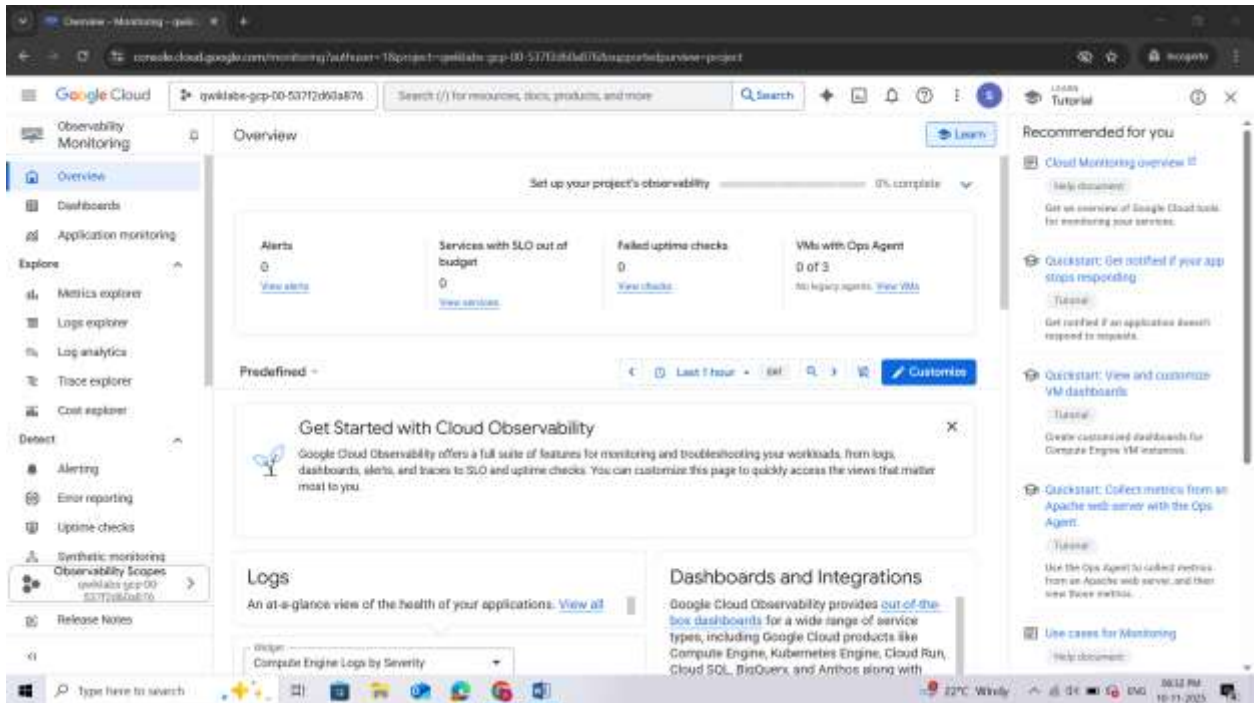
1. In the Google Cloud console, in the **Navigation menu** (☰), click **View all products >**



Observability >



Monitoring



When the Monitoring dashboard opens, your workspace is ready.

Why is monitoring important to Google?



Google uses monitoring to ensure they have all the important metrics for reporting purposes to customers and the other interested bodies. The number of reports requires the collection and reporting to be both broad and deep.

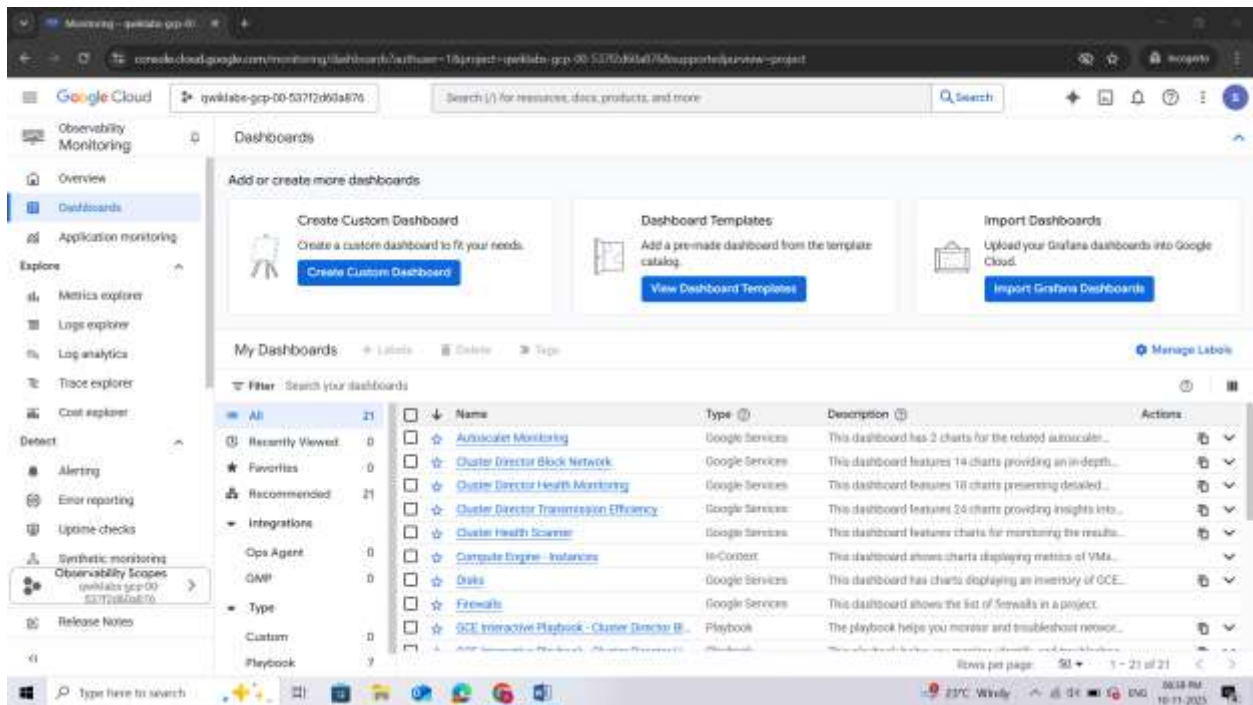


Monitoring is important to ensure that Google complies with regulatory requirements defined by both government and industry security bodies.



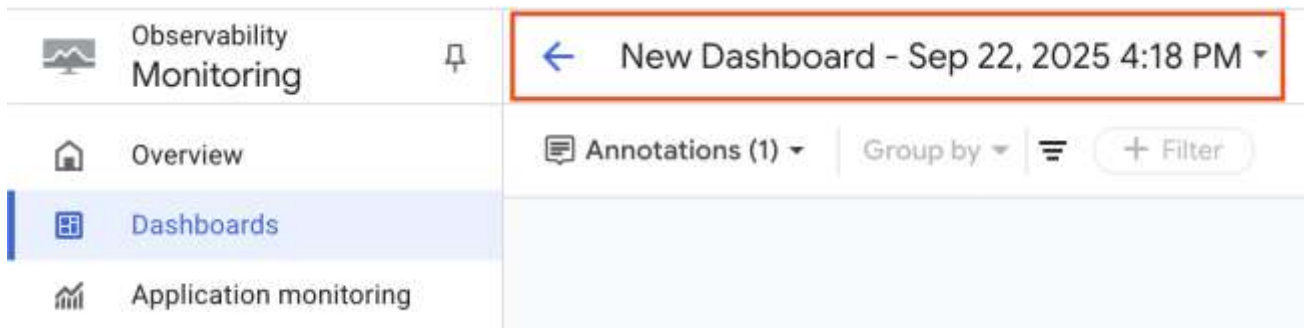
It is at the base of site reliability which incorporates aspects of software engineering and applies that to operations whose goals are to create ultra-scalable and highly reliable software systems.

2. In the left pane, click **Dashboards**.
3. Click **Create Custom Dashboard**.

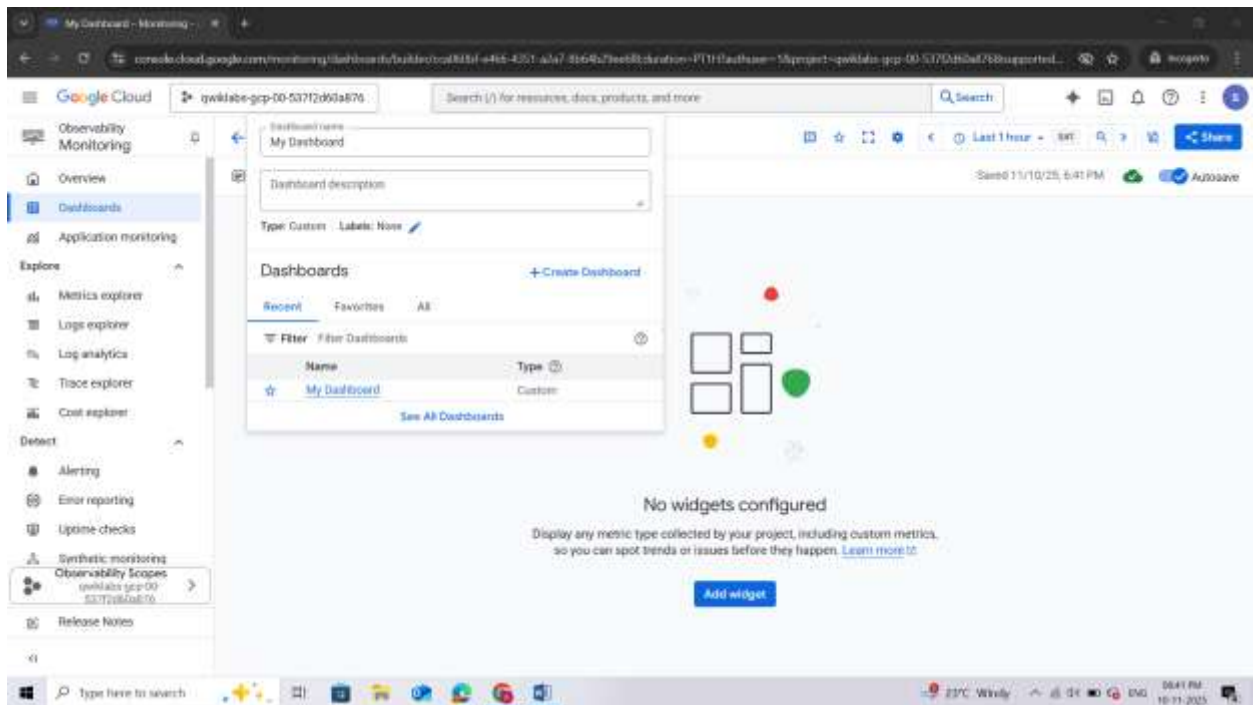


4. In the top pane, click **New Dashboard**.

Note: The name will also contain the time and date you created the dashboard.

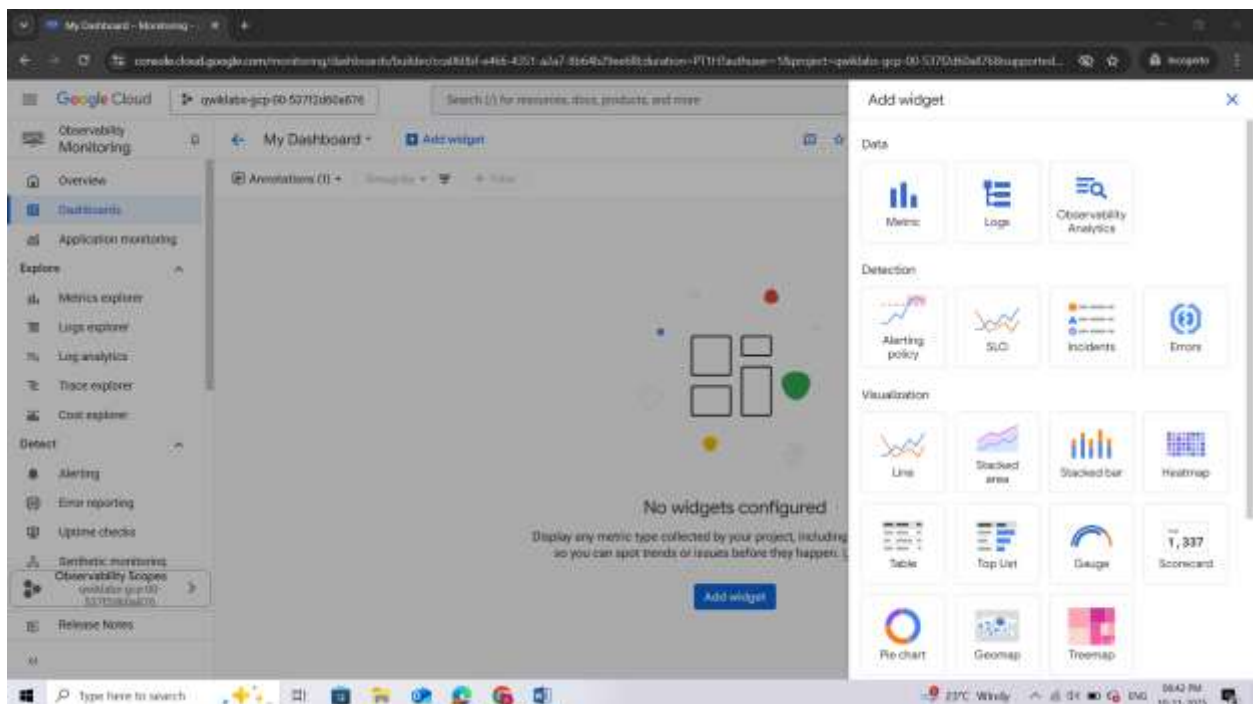


5. For **Dashboard name**, type **My Dashboard**. Click outside of the box to continue.

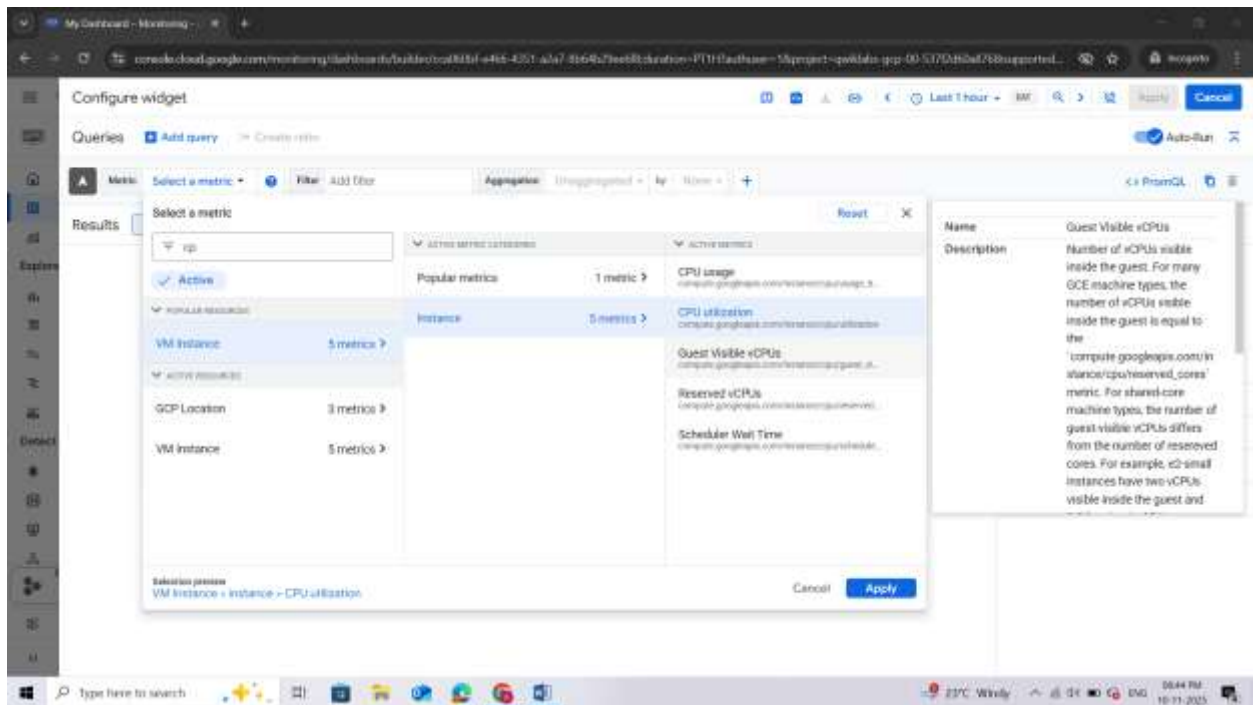


Add a chart

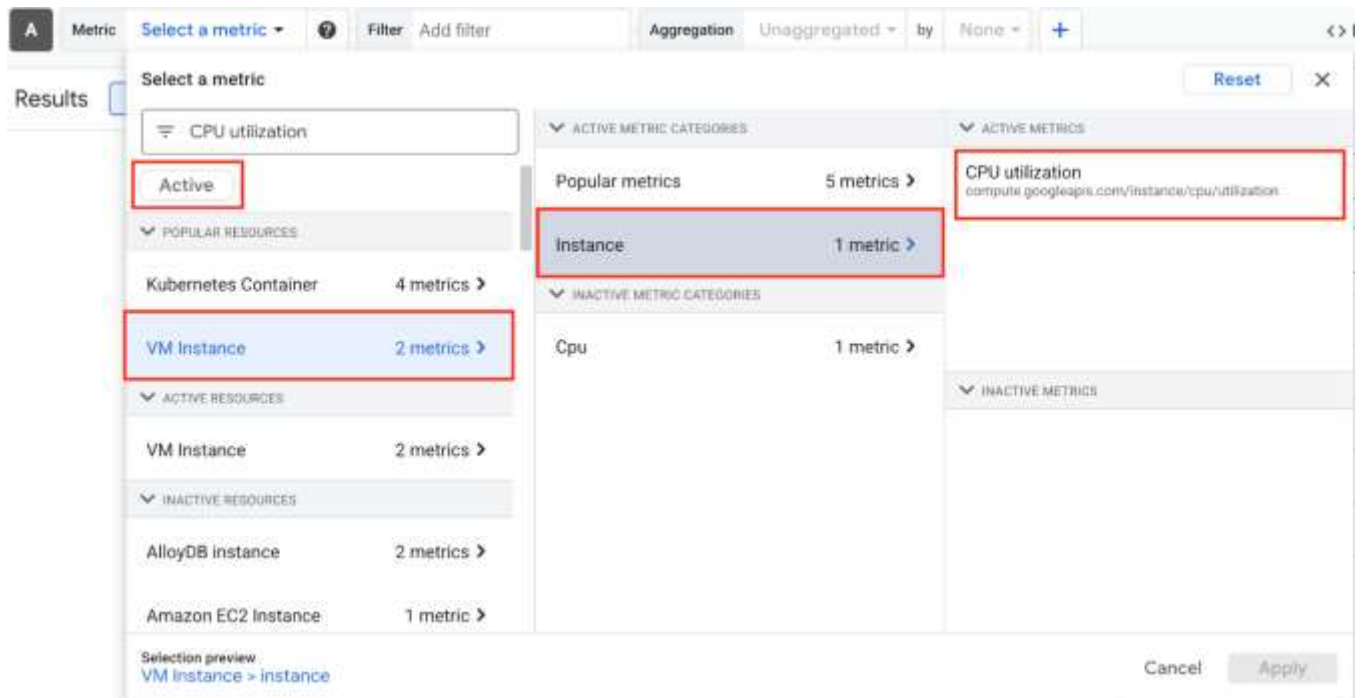
1. Click **Add widget**, and then select **Line**.



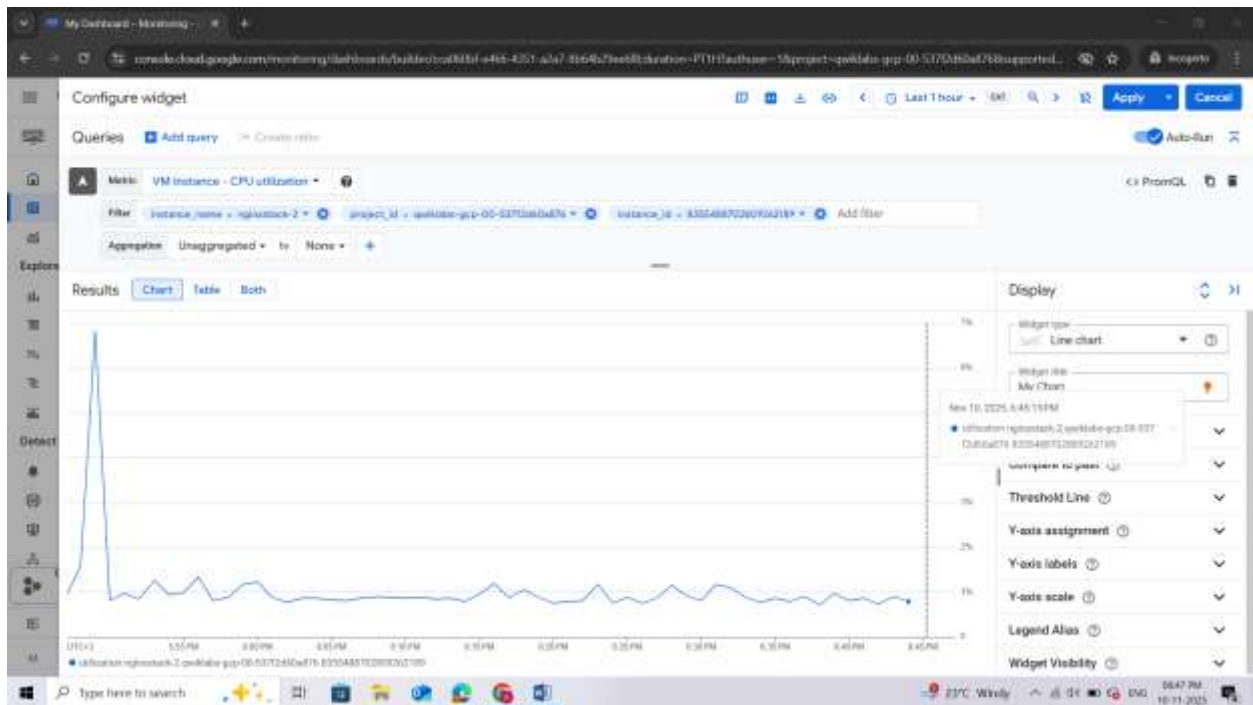
2. In **Widget title**, type **My Chart**.
3. In the **Select a metric** dropdown, type **CPU utilization**. Then, click **VM Instance > Instance**, select **CPU utilization**, and click **Apply**.



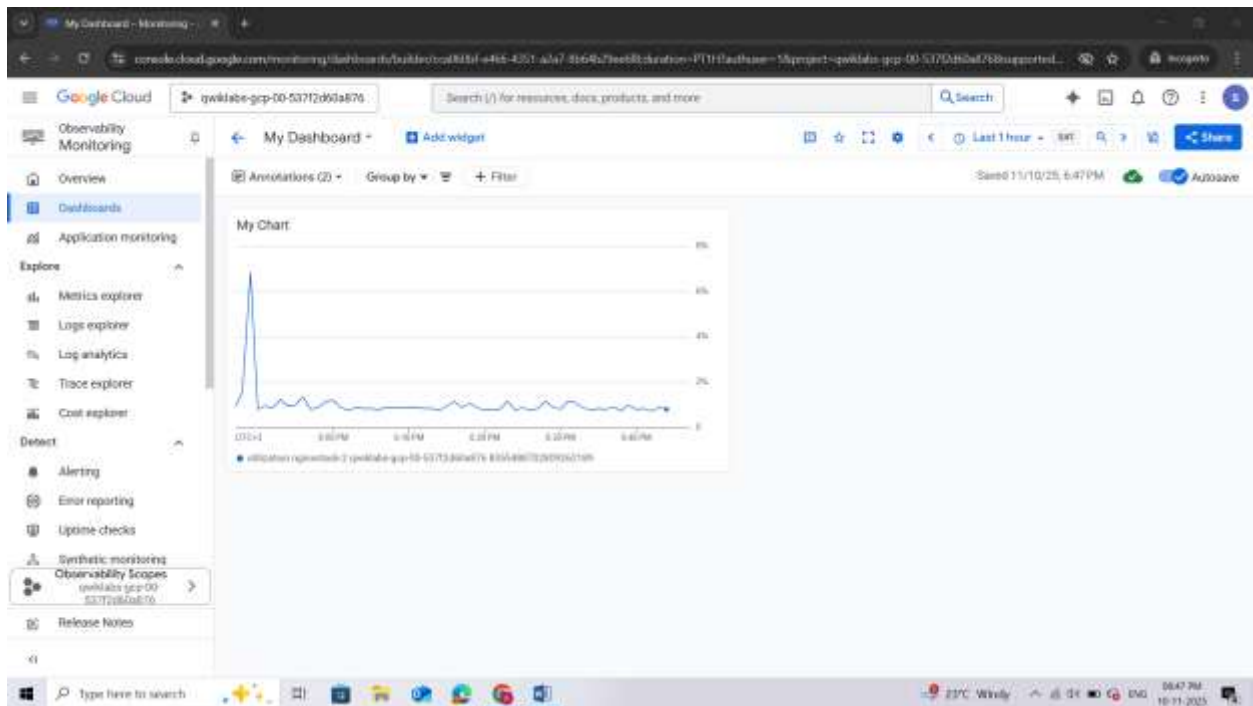
Note: Uncheck Active if you are unable to find the CPU utilization



- Click + **Add Filter** and explore the various options.



5. Click **Apply** to create the chart.

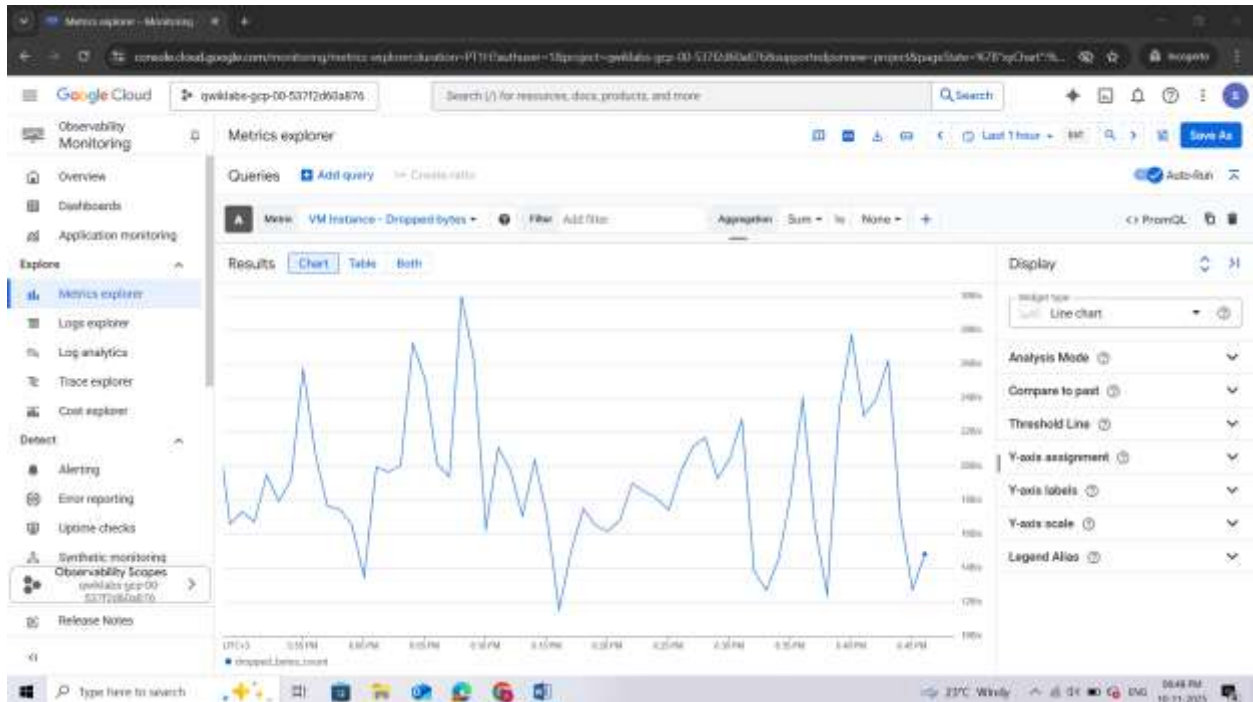


Metrics Explorer

The **Metrics Explorer** allows you to examine resources and metrics without having to create a chart on a dashboard. Try to recreate the chart you just created using the **Metrics Explorer**.

1. In the left pane, click **Metrics explorer**.

2. Click **Select a metric**.
3. Select a Metric from the dropdown, explore the various options, and then try to recreate the chart you created earlier.



Note: Not all metrics are currently available on the Metrics Explorer, so you might not be able to find the exact metric you used on the previous step.

Task 3. Alerting policies

In this task, you create an alert policy in Cloud Monitoring to notify you when CPU usage exceeds a specified threshold.

What is not a recommended best practice for alerts?

☐

Use multiple notification channels so you avoid a single point of failure.

☐

Customize your alerts to the audience need.

☒

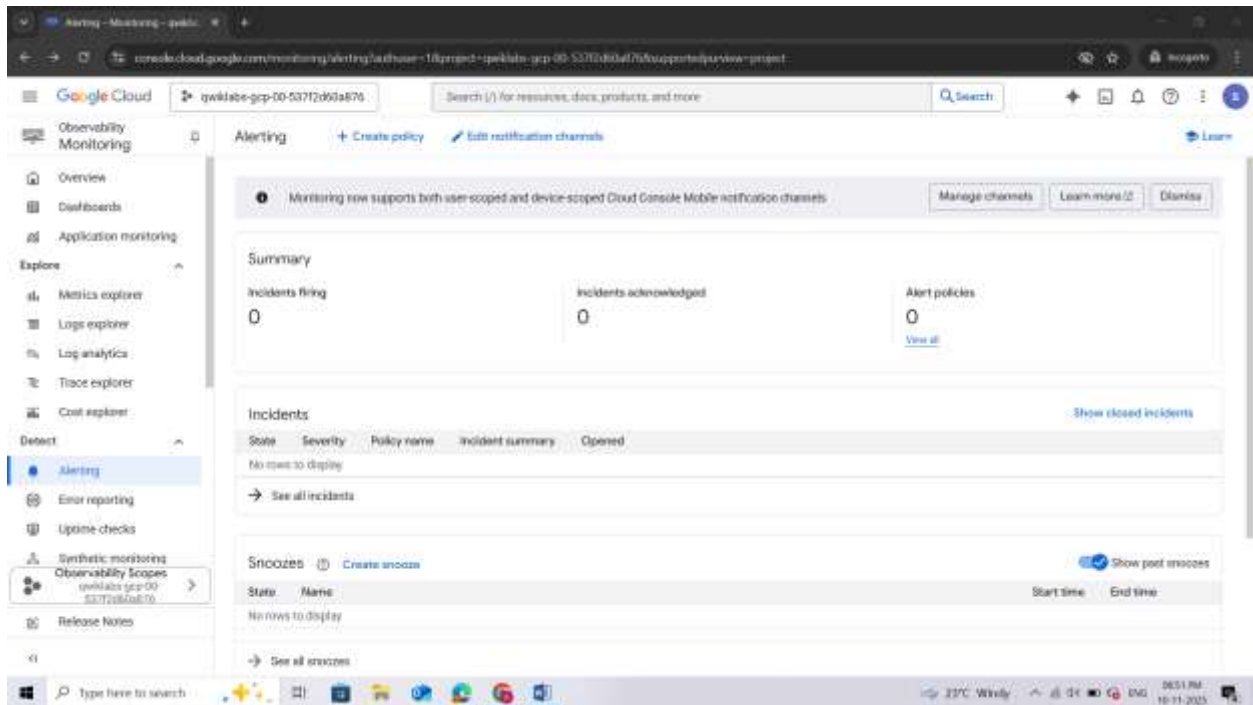
Report all noise to ensure all data points are presented.

☐

Configure alerting on symptoms and not necessarily causes.

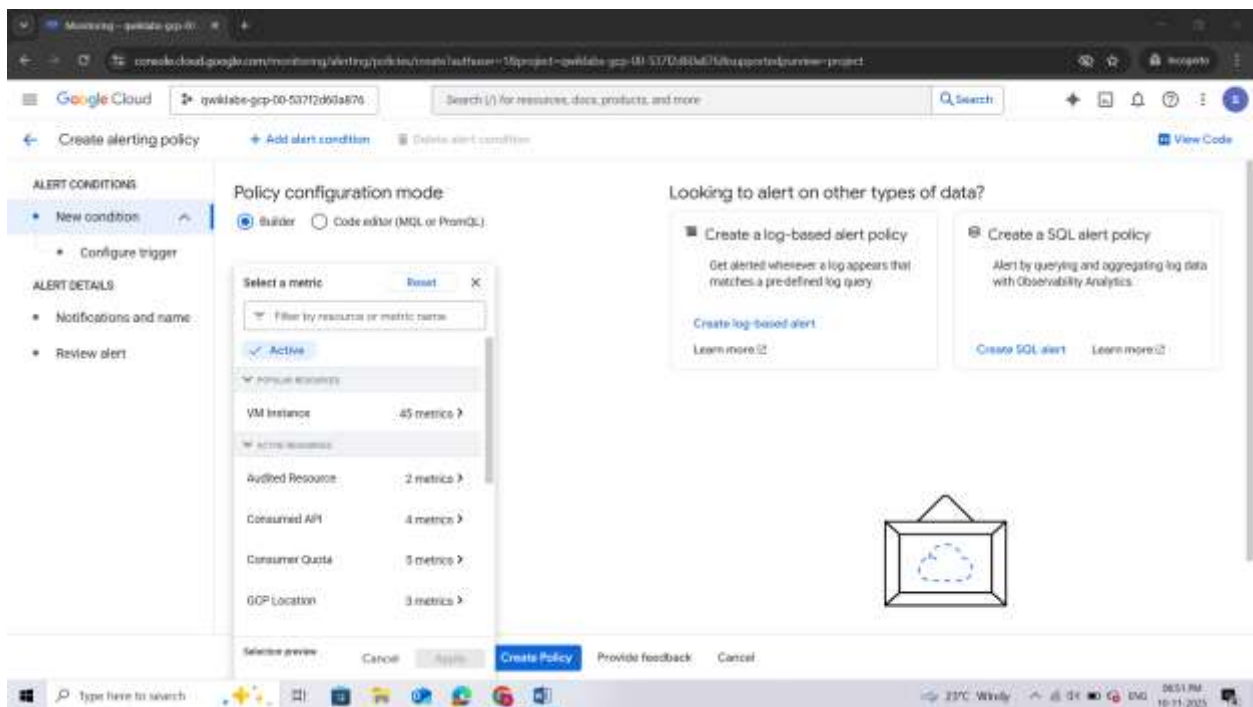
Create an alert and add the first condition

1. In the left pane, select **Alerting**.



2. Click **+ Create policy**.

3. Click on **Select a metric** dropdown. Uncheck the **Active** option.



4. Type **VM Instance** in filter by resource and metric name and click on **VM Instance > Instance**. Select **CPU usage** and click **Apply**.

The screenshot shows the Google Cloud Monitoring console in 'Policy configuration mode'. The left sidebar has 'ALERT CONDITIONS' with 'New condition' selected. The main area shows 'Looking to alert on other types of data?' with options for 'a log-based alert policy' and 'Create a SQL alert policy'. The 'VM Instance' resource type is selected, and the 'CPU usage' metric is chosen. The 'Apply' button is visible at the bottom right.

Note: If you cannot locate the **VM Instance** resource type, you might have to refresh the page.

5. Set **Rolling windows** to 1 min.
6. Click **Next**.

The screenshot shows the Google Cloud Monitoring console in 'Policy configuration mode'. The left sidebar has 'ALERT CONDITIONS' with 'VM Instance - CPU usage' selected. The main area shows 'Add filters' and 'Transform data' sections. The 'Rolling window' is set to 1 min. The 'Next' button is visible at the bottom right.

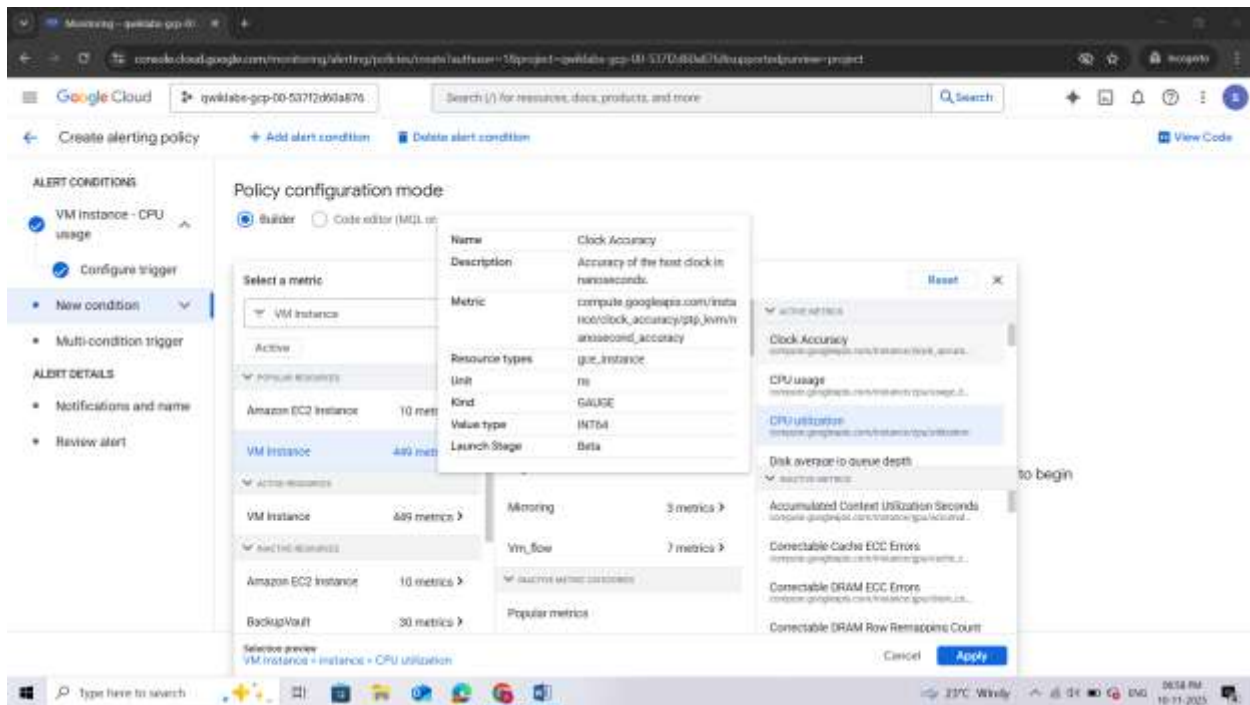
- Set **Threshold** position to **Above Threshold** and set **20** as your **Threshold value**.
- Click **Next**.

The screenshot shows the 'Configure alert trigger' interface in the Google Cloud Monitoring console. The 'Condition Types' section has 'Threshold' selected. The 'Alert trigger' is set to 'Any time series violates'. The 'Threshold position' is 'Above threshold' and the 'Threshold value' is '20'. The 'Advanced Options' section is expanded, showing a table of instances and their CPU usage values.

Name (from instance_id)	Instance name	Value
nginxstack-1 nginxstack-1	nginxstack-1	0.022
nginxstack-2 nginxstack-2	nginxstack-2	0.002
nginxstack-3 nginxstack-3	nginxstack-3	0.033

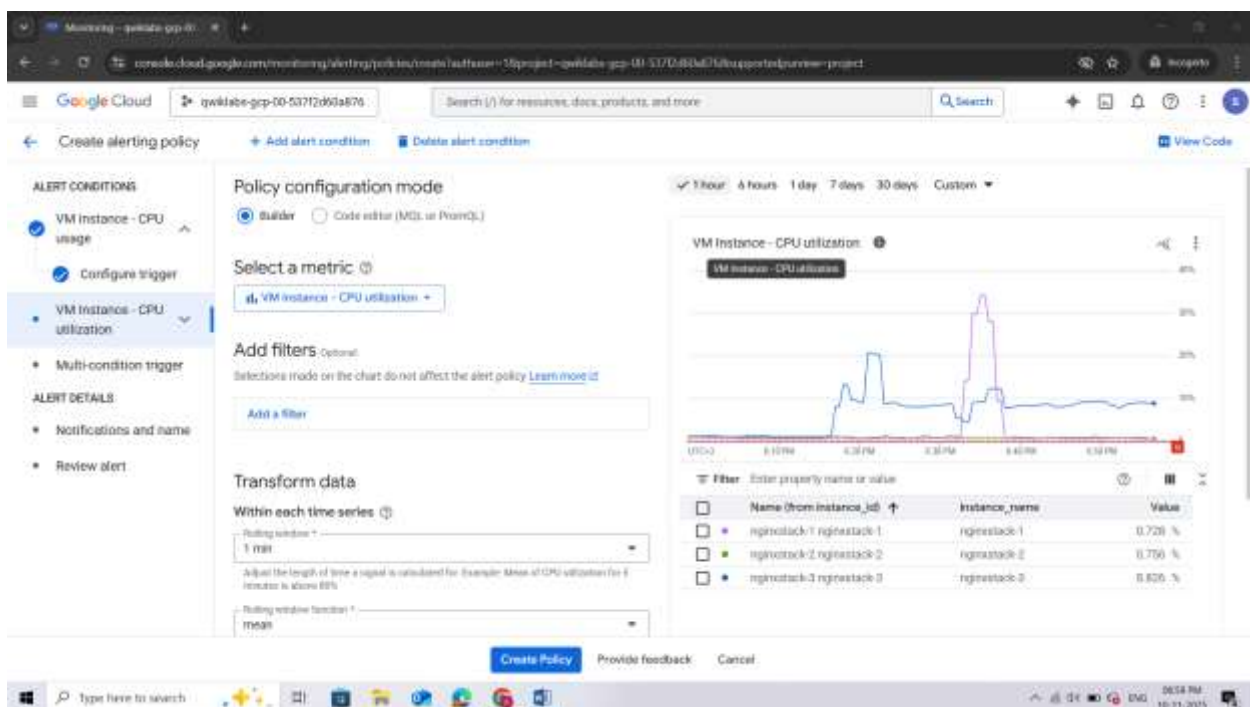
Add a second condition

- Click **+Add alert condition**.
- Click on **Select a metric** dropdown. Uncheck the **Active** option.
- Type **VM Instance** in filter by resource and metric name and click on **VM Instance > Instance**. Select **CPU utilization** and click **Apply**.



4. Set **Rolling windows** to 1 min.

5. Click **Next**.



6. Set **Threshold** position to **Above Threshold** and set **20** as your **Threshold value** and click **Next**.

Google Cloud Monitoring - Create alerting policy

Alert conditions:

- VM Instance - CPU usage
- Configure trigger
- VM Instance - CPU utilization
- Multi-condition trigger

Alert details:

- Notifications and name
- Review alert

Alert trigger: Any time series violates

Threshold position: Above threshold

Threshold value: 20%

Advanced Options:

Condition name: VM Instance - CPU utilization

VM Instance - CPU utilization graph

Filter:

Name (from instance_id)	Instance name	Value
nginxstack-1 nginxstack-1	nginxstack-1	0.726 %
nginxstack-2 nginxstack-2	nginxstack-2	0.756 %
nginxstack-3 nginxstack-3	nginxstack-3	0.826 %

Estimated monthly cost: \$0.10 \$0.00

Create Policy

7. In **Multi-condition trigger**, select **All conditions are met**.

8. Click **Next**.

Google Cloud Monitoring - Create alerting policy

Alert conditions:

- VM Instance - CPU usage
- Configure trigger
- VM Instance - CPU utilization
- Multi-condition trigger

Alert details:

- Notifications and name
- Review alert

Multi-condition trigger

You must specify when to create an incident when multiple conditions are met.

Any condition is met

All conditions are met

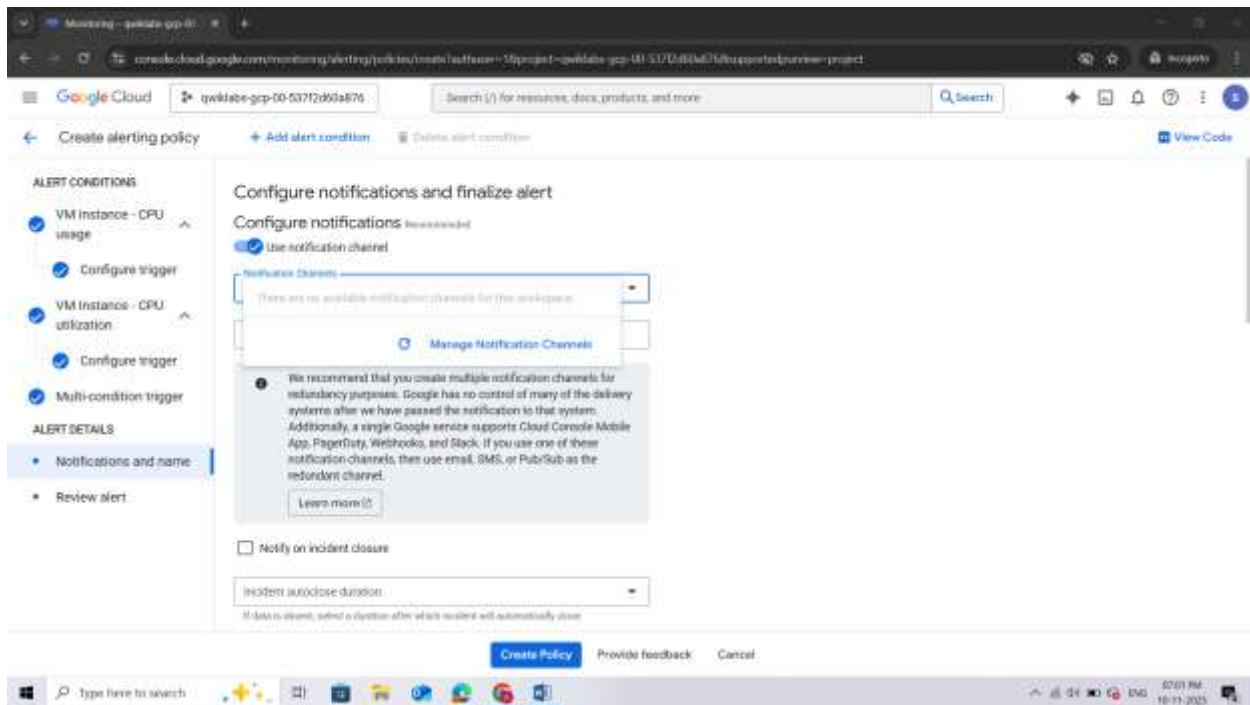
All conditions are met even for different resources for each condition

Next

Create Policy

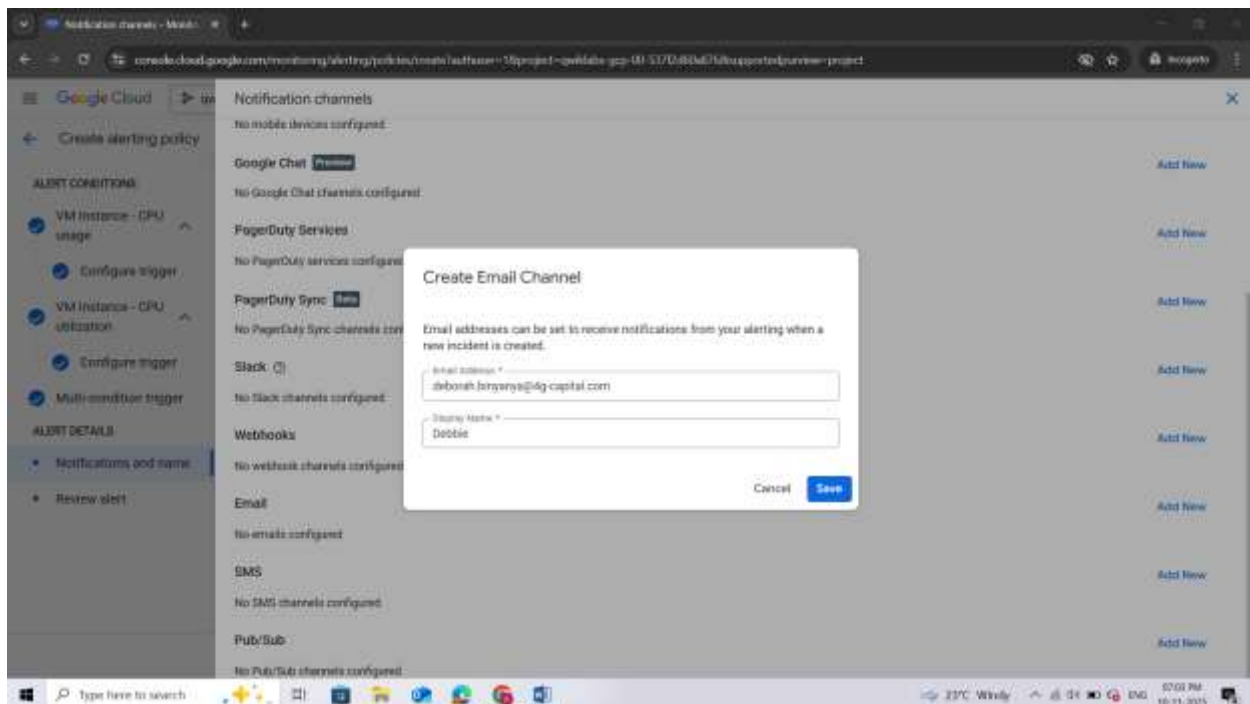
Configure notifications and finish the alerting policy

1. Click on **Notification Channels** dropdown list, then click on **Manage Notification Channels**.

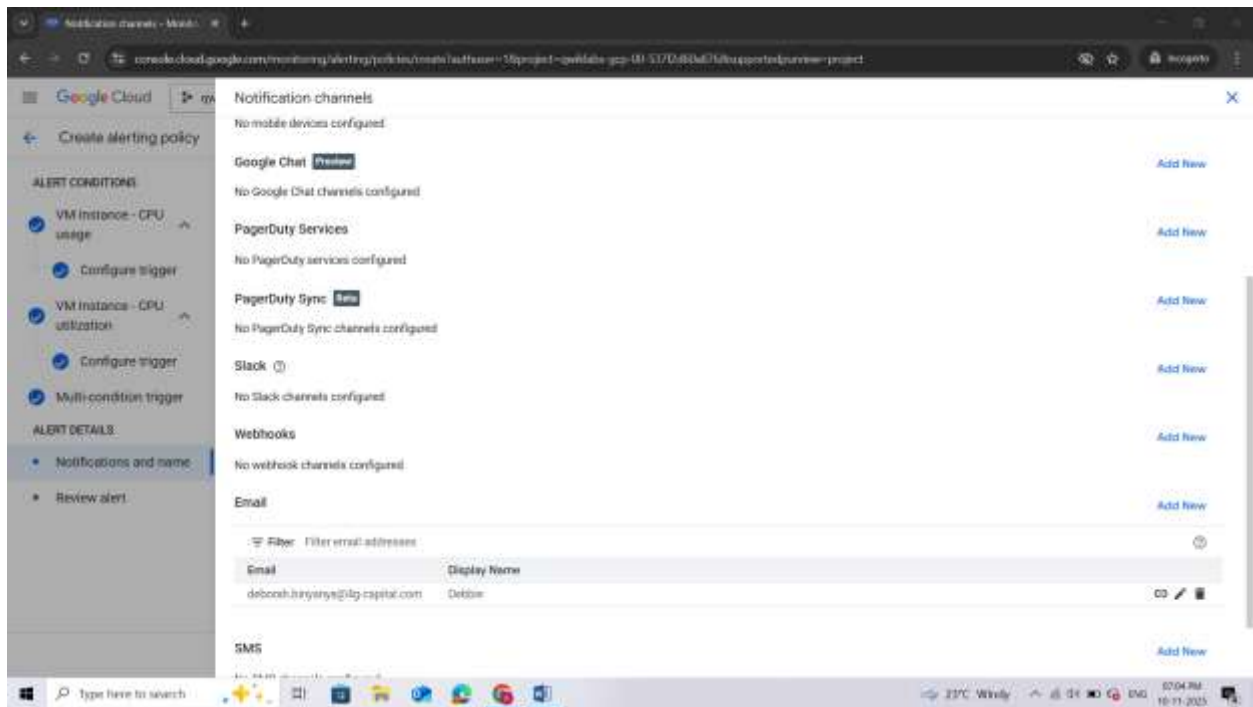


The **Notification channels** page will open in a new tab.

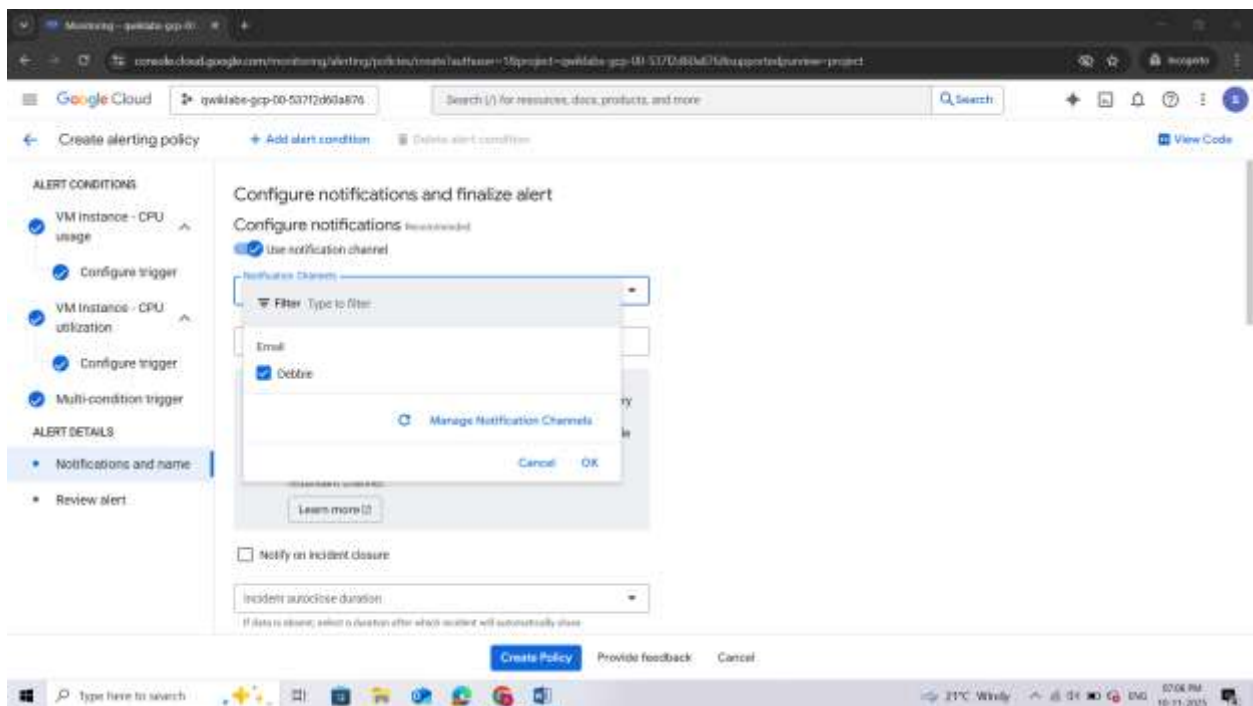
2. Scroll down the page and click on **Add New** for **Email**.
3. Enter your personal email in the **Email Address** field and a **Display name**.
4. Click **Save**.



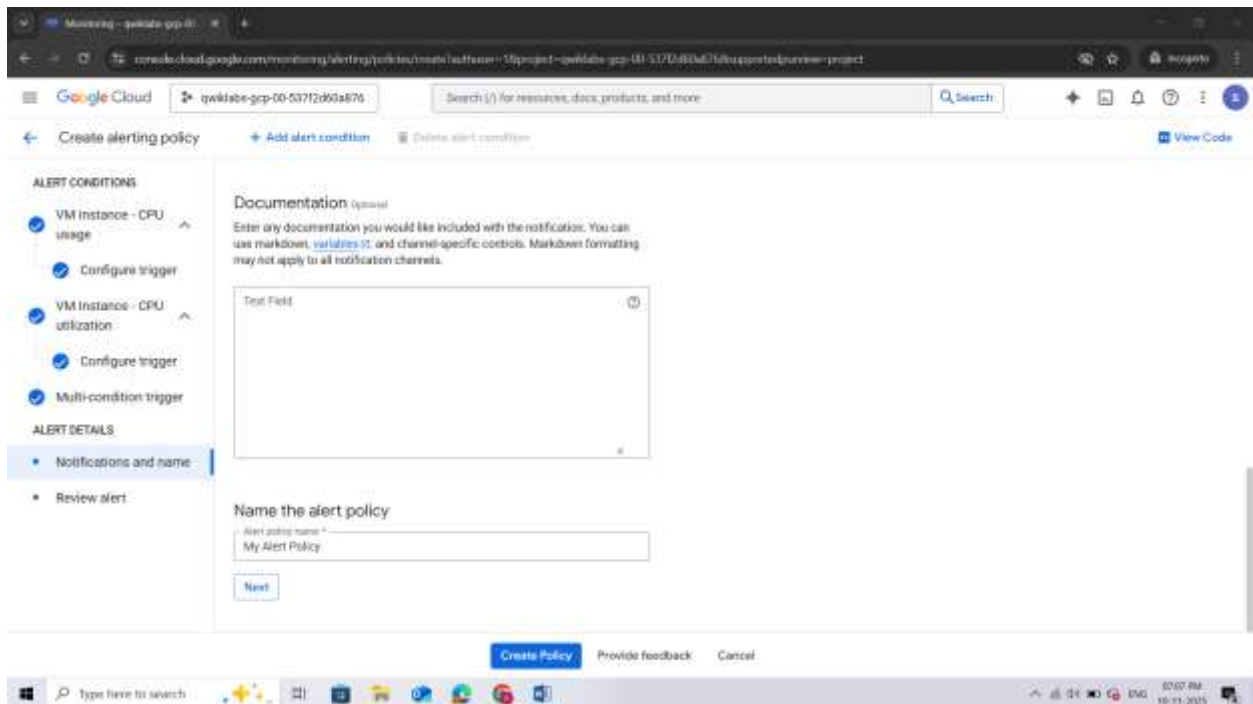
- Go back to the previous **Configure notifications and finalize alert** tab.



- Click on **Notification Channels** again, then click on the **Refresh** icon to get the display name you mentioned in the previous step. Click **Notification Channels** again if needed.
- Now, select your **Display name** and click **Ok**.



8. In **Alert policy name** type **My Alert Policy**. You may need to scroll down to the bottom of the page to see this.
9. Click **Next**.

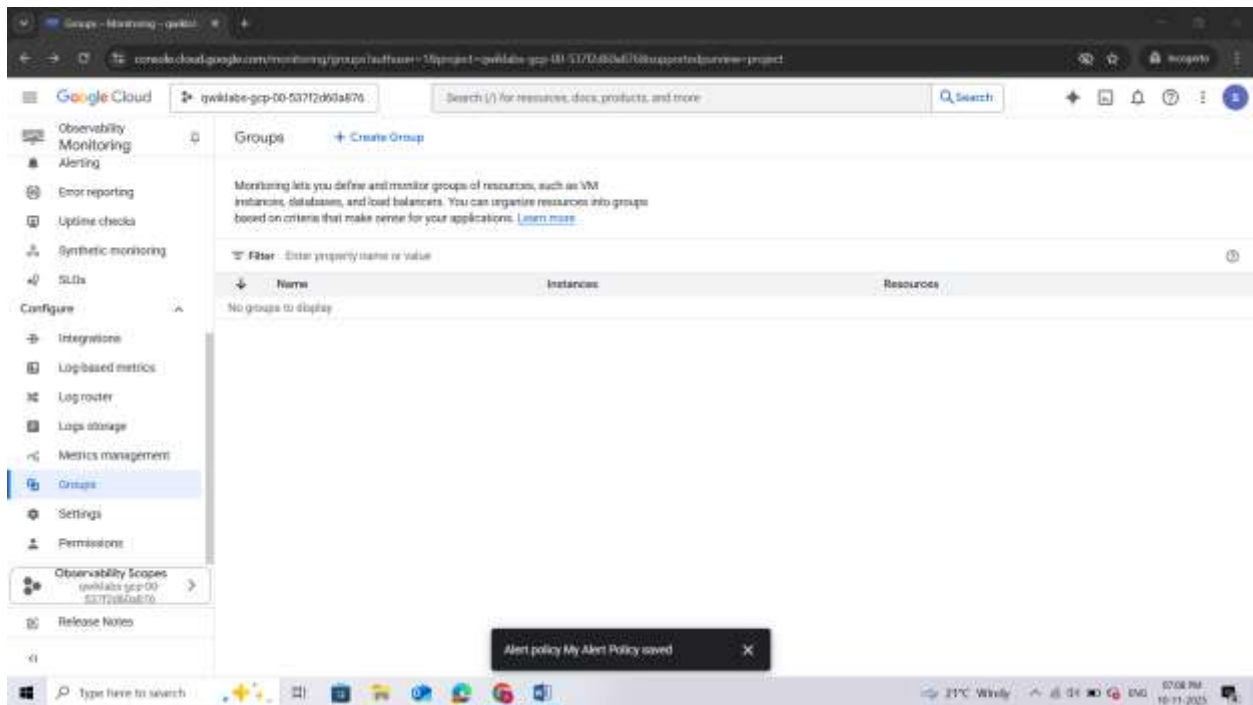


10. Review the alert and click **Create Policy**.

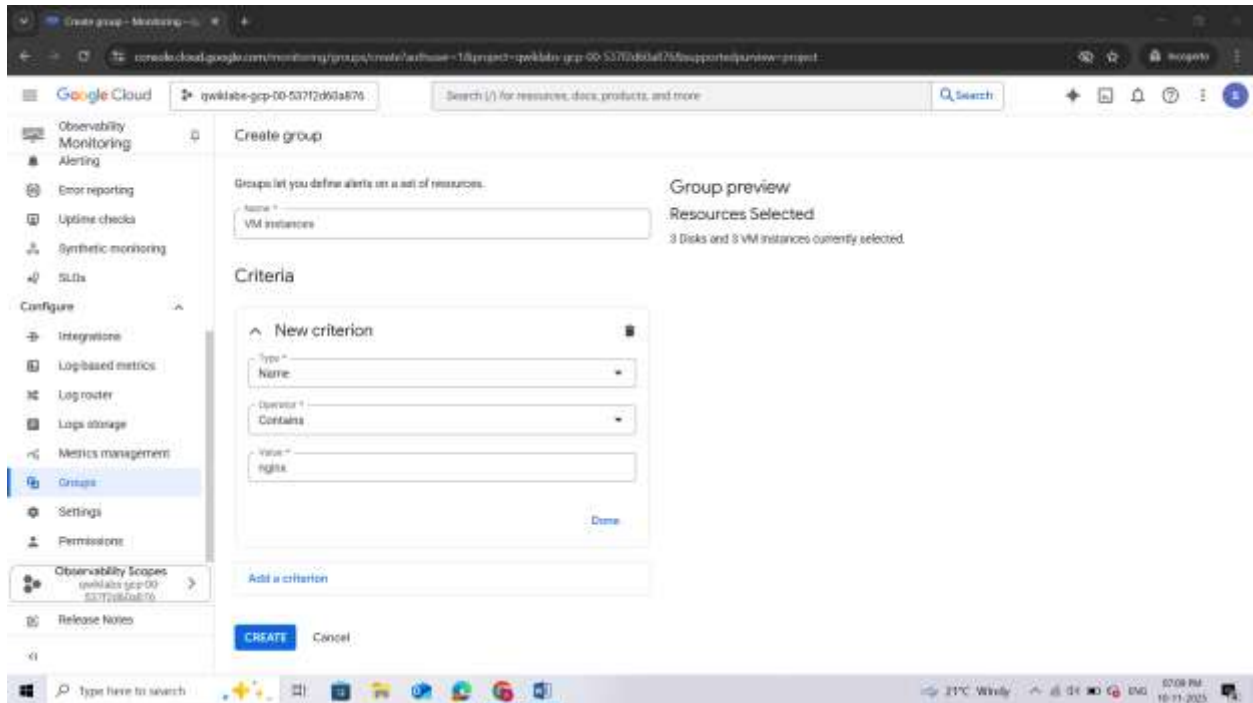
Task 4. Resource groups

In this task, you create a group in Cloud Monitoring to organize your resources.

1. In the left pane, click **Groups**.



2. Click **+ Create Group**.
3. In the **Name** field, type **VM instances**.
4. In the **Criteria** section, type **nginx** in the **Value** field below Contains.
5. Click **Done**.
6. Click **Create**.



7. Review the dashboard Cloud Monitoring created for your group.

Task 5. Uptime monitoring

In this task, you create an uptime check in Cloud Monitoring to monitor the availability of your resources.

Select all valid targets for Cloud Monitoring uptime alert notifications.



3rd party service



EC2 service



email



SMS

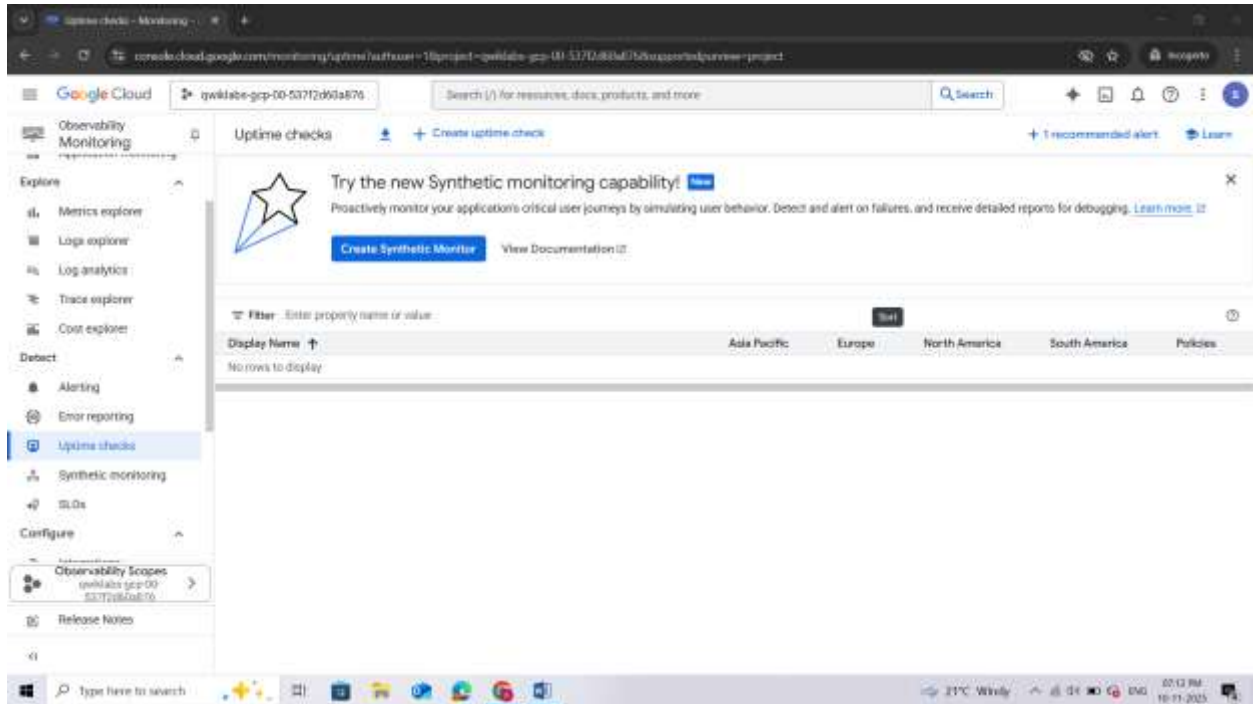


Pub/sub



webhook

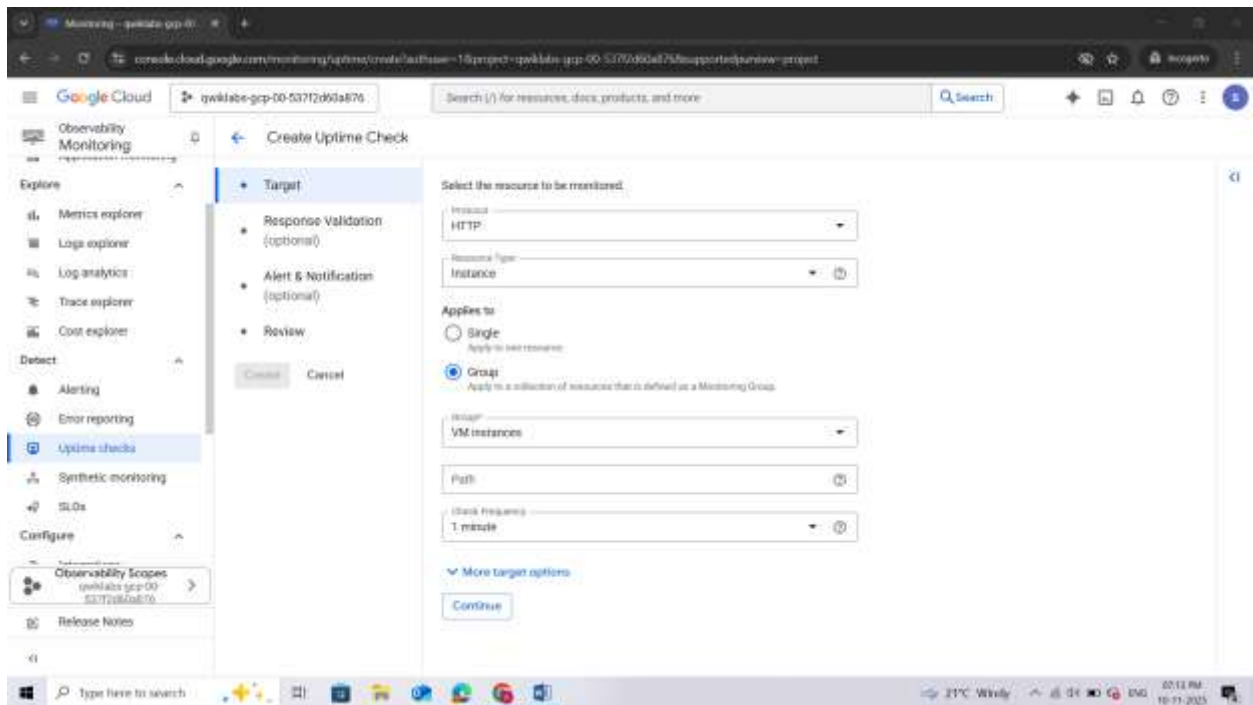
1. In the Monitoring tab, click on **Uptime checks**.
2. Click **+ Create uptime check**.



3. Specify the following, and leave the remaining settings as their defaults:

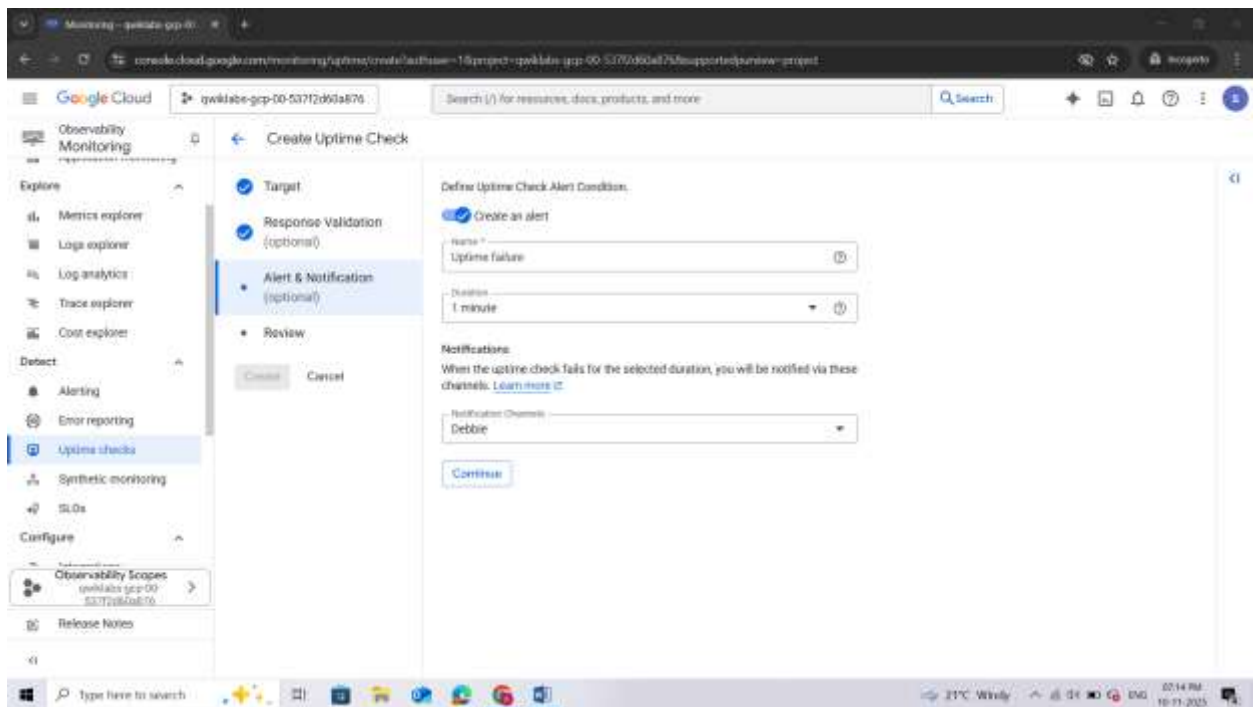
Property	Value (type value or select option as specified)
Protocol	HTTP
Resource Type	Instance
Applies To	Group
Group	VM instances
Check Frequency	1 minute

4. Click on **Continue** to leave the other details to default.



5. Click **Alert & Notification** , and then select your notification channel from the **Notifications Channel** dropdown.

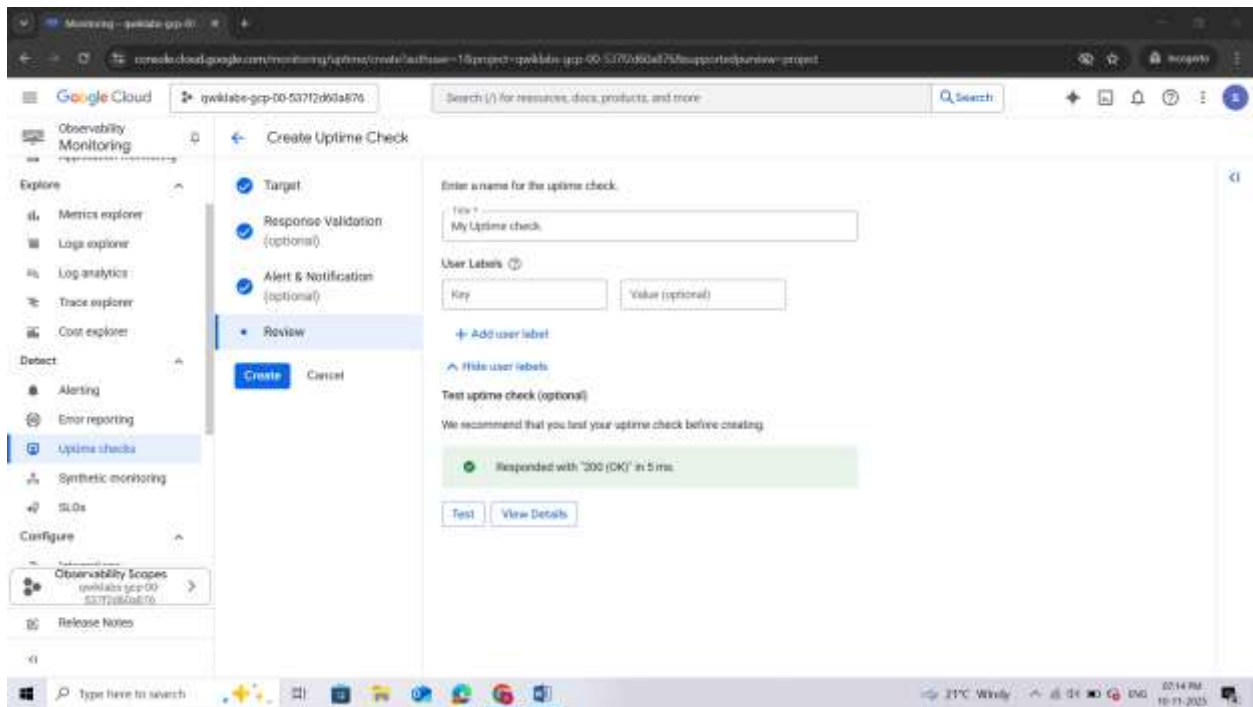
6. Click **Continue**.



7. For **Title**, type **My Uptime check**.

8. Click **Test** to verify that your uptime check can connect to the resource.

9. When you see a green check mark everything can connect. Click **Create**.



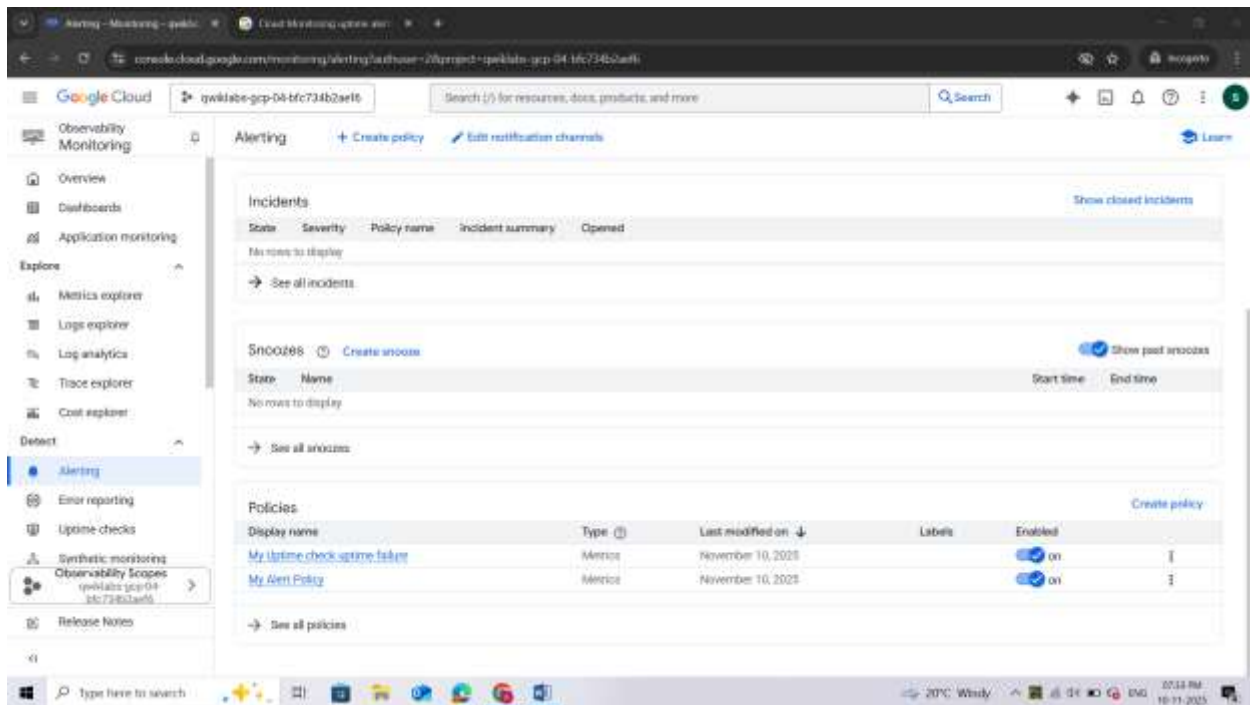
The uptime check you configured takes a while for it to become active.

Task 6. Disable the alert

In this task, you disable an existing alert policy in Cloud Monitoring.

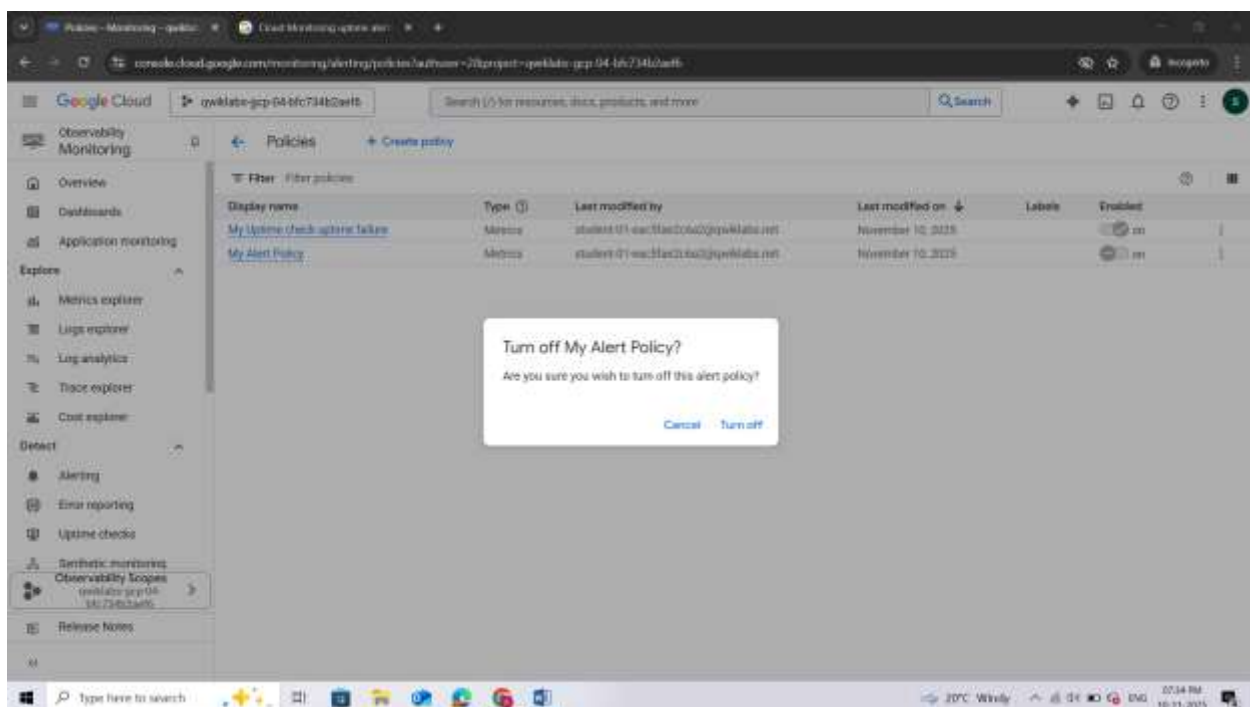
Alerting policies stay active for a while after a project is deleted, just in case it needs to be reinstalled. Since this is a lab, and you will not have access to this project again, remove the alerting policy you created.

1. In the Monitoring tab, click on **Alerting**.
2. Click **See all policies**,



on your **Policies** details page, and then click on the **Enabled** toggle beside your created alert.

3. You will be asked to confirm that you want to turn off the alerting policy - click **Turn off**.



Review

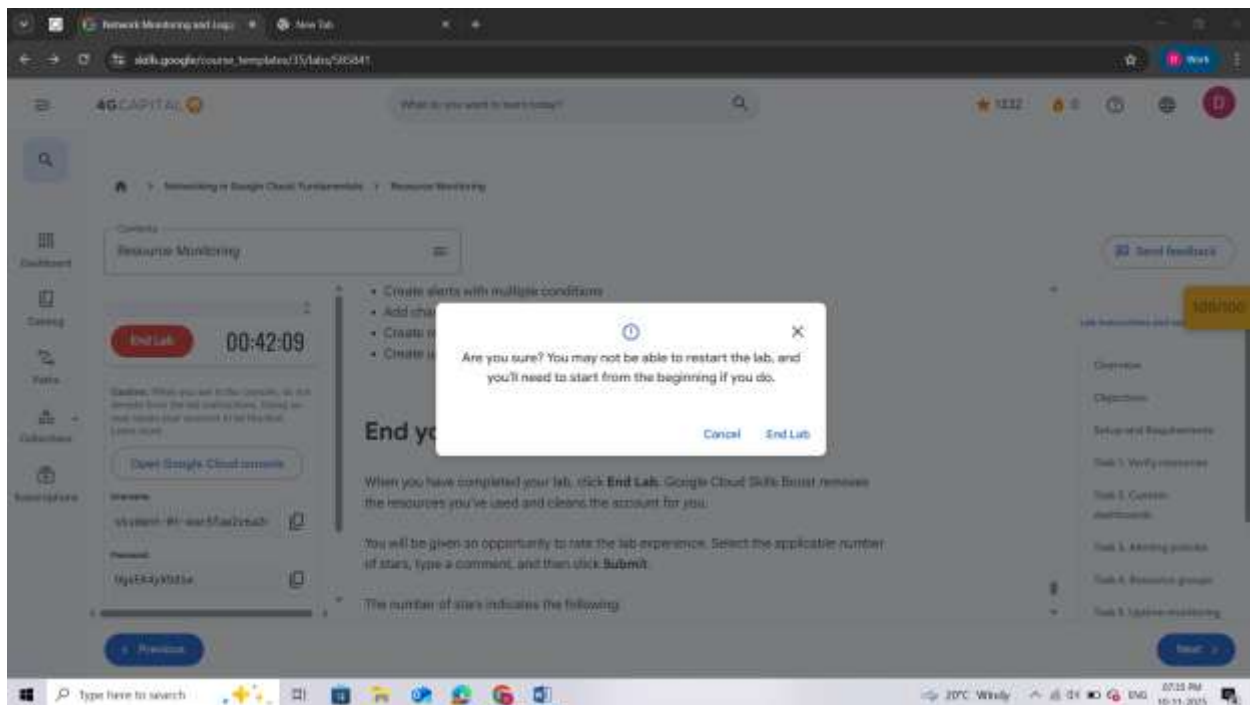
In this lab, you learned how to:

- Explore Cloud Monitoring

- Create a custom dashboard with charts
- Create alerts with multiple conditions
- Add charts to dashboards
- Create resource groups
- Create uptime checks for your services

End your lab

When you have completed your lab, click **End Lab**. Google Cloud Skills Boost removes the resources you've used and cleans the account for you.



LAB 4: ANALYZING NETWORK TRAFFIC WITH VPC FLOW LOGS

Overview

In this lab, you will configure a network to record traffic to and from an Apache web server using VPC Flow Logs. You will then export the logs to BigQuery to analyze them.

Objectives

In this lab, you will learn how to perform the following tasks:

- Configure a custom network with VPC flow logs.
- Create an Apache web server.
- Verify that network traffic is logged.
- Export the network traffic to BigQuery to further analyze the logs.

- Setup VPC flow log aggregation.

Setup and requirements

For each lab, you get a new Google Cloud project and set of resources for a fixed time at no cost.

1. Click the **Start Lab** button. If you need to pay for the lab, a pop-up opens for you to select your payment method. On the left is the **Lab Details** panel with the following:
 - The **Open Google Cloud console** button
 - Time remaining
 - The temporary credentials that you must use for this lab
 - Other information, if needed, to step through this lab
2. Click **Open Google Cloud console** (or right-click and select **Open Link in Incognito Window** if you are running the Chrome browser).

The lab spins up resources, and then opens another tab that shows the **Sign in** page.

***Tip:** Arrange the tabs in separate windows, side-by-side.*

***Note:** If you see the **Choose an account** dialog, click **Use Another Account**.*

3. If necessary, copy the **Username** below and paste it into the **Sign in** dialog.

"Username"

You can also find the **Username** in the **Lab Details** panel.

4. Click **Next**.
5. Copy the **Password** below and paste it into the **Welcome** dialog.

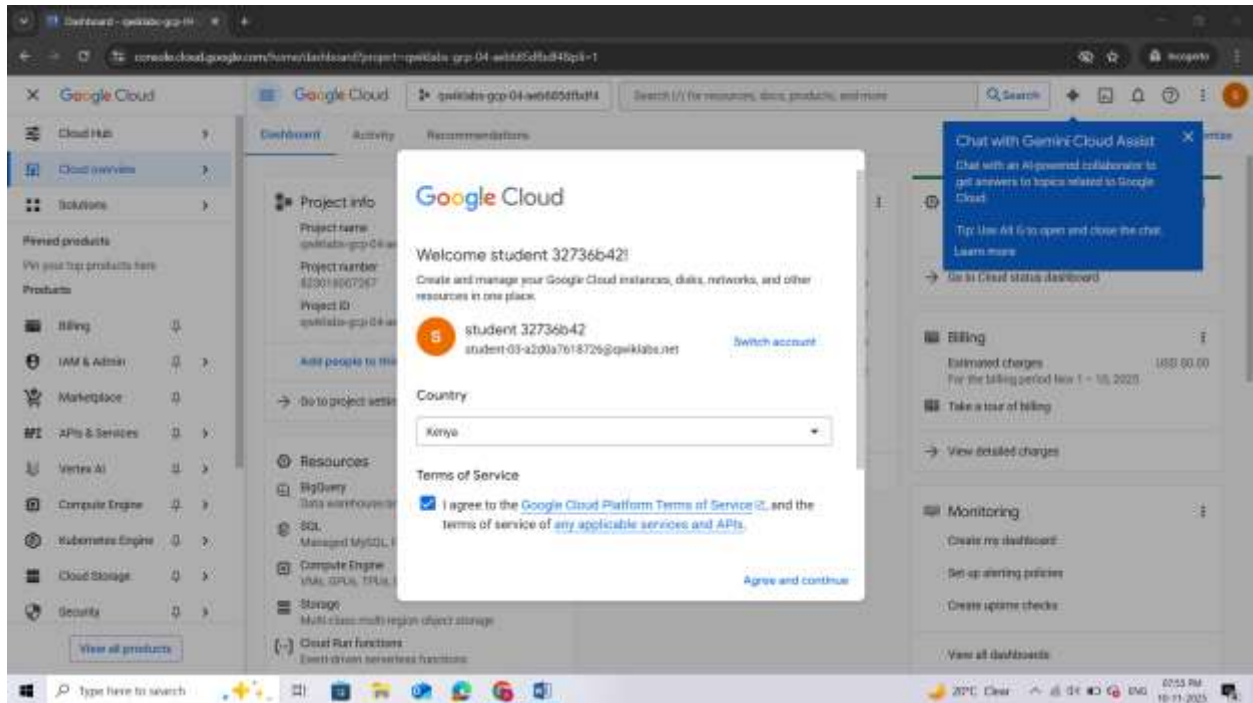
"Password"

You can also find the **Password** in the **Lab Details** panel.

6. Click **Next**.

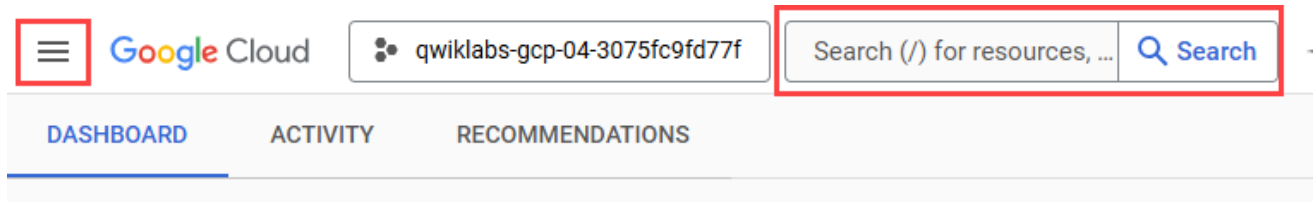
***Important:** You must use the credentials the lab provides you. Do not use your Google Cloud account credentials. **Note:** Using your own Google Cloud account for this lab may incur extra charges.*

7. Click through the subsequent pages:
 - Accept the terms and conditions.
 - Do not add recovery options or two-factor authentication (because this is a temporary account).
 - Do not sign up for free trials.



After a few moments, the Google Cloud console opens in this tab.

Note: To view a menu with a list of Google Cloud products and services, click the **Navigation menu** at the top-left, or type the service or product name in the **Search** field.

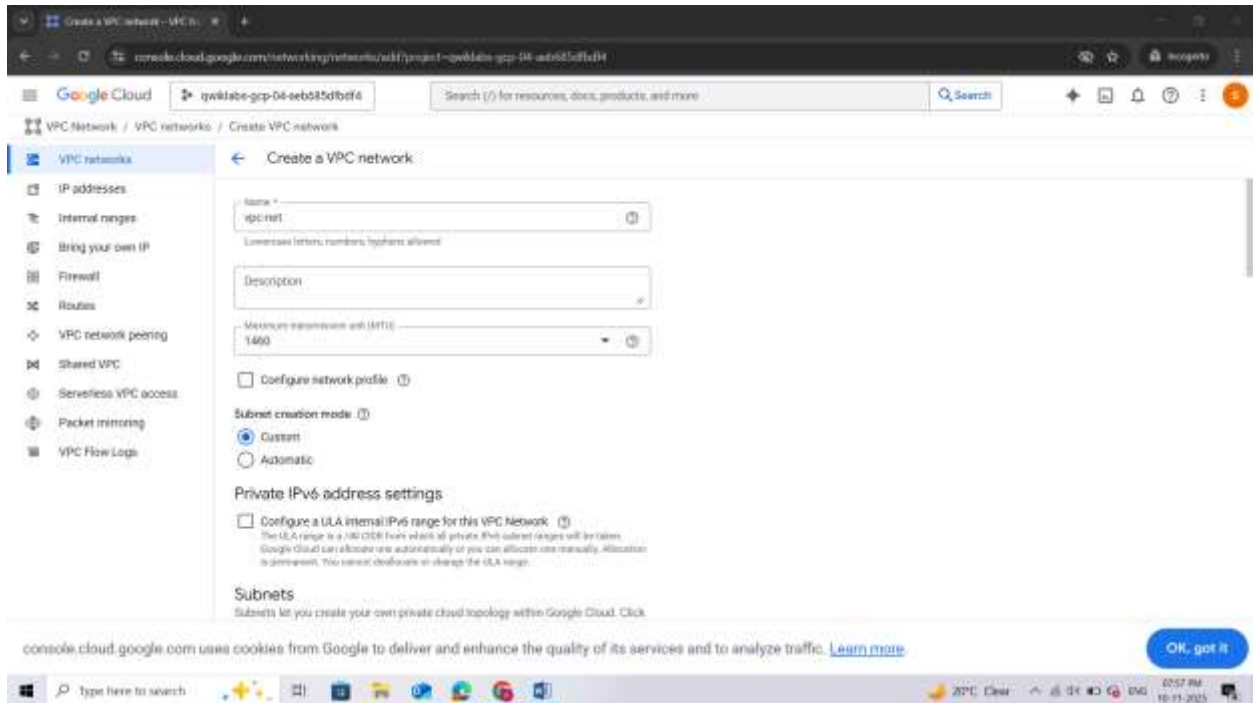


Task 1. Configure a custom network with VPC flow logs

Create the custom network

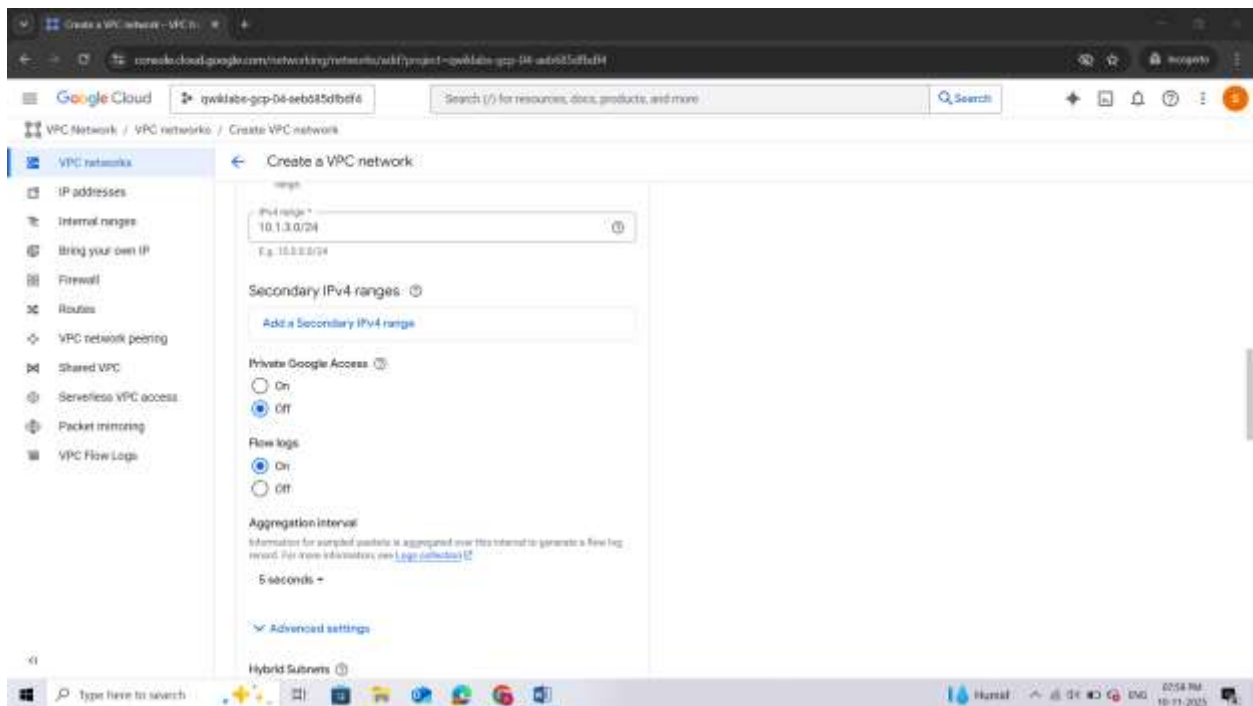
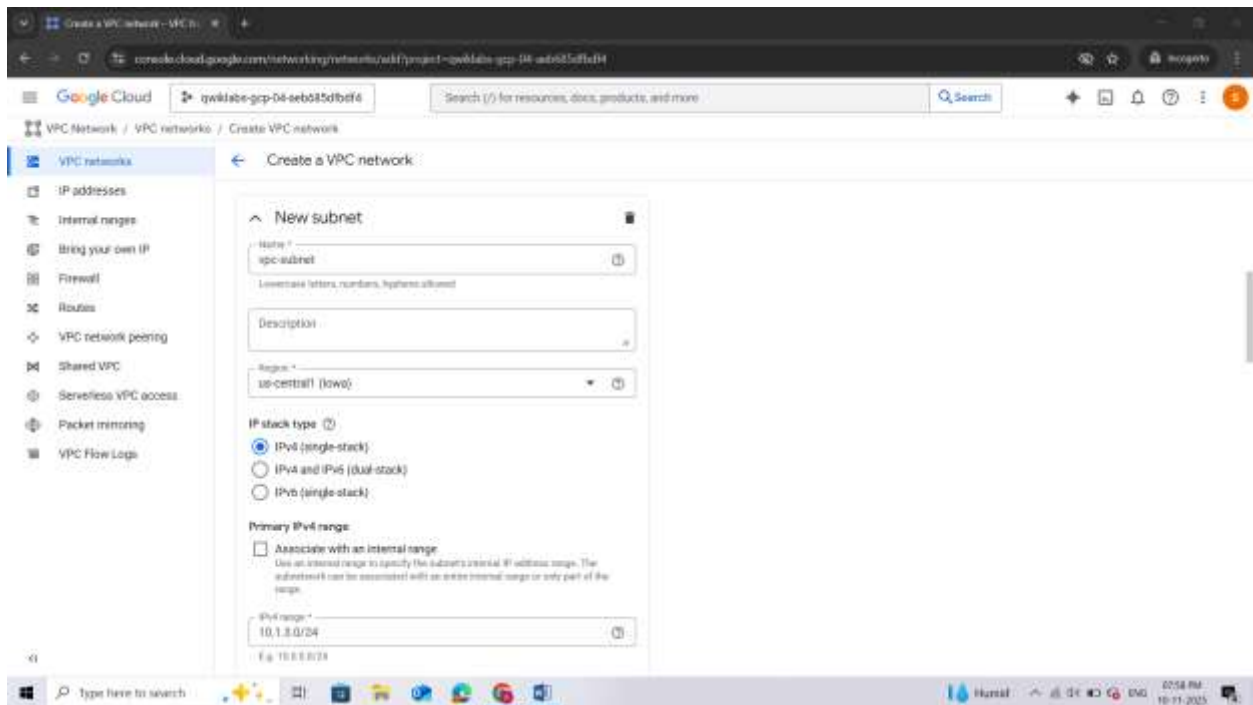
By default, VPC Flow Logs are disabled for a network. Therefore, you will create a new custom-mode network and enable VPC flow logs.

1. In the Google Cloud console, in the **Navigation menu** () , select **VPC network > VPC networks**.



- Specify the following and leave the remaining settings as their defaults:

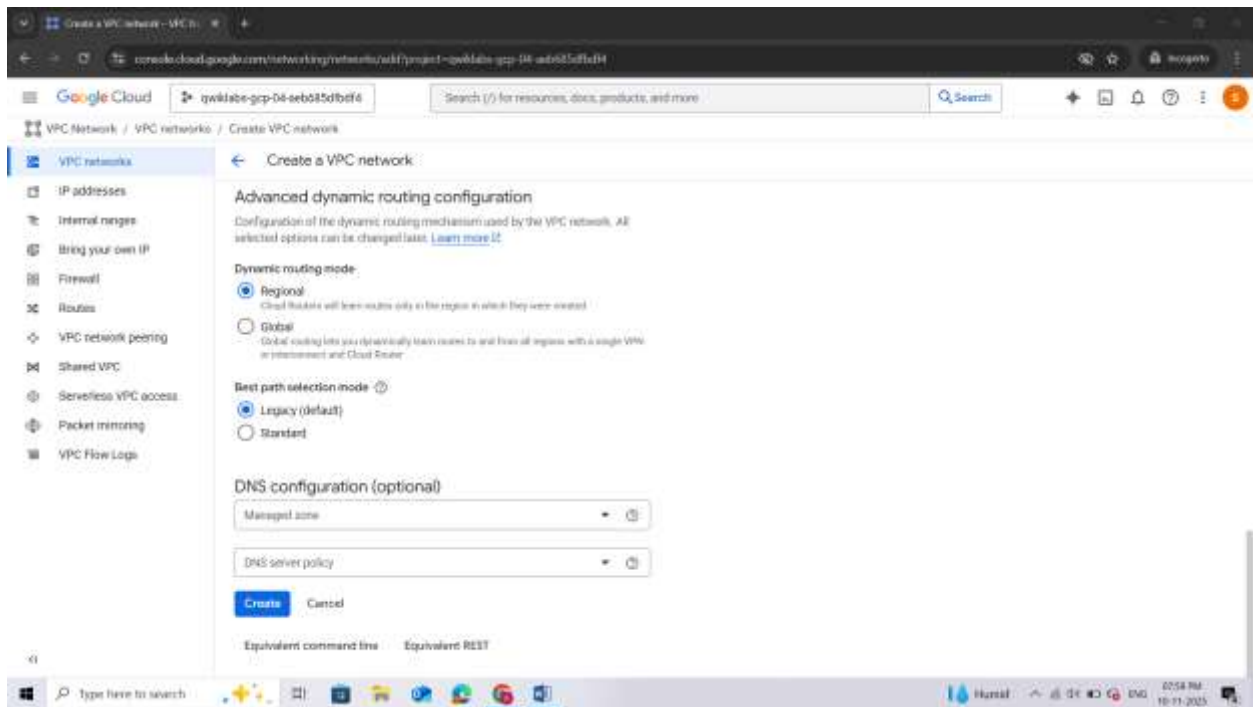
Property	Value (type value or select option as specified)
Name	vpc-subnet
Region	REGION
IPv4 ranges	10.1.3.0/24
Flow Logs	On



Note: Turning on VPC flow logs doesn't affect performance, but some systems generate a large number of logs, which can increase costs. If you click on **Configure logs** you'll notice that you can modify the aggregation interval and sample rate. This allows you to trade off longer interval updates for lower data volume generation which lowers logging costs. For more information on this, refer to the [log sampling and processing](#) documentation.

6. Click **Done**.

7. Click **Create**.

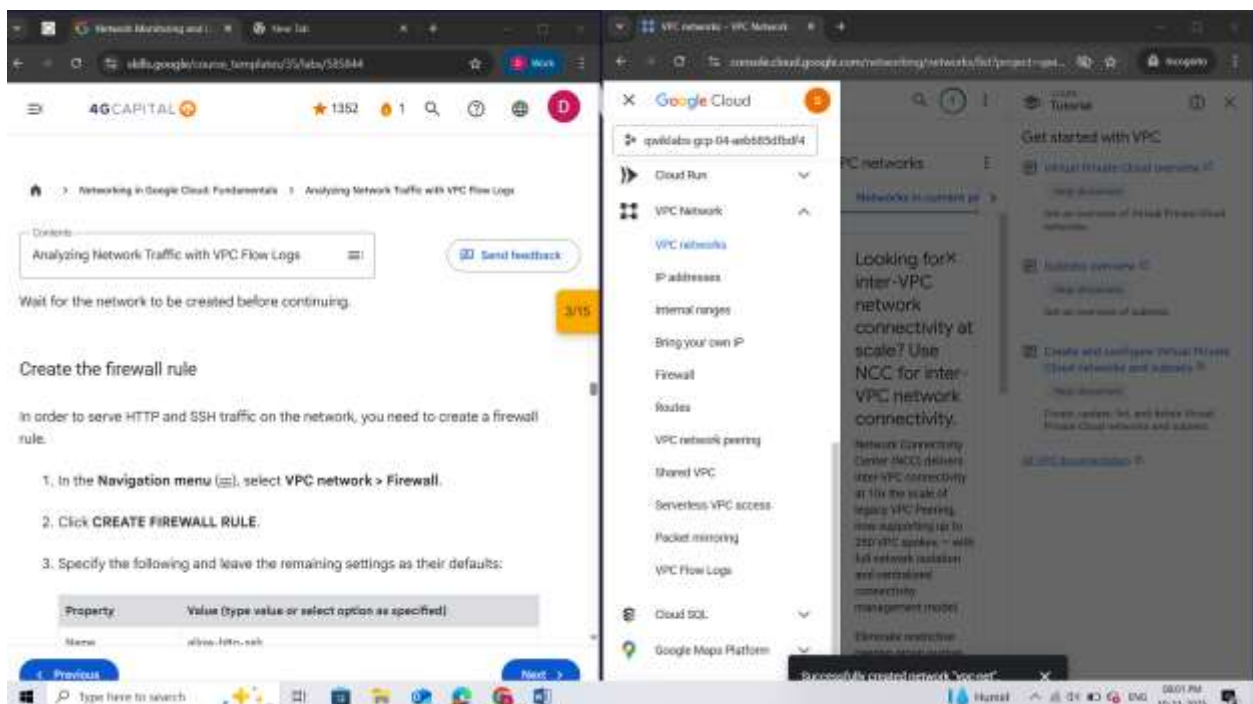


Wait for the network to be created before continuing.

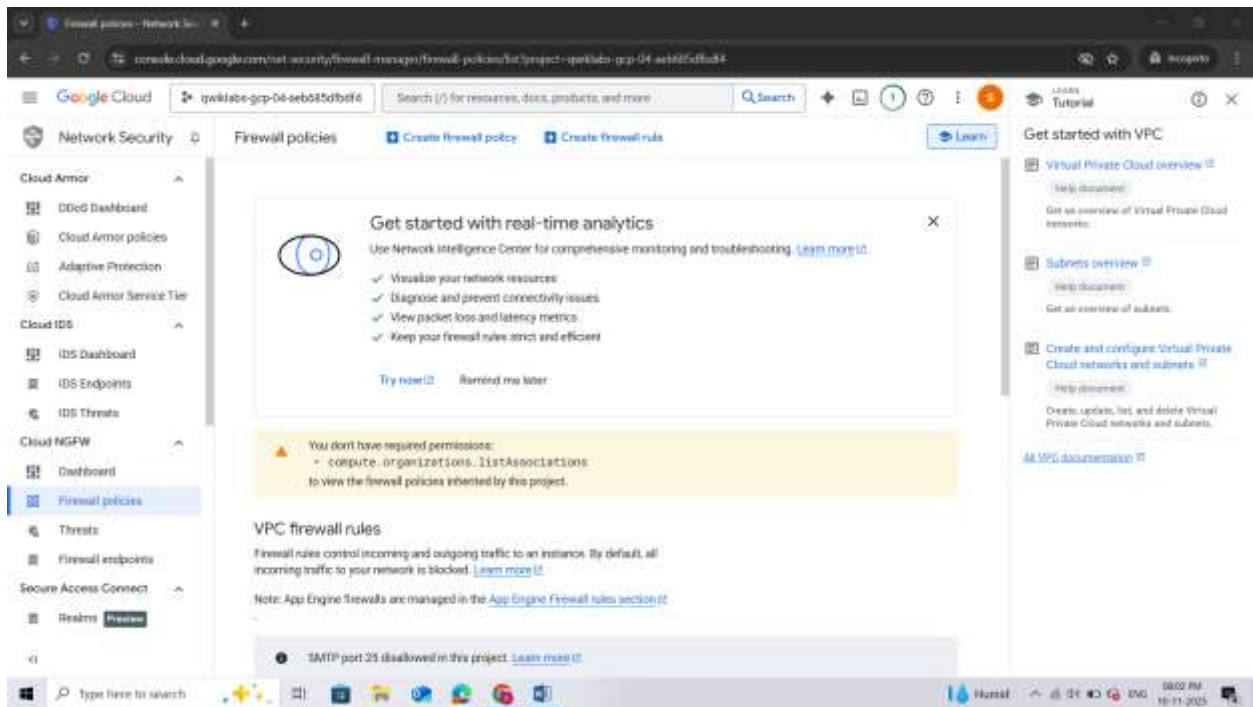
Create the firewall rule

In order to serve HTTP and SSH traffic on the network, you need to create a firewall rule.

1. In the **Navigation menu** () , select **VPC network > Firewall**.



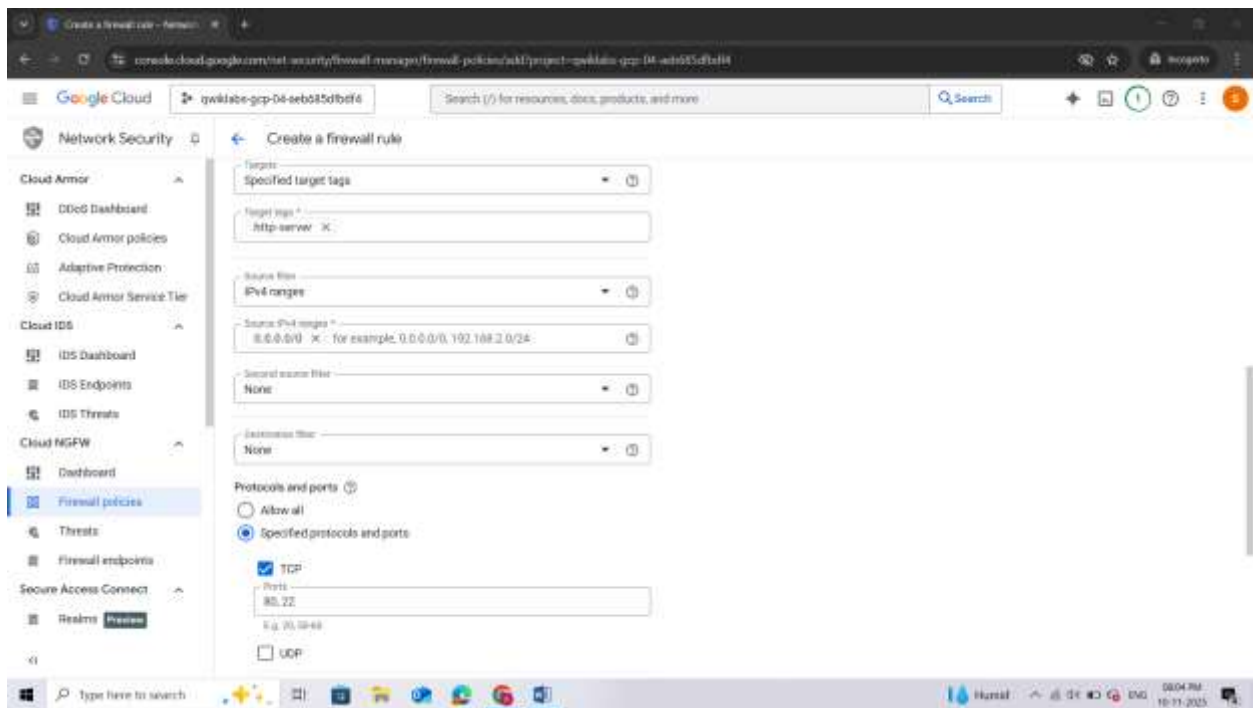
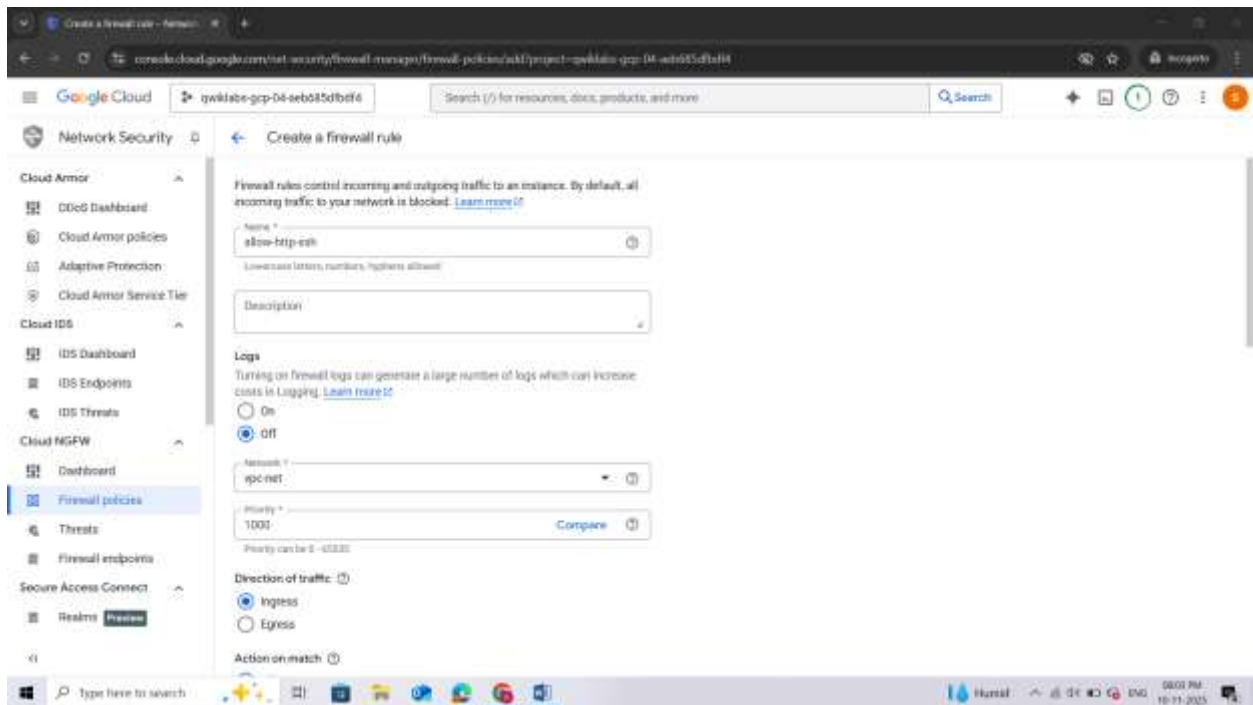
2. Click **CREATE FIREWALL RULE**.



3. Specify the following and leave the remaining settings as their defaults:

Property	Value (type value or select option as specified)
Name	allow-http-ssh
Network	vpc-net
Targets	Specified target tags
Target tags	http-server
Source filter	IPv4 Ranges
Source IPv4 ranges	0.0.0.0/0
Protocols and ports	Specified protocols and ports, and then <i>check tcp, type: 80, 22</i>

Note: Make sure to include the /0 in the **Source IPv4 ranges** to specify all networks.

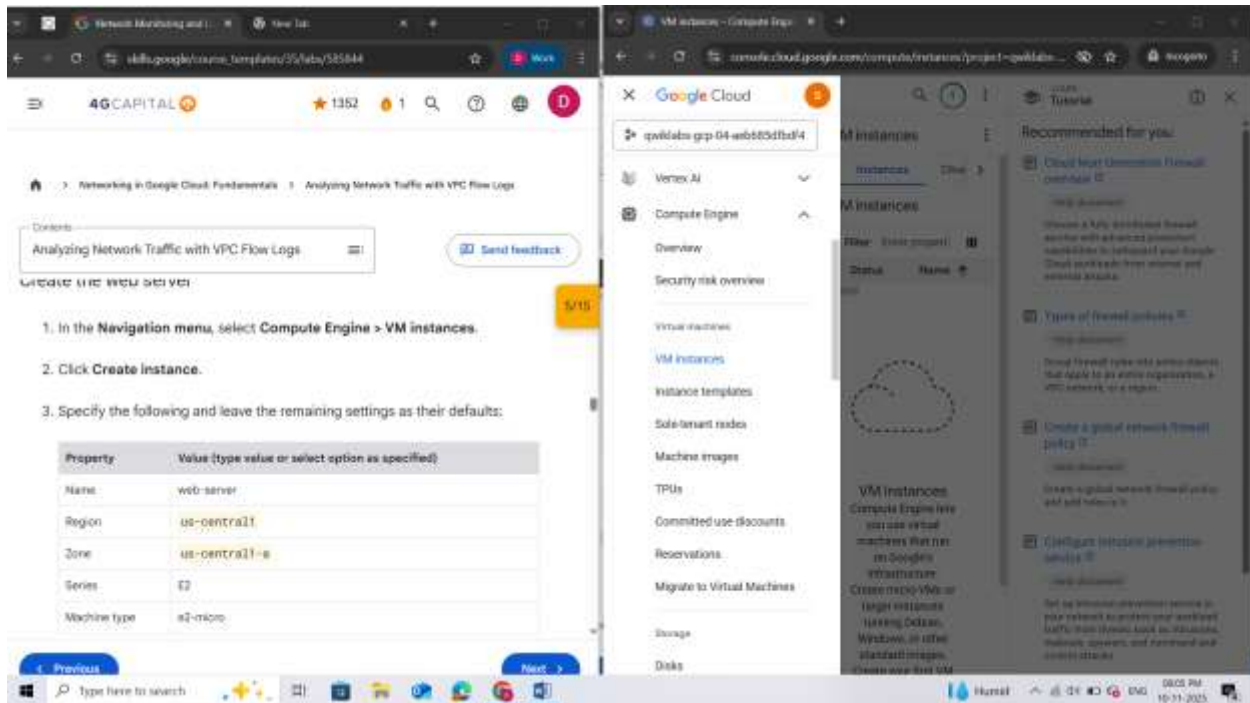


4. Click **Create**.

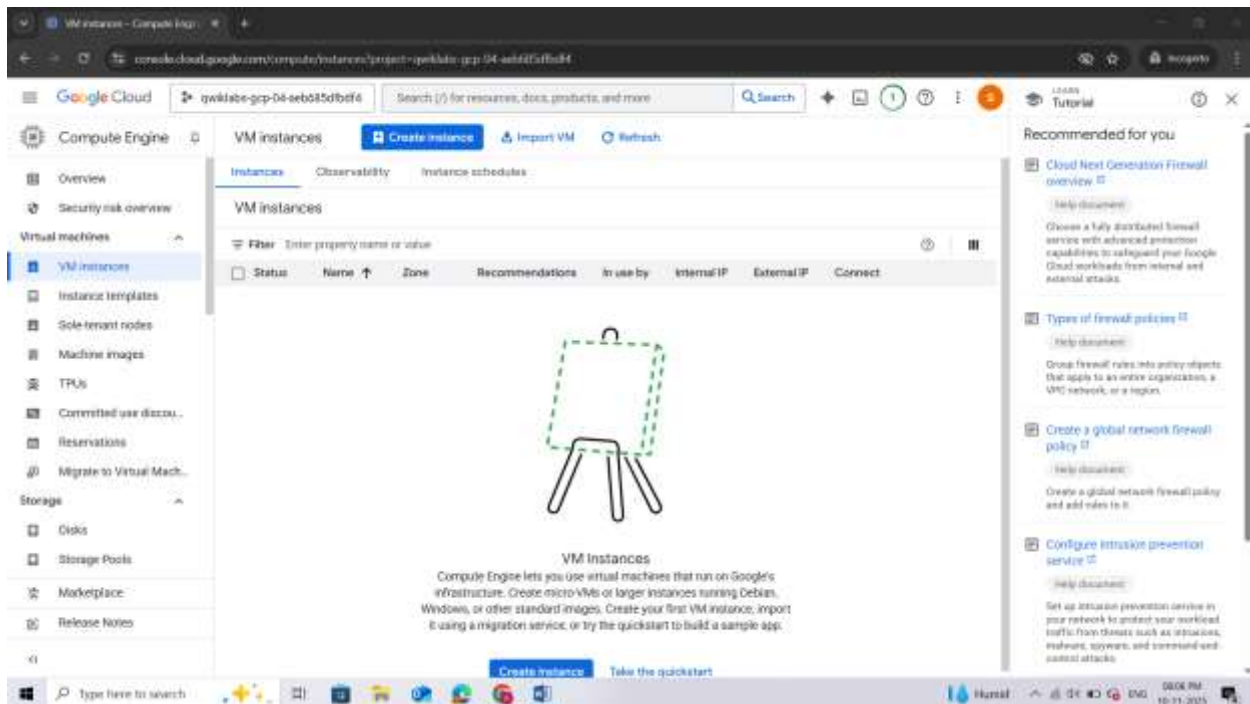
Task 2. Create an Apache web server

Create the web server

1. In the **Navigation menu**, select **Compute Engine > VM instances**.



2. Click **Create instance**.



3. Specify the following and leave the remaining settings as their defaults:

Property	Value (type value or select option as specified)
----------	--

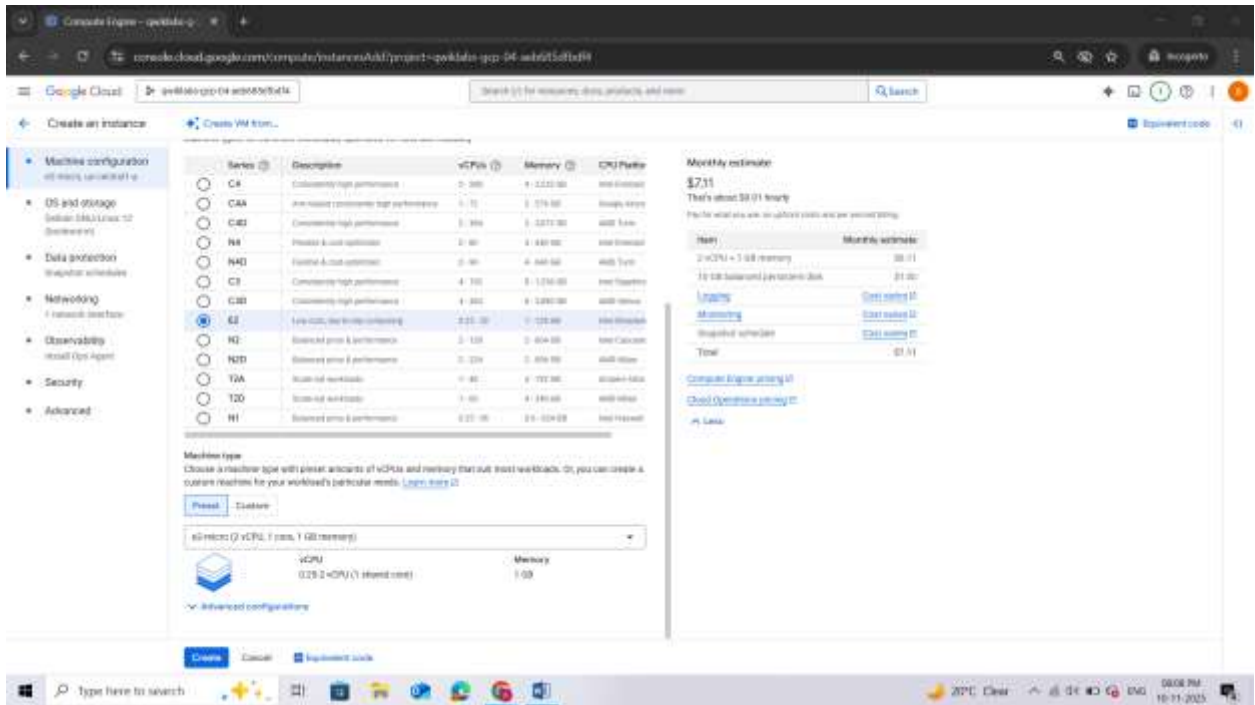
Name	web-server
Region	REGION
Zone	ZONE
Series	E2
Machine type	e2-micro

The screenshot shows the Google Cloud Platform console's 'Create an instance' wizard. The 'Machine configuration' tab is selected, showing a list of machine series. The 'E2' series is highlighted, which is a Linux, x86-64, AMD-based series. The 'Monthly estimate' on the right indicates a total cost of \$25.46, broken down into \$24.40 for the instance and \$1.00 for the disk.

Series	Description	vCPUs	Memory	CPU Platform
C4	Consistently high performance	2, 3, 4	4, 8, 16, 32 GB	Intel Cascade Lake
C4A	General-purpose high performance	1, 2, 4	4, 8, 16, 32 GB	Intel Cascade Lake
C4D	Consistently high performance	2, 4, 8	8, 16, 32, 64 GB	Intel Cascade Lake
H4	Flexible & cost-optimized	2, 4, 8	4, 8, 16, 32 GB	Intel Cascade Lake
H4D	Flexible & cost-optimized	2, 4, 8	4, 8, 16, 32 GB	AMD Zen
C3	Consistently high performance	4, 8, 16	8, 16, 32, 64 GB	Intel Cascade Lake
C3D	Consistently high performance	4, 8, 16	8, 16, 32, 64 GB	AMD Zen
E2	Linux, x86-64, AMD-based series	0.5, 1, 2, 4	1, 2, 4, 8 GB	Intel Cascade Lake
N2	Network-optimized performance	2, 4, 8	8, 16, 32, 64 GB	Intel Cascade Lake
N2D	Network-optimized performance	2, 4, 8	8, 16, 32, 64 GB	AMD Zen
T2A	General-purpose workload	1, 2, 4	4, 8, 16, 32 GB	Intel Cascade Lake

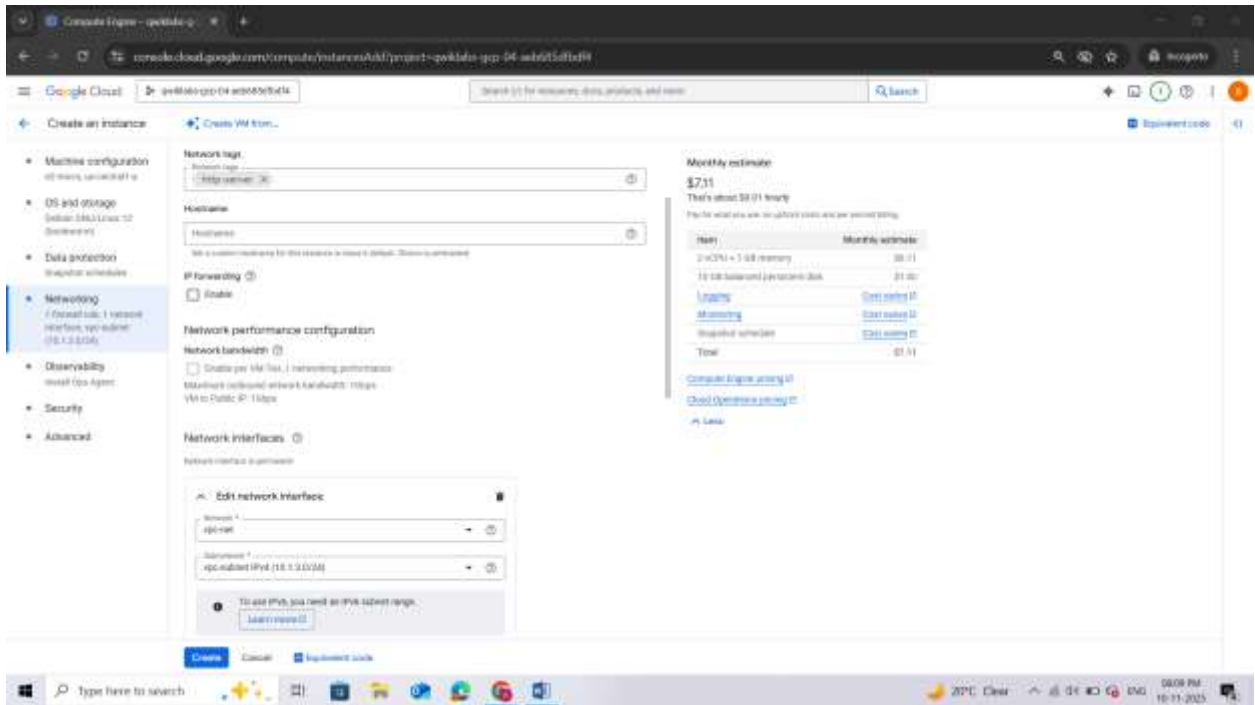
Monthly estimate
\$25.46
 That's about \$0.01 hourly
 Pay for what you use, no upfront costs and no reserved pricing.

Item	Monthly estimate
2 vCPUs + 4 GB memory	\$24.40
10 GB standard persistent disk	\$1.00
Total	\$25.40



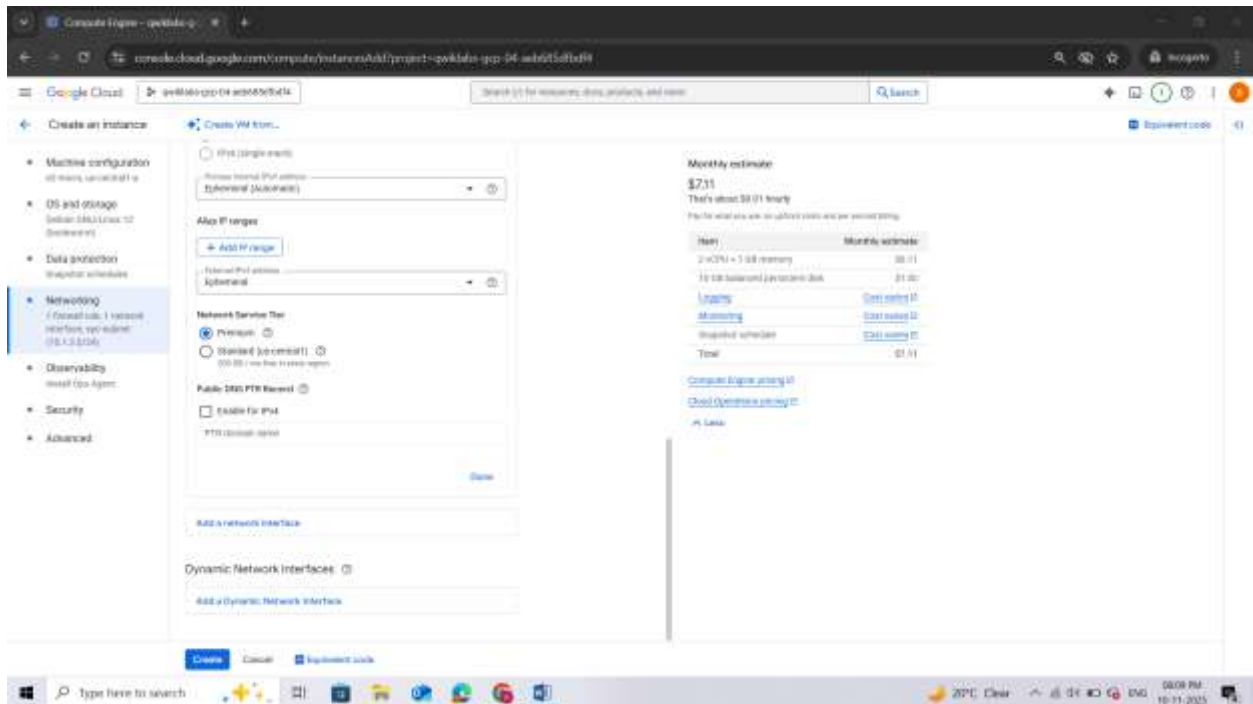
- Click **Networking**.
- For **Network tags**, type http-server.
- For **Network interfaces**, click **default**.
- Specify the following and leave the remaining settings as their defaults:

Property	Value (type value or select option as specified)
Network	vpc-net
Subnetwork	vpc-subnet (10.1.3.0/24)



8. Click **Done**.

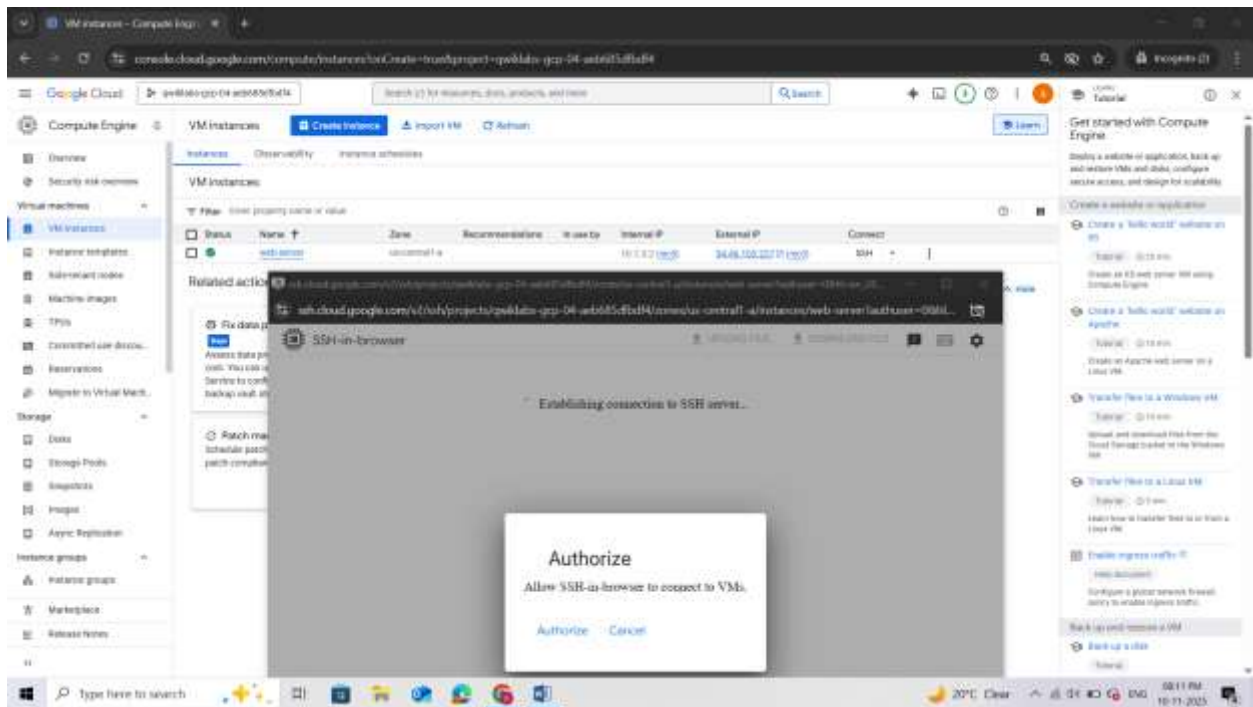
9. Click **Create**.



Install Apache

Configure the VM instance that you created as an Apache web server and overwrite the default web page.

1. For **web-server**, click **SSH** to launch a terminal and connect.



2. In the **web-server** SSH terminal, update the package index:

```
sudo apt-get update
```

3. Install the apache2 package:

```
sudo apt-get install apache2 -y
```

[illegible]

Task 3. Verify that network traffic is logged

Generate network traffic

1. Return to the **VM instances** page in the Cloud console.
2. For **web-server**, click on the **External IP** to access the server.

Note: You should see the **Hello World!** welcome page that you configured. Alternatively, you can access the server in a new tab by navigating to `http://Enter the external IP Address`.



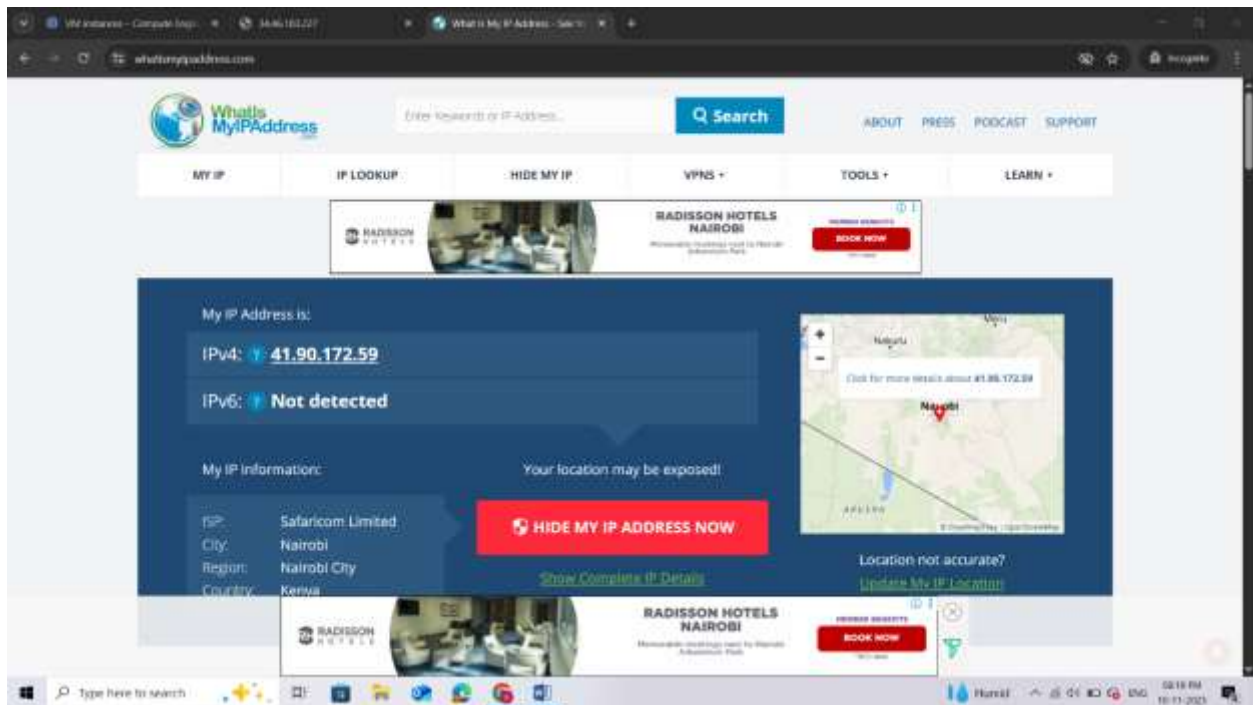
Hello World!



Find your IP address

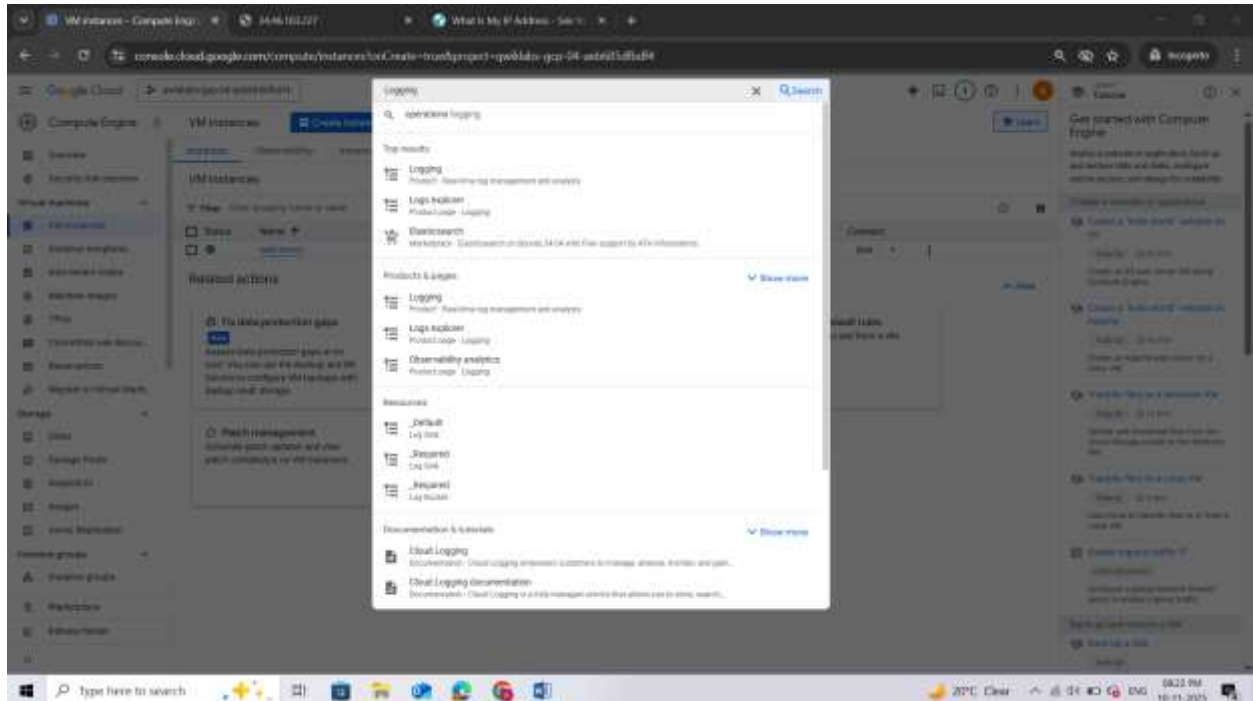
Find the IP address of the computer you are using. One easy way to do this is to go to a website that provides this address.

1. Open a browser in a new tab.
2. Go to [Google.com](https://www.google.com) and search for **what's my IP**. It will either directly reply with your IP or give you a list of sites that perform this service.
3. Ensure that the IP address only contains numerals (IPv4) and is not represented in hexadecimal (IPv6).
4. Copy your IP address. It will be referred to as **YOUR_IP_ADDRESS**. **41.90.172.59**

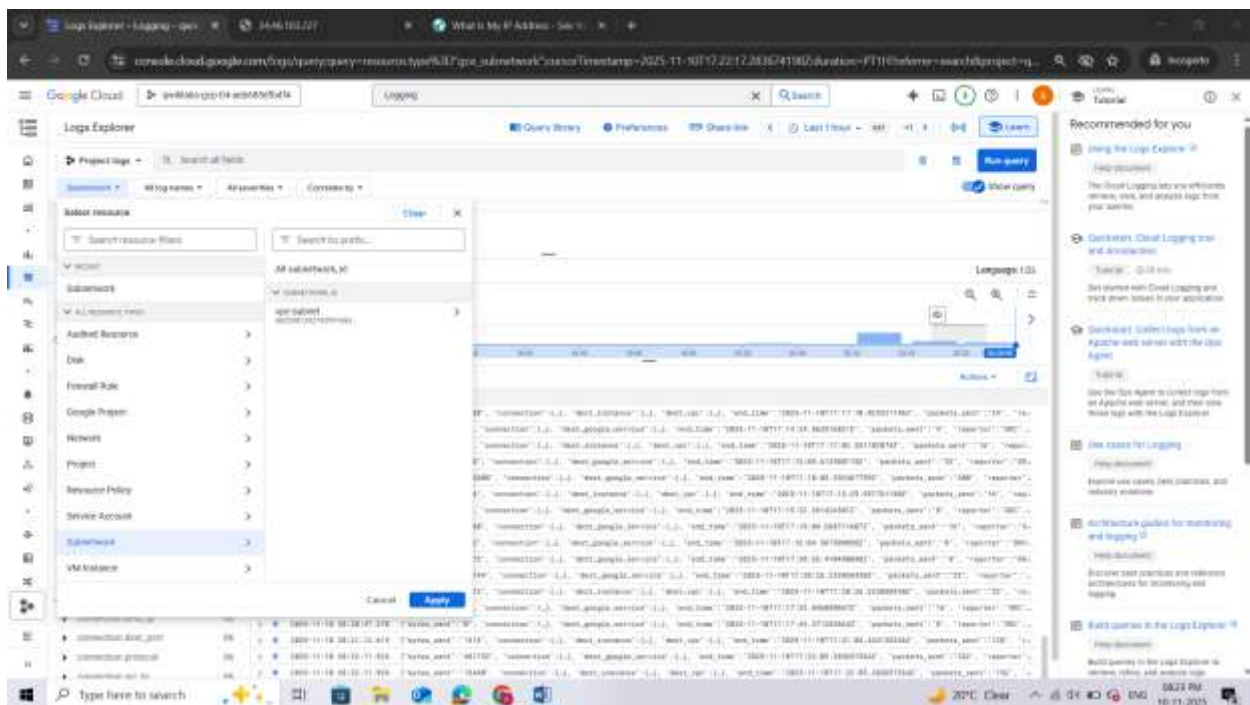


Access the VPC flow logs

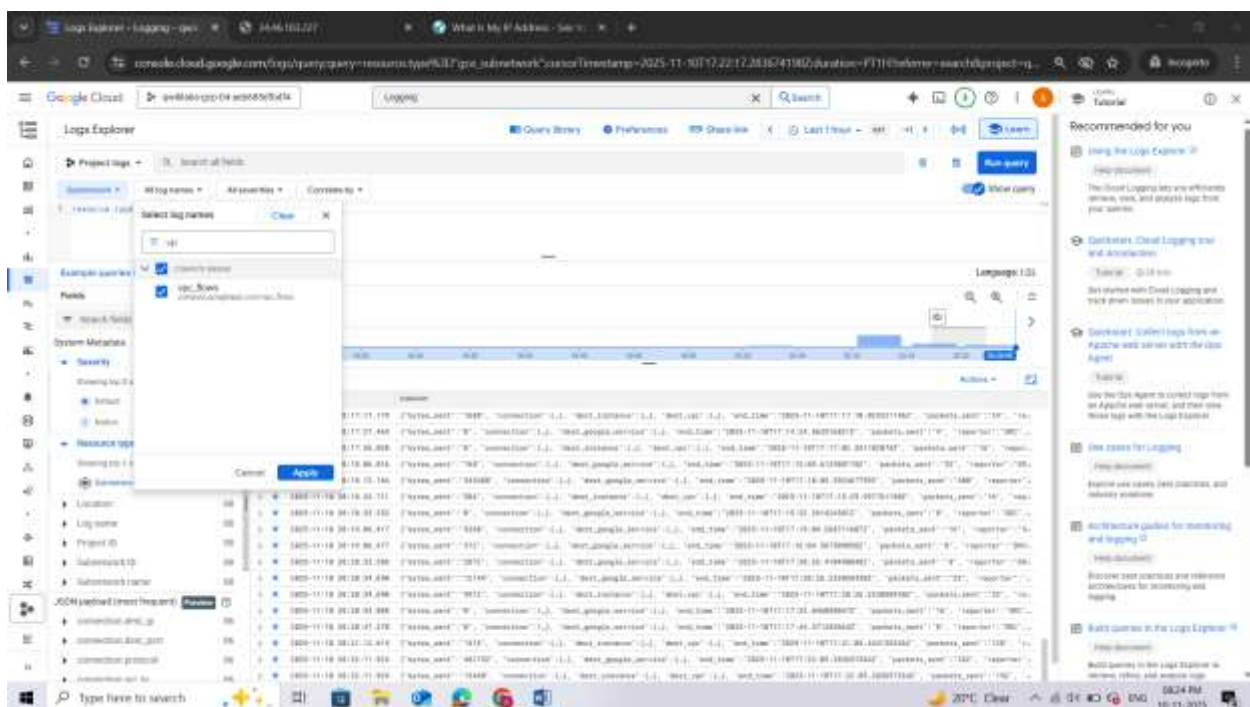
1. On the Google Cloud console title bar, in the Search field, type **Logging**, click Search, and then click **Logging**.



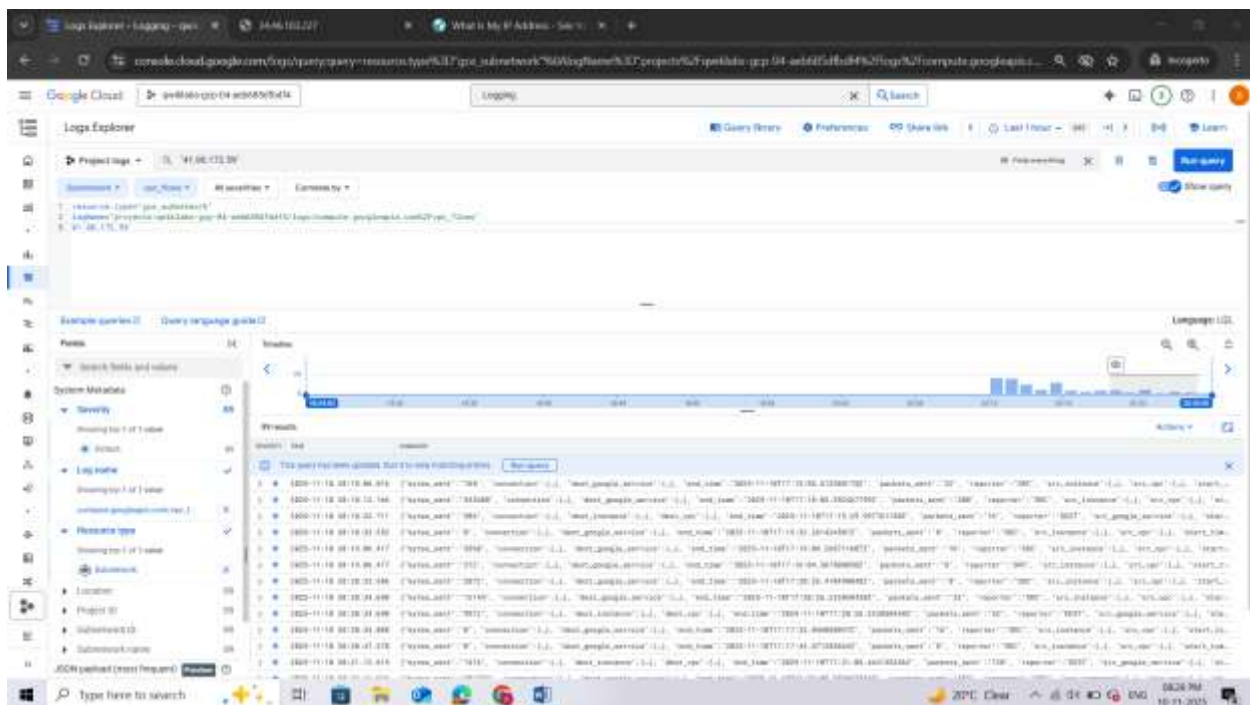
2. In the **All resources** selector dropdown, select **Subnetwork** and click **Apply**. In the Query results pane, entries from the subnetwork logs appear.



3. In the **All log names** selector dropdown, select **vpc_flows** and click **Apply**. In the Query results panel, entries from the VPC flow logs appear. If you do not see **vpc_flows**, wait a few minutes for this log type to show up.

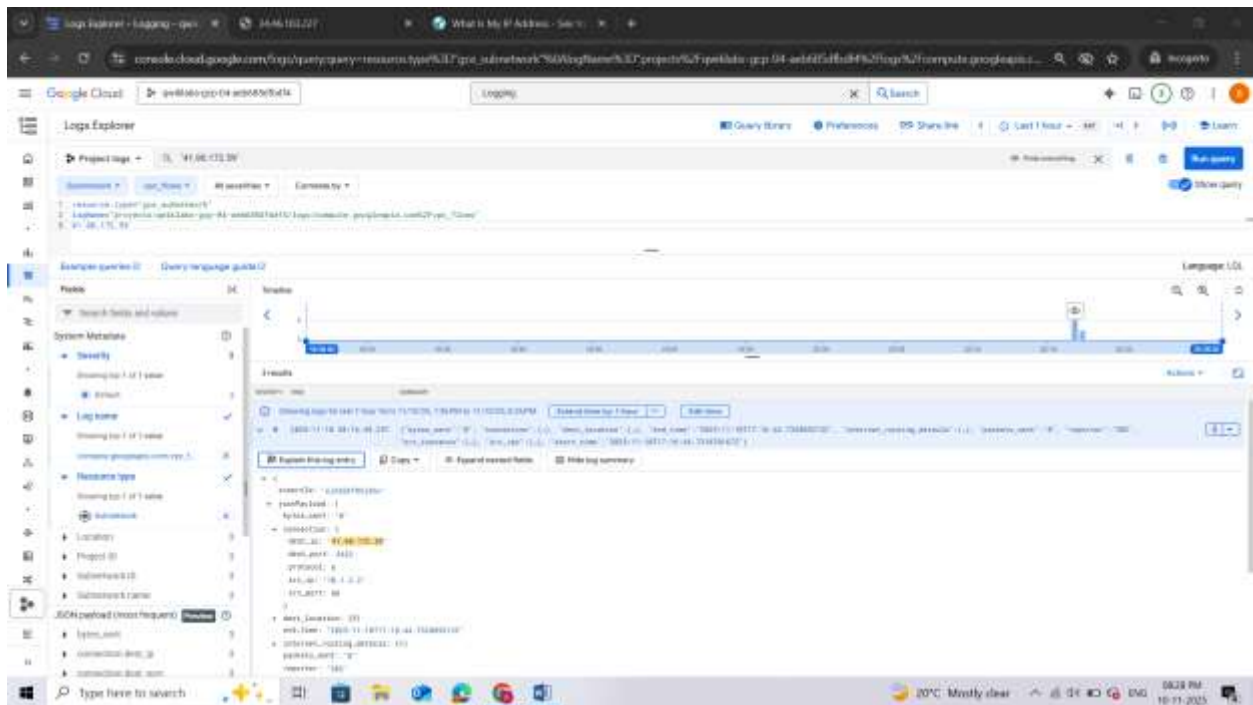


4. In the Query builder box, at the end of line 2, press **Enter** to create a new line.
5. On line 3, enter **YOUR_IP_ADDRESS** and click **Run Query**.



Note: If you do not see the **vpc_flows** filter option or any logs after running the query, you might have to wait a few minutes and refresh. If the **vpc_flows** filter option or logs still do not appear, click on the **External IP** of the **web-server** a few times to generate more traffic. Then, check the **vpc_flows** filter option again and rerun the query.

6. Expand one of the log entries.
7. Within the entry, expand the **jsonPayload** and then expand the **connection**.



Which fields does the connection contain?



Source IP address



Source port



Destination IP address



Destination port



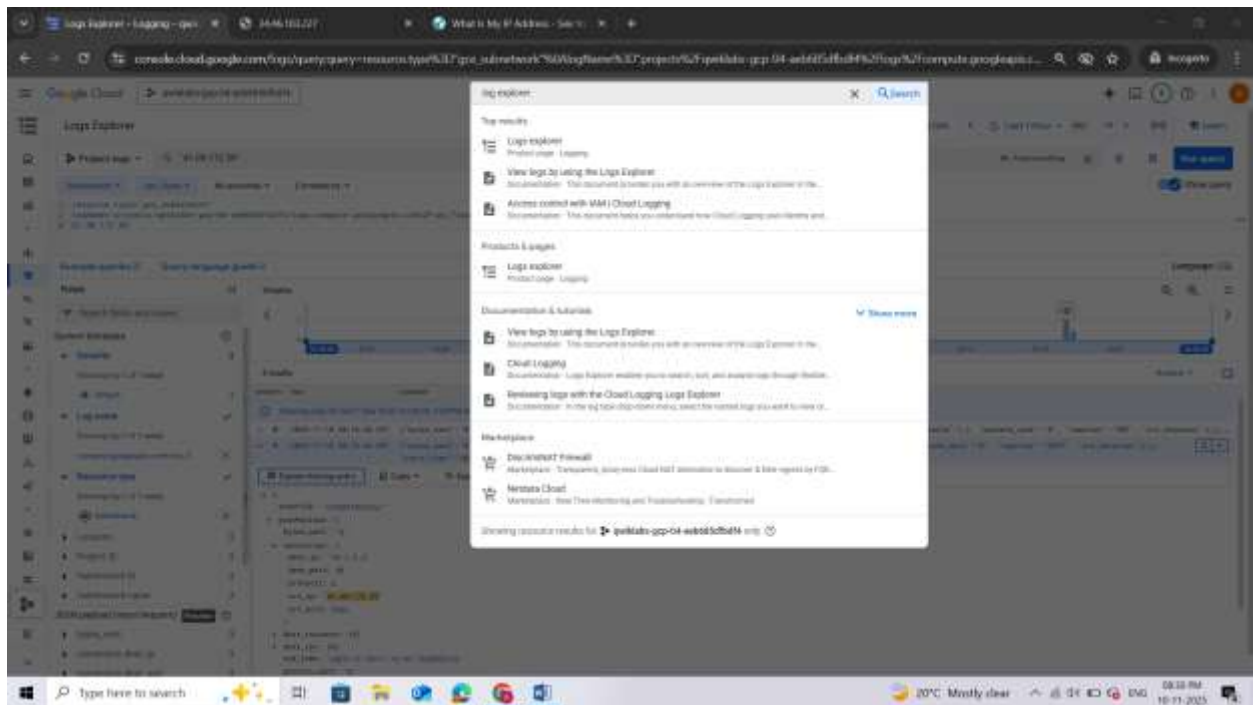
The IANA protocol number

You can explore other fields within the log entry before continuing to the next task.

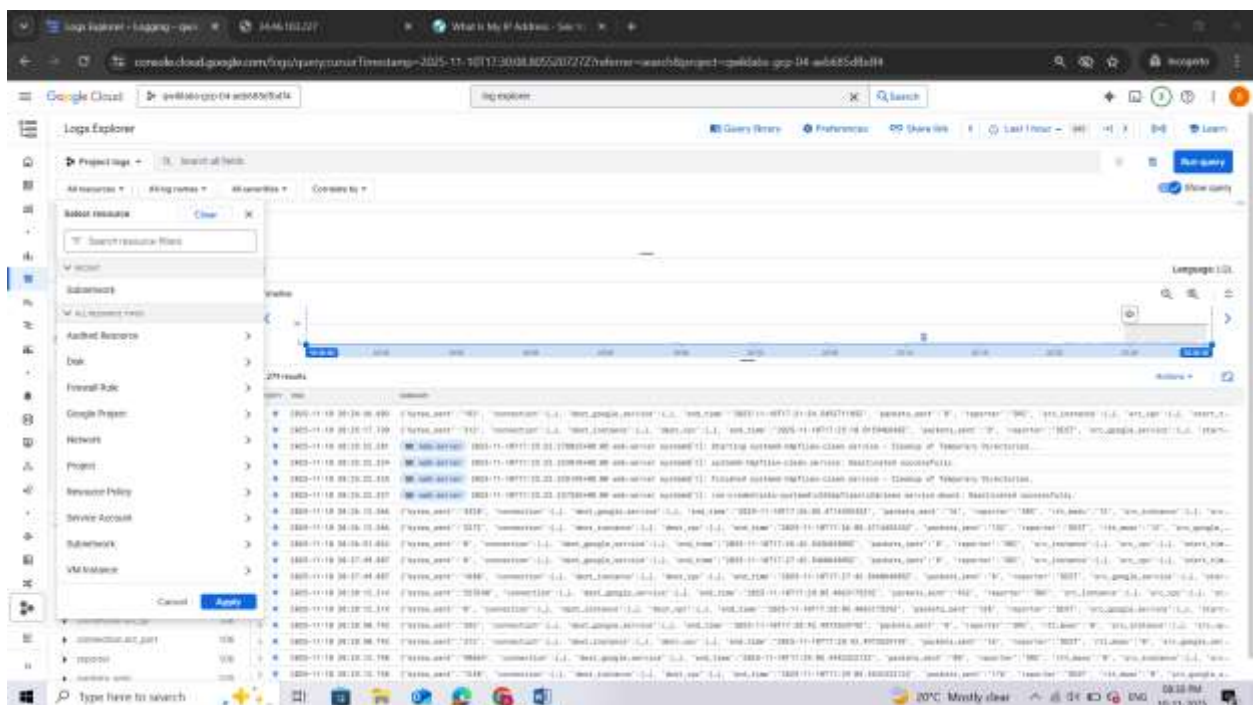
[Task 4. Export the network traffic to BigQuery to further analyze the logs](#)

Create an export sink

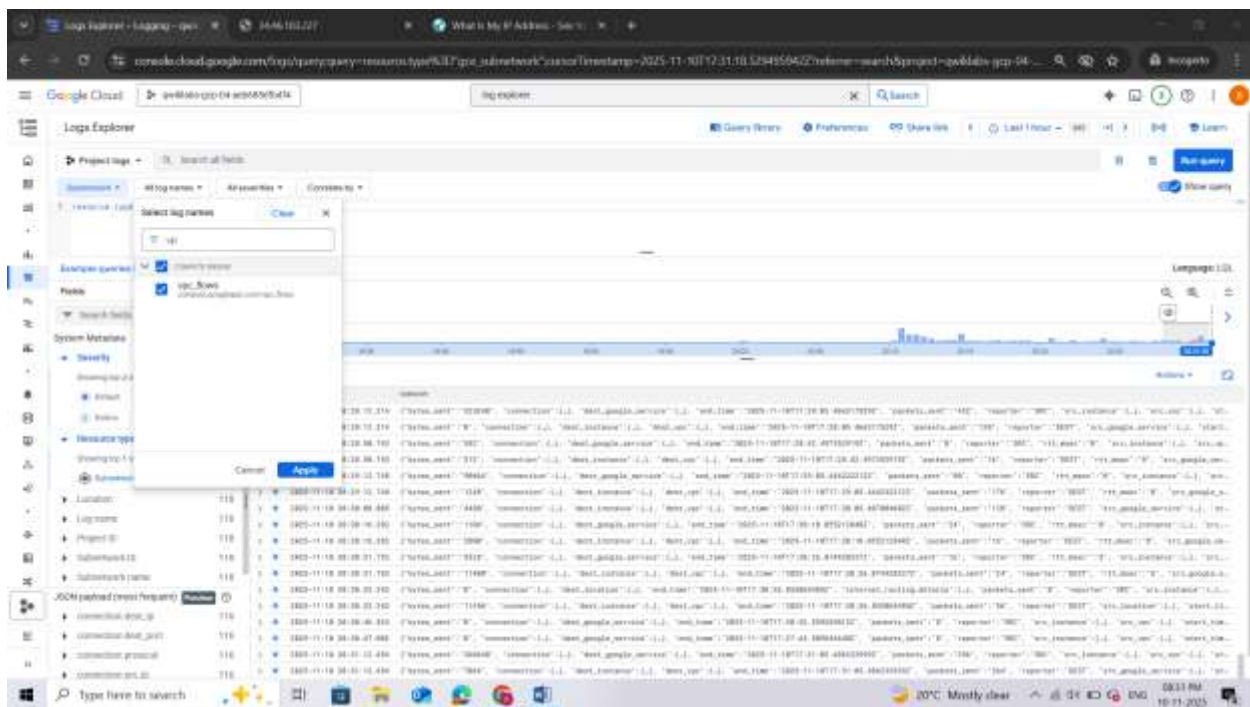
1. On the Google Cloud console title bar, type **Logs explorer** in the **Search** field, then select **Logs explorer** from **Search Results**.



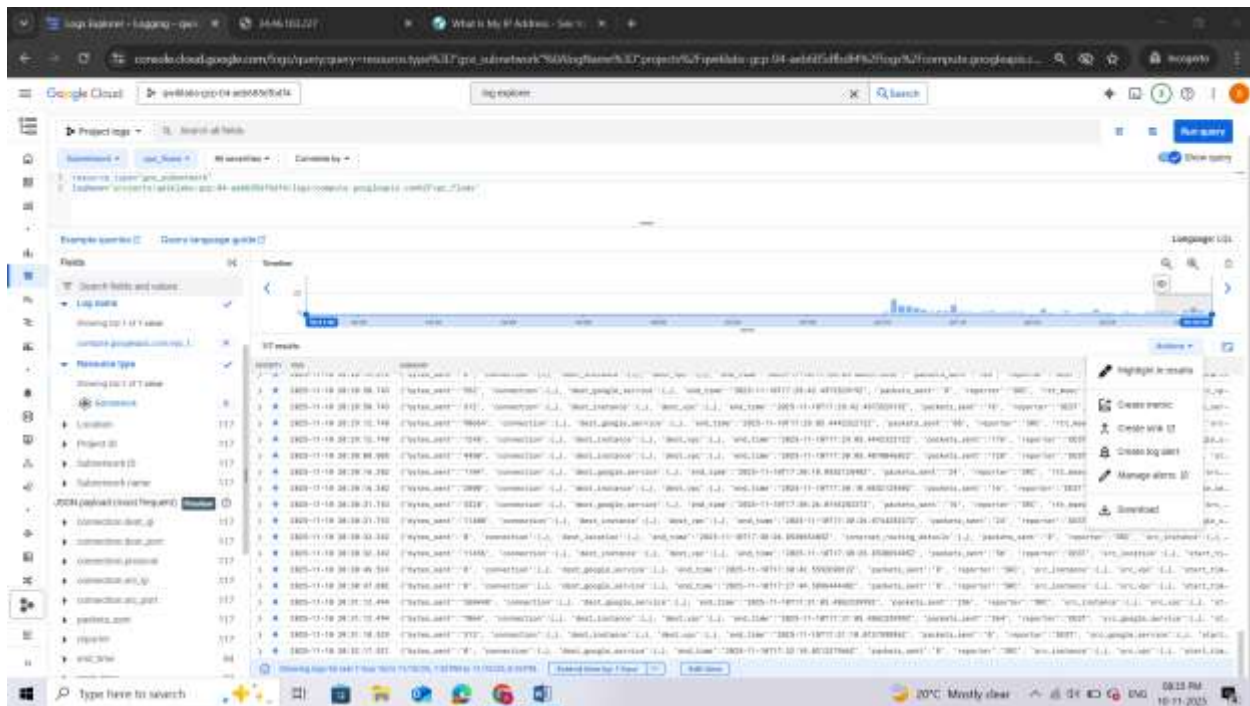
2. In the **All resources** selector dropdown, select **Subnetwork** and click **Apply**. In the Query results pane, entries for all available subnetworks appear.



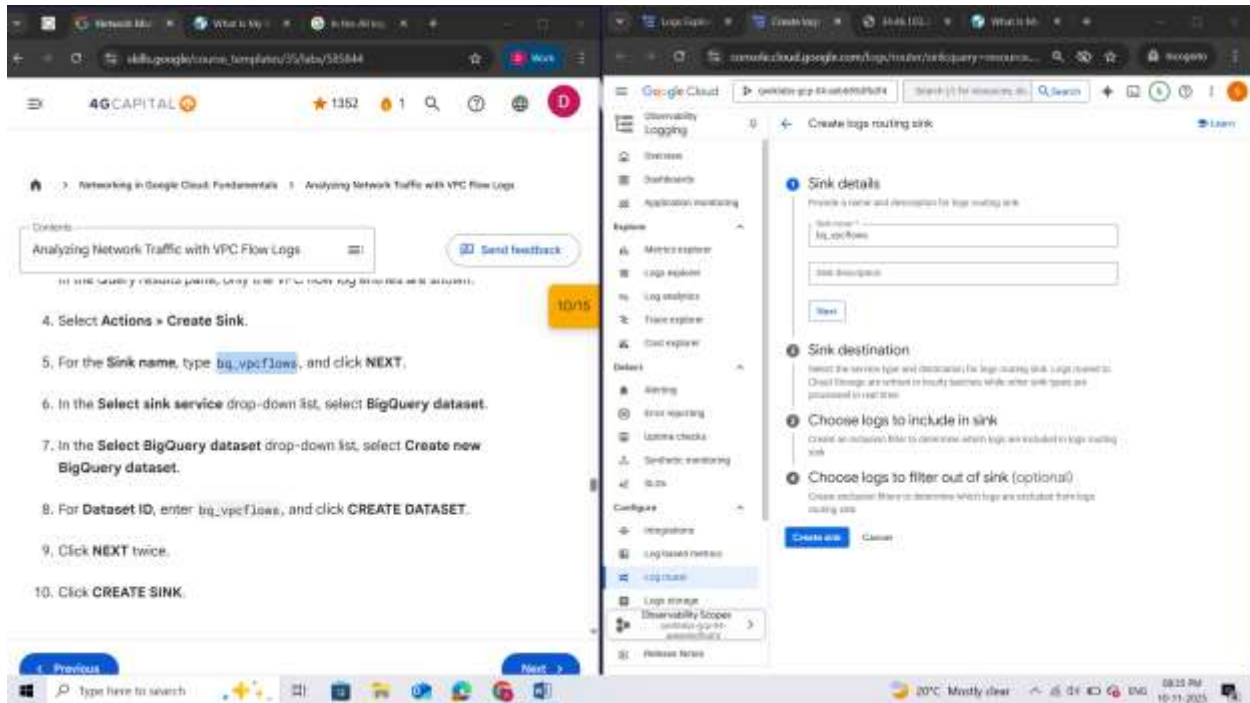
3. In the **All log names** selector dropdown, select **vpc_flows** and click **Apply**. In the Query results pane, only the VPC flow log entries are shown.



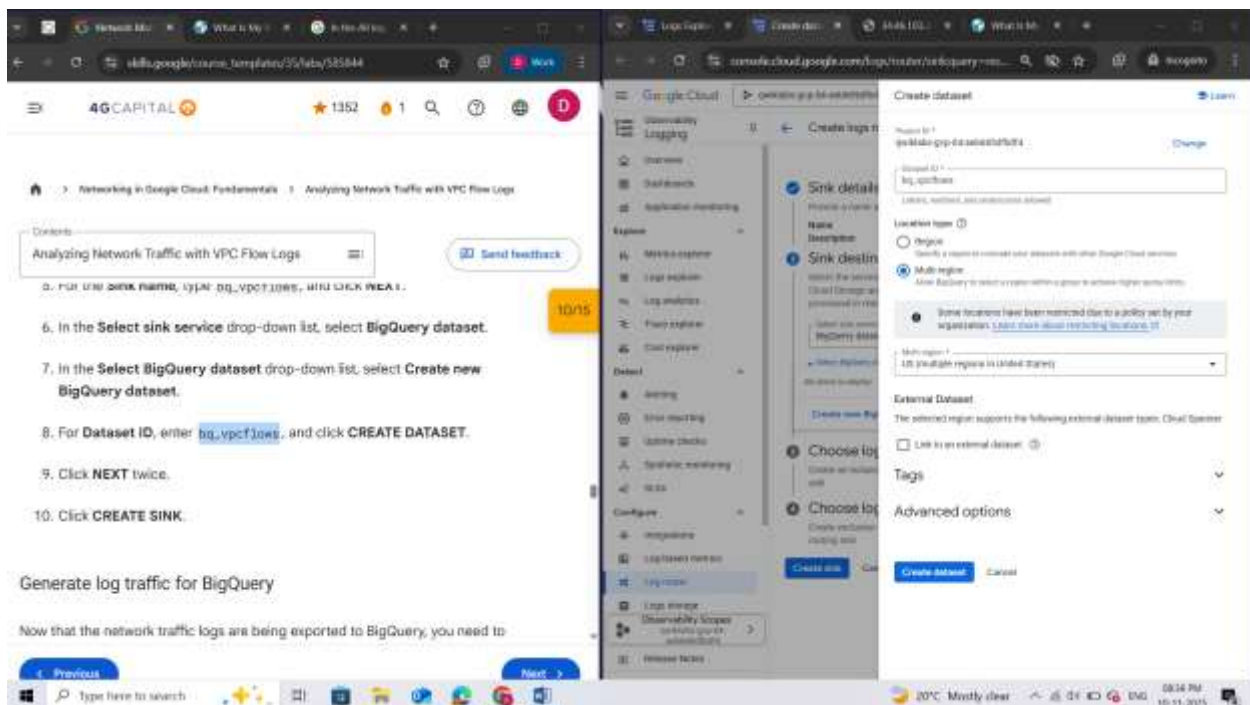
4. Select Actions > Create Sink.



5. For the Sink name, type bq_vpcflows, and click NEXT.



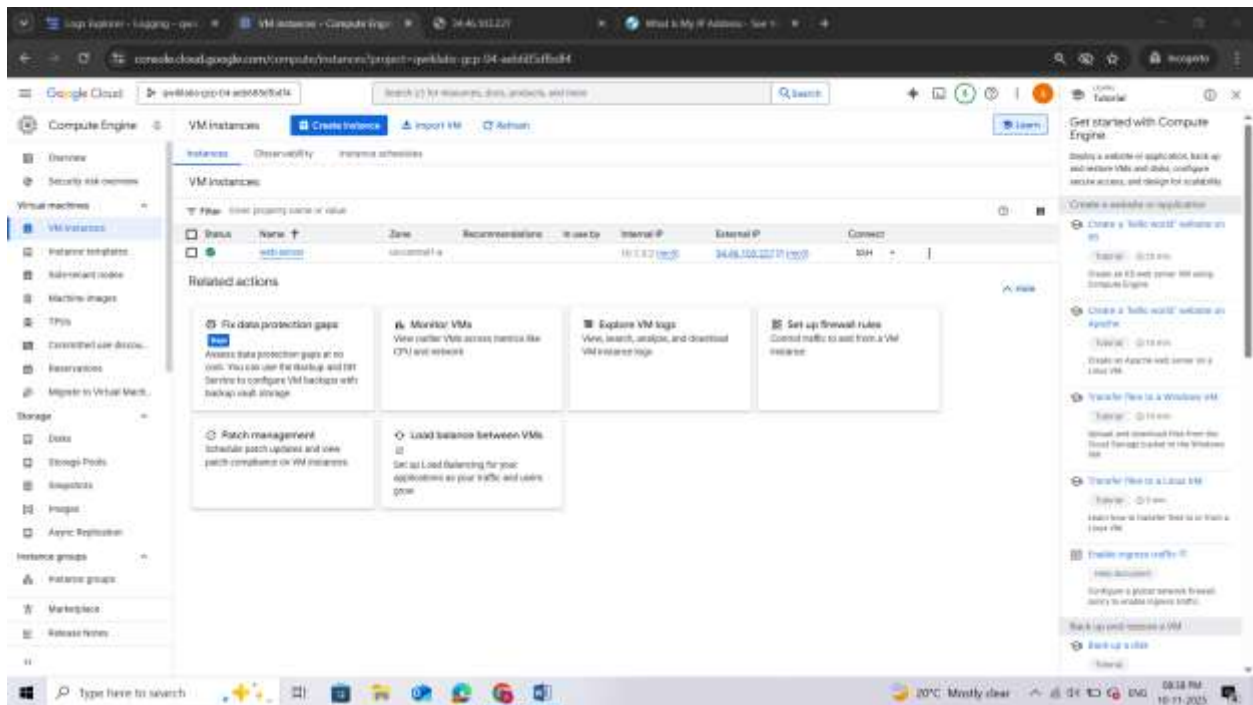
6. In the **Select sink service** drop-down list, select **BigQuery dataset**.
7. In the **Select BigQuery dataset** drop-down list, select **Create new BigQuery dataset**.
8. For **Dataset ID**, enter **bq_vpcflows**, and click **CREATE DATASET**.



9. Click **NEXT** twice.

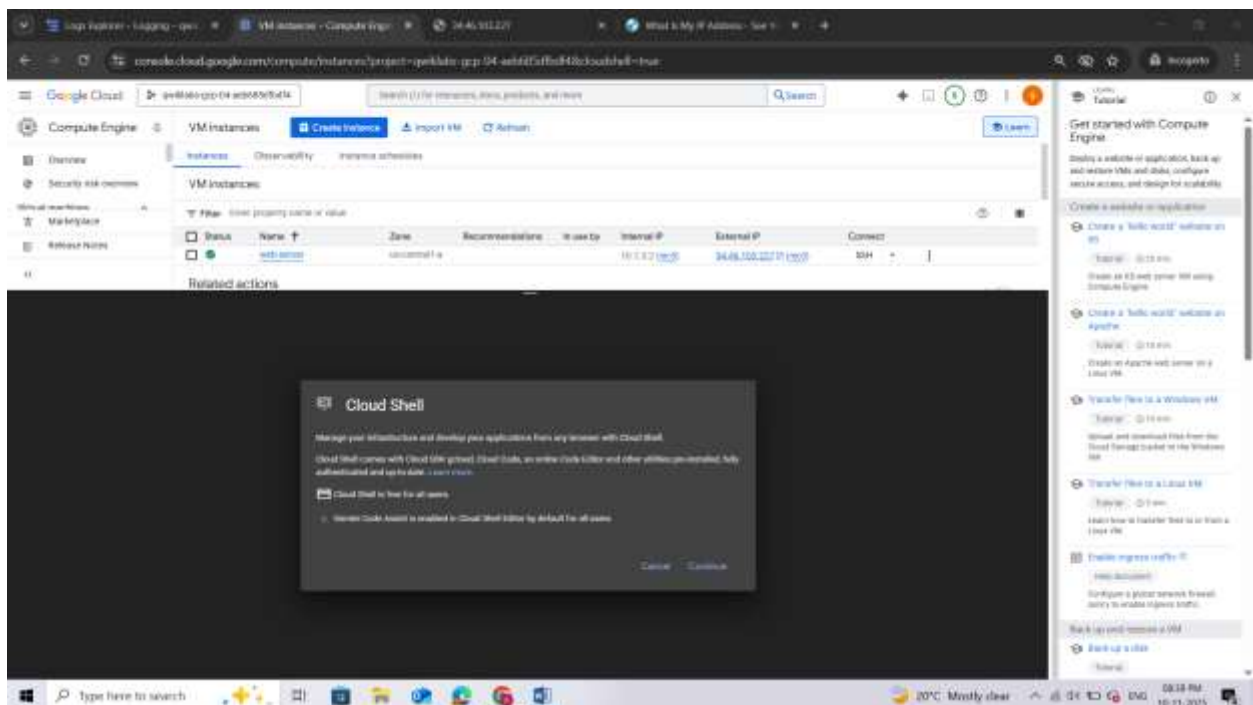
Now that the network traffic logs are being exported to BigQuery, you need to generate more traffic by accessing the web-server several times. Using Cloud Shell, you can curl the IP address of the web-server several times.

1. In the **Navigation menu**, select **Compute Engine > VM instances**.
2. Note the **External IP** address for the **web-server** instance. It will be referred to as *EXTERNAL_IP*.
34.186.59.203



3. Click **Activate Cloud Shell** (▶).

4. If prompted, click **Continue**.

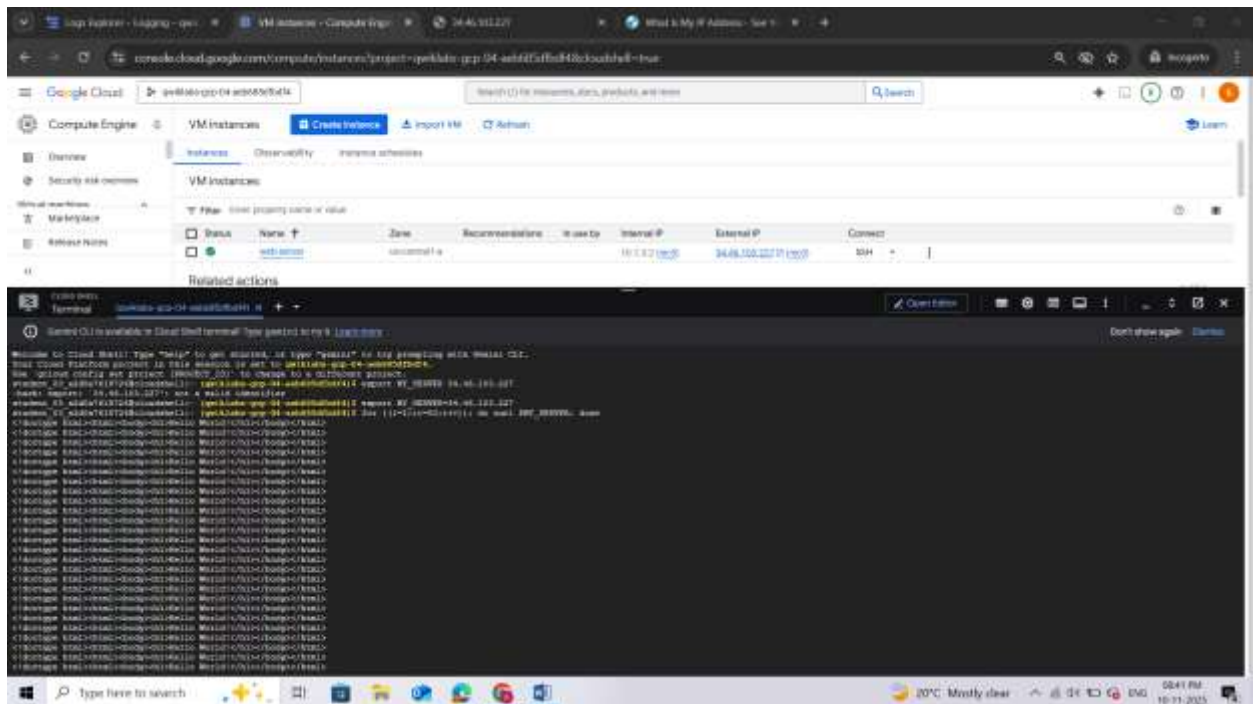


5. Store the *EXTERNAL_IP* in an environment variable in Cloud Shell:

export MY_SERVER=<Enter the EXTERNAL_IP here>

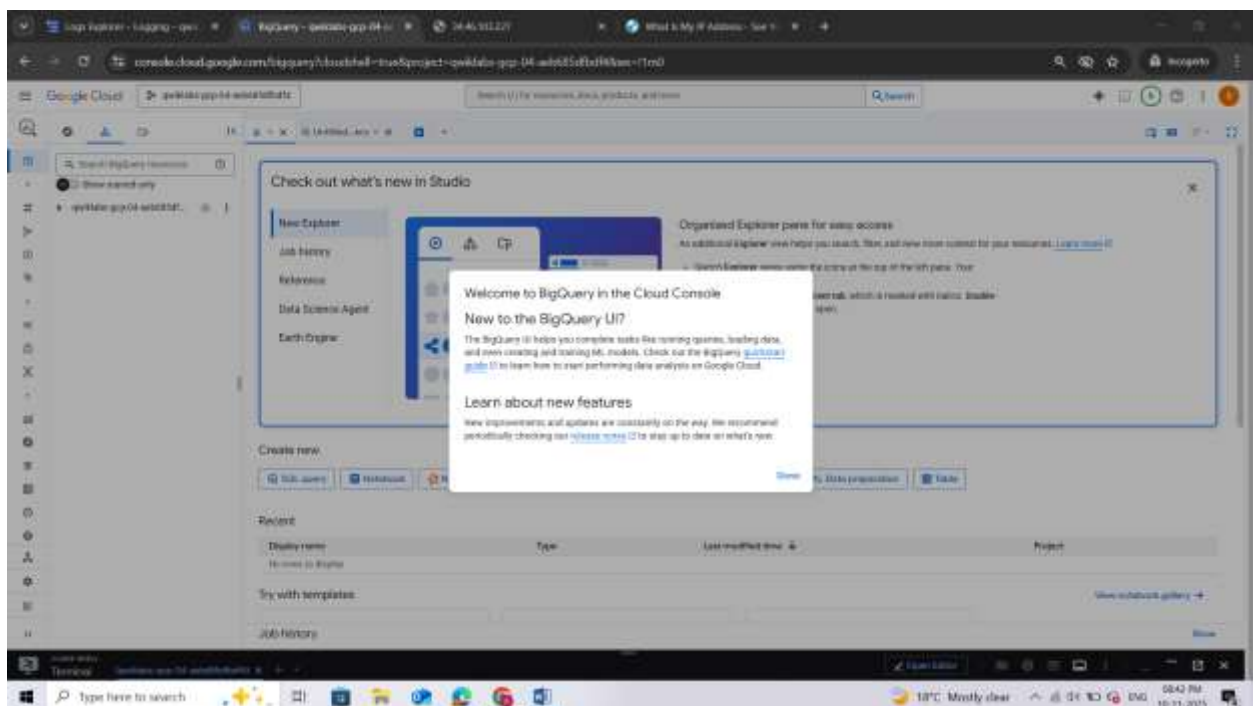
6. Access the web-server 50 times from Cloud Shell:

for ((i=1;i<=50;i++)); do curl \$MY_SERVER; done

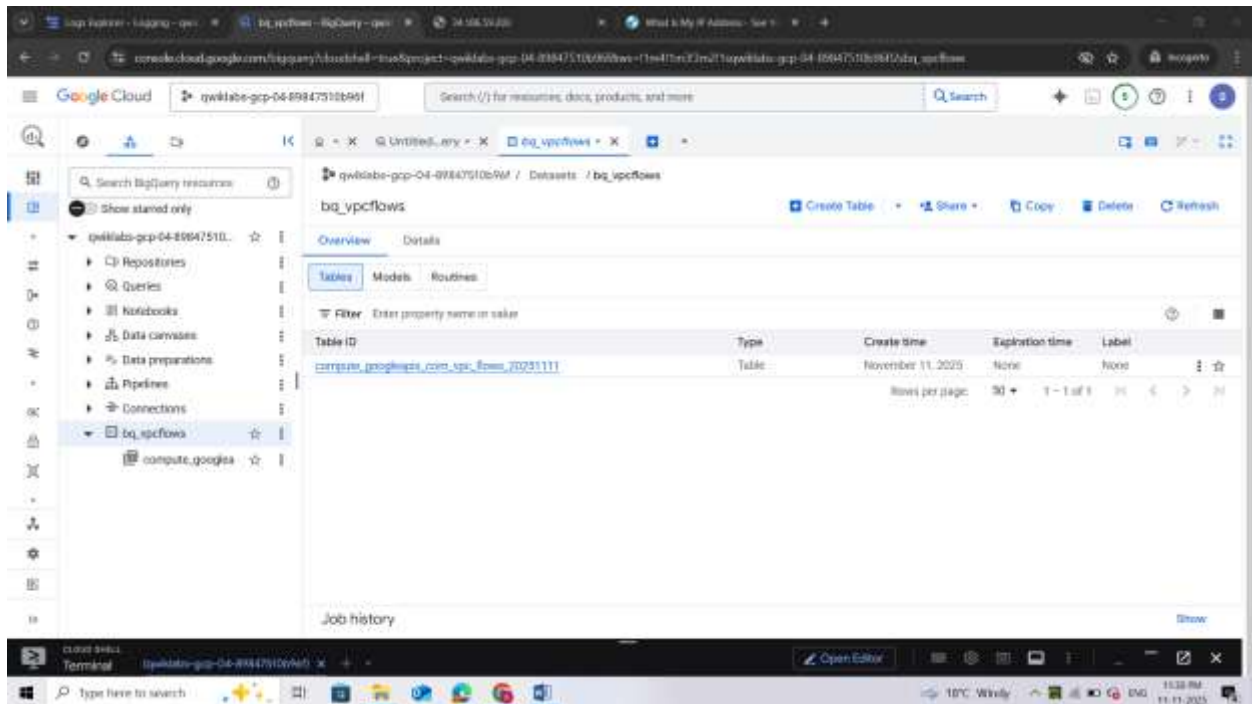


Visualize the VPC flow logs in BigQuery

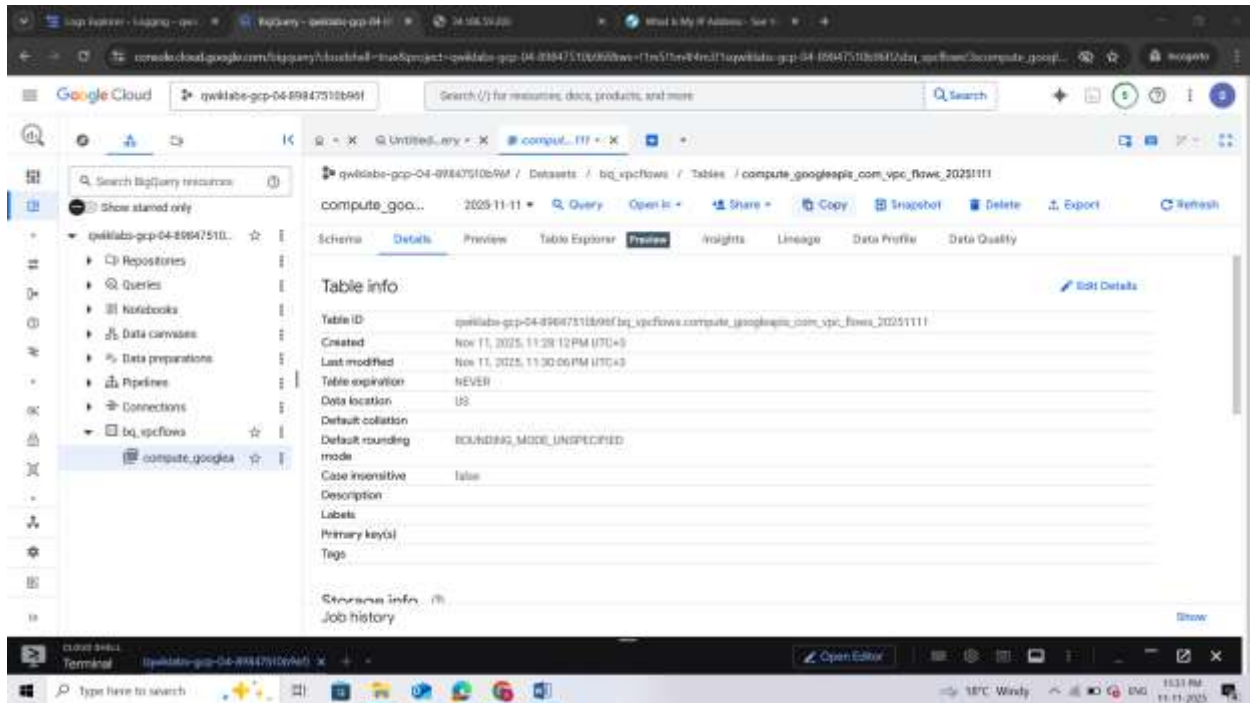
1. In the Cloud console, in the **Navigation menu**, click **BigQuery**.
2. If prompted, click **Done**.



3. In the left pane, expand the **bq_vpcflows** dataset to reveal the table. You might have to first expand the **Project ID** to reveal the dataset.



4. Click on the name of the table. It should start with **compute_googleapis**.
5. Click on the **Details** tab.
6. Copy the *Table ID* value under Table info. **qwklabs-gcp-04-89847510b96f.bq_vpcflows.compute_googleapis_com_vpc_flows_20251111**



Note: If you do not see the **bq_vpcflows** dataset or if it does not expand, wait and refresh the page.

7. Click the + icon to open a new BidQuery Editor tab.
8. Add the following to the BigQuery **Editor** and replace **your_table_id** with **TABLE_ID** while retaining the accents (') on both sides:

#standardSQL

SELECT

```
jsonPayload.src_vpc.vpc_name,
SUM(CAST(jsonPayload.bytes_sent AS INT64)) AS bytes,
jsonPayload.src_vpc.subnetwork_name,
jsonPayload.connection.src_ip,
jsonPayload.connection.src_port,
jsonPayload.connection.dest_ip,
jsonPayload.connection.dest_port,
jsonPayload.connection.protocol
```

FROM

`your\_table\_id`

GROUP BY

jsonPayload.src_vpc.vpc_name,

jsonPayload.src_vpc.subnetwork_name,

jsonPayload.connection.src_ip,

jsonPayload.connection.src_port,

jsonPayload.connection.dest_ip,

jsonPayload.connection.dest_port,

jsonPayload.connection.protocol

ORDER BY

bytes DESC

LIMIT

15

9. Click Run.

The screenshot shows the Google Cloud BigQuery console interface. The query editor at the top contains a SQL query with a SELECT statement listing various fields from the jsonPayload table, grouped by these fields and ordered by bytes in descending order, with a limit of 15. The query has been executed successfully, as indicated by the 'Query completed' status. Below the query editor, the 'Query results' section is displayed, showing a table with 15 columns: id, src_ip, bytes, src_vpc_name, src_vpc_subnetwork_name, src_ip, src_port, dest_ip, dest_port, and protocol. The table contains 15 rows of data. The interface also includes a sidebar with navigation options like Repositories, Queries, Notebooks, Data console, Data preparations, Pipelines, Connections, and bq_spcflows. The bottom of the screen shows a terminal window and a taskbar with various application icons.

id	src_ip	bytes	src_vpc_name	src_vpc_subnetwork_name	src_ip	src_port	dest_ip	dest_port	protocol
1	10.1.2.2	1672	vpc-subnet	vpc-subnet	10.1.2.2	34179.0	162.201.36.95		
2	10.1.2.2	608	vpc-subnet	vpc-subnet	10.1.2.2	58350.0	10.1.3.3		
3	10.1.2.2	608	vpc-subnet	vpc-subnet	10.1.2.2	58344.0	10.1.3.2		
4	10.1.2.2	608	vpc-subnet	vpc-subnet	10.1.2.2	58424.0	10.1.3.2		

Note: If you get an error, ensure that you did not remove the **#standardSQL** part of the query. If it still fails, ensure that the **TABLE_ID** did not include the Project ID.

Which columns does the results table contain?



Protocol



VPC name



Subnet name



Source IP address and port



Sum of bytes sent



Destination IP address and port

[Analyze the VPC flow logs in BigQuery](#)

The previous query gave you the same information that you saw in the Cloud console. Now, you will change the query to identify the top IP addresses that have exchanged traffic with your **web-server**.

1. Create a new query in the BigQuery **Editor** with the following and replace **your_table_id** with *TABLE_ID* while retaining the accents (') on both sides:

#standardSQL

SELECT

jsonPayload.connection.src_ip,

jsonPayload.connection.dest_ip,

SUM(CAST(jsonPayload.bytes_sent AS INT64)) AS bytes,

jsonPayload.connection.dest_port,

jsonPayload.connection.protocol

FROM

`your\_table\_id`

WHERE jsonPayload.reporter = 'DEST'

GROUP BY

jsonPayload.connection.src_ip,

jsonPayload.connection.dest_ip,

jsonPayload.connection.dest_port,

jsonPayload.connection.protocol

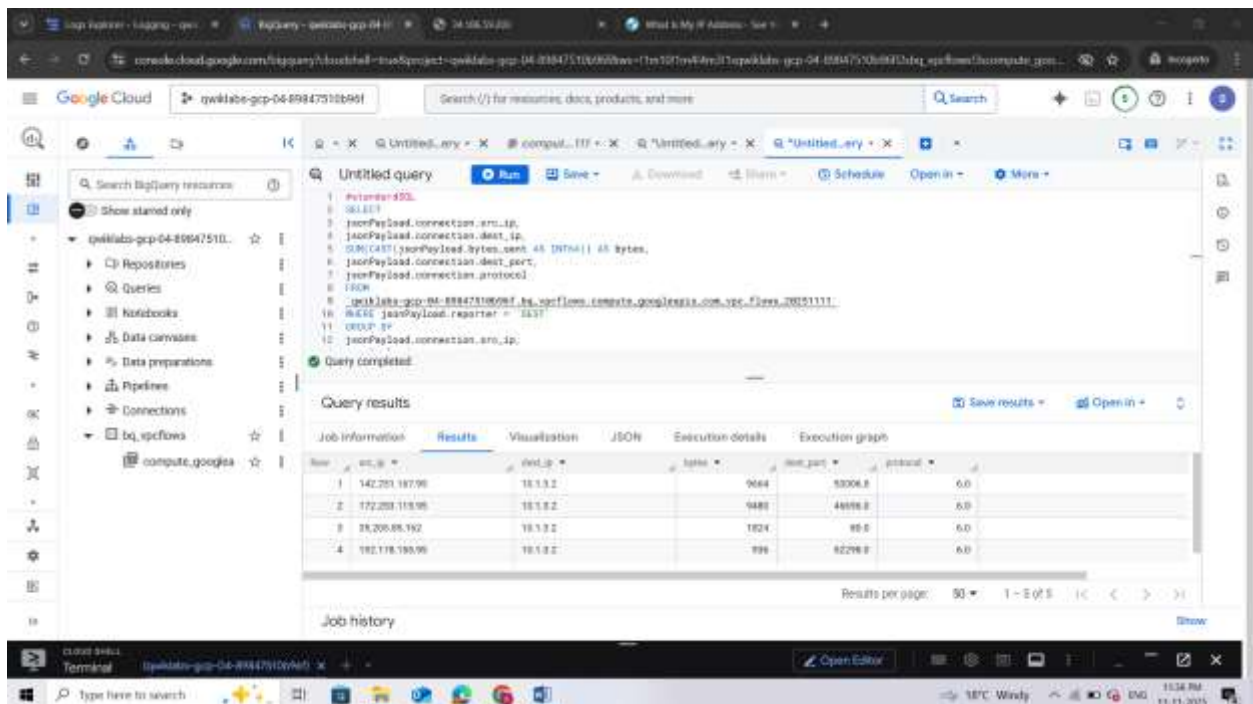
ORDER BY

bytes DESC

LIMIT

15

2. Click **Run**.



The screenshot shows the Google Cloud BigQuery console. A query has been executed, and the results are displayed in a table. The table has the following columns: row, src_ip, dest_ip, bytes, dest_port, and protocol. The results are sorted by bytes in descending order.

row	src_ip	dest_ip	bytes	dest_port	protocol
1	142.251.187.96	10.1.3.2	9684	53008.0	6.0
2	172.253.118.96	10.1.3.2	9480	48916.0	6.0
3	35.208.85.192	10.1.3.2	1824	80.0	6.0
4	192.178.190.96	10.1.3.2	896	82296.0	6.0

Note: The results table now has a row for each source IP and is sorted by the highest amount of bytes sent to the **web-server**. The top result should reflect your Cloud Shell IP address.

Note: Unless you accessed the **web-server** after creating the export sink, you will not see your machine's IP address in the table.

You can generate more traffic to the **web-server** from multiple sources and query the table again to determine the bytes sent to the server.

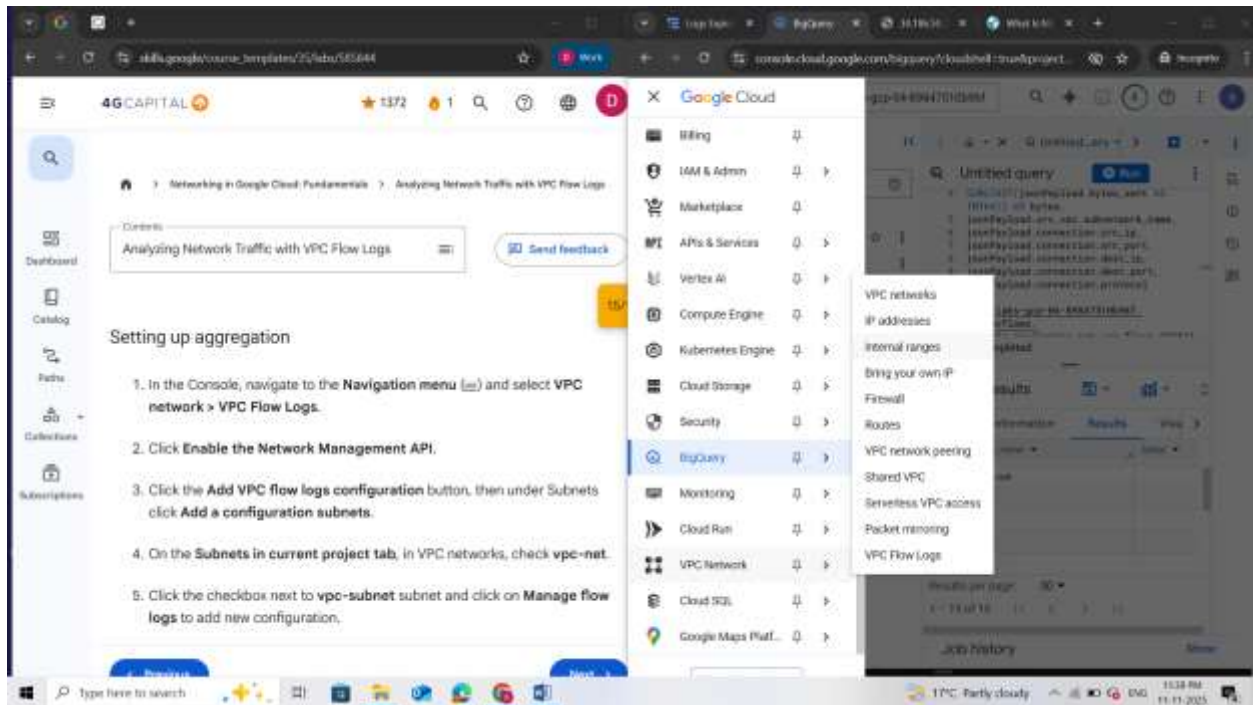
Task 5. Add VPC flow log aggregation

In this task, you will now explore a new release of [VPC flow log volume reduction](#). Not every packet is captured into its own log record. However, even with sampling, log record captures can be quite large.

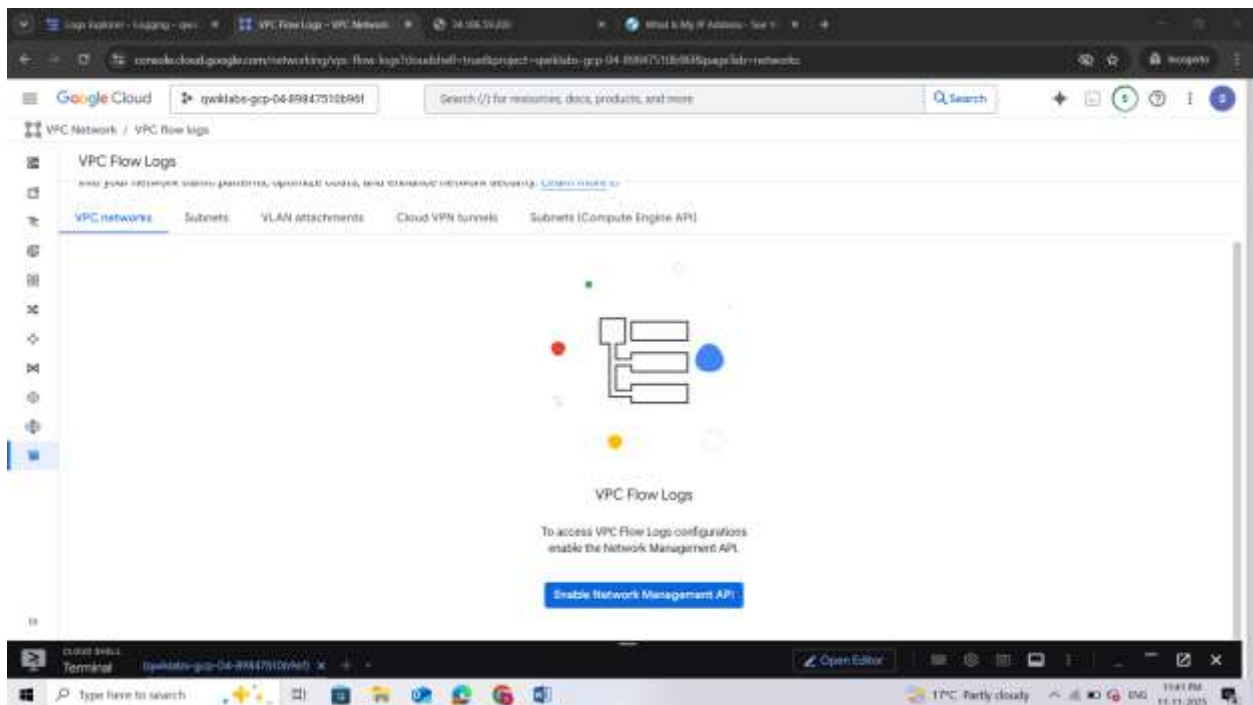
You can balance your traffic visibility and storage cost needs by adjusting specific aspects of logs collection, which you will explore in this section.

Setting up aggregation

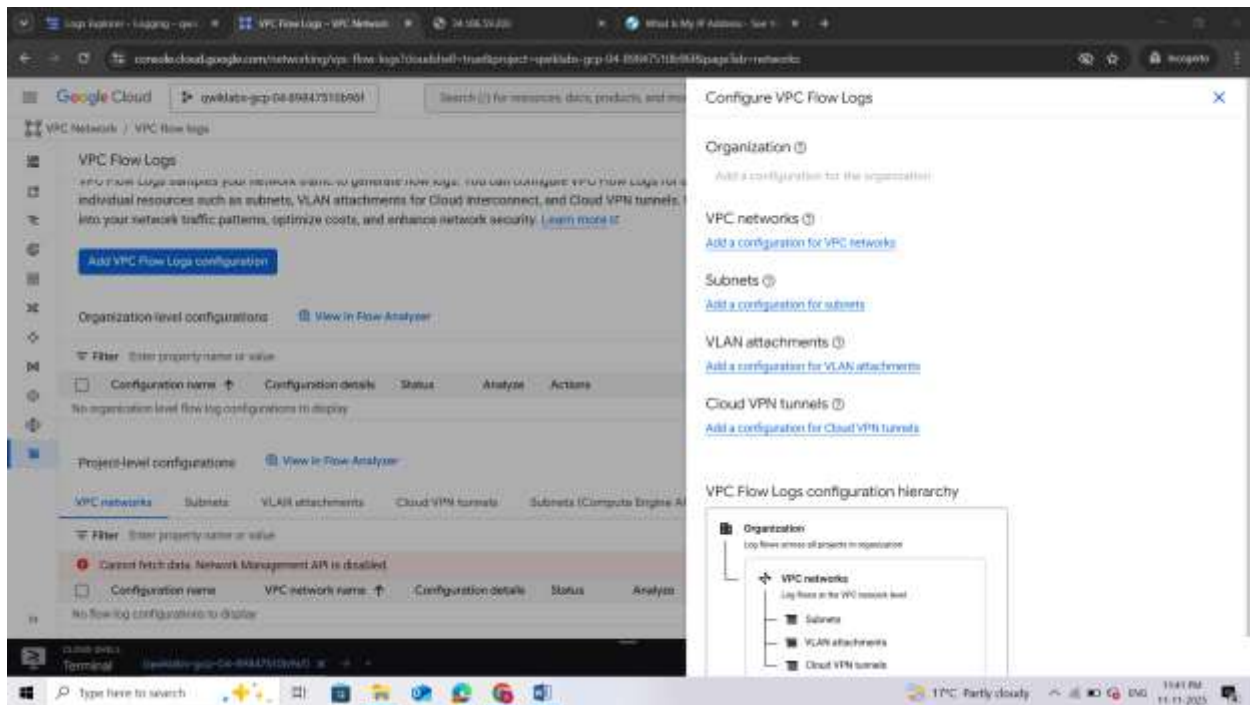
1. In the Console, navigate to the **Navigation menu** (☰) and select **VPC network > VPC Flow Logs**.



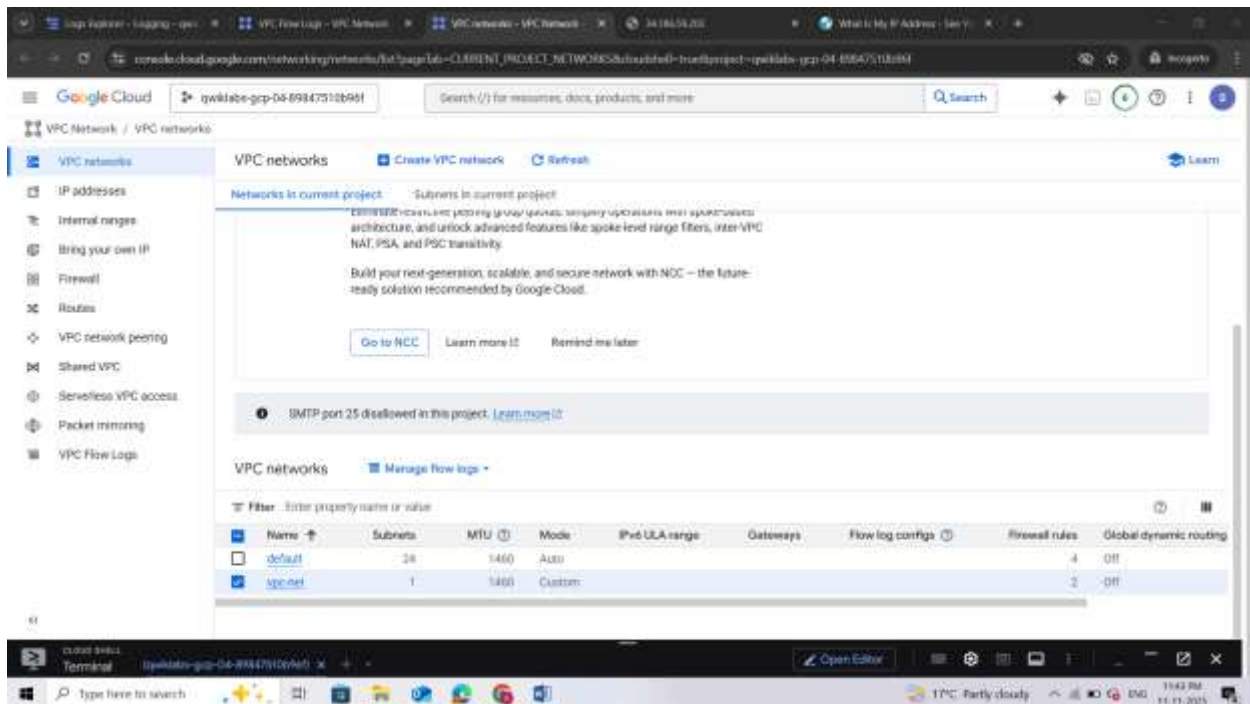
2. Click **Enable the Network Management API**.



3. Click the **Add VPC flow logs configuration** button, then under Subnets click **Add a configuration subnets**.



4. On the **Subnets in current project tab**, in VPC networks, check **vpc-net**.



- Click the checkbox next to **vpc-subnet** subnet and click on **Manage flow logs** to add new configuration.
- Under **Configurations - Subnets (Compute Engine API)**, click the dropdown **Compute Engine API** configuration.

7. Click **Advanced Settings**.

Configurations - Subnets (Compute Engine API)

i VPC Flow Logs is transitioning from the Compute Engine to Network Management API. Configurations in this list are created and managed by the Compute Engine API. For all other configurations, VPC Flow Logs uses the Network Management API. [Learn more](#)

^ Edit configuration

Aggregation interval

Information for sampled packets is aggregated over this interval to generate a flow log record. For more information, see [Logs collection](#)

✓ 5 seconds

30 seconds

1 minute

5 minutes

10 minutes

15 minutes

^ Advanced settings

☐ Keep only logs that match a filter

For details, see [Log filtering](#)

☒ Metadata annotations

Additional information that you can include in flow log records. For a list of available metadata annotations, see [Record format](#)

Secondary sampling rate *

50

%

?

Done

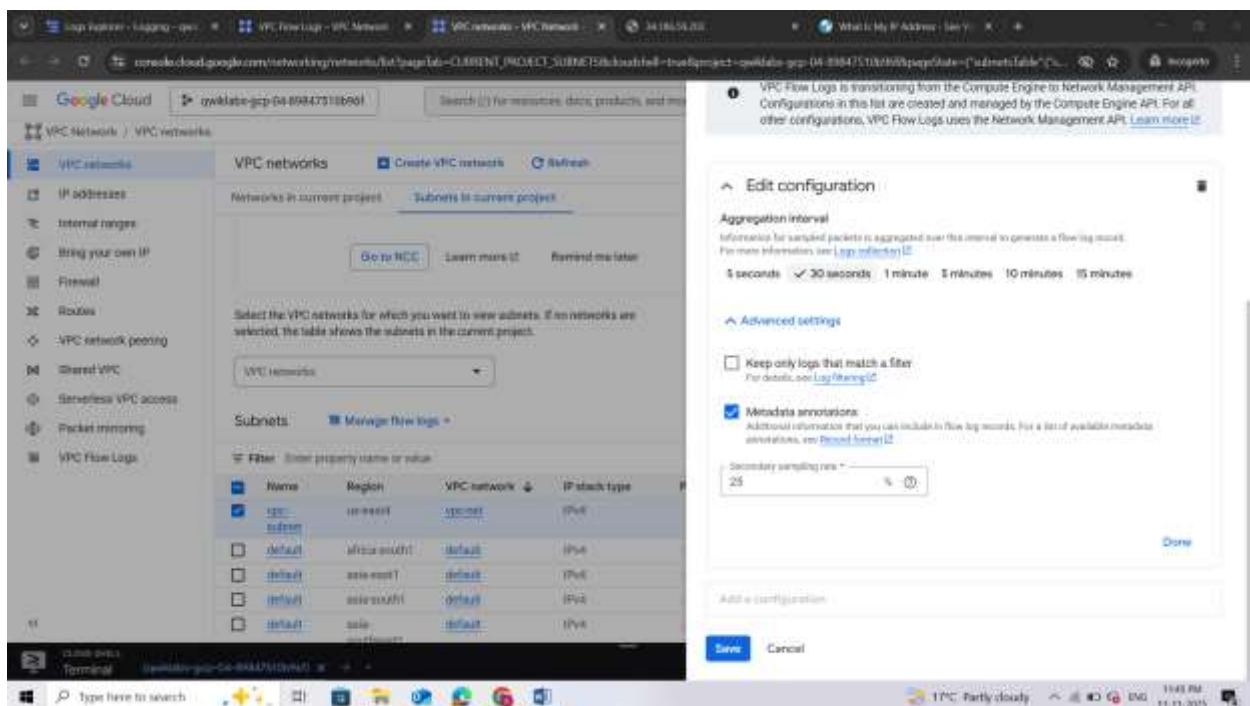
Add a configuration

You can't have more than one Compute Engine API configuration

The purpose of each field is explained below.

- **Aggregation time interval:** Sampled packets for a time interval are aggregated into a single log entry. This time interval can be 5 sec (default), 30 sec, 1 min, 5 min, 10 min, or 15 min.

- **Metadata annotations:** By default, flow log entries are annotated with metadata information, such as the names of the source and destination VMs or the geographic region of external sources and destinations. This metadata annotation can be turned off to save storage space.
 - **Log entry sampling:** Before being written to the database, the number of logs can be sampled to reduce their number. By default, the log entry volume is scaled by 0.50 (50%), which means that half of entries are kept. You can set this from 1.0 (100%, all log entries are kept) to 0.0 (0%, no logs are kept).
8. Set the **Aggregation Interval** to **30 seconds**.
 9. Set the **Secondary sampling rate** to **25%**.
 10. Click **Done** and then click **Save**.



Setting the aggregation level to 30 seconds can reduce your flow logs size by up to 83% compared to the default aggregation interval of 5 seconds. Configuring your flow log aggregation can seriously affect your traffic visibility and storage costs.

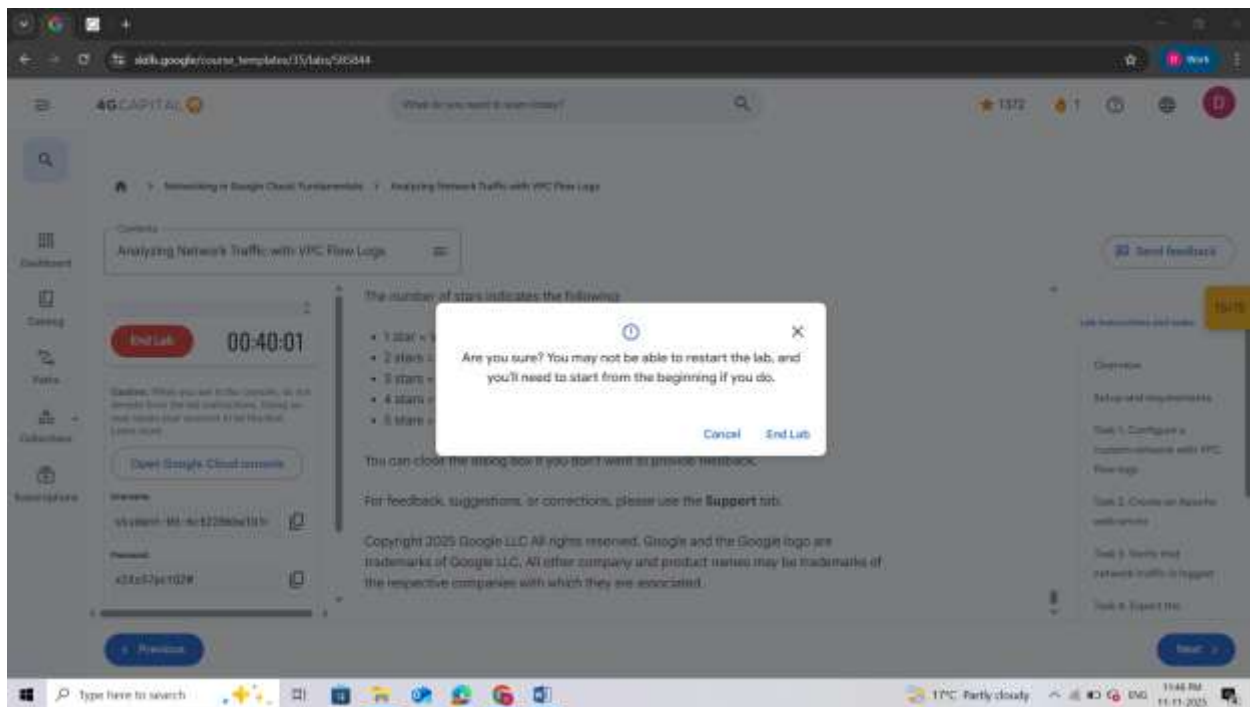
Review

You have configured a VPC network, enabled VPC Flow Logs, and created a web server in that network. Then, you generated HTTP traffic to the web server, viewed the traffic logs in the Cloud console, and analyzed the traffic logs in BigQuery. Finally, you used VPC Flow Log aggregation for balancing your traffic visibility and storage cost.

There are multiple use cases for VPC Flow Logs. For example, you might use VPC Flow Logs to determine where your applications are being accessed from in order to optimize network traffic expense, to create HTTP load balancers to balance traffic globally, or to deny unwanted IP addresses with Cloud Armor.

End your lab

When you have completed your lab, click **End Lab**. Google Cloud Skills Boost removes the resources you've used and cleans the account for you.



QUIZ

- ✓ 1. Which of the following two Google Cloud Monitoring features will notify you through email, SMS, or other channels when your web server cannot be reached?

☐ Ops Agent

✓ Alerting policies

Correct! Alerting policies can email you when your web server cannot be reached.

☐ Dashboards

✓ Uptime checks

Correct! Uptime checks test the availability of your web server and allow you to specify alerting policies.

- ✓ 2. In regards to VPC Flow Logs, which of the following statements is correct?
- ☐ There is a delay and performance penalty in routing logged IP packets.
 - ☐ Logs cannot be analyzed in BigQuery or visualized in Looker Studio.
 - ☐ Log updates are provided every 5 minutes.
 - ✓ ☒ Logs can be used for network monitoring, forensics, real-time security analysis, and expense optimization.

Correct! VPC Flow Logs can be used for network monitoring, forensics, real-time security analysis, and expense optimization.